



(12) **United States Patent**
Edwards et al.

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- (54) **METHODS OF DETECTING INHERITED MYOPATHIES IN HORSES**
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- (73) Assignee: **UNM RAINFOREST INNOVATIONS**, Albuquerque, NM (US)
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C12Q 1/68 (2018.01)
C12P 19/34 (2006.01)
C12Q 1/6883 (2018.01)
- (52) **U.S. Cl.**
CPC **C12Q 1/6883** (2013.01); **C12Q 2600/156** (2013.01)
- (58) **Field of Classification Search**
CPC **C12Q 1/6883**
See application file for complete search history.
- (56) **References Cited**

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(57) **ABSTRACT**

This disclosure describes detecting four genetically distinct kinds of inherited myopathy in horses, referred to as Polysaccharide Storage Myopathy type 2 (PSSM2), or in some cases as Myofibrillar Myopathy (MFM).

28 Claims, 36 Drawing Sheets

Specification includes a Sequence Listing.

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FIG. 1

31,307,949

TTCAGTCTA AATCATGATA ATTTAAGGAA AAGTGCCAGC TATAACCACT
ATGTTTACCC GAAAGACAAA ATTCTAATTT ATTGTTATTA TTATTATTCC
CAAGGCATGT TTTTATTGTC TTTTTCCTCC TTAGAAAATT AATATATATT
TATTGTAGAA AACTTGGAAA ATACAGCACA ATGAAGAGAA TAAAAATCAC
TCACCCACCT TCCGATTGTT ATTAACAAAA TAACATTCTT CTATTTTAGT
ATGCTGTFTT TTTTAAAAAG AGTTAATCCA TAAAGAAATA ACTGTTGATT
AGAAATTAGA ACCAGTAAAT ATCTTTTTTC CATGAAACTT GTTTTAATAT
TTTTAAATTT CATCTTTTAA AAAATATTTT GCCTACATTT AAGAAGCACA
GATATTTACG AAACCTTCATT TACTTAAAGT TCAATGGAGA AAAGTCTTAA
TTTATTCTC AGAATGTGAG AAATCCTTAC ATCTTATAAA GAATAAAGTG
GGGTCAATGC GCCACGTAAG TTFGACCTCT GAGGGAATTG TTAGTAAAGA
ATGTTCCGAGC CACTTCAAAA CGTCAGCCTC CAGGGTCCGA GGTGGGGATT
GTGCCAACGG CTGCATGAAG TGAACCGGGC TGCATCCTTG CCCCAGCTCT
GGGATCTAAC ACTGGTGGCT TATTTGGGGC AGCCTGGCAG CTCTTTTGCA
GAAGTGTGGG ATTTTACAGC TGCAAGGTAC CTTGATGGGA AAATATGACT
GTTCTTACAA TGCTCTCCTT AATTCTTTTT TAATTTCTCT CCAAGAAGC
CAGCAGGGAG GGTGGTCCCC GAGATTTCTG GCTTTCTTTG TGGCAGGTG
GGGGGAGGTG GGTGCTGAC TGFTTTCCCC TTCCTCCTTC TCATGCCCT
TCCTGGCTTT CAGATCAGGG TCCAGGTGAT CGAGGGGGCC CAGCTGCCAG
GGGTGAACAT CAAGCCTGTG GTCAAGGTCA CCGCGGCCAG GCAGACCAAG

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(C/T)
GGACTCGGAT TCACAGGGGA AACAGCCCGC TCTTCAACGA GGTGGGAGAC
ATGGCGTTT AGGGCTGGTA GCTTGGTGGG CCTTCCAGAT TGGGAGCACC
CGGCAGATAC CTGGCAATTC TTTCAGTTTT TGTTCATGGC GCTAACTTTG
GTTTGAGAGG TGTGCCAGGT CCTGAGTACG TTATCTGAGG AACTGGAGAG
AGGCTTCTAG TTCTTATTC TGTCCCGGGC TCCTGCTGCT CCACATCCCT
GTCTTCTCTT GGGGCCAGCC ACCCATGCTG TCCTGGAGAT GACAACCTCT
GAGAGGTGAG GGGTGGAAACA CCCCAGAACT TGTGAGTTC TCAGAGCTCA
GGGCCGGGCA GAGTCACTC TGTGCTTTCC GTGTGGACCA GACCTGAAGG
CTGTGGGTGT GGGCGACCTT CTTCACAGCC TGGGGGTCAA CAGGCTCTCG
TTTATCTTCC TTTCGCCCTG AACCAACAGA CTCTCTTCTT CAACGTGTTT
GACTCTCCCT CGGAGCTGTT TGACGAGGCC GTCTTTATCA CGGTACGTCT
CAGGGATCAA GGCCTGCTCT GTGGGCCGTG TGTACACACA TGCATTCAGT
GTGCATGTGT GTGTATGCAC GTAGGGGTGT GAGTGTGAGA GTGTGTAGGA
GAAGCCTTAG GGGCCGGGGC CTGGGTGATG TGGGGAGCTC GCTGCTAAGC
TCTGCTGGTC ACAAAGTGC CTTCAGCAGC TCAGATGAGG CAGCAGCCCA
GTGGGGAGAC CCCCCTGCTC TCAGCACCCC CAGGTGCCCT AGGCCAGGTC
CTGATATCAT TCCTGCAGGG GATACTTTGA TTTTCTCTT TTTTCTTTG
TCCACTTGGC TGCTTCTGGC CAAGATTGCC TCTCTCTGAG CCTCAGCTCT
TTGTTTCTT TCCCTCTGAA AGGCAGTATA GTTTGGTGAA AAGCSTGGGC
TTTGGAGCCA AACTGTCTGG ATTCTAGTCC TGCCTGTGCC ATGTAGGGGC

31,305,949

(SEQ ID NO:1)

FIG. 2

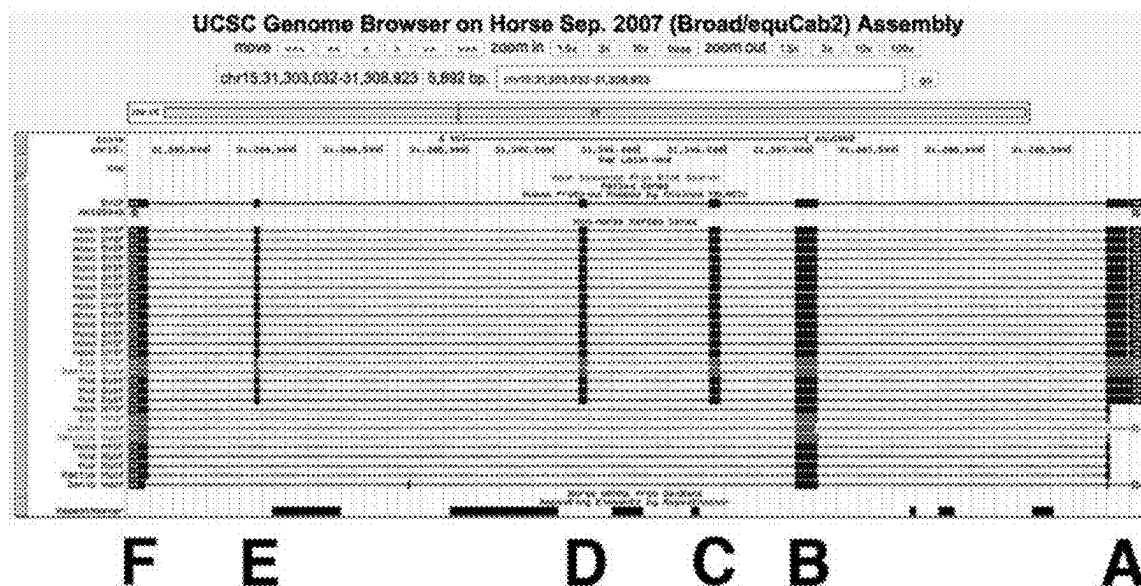
764 ATACAGGAGG AGAGGAGGAC ACCGAGGACC AGGGGCTCAC GGGAGATGAG
GCAGAGCCAT TCTTGGATCA GAACGGAGCC CCAGGCCCCG GGGCTCCCAC
CACCCCTGAAG AAGCCACCTT CCCATCCTCC CCCCTACCAT CCTGGGGGGA
AAAGGAAGAG AAGCAAGCCT GCGCCAGAA AGCTGCTTTC GGATAAACCA
CAGGACTTCC **AGATCAGGGT** **CCAGGTGATC** **GAGGGGCGCC** **AGCTGCCAGG**
GGTGAACATC **AAGCCTGTGG** **TCAAGGTCAC** **CGCGGCCAGG** **CAGACCAAGC**
GGACTCGGAT **TCACAGGGGA** **AACAGCCCCG** **TCTTCAACGA** GACTCTCTTC
TTCAACGTGT TTGACTCTCC CTCGGAGCTG TTTGACGAGG CCGTCTTTAT
CACGGTGGTA GACTCCTGTT CGCTCCGGAC AGATGCCCTC ATCGGGGAGT
TCCGGATGGA TGTGGGTACC ATCTACAGAG AGCCCCGACA CGCCTATCTC
AGGAAGTGGC TGCTGCTCTC GGACCCCTGAC GATTTCTCTG CTGGGCCCAA
AGGCTACCTG AAAGCAAGCC TTTGTGTGCT GGGGCCCTGGA GACGAAGCTC
CGCTGGAGAG AAAGGACCCC TCTGAAGACA AGGAGGACAT TGAAAGCAAT
CTGCTCAGGC CAACTGGCAT GGCCCTTCGA GGAGCGCACT TCTGTCTGAA
GGTCTTCAGG GCTGAGGACT TACCACAGAT GGACGATGCC GTGGTGGACA
GCGTGAAGCA GATCTTCGGC TTTGACAGTA ACAAGAAGAA CCTGGTGGAT
CCCTTCGTG AGGTCAGCTT TGCGGGGAAA ATGCTCTGCA GCAAGATCCT
GGAGAAGATG GCCAACCCCTC AGTGGAAACCA GAGCATCACG CTGCCTGTCA
TGTTCCTCCTC CATGTCTGAA AAAATGAGGA TTCGTGTCTAT AGACTGGGAC
CGCCTCACCC ACAATGACAT CGTGGCCACC ACCTACCTGA ATATGTGCAA
AATCTCTGCC CCTGGAGGAG AAATAGCAG 1792

(SEQ ID NO:2)

764 ATACAGGAGG AGAGGAGGAC ACCGAGGACC AGGGGCTCAC GGGAGATGAG
GCAGAGCCAT TCTTGGATCA GAACGGAGCC CCAGGCCCCG GGGCTCCCAC
CACCCCTGAAG AAGCCACCTT CCCATCCTCC CCCCTACCAT CCTGGGGGGA
AAAGGAAGAG AAGCAAGCCT GCGCCAGAA AGCTGCTTTC GGATAAACCA
CAGGACTTCC **AGATCAGGGT** **CCAGGTGATC** **GAGGGGCGCC** **AGCTGCCAGG**
GGTGAACATC **AAGCCTGTGG** **TCAAGGTCAC** **CGCGGCCAGG** **CAGACCAAGC**
GGACTCGGAT **TCACAGGGGA** **AACAGCCCCG** **TCTTCAACGA** GACTCTCTTC
TTCAACGTGT TTGACTCTCC CTCGGAGCTG TTTGACGAGG CCGTCTTTAT
CACGGTGGTA GACTCCTGTT CGCTCCGGAC AGATGCCCTC ATCGGGGAGT
TCCGGATGGA TGTGGGTACC ATCTACAGAG AGCCCCGACA CGCCTATCTC
AGGAAGTGGC TGCTGCTCTC GGACCCCTGAC GATTTCTCTG CTGGGCCCAA
AGGCTACCTG AAAGCAAGCC TTTGTGTGCT GGGGCCCTGGA GACGAAGCTC
CGCTGGAGAG AAAGGACCCC TCTGAAGACA AGGAGGACAT TGAAAGCAAT
CTGCTCAGGC CAACTGGCAT GGCCCTTCGA GGAGCGCACT TCTGTCTGAA
GGTCTTCAGG GCTGAGGACT TACCACAGAT GGACGATGCC GTGGTGGACA
GCGTGAAGCA GATCTTCGGC TTTGACAGTA ACAAGAAGAA CCTGGTGGAT
CCCTTCGTG AGGTCAGCTT TGCGGGGAAA ATGCTCTGCA GCAAGATCCT
GGAGAAGATG GCCAACCCCTC AGTGGAAACCA GAGCATCACG CTGCCTGTCA
TGTTCCTCCTC CATGTCTGAA AAAATGAGGA TTCGTGTCTAT AGACTGGGAC
CGCCTCACCC ACAATGACAT CGTGGCCACC ACCTACCTGA ATATGTGCAA
AATCTCTGCC CCTGGAGGAG AAATAGCAG 1792

(SEQ ID NO:3)

FIG. 3



Exon Sequences

A

31,308,932 ctctctccag ATACAGGAGS AGAGGAGSAC ACCGAGGACC AGGGGCTCAC
 GGGAGATGAG GCAGAGCCAT TCTTGGATCA GAACGGAGCC CCAGGCCCCG
 GGGCTCCAC CACCTGAAG AAGCCACCT CCCATCCTCC CCCTACCAT
 CCTGGGGGGA AAAGGAAGAG AAGCACGCT CCGCCAGAA AGCTGCTTC
 GGATAAACCA CAGGACTTCC AGgtgatgca cc 31,308,701
 (SEQ ID NO:4)

B

31,307,046 tggctttcag ATCAGGGTCC AGGTGATCGA GGGGCCCCAG CTGCCAGGGG
 TGAACATCAA GCTGTGGTC AAGGTACCCG CGGCCAGCA GACCAAGCGG
 ACTCGGATTC ACAGGGGAAA CAGCCGCTC TTCAACGAGg tgggagaca 31,306,989
 (SEQ ID NO:5)

C

31,306,479 gaaccaacag ACTCTCTTCT TCAACGTGTT TGACTCTCCC TCGGAGCTGT
 TTGACGAGGC CGTCTTATC ACGgtacgtc tca 31,306,397
 (SEQ ID NO:6)

D

31,305,709 ttccctccag GTGGTAGACT CTTGTTGCT CCGACAGAT GCCTCATCG
 GGGAGTTCCG Ggtaattagt t 31,305,639
 (SEQ ID NO:7)

E

31,303,803 ctgattgcag ATGGATGTGG GTACCATUTA CAGAGAGCCC Cgtgagtcatt
 att 31,303,753
 (SEQ ID NO:8)

F

31,303,157 tctctcttag GACACGCCTA TCTCAGGAAG TGGCTGCTGC TCTCGGACCC
 TGACGATTC TCTGCTGGC CCAAAGCTA CCTGAAAGCA AGCCTTTGTG
 TGCTGGGGCC TGGAGACGAA GCTCCGgtga gtcatt 31,303,022
 (SEQ ID NO:9)

FIG. 4

154 TGGEEDTEDQ GLTGDEAEFF LDQNGAPGPG APTTLKKPPS HPPPYHPGGK
 RKRSTFAPRK LLSDKPQDFQ **IRVQVIEGRQ** **LPGVNIKEVV** **KVTAARQTKR**
 TRIHRGNSPL FNETLFFNVF DSPSELEDEA VFITVVDSCS LRTDALIGEF
 RMDVGTIYRE PRHAYLRKWL LLSDPDDFSA GPKGYLKASL CVLGPGDEAP
 LERKDPSEDK EDIESNLLRP TGMALRGAHF CLKVFRAEDL PQMDDAVVDS
 VKQIFGFDSN KKNLVDPFVE VSFAGKMLCS KILEKMANPQ WNQSITLPEM
 FPSMSEKMRI RVIDWDRLTH NDIVATTYLN MSKISAPGGE IA 495

(SEQ ID NO:10)

154 TGGEEDTEDQ GLTGDEAEFF LDQNGAPGPG APTTLKKPPS HPPPYHPGGK
 RKRSTFAPRK LLSDKPQDFQ **IRVQVIEGRQ** **LPGVNIKEVV** **KVTAARQTKW**
 TRIHRGNSPL FNETLFFNVF DSPSELEDEA VFITVVDSCS LRTDALIGEF
 RMDVGTIYRE PRHAYLRKWL LLSDPDDFSA GPKGYLKASL CVLGPGDEAP
 LERKDPSEDK EDIESNLLRP TGMALRGAHF CLKVFRAEDL PQMDDAVVDS
 VKQIFGFDSN KKNLVDPFVE VSFAGKMLCS KILEKMANPQ WNQSITLPEM
 FPSMSEKMRI RVIDWDRLTH NDIVATTYLN MSKISAPGGE IA 495

(SEQ ID NO:11)

FIG. 5

5'-cccgag atttcttggt ttct-3' (SEQ ID NO:14)
 31,307,136 GGTCCCGAG ATTCTTGCT TTCTTTGTGG CGAGGTGGG GGAGGTGGT
 GGCTGACTGT TTCCCGCTTC CTCTGCTCA TGCCCGTTCC TGGCTTTCAG
 31,307,036 ATC AGG GTC CAG GTG ATC GAG GGG CGC CAG CTG CCA GGG
 31,307,036 ATC AGG GTC CAG GTG ATC GAG GGG CGC CAG CTG CCA GGG
 GTG AAC ATC AAG CTT GTG GTC AAG GTC ACC GCG GCC AGG
 GTG AAC ATC AAG CTT GTG GTC AAG GTC ACC GCG GCC AGG
 CAG ACC AAG CGG ACT CGG ATT CAC AGG GGA AAC AGC CCG
 CAG ACC AAG TGG ACT CGG ATT CAC AGG GGA AAC AGC CCG
 CTC TTC AAC GAG (SEQ ID NO:12) 31,306,908
 CTC TTC AAC GAG (SEQ ID NO:13) 31,306,908
 GTGGGAGACA TGGCTTTTA GGGCTGGTAG CTTGGTGGC CTTCCAGATT
 GGGAGCAGCC GGCAGATACC TGGCAATTCT TTCAGTTTTC GTTCATGGCG
 CTAACCTTGG TTGAGAGGT GTGCCAGTTC CTGAGTACGT TATCTGAGGA
 ACTGGAGAGA GGGTCTAGT TCTTATTCCT GTCCCGGCT CCTGGTGCTC
 CACATCCCTG TCTTCCTGTG GGCCAGCCA CCCATGCTGT CCTGGAGATG
 ACAACCTCTG AGAGGTGAG GGTGGAACAC CCCAGACTT GTCGACTCCT 31,306,608
 (SEQ ID NO:15) 3'-tgtg gggctcttgsa cagctc-5'

FIG. 6

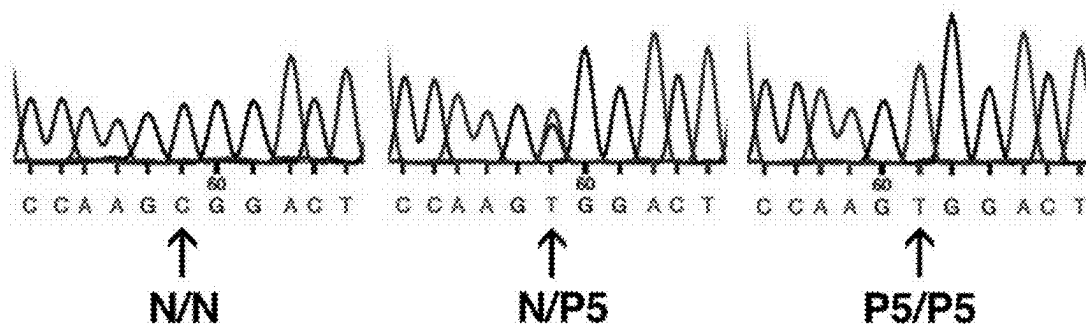


FIG. 7

377 ATGCTGAGGG TCTTCATCCT CTATGCCGAG AACCTCCACA CACCCGACAC
CGACATCAGC GATGCCCTACT GCTCCGCGGT GTTTGCGAGG GTGAAGAAGA
GAACCAAGT CATCAAGAAC AGCGTGAACC CTGTATGGAA TGAGGGATTT
GAATGGGACC TCAAGGGCAT CCCCCTGGAC CAGGGCTCTG AGCTTCATGT
GGTGGTCAAA GACCATGAGA CGATGGGGAG GAACAGGTTT CTGGGGGAAG
CCAAGGTCCC ACTCCGAGAG STCCTCGCCA CCCCTASTCT GTCCGCCAGC
TTCAATGCCC CCCTGCTGGA CACCAAGAAG CAGCCACAG GGGCCTCGCT
GGTCTGCGAG GTGTCTTACA CACCGCTGCC TGGAGCTGTG CCCCTTTTCC
CGCCCCCTAC TCCTCTGGAG CCCTCCCGCA CTCTGCCCTGA CCTGGATGTA
GTGGCAGACA CAGGAGGAGA GGAAGACACA GAGGACCAGG GACTCACTGG
AGATGAGGCG GAGCCATTCC TGGATCAAAG CGGAGGCCCG GGGGCTCCCA
CCACCCCAAG GAACTACCT TCACGTCTTC CGCCCCACTA CCCCAGGATC
AAAAGAAAGC GAAGTGGGCC TACATCTAGA AAGCTGCTGT CAGACAAACC
GCAGGATTTT CAGATCAGGG TCCAGGTGAT CGAGGGGCGC CAGCTGCCGG
GGGTGAACAT CAGCCTGTG GTCAAGGTTA CCGCTGCRGG GCAGACCAAG
CGGACGCGGA TCCACAAGGG AAACAGCCCA CTCTTCAATG AGACTCTTTT
CTTCAACTTG TTTGACTCTC CTGGGGAGCT GTTTGATGAG CCCATCTTTA
TCACGGTGGT AGACTCTCGT TCTCTCAGGA CAGATGCTCT CCTCGGGAG
TTCCGGATGG ACGTGGGCAC CATTACAGA GAGCCCGGGC ACGCCTATCT
CAGGAAGTGG CTGCTGCTCT CAGACCTGA TGACTTCTCT GTGGGGCCA
GAGGCTACCT GAAACAAGC CTTTGTGTGC TGGGGGCTGG GGACGAAGCG
CCTCTGGAGA GAAAGACCC CTCTGAAGAC AAGGAGGACA TTGAAAGCAA
CCTGCTCCGG CCCACAGGCG TAGCCCTGCG AGGAGCCAC TTCTGCCTGA
AGGTCTTCCG GGCCGAGGAC TTGCCGAGA TGGACGATGC CGTGATGGAC
AACGTGAAAC AGATCTTTGG CTTCGAGAT AACAAGAAGA ACTTGGTGGG
CCCTTTTGTG GAGGTCAGCT TTGCGGGGAA AATGCTGTGC AGCAAGATCT
TGGAGAAGAC GCGCAACCT CAGTGAACC ACAACATCAC ACTGCTGCC
APGTTTCCCT CATGTGCGA AAAATGAGG ATTOGTATCA TAGACTGGG
CCGCTGACT CACAATGACA TCGTGGCTAC CACCTACCTG AGTATGTGCA
AAATCTCTGC CCTGGAGGA GAAATAGAAG AGGAGCCTGC AGGTGCTGTC
AAGCCTTCGA AAGCCTCAGA CTTGGATGAC TACCTGGGCT TCCTCCCCAC
TTTTGGGCCC TGCTACATCA ACCTCTATGG CAGTCCAGA GAGTTACAG
GCTTCCAGA CCGCTACACA GAGCTCAACA CAGGCAAGGG GGAAGGTGTG
GCTTATCGTG GCGGCTTCT GCTCTCCCTG GAGACCAAGC TGGTGGASCA
CAGTGAACAG AAGGTGGAGG ACCTTCTTC GAGTGACATC CTCGGGGTGG
AGAAGTACCT TAGGAGGCGC AAGTACTCCC TGTTTGCSSC CTTCTACTCA
GCCACCATGC TGCAGGATGT GGATGATGCC ATCCAGTTTG AGGTGAGCAT
CGGSACTAC GGGAACTAGT TCGACATGAC CTGCTGCCG CTGGCCTCCA
CCACTCAGTA CAGCCGTGCA GTCTTTGACG GGTGCCACTA CTACTACCTA
CCCTGGGGTA ACGTGAACC TGTGGTGGTG CTGTCTCTCT ACTGGGAGGA
CATCAGCCAT AGAATCGAGA CTCAGAACCA GCTGCTTGGG ATTGCTGACC
GGCTGGAAGC TGGCTTGGAG CAGGTCCACC TGGCCCTGAA GCGCAGTGC
TCCACGGAGG ACGTGGACTC GCTGGTGGCT CAGCTGACCG ATGAGCTCAT
CGCAGGCTGC AGCCAGCCTC TGGGTGACAT CCATGAGACA CCCTCTGCCA
CCACCTGGA CCACTACCTG TACCAGCTGC GCACCCATCA CCGAGCCAA
ATCACTGAGG CTGCCCTGGC CCTGAAGCTC GGCCACAGTG AGCTCCCTGC
AGCTCTGGAG CAGGCGGAGG ACTGGCTCCT GCGTCTGCTT GCGCTGGCAG
AGGAGCCCCA GAACAGCCTG CCGGACATCG TCATCTGGAT GGTGCAGGGA
GACAAGCGTG TGGCATACCA GCGGGTGCCG GCGCACCAAG TCCTCTTCTC
CCGGCGGGGT GCCAATACT GTGGCAAGAA TTGTGGGAAG CTACAGACAA
TCTTTCTGAA ATATCCGATG GAGAAGGTGC CTGGCGCCCG GATGCCAGTG
CAGATACGGG TCAAGCTGTG GTTTGGGCTC TCAGTGGATG AGAAGGAGTT
CAACCAAGTTT GCTGAGGGGA AGCTGTCTGT CTTTGTGTA ACCTATGAGA
ACGAGACTAA GTTGGCCCTT GTTGGGAACT GGGGCACAA CCGCCTCACC
TACCCCAAGT TTTCTGACCT CACGGGCAAG ATCAAGCTAC CCAAGGACAG
CTTCCGCCCC TCGGCCGGCT GGACCTGGGC TGGAGATTGG TTCGTGTGTC
CGGAGAAGAC TCTGCTCCAT GACATGGACG CCGGTCACTT GAGCTTCTGT
GAAGAGGTGT TTGAGAACCA GACCCGGCTT CCGGAGGCC AGTGGATCTA
CATGAGTGAC AACTACACCG ATGTGAACCG GGAGAAGGTG CTTCCCAAGG
APGACATTGA GTGCCACTG GGCTGGAAGT GGAAGATGA GGAATGCTCC
ACAGACCTCA ACCGGGCTGT CGATGAGCAA GGCTGGGAGT ATAGCATCAC

Fig. 7 (continued)

CATCCCCCG	GAGCGGAAGC	CGAAGCACTG	GGTCCCTGCT	GAGAAGATGT
ACTACACACA	CCGACGGCGG	CGCTGGGTGC	GCCTGCGCAG	GAGGGATCTC
AGCCAAATGG	AAGCACTGAA	AAGGCACAGG	CAGGCGGAGG	CGGAGGSCGA
GGGCTGGGAG	TACGCTCTC	TTTTTGGCTG	GAAGTTCAC	CTCGAGTACC
GCAAGACAGA	TGCCTTCCGC	CGCCGCCGCT	GGCGCCGTGC	CATGGAGCCA
CTGGAGAAGA	CGGGGCCTGC	AGCTGTGTTT	GCCCTTGAGG	GGGCCCTGGG
CGGCTGATG	GATGACAAGA	GTGAAGATTG	CATGTCCGTC	TCCACCTTGA
GCTTCGGTGT	GAACAGACCC	ACGATTTCTT	GCATATTGGA	CTATGGGAAC
CGCTACCATC	TACGCTGCTA	CATGTACCAG	GCCCAGGACC	TGGCTGCGAT
GGACAAGGAC	TCTTTTCTG	ATCCCTATGC	CATCGTCTCC	TTCTTGACCC
AGAGCCAGAA	GACGGTGGTG	GTGAAGAACA	CCCTTAACCC	CACCTGGGAC
CAGACGCTCA	TCTTCTACGA	GATCGAGATC	TTTGGCGAGC	CGGCCACAGT
TGCTGAGCAA	CCGCCACGCA	TTGTGGTGGG	GCTGTACGAC	CATGACACTT
ATGGTGACGA	CGAGTTTATG	GGTGGCTGCA	TCTGTCAACC	GAGTCTGGAA
CGGATGCCAC	GGCTGGGCTG	GTTCCCACTG	ACGAGGGGCA	GCCAGCGCTC
GGGGGAGCTG	CTGGCCTCTT	TTGAGCTCAT	CCAGAGAGAG	AAGCCGGCCA
TCCACCATAT	TCTTGGTTTT	GAGGTGCAGG	AGACATCAAG	GATCCTGGAT
GAGTCTGAGG	ACACAGACCT	GCCCTACCCA	CCACCCAGAG	GGGAGGCCAA
CATCTACATG	GTTCCTCAGA	ACATCAAGCC	AGCGCTCCAG	CGTACCGCCA
TCGAGATCCT	GGCATGGGGC	CTGCGGAACA	TGAAGAGTTA	CCAGCTGGCC
AACATCTCCT	CCCCCAGCCT	CGTGGTAGAG	TGTGGGGGCC	AGACGGTGCA
GTCTGTGTG	ATCAGGAACC	TCCGGAAGAA	CCCCAACTTT	GACATCTGCA
CCCTCTTCAT	GGAAGTGATG	CTGCCCAGGG	AGGAGCTCTA	CTGCCCCCCC
ATCACCGTCA	AGGTCATCGA	TAACCGCCAG	TTTGGCCGCC	GGCCTGTGCT
GGGCCAGTGT	ACCATCCGCT	CCCTGGAGAG	CTTCTGTGT	GACCCCTACT
CGGCGGAGAG	TCCATCCCCA	CAGGGTGGCC	CAGACGATGT	GAGCCTACTC
AGTCTGGGG	AAGACGTGCT	CATCGACATT	GATGACAAGG	AGCCCTCAT
CCCCATCCAG	GAGGAAGAGT	TCATCGATTG	TGTGGAGCAA	TTCTTTGCTT
CCATAGGGGA	GAGGGAAAAG	TGCGGCTCCT	ACCTGGAGAA	GGATTTTGAC
ACCTTGAAGG	TCTATGACAC	ACAGCTGGAG	AATGTGGAGG	CCTTTGAGGG
CCTGTCTGAC	TTTTGTAAACA	CCTTCAAGCT	GTACCGGGGC	AAGACGAGGG
AGGAGACAGA	AGATCCATCT	GTGATTGGTG	AATTTAAGGG	CCTCTTCAAA
ATTTATCCCC	TCCCAGAAGA	CCCAGCCATC	CCCATGCCCC	CAAGACAGTT
CCACCAGCTG	GCCGCCCAGG	GACCCACAGG	GTGCTTGGTC	CGTATCTACA
TTGTCCGAGC	ATTTGGCCTG	CAGCCCAAGG	ACCCCAATGG	AAAGTGTGAT
CCTTACATCA	AGATCTCCAT	AGGGAAGAAA	TCAGTGAGTG	ACCAGGATAA
CTACATCCCC	TGCACGCTGG	AGCCCGTATT	TGGAAAGATG	TTTCGAGCTG
CCTGCACTCT	GCCTCTGGAG	AAGGACCTAA	AGATCACTCT	CTATGACTAT
GACCTCCTCT	CCAAGGACGA	AAAGATCGGT	GAGACGGTCG	TCGACCTGGA
GAACAGGCTG	CTGTCCAAGT	TTGGGGCTCG	CTGTGGACTC	CCACAGACCT
ACTGTGTCTC	TGGACCGAAC	CAGTGGCGGG	ACCAGCTCCG	CCCCTCCAG
CTCCTCCACC	TCTTCTGCCA	GCAGCATAGA	GTCAAGGCAC	CTGTGTACCG
GACAGACCGT	GTAATGTTTC	AGGATAAAGA	ATATTCCATT	GAAGAGATAG
AGGCTGGCAG	GATCCCAAAC	CCACACCTGG	GCCCAGTGGA	GGAGCGTCTG
GCTCTGCATG	TGCTTCAGCA	GCAGGGCCTG	GTCCCGGAGC	ACGTGGAGTC
ACGGCCCTCT	TACAGCCCCC	TGCAGCCAGA	CATCGAGCAG	GGGAAGCTGC
AGATGTGGGT	CGACCTATTT	CCGAAGGCCC	TGGGGCGGCC	TGGACCTCCC
TTCAACATCA	CCCCACGGAG	AGCCAGAAGG	TTTTTCTGTC	GTTGTATTAT
CTGGAATACC	AGAGATGTGA	TCTTGGATGA	CCTGAGCCTC	ACGGGGGAGA
AGATGAGCGA	CATTTATGTG	AAAGGTTGGA	TGATTGGCTT	TGAAGAACAC
AAGCAAAAGA	CAGACGTGCA	TTATCGTTCC	CTGGGAGGTG	AAGGCAACTT
CAACTGGAGG	TTCATTTTCC	CCTTCGACTA	CCTGCCAGCT	GAGCAAGTCT
GTACCATTGC	CAAGAAGGAT	GCCTTCTGGA	GGCTGGACAA	GACTGAGAGC
AAAATCCCAG	CACGAGTGGT	GTTCCAGATC	TGGGACAAATG	ACAAGTCTCT
CTTTGATGAT	TTTCTGGGCT	CCCTGCAGCT	CGATCTCAAC	CGCATGCCCA
AGCCAGCCAA	GACAGCCAAG	AAGTGCTCCT	TGGACCAAGT	GGATGATGCT
TTCCACCCAG	AATGGTTTGT	GTCCCTTTTT	GAGCAGAAAA	CAGTGAAGGG
CTGGTGGCCC	TGTGTAGCAG	AAGAGGGTGA	GAAGAAAATA	CTGGCGGGCA
AGCTGGAAAT	GACCTTGGAG	ATTGTAGCAG	AGAGTGAGCA	TGAGGAGCGG
CCTGCTGGCC	AGGGCCGGGA	TGAGCCCAAC	ATGAACCCTA	AGCTTGAGGA
CCCAAGGCGC	CCCGACACCT	CCTTCCTGTG	GTTTACCTCC	CCATACAAGA
CCATGAAGTT	CATCCTGTGG	CGGCGTTTCC	GGTGGGCCAT	CATCCTCTTC
ATCATCCTCT	TCATCCTGCT	GCTGTTCTCT	GCCATCTTCA	TCTACGCTTT

Fig. 7 (continued)

CCCGAACTAT GCTGCCATGA AGCTGGTGAA GCCCTTCAGC TGA

6619

(SEQ ID NO:16)

FIG. 8

1 MLRVFILIYAE NVHTFDTDIS DAYCSAVFAG VKKRTKVIKN SVMFVWNEGF
EWDLKGIFLD QGSELHVVK DHETMGRNRF LGEAKVFLRE VIATFSLSAS
FNAPLLDTKK QPTGASLVVQ VSYTPLPGAV PLFPPPTFLE PSPTLPDLV
VADTGGEEDT EDQGLTGDEA EFFLDQSGGP GAPTTPRKLP SRPPPHYPGI
KRKRSAPTSR KLLSDKPQDF **QIRVQVIEGR QLPGVNIKPV VKVTAAGQTK**
RTRIHKGNSP LFNETLFFNL FDSFGELFDE PIFITVVDNR SLRTDALLGE
FRMDVGTIYR EPRHAYLRKW LLLSDPDDFS AGARGYLKTS LCVLGPDEA
PLERKDPSED KEDIESNLLR PTGVALRCAH FCLKVFRAED LPQMDDAVMD
NVKQIFGFES NKKNLVDPFV EVSFAGKMLC SKILEKTANP QWNQNTLPA
MFPSMCEKMR IRIIDWDRIT HNDIVATTYL SMSKISAPGG EIEEEPAGAV
KPSKASDLDD YLGFLPTFGP CYINLYGSPR EFTGFDPDYT ELMTGKGEGV
AYRGRLLLSL ETKLVEHSEQ KVEDLPADDI LRVEKYLRRR KYSLEAFYFS
ATMLQDVDDA IQFEVSIGNY GNKFDMTCLF LASTTQYSRA VFDGCHYYL
PWGNVFPVVV LSSYWEDISH RIETQNLG IADRLEAGLE QVHLAKAQC
STEDVDSLVA QLTDELIAGC SQPLGDIHET PSATHLDQYL YQLRTHHLSQ
ITEAALAKL GHSELPAAL QAEDWLLRLR ALAEPQNSL PDIVINMLQG
DKRVAYQRPV AHQVLFSSRG ANYCGKNCCK LQTI FLKYPM EKVPGARMEV
QIRVKLWFGL SVDEKEFNQF AEGKLSVFAE TYENETKLAL VGNWGTGLT
YPKFSDVTGK IKLPKDSFRP SAGWTWAGDW FVCPEKTLH DMDAGHLSFV
EEVFENQTRL PGGQWYMSD NYTDVNGEKV LPKDDIECEL GKWWEDEWS
TDLNRAVDEQ GWEYSITIPP ERKPKHWVPA EKMYTHRRR RMVRLRRRDL
SQMEALNRHR QAEAEGEWE YASLFGWKFH LEYRKTDAR RRRWRRRMEP
LEKTGPAAVF ALEGALGGVM DDKSEDSMSV STLSEGVNRP TISCIFDYGN
RYHLRCYMYQ ARDLAAMDKD SPSDPYATVS FLHQSQKTVV VKNTLNPTWD
QTLIFYEIEI FGEPATVAEQ PPSIVVELYD HDTYGADEFM GRCICQPSLE
RMPRLAWFPL TRGSQPSGEL LASFELTQRE KPAIHHPGF EVQETSRILO
ESETDLPPY FPQREANIYM VPQNIKPAQ RTAIEILAWG LRNMKSYQLA
NISSPSLVVE CGGQTVQSCV IRNLRRKNPF DICTLFMEVM LPREELNCFP
ITVKVIDNRQ FGRRPVVGQC TIRSLESFLC DPYSAESFSP QGGPDDVSL
SPGEDVLIDI DDKEPLIPIQ EEEFIDWWSK FFASIGEREK CGSYLEKDFD
TLKVYDTQLE NVEAFEGLSF FCNTFKLYRG KTQEETEDPS VIGEFKGLFK
IYPLPEDPAI PMFPRQFHQL AAQGPQECV RIYIVRAFGL QPKDFNGKCD
FYIKISIGKK SVSDQDNYP CTLEPVFGKM FELTCTFLP KDLKITLYDY
DLLSKDENIG ETVVDLENRL LSKFGARCGP PQTYCVSGPN QWRDQLRPSQ
LLHLFCQQHR VKAPVYRTDR VMFQDKEYSI EEIEAGRIPN PHLGPVEERL
ALHVLQQQGL VPEHVESRPL YSPLQFDIEQ GKLQMWVDLF PKALGRPGFP
FNITFRARR FFLRCIIWNT RDVLEDDL SL TGERMSDIYV KGMIGFEH
KQKTGVHYRS LGGEGNFNR FIFFFDYLPV EQVCTIAKAD AFWRLDKTES
KIPARVVFQI WDNDKESFDD FLGSLQLDLN RMPKPAKTAK KCSLDQLDDA
FHPENFVSLF EQKTVKGWMP CVAEEGEKKI LAGKLEMTLE IVAESEHEER
PAGQGRDEPN MNPKLEDPRR PDTSFLWFTS PYKTMKFIW RFRFWAILF
IILFILLFL AIFIYAFPNY AAMKLVPKPS

2080

(SEQ ID NO:17)

FIG. 9

NP_003485.1	220	FQIRVQVIEGRQLPGVNIKPVVKVTAAGQTKR	TRIHKGNSPLFNE	264 (SEQ ID NO:18)
		*****4*****		
XF_023474694.1	222	FQIRVQVIEGRQLPGVNIKPVVKVTAARQTKR	TRIHRGNSPLFNE	266 (SEQ ID NO:19)
R253W	222	FQIRVQVIEGRQLPGVNIKPVVKVTAARQTKW	TRIHRGNSPLFNE	266 (SEQ ID NO:20)

FIG. 10

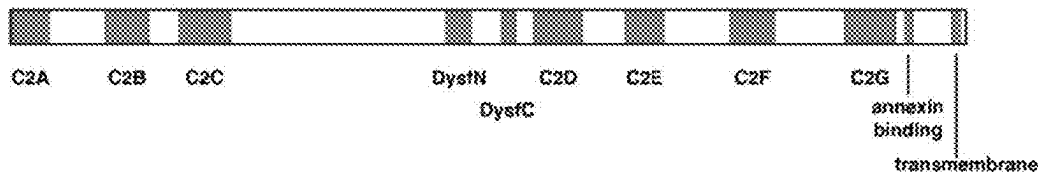
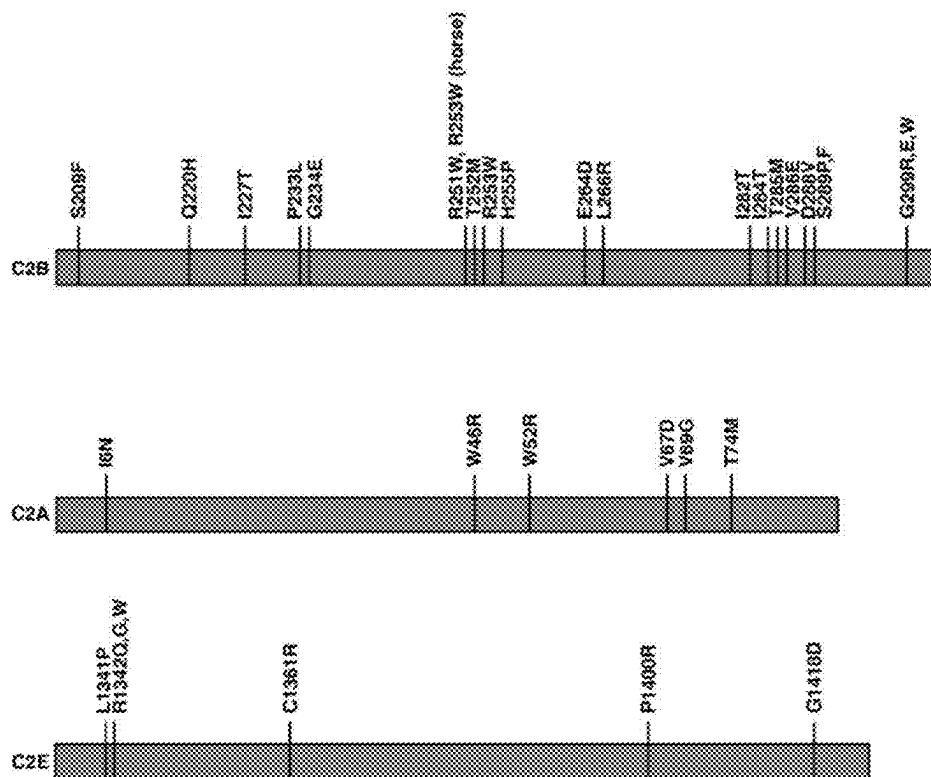
A**B**

FIG. 11

```
SEQ ID NO: 21      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 22      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 23      ----SKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 24      ----SKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 25      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 26      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 27      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 28      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 29      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 30      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 31      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 32      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 33      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 34      ----SKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 35      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 36      ----SKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 37      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 38      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 39      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 40      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 41      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 42      -----LSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 43      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 44      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 45      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 46      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 47      PPSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 48      -----SSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 49      -----SLSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
SEQ ID NO: 50      -----SLSKLLSDEKPDQFQIRVQVIEGRQLPGVNIKPVVYVTAAGQTKRTHIRKGNHPLFNETLFTNVDSPAEFLFDEPIFTVVDSSRLTQALLGEFR
CLUSTAL
R2S3W
      1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
      *
      1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
      *
      1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50
      *
```

FIG. 12

A

Human DYSF C2 domain

C2A1	3	KLAVPLIADNNTPTSTDISDAIKUAVFAZVKRTTVIKMSYHPVWNSGFENDLGPIFDQSSSLVYVYKHETHGNRHFLOEAK (SEQ ID NO:51)	85
C2A2	3	KLCMLVWASHLPAAKNDRSDFVSLTPFSVCKRTTVIKNSVHPVWNSGFENDLGPIFDQSSSLVYVYKHETHGNRHFLOEAK (SEQ ID NO:52)	86
C2B	267	PTSKHLLEKPKGDFQINQVVISRQLPGVNIKPVVYVTAAGQKRTKIKRGNSELPFNETLFPFSLFPGSPKLFNKEIFLYVDSRSLRTUALLSEK (SEQ ID NO:53)	302
C2B	1334	ILAKGLMKKSYGLANTSSPFLYVEGGQTVQSCVIRMLRNPWFOTCTLFNEVMLPRELYCPPTTVFVIDNRQPGRRPVVQQTIR (SEQ ID NO:54)	1427
New DYS128			
C2B	269	BAKRTLEKPKGDFQINQVVISRQLPGVNIKPVVYVTAAGQKRTKIKRGNSELPFNETLFPFSLFPGSPKLFNKEIFLYVDSRSLRTUALLSEK (SEQ ID NO:55)	304

B[illegible]

FIG. 13

31,226,630

```
GACTCTTTTT TCTCCACGTC GGGTCTAGGG TGCAGATGAG TTTATGGGAC
GCTGCACTCTG TCAGCCGACT CTGGAACGGA TGCCCCGGCT GACCTGGTTC
CCACTGACTA GGGGSCAGCCA GCCGGCGGGC GAGCTGCTGG CCTCTTTTGA
GCTCATCCAG AGAGAGAAGG TGAGGCTGGT CCCCATCCTG ATCCAGGAAG
CCCAGACAGG GACTACATGG TAGGAGAGGG TAGGGGGCAC AGGCAGGAT
GCCCCGAAGG AGGGGAGGGG CTGCGGATAC ATCCTTCTCC CCCCACCAGG
CCTGCTCAGT GGGGAGCACT GAACCGCCAC CTCTGGAATF CAGGCTTCTT
GCTGAGACTG GTTTTSCCCC TCTTTTCTGA ACCCCAAGTT ASSCTACTGT
CTAAACATTG GGGTAGAATG TTGGAATCTG GGCTGTGCTG GAATGTCTGA
GTCCTCTGGC CATTGGATGT ATTTTCCCT GGGTCATATC CTGACGTGGT
TTCTGAGAGG AGGACCTCCG TGATCTGGTC GGGGAGACTT AATTGTGATG
GTGGCACCCA GGGACAGGTA AGGTGAGGAC CTCCGGTSTA GTTTTGGGTC
TGCCATGGCC TGGCTGTGCG ACCCTGGGCC AGGCACTTCC CTGCTGTGAA
ATAGGGGGTT TGGGTTAGAA CCATATCTTC AGCTTTGCTC TTAGAGTGA
GTGAACCCCA GGTGTAAGCT GGATGGAAGT CAGGCTACTC TGGGTCAAGT
GTCTGGAGGG GGCACACAGT ATGCTGCAAC CTGGCTTCCC CTGGCACCTG
GCTGCCAACT CCGGGGCACC TCCAAGGAGT CCTAGGTTG CAAACTCCCA
ACTGTGATCA CTGACATATT CCATGAGTGT CATTTTGTGG CTGAAGGTGT
GTTTGGCTTG CCGGACAGA AAGGAGGGTG ATGGGGCTT GGGTGACAGG
CACTGACCAA AGCTCTTTTT TCTACCCCGC AGCGGGCCAT CTACCACATT
```

31,225,630

```
(C/A)
CTGGTTTTGA GGTAASTCTT GCTCTTCCCT CTCTTCTTC AAACTCATGG
CCCGCCTCTG TGTGTTTCCA GCCCCTATG AGCTAGGAAG GGCAGTCAGG
TGTATACCAC CACGCGCCCG CTTTCAGGGA TGGGGTGGGT GAGGGCCGGG
GTCCCTGGCC AGCTCAGGCC TTTCCCGCCC TTCTTCCCTG AGACTCAGAC
TGTAOCCACC TCATGCTCAH CCAGCCACG GCTCTCAGC TGGGCTGCTAT
GTTAGAATCC CTTGCGCAGC TTGAAAAATC ACCAGTGCTC AGGCCCCACC
CCAGACCAGA TGGAAACACG TCTCTGGGGA TGTGGCCGAG GCATTGGTAT
CTTTTCAGAG TTCCAGGGG ATTTTAATGT GCTGCTAGGG CTGAGCACCA
TCGACTTAGC CCAGTGCTTC TCAGTGTGAT ATCACCTGGG AIGTTGTTAG
AAATGCCGAT TCTTGGCTCC TGCTTCACAC CTACTGAAGC AAACCTCTGAG
GGGCAGTGCA GCAACCTGCA AATTAACAAG CCTCCAGGGG ATTCTGATGC
CCACTCGAGT TTCAGAAAGT CTGGCTTAGA CCATCTTCCC CAACTCATTG
TTCCCCCAT GACCGGTTTT TCCCTCTCTA CTAGTGATGG GGGCTCTGTT
CCCTTCTCCC TGCTCTCCTT CCTTCTCCTG GGGGCATGGT AGAAAATGCA
GACCCTTTCC TGGTACCCAG GCTAATTTGA TTCAGGAGAT ACTGTTCTCA
GTCAGGTTCA GCTGCATCTT GGAGGAGCGG TCATCCGGTC TGTGCTTGGC
TGCTTTCCCC CTTACAAGGC AGTCCACGCA ATCTGCAGAC AGTTTTTACC
ATGCAGGCTT TCCTTCACCC CTGTCTCTGC TCCACCCCGG GCATCCCTCA
GCCCCCTCTC TCTCCTGCTC CCTCTCCTGT CCAGCCCAAG CTCCCGCTCT
CAGGTCTGCC AGCCCTGGG GATCCAGATT CTTTCTCCCT GGCTCTCCAA
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31,224,630

(SEQ ID NO:56)

FIG. 14

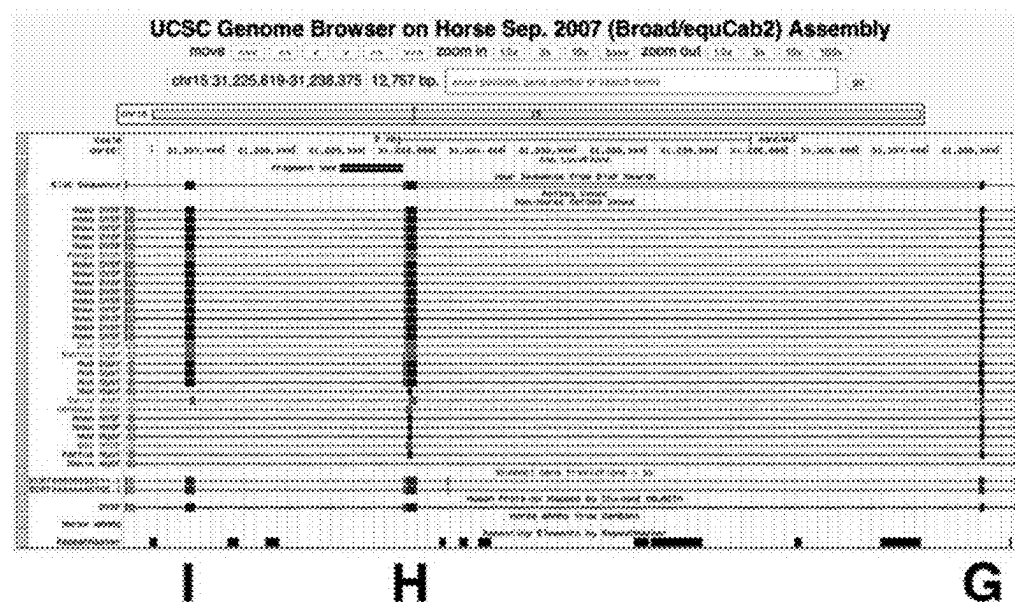
3,727 AGACCCACGA TTTCCTGCA TCTCGACTAC GGGAAACCGCT ACCATCTACG
CTGCTACATG TACCAGGCCC GGGACCTGCC CGCCATGGAC AAGGACTCTT
TTTCTGATCC CTACGCAATC GTCTCCTTCC TGCACCAGAG CCAGAAGACA
GTGGTGGTGA AGAACACCCT GAACCCGACC TGGGACCAGA CACTCATCTT
CTATGAGATC GAGATCTTTG GTGAGCCGCC CAGCATCGCC GAGCAGCCGC
CCAGCATCGT GGTGGAGCTA TACGACCATG ACACCTACGG TGCAGATGAG
TTTATGGGAC GCTGCATCTG TCAGCCGACT CTGGAACGGA TGCCCCGGCT
GACCTGGTTC CCACTGACTA GGGGCAGCCA GCGGGCGGGC GAGCTGCTGG
CCTCTTTTGA GCTCATCCAG AGAGAGAAGC **CGGCCATCTA CCACATTCT**
GGTTTTGAG 4,185

(SEQ ID NO:57)

3,727 AGACCCACGA TTTCCTGCA TCTCGACTAC GGGAAACCGCT ACCATCTACG
CTGCTACATG TACCAGGCCC GGGACCTGCC CGCCATGGAC AAGGACTCTT
TTTCTGATCC CTACGCAATC GTCTCCTTCC TGCACCAGAG CCAGAAGACA
GTGGTGGTGA AGAACACCCT GAACCCGACC TGGGACCAGA CACTCATCTT
CTATGAGATC GAGATCTTTG GTGAGCCGCC CAGCATCGCC GAGCAGCCGC
CCAGCATCGT GGTGGAGCTA TACGACCATG ACACCTACGG TGCAGATGAG
TTTATGGGAC GCTGCATCTG TCAGCCGACT CTGGAACGGA TGCCCCGGCT
GACCTGGTTC CCACTGACTA GGGGCAGCCA GCGGGCGGGC GAGCTGCTGG
CCTCTTTTGA GCTCATCCAG AGAGAGAAGC **CGGCCATCTA CCACATTACT**
GGTTTTGAG 4,185

(SEQ ID NO:58)

FIG. 15



I 31,225,658 tacccgcag CCGGCCATCT ACCACATTCC TGGTTTTGAG gtaagtcttg 31,225,609
(SEQ ID NO:59)

FIG. 16

1,141 RPTISCIFDY GNRYHLRCYM YQARDLPAMD KDSFSDPYAI VSFLHQSQKT
VVVKNLTNPT WDQTLIFYEI EIFGEPPSIA EQPPSIVVEL YHDTYGADE
FMGRICQPT LERMPRLTWF PLTRGSQPAG ELLASFELIQ REKPAIYHIT
GFE 1,293
(SEQ ID NO:60)

1,141 RPTISCIFDY GNRYHLRCYM YQARDLPAMD KDSFSDPYAI VSFLHQSQKT
VVVKNLTNPT WDQTLIFYEI EIFGEPPSIA EQPPSIVVEL YHDTYGADE
FMGRICQPT LERMPRLTWF PLTRGSQPAG ELLASFELIQ REKPAIYHIT
GFE 1,293
(SEQ ID NO:61)

FIG. 17

5'-ggttgca aactcccaac tgt-3' (SEQ ID NO:64)
ACTGGTTGCA AACTCCCAAC TGTGATCACT GACATATTCC ATGAGTGTCA
TTTTGTGGCT GAAGGTGTGT TTGCTGGCC CGGACAGAAA GGAGGTGAT
GGGGCCTTGG GTGACAGGCA CTGACCAAAG CTCTTTTTC TACCCCGCAG
31,225,648 CCG GCC ATC TAC CAC ATT CCT GGT TTT GAG (SEQ ID NO:62) 31,225,619
31,225,648 CCG GCC ATC TAC CAC ATT ACT GGT TTT GAG (SEQ ID NO:63) 31,225,619
GTAAGTCTTG CTCTCCCTC TTCTCTTCA AACTCATGGC CCGCCTCTGT
GTGTTTGAG CCCCTATGA GCTAGGAAGG GCAGTCAGCT GTATACCACC
ACCGSCCSC CTTAGGGAT GGGGTGGGTG AAGGCGGGG TCCCTGGCA
GTCAGGCCT TTCCCGCCT TCTTCTCTGA GACTCAGACT GTAGGGACCT
CATGGTCATC CAGCCACGG GGTTCAGCT GGGCTGCATG TTAGAATCC
3'-g
TTOGSCAGT TAAAAATCA
aagcgcgcga acttttttag-5' (SEQ ID NO:65)

FIG. 18

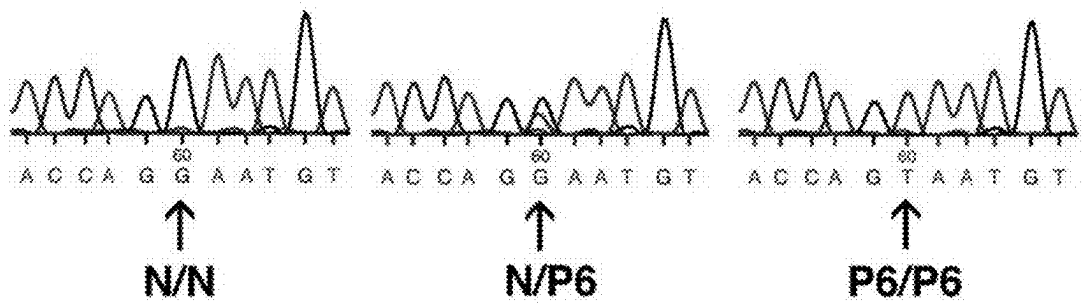


FIG. 19

human	1261	TRGSQPSGELLASFELIQREKPAIHNI	<u>P</u> GFEVQETSRIIDSESDTDLPPFPQREANIYM	1320
(SEQ ID NO:66)				
horse	1263	TRGSQPSGELLASFELIQREKPAIYHI	<u>P</u> GFEVQDTSGILEESEDTLPPFPQREANIYM	1322
(SEQ ID NO:67)				
P1290T	1263	TRGSQPSGELLASFELIQREKPAIYHI	<u>T</u> GFEVQDTSGILEESEDTLPPFPQREANIYM	1322
(SEQ ID NO:68)				

FIG. 20

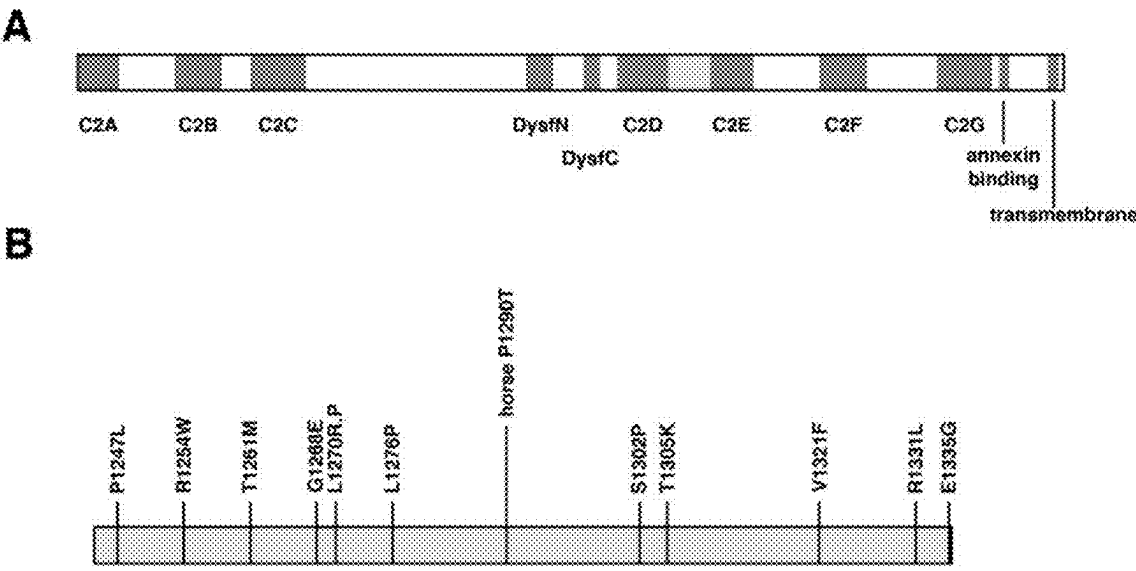


FIG. 21

```
SEQ ID NO:69 TRGSQPSGELLASFELIQREKPAINHNI-PGFEEVQ-ETSRILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:70 -RGSQPSGELLASFELIQREKPAINHNI-PGFEEVQ-ETSRILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:71 TRGSQFAGELLAAFELIQREKPAINHNI-PGFEMH-ETSRILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:72 TRGSQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DTSRILEESED TDLFYPPFPQREANIYM
SEQ ID NO:73 TRGNQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DTAGILEESED TDLFYPPFPQREANIYM
SEQ ID NO:74 -RGSQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DTSGILEESED TDLFYPPFPQREANIYM
SEQ ID NO:75 TRGSQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DTSGILEESED TDLFYPPFPQREANIYM
SEQ ID NO:76 TRGSQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DTSGILEESED TDLFYPPFPQREANIYM
SEQ ID NO:77 TRGSQPTGELLASFELIQREKPAINHNI-PGFEEVQ-ETSRILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:78 TRGSQPTGELLASFELIQREKPAINHNI-PGFEEVQ-ETSRILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:79 TRGSQPMGELLASFELIQREKPAINHNI-PGFEEVQ-ETSRILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:80 -RGSQPSGELLASFELIQREKPAIYHI-PGFEEVQ-DTSRILDESEDTDLFYPPFPQREANVYM
SEQ ID NO:81 -RGGQPSGELLAAFELIQREKPAINHNI-PGFEEVQ-DASRILDEAED TDLFYPPFPQREANVYM
SEQ ID NO:82 -RGSQPSGELLASFELIQREKPAINHNI-PGFEEVQ-DTARILDESEDTDLFYPPFPQREANIYV
SEQ ID NO:83 -RGSQASGELLASFELIQREKPAIYHI-PGFEEVQ-DTSRILDESEDTDLFYPPFPQREANVYM
SEQ ID NO:84 TRGGQFAGELLAAFELIQREKPAINHNI-PGFEMH-ETSRILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:85 TRGSQFAGELLAAFELIQREKPAINHNI-PGFEMH-ETSRILEESED TDLFYPPFPQREANIYM
SEQ ID NO:86 TRGSQFAGELLAAFELIQREKPAINHNI-PGFEEVQ-ETSRILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:87 -RGSQFAGELLASFELIQREKPAIYHI-PGFEEVQ-DTSRILDESEDTDLFYPPFPQREANIYV
SEQ ID NO:88 TRGNQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DTSRILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:89 -RGSQFAGELLASFELIQREKPAIYHI-PGFEEVQ-DTSRILDESEDTDLFYPPFPQREANVYV
SEQ ID NO:90 -RSSQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DTSRILDESEDA DLFYPPFPQREVNIM
SEQ ID NO:91 -RGSQFAGELLASFELIQREKPAINHNI-PGFEEVH-DTSGILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:92 TRGSQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DTSRILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:93 TRGSQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DASRILEESED TDLFYPPFPQREANIYM
SEQ ID NO:94 TRGSQFAGELLASFELIQREKPAIYHI-PGFEEVQ-DTSRILEESED TDLFYPPFPQREANIYI
SEQ ID NO:95 TRGSQFAGELLASFELIQREKPAIYHI-PGFEEVQ-DTSGILEESED TDLFYPPFPQREANIYM
SEQ ID NO:96 TRGSQLAGELLASFELIQREKPAIYHI-PGFEEVQ-DTSGILEESED TDLFYPPFPQREANIYM
SEQ ID NO:97 TRGSQAAGELLASFELIQREKPAINHNI-PGFEEVQ-DTSRILEESED TDLFYPPFPQREANIYM
SEQ ID NO:98 TRGNQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DTSGILEESED TDLFYPPFPQREANIYM
SEQ ID NO:99 VRGSQFAGELLASFELIQREKPAINHNI-PGFEEVQ-DTSGILDESEDTDLFYPPFPQREANIYM
SEQ ID NO:100 TRGNQFAGELLASFELIQREKPAIYHI-PGFEEVQ-DTSGILEESED TDLFYPPFPQREANIYM
SEQ ID NO:101 VRGSRFAGELLAAFELIMREKPAITHHI-PGFEEF-EISGVADEIGD TDLFYPPFPQREANIYV
SEQ ID NO:102 TKKNKPAGDLLAAFELIQREKPFSTHHSDPGFEEAVRTSRPLGKAGDS DLFYPPFPQREPNV---
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CLUSTAL
P1290T

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FIG. 22

47,660,977 CTTCCAGATT TGCTCCCTGT TACTGCATTT CTAAATTTTA CTTGAGATTG
ATTCTTTTCTT TGTGAGGGCA ACTATCAGCT ATCATACTCA AGAATGAAGC
AAGTGCTTTC CCACATGTAT TATCTTAATC CTCTACTTTG AGGCAAGTAT
TTTCTCCCTT ATTTACTGAT AAGTAGAAAC ACACACATAC AAACACGTGC
ACAAGGTTCA AGGAGAGTAA ATAATTTGUC CATGAAAGAT CTAACCTCAGT
TCCTTATTAA ATCTTTAAAT CATGACTCCA GGCTAATGTA GTCTACATCC
TACATCTACA TATTTGTATA TGCCCTATAT TTTTTTTTTT TTTTGGAGGA
AGATTGTGAG CTSACGTCTA TTGCCAATCT TCCTCTTTT GCTTAAGGAA
GACCTGAGCT AACATCTGTG CCATCTTCCT GTATTTTGTA TGTGGGCCAC
TGCCACAGCA TTACTTGATG ACCGGTGTGT AGGTCTGTGC CTGGGATCCG
AACCCAGGC CACCAAAGCG GAGCCTGTGA ACTTAACCAC TACGCCACCA
GGCCGGCCCC TCCTTAGATA TTTTAAATGT TCATATAGAT TATGTGGTAT
TATCATCTGT AGGATAAATG TATGCCATTC AAATGTTTTT TTTGGAATC
TCTCCAGTGA CACTTTTAGC TGTAATAATTT TGTGAACAGA TTTTCTGCTG
GCCATTCTAA ATAACACAAA ATAACATATAT ATTGTGTTTC CTTTTATAC
TCACTAGAAAT AACTTCTTTA TCGCCTTTTA TATATAGGTT GTATTGCTGG
GAAATACAA TSCACAGGGC TTAGGTTTCAG ATCATGAATT AATGCTGAGA
TGTACCAAG GACAAGAATA CGTCAAAGTC GTCATGCAAA ATGGGCGAAT
GATGGGAGCT STCTTAATTG GTGAAACCGA TTTAGAAGAA ACATTTGAAA
ACTTGATTTT AAACCAGATG AATCTTTCAG CATACGGAGA AGATCTGCTG

47,661,977 (G/C)
ATCCAGATAT TGACATAGAA GATTATTTTG ACTAAAAAGG TCATTCCAAG
AACCACATAA AGTTCCAAAT AAGACAAAAA AGTCACACAT CAATAAAGTA
AATGATTGCA CTGATTTAAT GATGACCACA TTGAAGTTAA AAGTACAGAA
GTGATAATGA TTTCAGTGGG AAAATATTAA AAATAAATF CTAAAGATAA
AATCAATTCA ASTAACTTAT TTACAGATTT TTTTCCTAAC ACAAATTGA
CCAATTATGT AAGAATTCTC AAGTTATTCA TTTCTGTGTT TTAACGTAC
ACACATTCAI TTGTGATTTT AGCTTTGGAG CACATTTAGC TAGGCTGTTA
TCTGCTCAGC CCACAACTTG GTCTTGTTA GTAGACCAGA GCCATTCTTT
GATTGGAAAA CGTCAAGAAG CTTGTAAATTT TATTTTACTT AGAGATGCCG
CAATATCTTC TTGCTAATAT TATTTGTATT GATACCTTAT TCCTTTTTTG
TTTGAATTCI ACAGCTATTA TTTTAACCTG AAATCATPCA CTTTCTGTCA
TGAGCCTGTG ACTTTTCAGT ACAATAACAC TACCATCAAA AGAGAGTTAT
GTGAAAAGAA AAAATATAGC TAGGTATATC ACGGCATAAA GAGCTTAAGA
CAGTTATGTG AACTTACTCT TTTAAGGAGT TTACATAAAA AAAGATTTAT
TTTCCCTTGT TTCATACTGT AAATTTATAC AATTCATGCT CTCAAATTTT
CAAAAGATCT CAAATTCTCA AGTCTCAAT TCTTTAGAGA TAAGTTCTTA
AAATATAGCT CCATTTGTAT AAAACATTAA TGAAATTGTC TTAAAAATTA
AACTTTGATT AACTACAAAA ATAATGGTGT AAACATGCAT AAAGTTCTT
GTTCTTGGAA AATTTAGATA CACTCATGCA TCATCAATTT TTCTTTCTT
AGCCCTGTG TTCTTAGCCT CAGATTGCTG AAGCATGTTT GCAGACTTCT

47,662,977

(SEQ ID NO:103)

FIG. 23

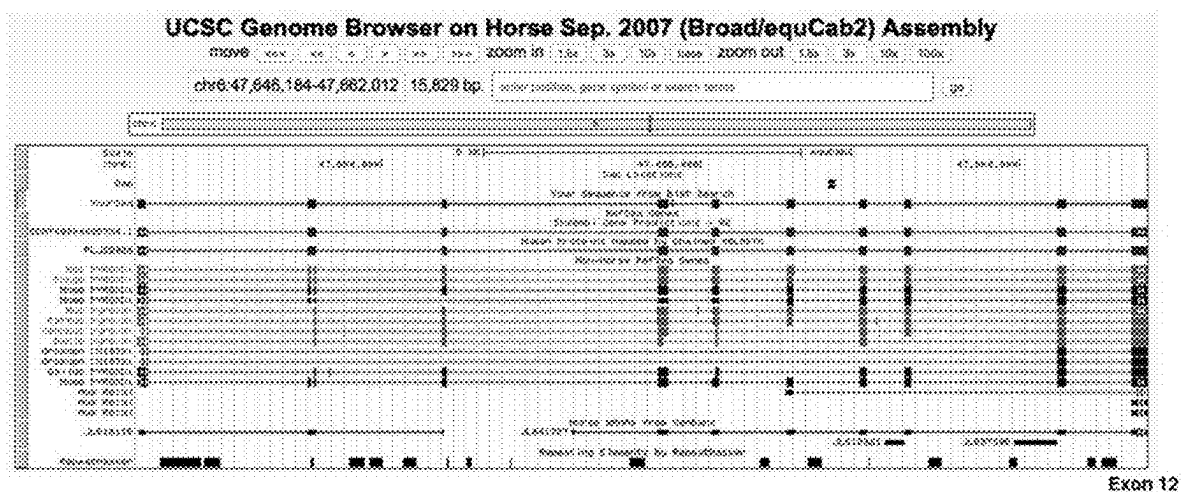
238 AGGTTTCCAG AGTATTGGAA GAATTTGATG TTGAAGAACA GCCAAATACC
ATGTTAGAAA ATCGCTTTCC CAACATTAAG GTTATAGAAT CTGGAGTAAA
GCAACTGAAG AGTGAAGAAC ACTGCATTTT AACAGAAGAT GGCAATCAGC
ATGTATATAA GAAACTCTGT CTGTGTGCTG GAGCTAAACC AAAGTTGATA
TGTGAAGGAA ATCCTTATGT ATTAGGAATC CGTGACACAG ACAGTGCTCA
GGAATTTTCAG AAACAGCTTA CTAAGGCTAA AAGAATAATG ATCATAGGCA
ATGGTGGGAT CGCACTTGAA TTAGTGTATG AAATTGAAGG CTGTGAAGTC
ATTTGGGTCA TTAAAGACAA AGCTATTGGG AATACGTTCT TCGATGCAGG
AGCAGCTGAA TTCTTGACTT CAAAGCTCAT CGCTGAAAAA CCAGAGGGTA
AAATTGCACA TAGAAGAACC AGATATACAA CTGAAGGAAG GAAAAAGGAA
GCACGAACCA AAGGTGATGC TGCTAATGTA GGCAGTGCCC TGGGACCTGA
CTGGCATGAA GCCTTGAATC TTAAAGGAAC AAAAGAGTTT TCTCATAAGA
TTCACATTGA AACTATGTGT GAAGTAAAGA AAATCTACCT TCAGGAAGAA
TTTAGAATTT CCAAGAAAAA GTCCCTGACT TTTCCAAGAG ACCATAATAA
TCACTCAGTT TCAACTGATA AAGAGATATG GCCTGTATAT GTGGAATTGA
CCAATGAAAA GATATATGGC TGCGATTTCA TTGTCACTGC TACAGGAGTT
ACACCAAATA TAGAACCTTT CCTCTGTGGC AACAAATTTG ATCTAGGAGA
AGATGGTGGC CTGAAAGTGA ATAATCATAT GCACACGTCC CTTCTTGACA
TCTATGCTGC AGGTGACATC TGCACTGCCT CCTGGGAACC CAGCCCACTG
TGGCAGCAGA TGAGGCTGTG GACGCAGGCT AGACAGATGG GATGGTATGC
AGCCAAGTGC ATGGCTGCAG CTACTTTAGG AGACTCCATT GACATGGATT
TCAGCTTCGA ACTGTTTGCT CATGTAACAA AATTTTPTAA CTATAAGGTT
GTATTGCTGG GAAAATACAA TGCACAGGGC TTAGGTTTCAG ATCATGAATT
AATGCTGAGA TGTACCAAAG GACAAGAATA CGTCAAAGTC GTCATGCAAA
ATGGGCGAAT GATGGGAGCT GTCTTAATTG GTGAACCGA TTTAGAAGAA
ACATTTGAAA ACTTGATTTT AAACCAGATG AATCTTTCAG CATATGGAGA
AGATCTGCTG CATCCAGATA TTGACATAGA AGATTATTTT GACTAA 1583

(SEQ ID NO:104)

238 AGGTTTCCAG AGTATTGGAA GAATTTGATG TTGAAGAACA GCCAAATACC
ATGTTAGAAA ATCGCTTTCC CAACATTAAG GTTATAGAAT CTGGAGTAAA
GCAACTGAAG AGTGAAGAAC ACTGCATTTT AACAGAAGAT GGCAATCAGC
ATGTATATAA GAAACTCTGT CTGTGTGCTG GAGCTAAACC AAAGTTGATA
TGTGAAGGAA ATCCTTATGT ATTAGGAATC CGTGACACAG ACAGTGCTCA
GGAATTTTCAG AAACAGCTTA CTAAGGCTAA AAGAATAATG ATCATAGGCA
ATGGTGGGAT CGCACTTGAA TTAGTGTATG AAATTGAAGG CTGTGAAGTC
ATTTGGGTCA TTAAAGACAA AGCTATTGGG AATACGTTCT TCGATGCAGG
AGCAGCTGAA TTCTTGACTT CAAAGCTCAT CGCTGAAAAA CCAGAGGGTA
AAATTGCACA TAGAAGAACC AGATATACAA CTGAAGGAAG GAAAAAGGAA
GCACGAACCA AAGGTGATGC TGCTAATGTA GGCAGTGCCC TGGGACCTGA
CTGGCATGAA GCCTTGAATC TTAAAGGAAC AAAAGAGTTT TCTCATAAGA
TTCACATTGA AACTATGTGT GAAGTAAAGA AAATCTACCT TCAGGAAGAA
TTTAGAATTT CCAAGAAAAA GTCCCTGACT TTTCCAAGAG ACCATAATAA
TCACTCAGTT TCAACTGATA AAGAGATATG GCCTGTATAT GTGGAATTGA
CCAATGAAAA GATATATGGC TGCGATTTCA TTGTCACTGC TACAGGAGTT
ACACCAAATA TAGAACCTTT CCTCTGTGGC AACAAATTTG ATCTAGGAGA
AGATGGTGGC CTGAAAGTGA ATAATCATAT GCACACGTCC CTTCTTGACA
TCTATGCTGC AGGTGACATC TGCACTGCCT CCTGGGAACC CAGCCCACTG
TGGCAGCAGA TGAGGCTGTG GACGCAGGCT AGACAGATGG GATGGTATGC
AGCCAAGTGC ATGGCTGCAG CTACTTTAGG AGACTCCATT GACATGGATT
TCAGCTTCGA ACTGTTTGCT CATGTAACAA AATTTTPTAA CTATAAGGTT
GTATTGCTGG GAAAATACAA TGCACAGGGC TTAGGTTTCAG ATCATGAATT
AATGCTGAGA TGTACCAAAG GACAAGAATA CGTCAAAGTC GTCATGCAAA
ATGGGCGAAT GATGGGAGCT GTCTTAATTG GTGAACCGA TTTAGAAGAA
ACATTTGAAA ACTTGATTTT AAACCAGATG AATCTTTCAG CATATGGAGA
AGATCTGCTG CATCCAGATA TTGACATAGA AGATTATTTT GACTAA 1583

(SEQ ID NO:105)

FIG. 24



Exon 12

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47,661,754      ttatatatag GTTGTATTGC TGGGAAAATA CAATGCACAG GGCTTAGGTT
                CAGATCATGA ATTATGCTG AGATGTACCA AAGGACAAGA ATACGTCAA
                GTCCTCATGC AAAATGGGCG AATGATGGGA GCTGTCTTAA TTGGTGAAAC
                CGATTAGAA  GAAACATTG  AAAACTTGAT TTTAAACCAG ATGAATCTTT
                CAGCATACGG AGAAGATCTG CTGGATCCAG ATATTGACAT AGAAGATTAT
                TTTGACTAAa aaggtcatt
                (SEQ ID NO:106)

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47,662,022

FIG. 25

56 VSRVLEEFDV EEQPNMTLEN RFPNIKVIES GVKQLKSEEH CILTEDGNQH
VYKKLCLCAG AKPKLICEGN PYVLGIRDTD SAQEFQKQLT KAKRIMLIGN
GGIALELVYE IEGCEVIWVI KDKAIGNTEF DAGAAEFLLS KLIAEKPECK
IAHRRTRYTT EGRKKEARTK GDAANVGSAL GPDWHEGLNL KGTKEFSHNI
HIETMCCEVKK IYLQEEFRIS KKSLLTFPRD HNNQSVSTDK EIWPVVVELT
NEKIYGCDFI VSATGVTPNI EPFLCGNNFD LGEDGGLKVN NHMHTSLPDI
YAAGDICTAS WEPSPVWQQM RLWTQARQMG WYAAKCMAAA TLGDSIDMDF
SFELFAHVTK FENYKVVLG KYNAQGLGSD HELMLRCTKG QEYVKVVMQN
GRMMGAVLIG ETDLEETFEN LILNQMNLSA YGEDLLDPDI DIEDYFD 502

(SEQ ID NO:107)

56 VSRVLEEFDV EEQPNMTLEN RFPNIKVIES GVKQLKSEEH CILTEDGNQH
VYKKLCLCAG AKPKLICEGN PYVLGIRDTD SAQEFQKQLT KAKRIMLIGN
GGIALELVYE IEGCEVIWVI KDKAIGNTEF DAGAAEFLLS KLIAEKPECK
IAHRRTRYTT EGRKKEARTK GDAANVGSAL GPDWHEGLNL KGTKEFSHNI
HIETMCCEVKK IYLQEEFRIS KKSLLTFPRD HNNQSVSTDK EIWPVVVELT
NEKIYGCDFI VSATGVTPNI EPFLCGNNFD LGEDGGLKVN NHMHTSLPDI
YAAGDICTAS WEPSPVWQQM RLWTQARQMG WYAAKCMAAA TLGDSIDMDF
SFELFAHVTK FENYKVVLG KYNAQGLGSD HELMLRCTKG QEYVKVVMQN
GRMMGAVLIG ETDLEETFEN LILNQMNLSA YGEDLLHPDI DIEDYFD 502

(SEQ ID NO:108)

FIG. 26

47,661,614 TTCTTTGGAA CTCTCTCCAG TGACACTTTT AGCTGTAAAA TTTTGTGAAC 5'--c
agatttttctg ctggccatt-3' (SEQ ID NO:111)
AGATTTTCTG CTGGCCATTC TAAATAACAC AAAATAACTA TATATTGTGT
TTCCTTTTTA TACTCACTAG AATAACTTCT TTATCGCCTT TTATATATAG
47,661,764 GTT GTA TTG CTG GGA AAA TAC AAT GCA CAG GGC TTA GGT
47,661,764 GTT GTA TTG CTG GGA AAA TAC AAT GCA CAG GGC TTA GGT
TCA GAT CAT GAA TTA ATG CTG AGA TGT ACC AAA GGA CAA
TCA GAT CAT GAA TTA ATG CTG AGA TGT ACC AAA GGA CAA
GAA TAC GTC AAA GTC GTC ATG CAA AAT GGG CGA ATG ATG
GAA TAC GTC AAA GTC GTC ATG CAA AAT GGG CGA ATG ATG
GGA GCT GTC TTA ATT GGT GAA ACC GAT TTA GAA GAA ACA
GGA GCT GTC TTA ATT GGT GAA ACC GAT TTA GAA GAA ACA
TTT GAA AAC TTG ATT TTA AAC CAG ATG AAT CTT TCA GCA
TTT GAA AAC TTG ATT TTA AAC CAG ATG AAT CTT TCA GCA
TAT GGA GAA GAT CTG CTG GAT CCA GAT ATT GAC ATA GAA
TAT GGA GAA GAT CTG CTG CAT CCA GAT ATT GAC ATA GAA
GAT TAT TTT GAC TAA (SEQ ID NO:109) 47,662,012
GAT TAT TTT GAC TAA (SEQ ID NO:110) 47,662,012
AAAGGTCATT CCAAGAACCA CATAAAGTTC CAATAAGAC AAAAAAGTCA
CACATCAATA AAGTAAATGA TTGCACTGAT TTAATGATGA CCACATTGAA 47,662,112
(SEQ ID NO:112) 3'-aacgtgacta aattactact ggt-5'

FIG. 27

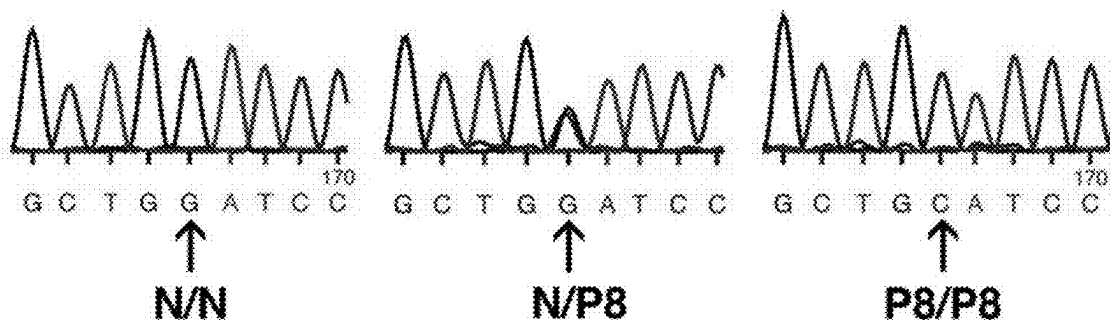


FIG. 28

619 TGAAATTGAA GGCTGTGAAG TGATTGGGGC CATTAAAGAT AAAGCTATAG
GGAACTACTTT CTTCGATGCA GGAGCAGCTG AATTCTTGAC TTCAAAAGCTC
ATTGCTGAAA AATCAGAGGC TAAAATTGCA CATAAAAGAA CCAGATATAC
AACTGAAGGA AGGAAAAAGG AAGCTAGAAG CAAATCTAAA GCAGATAATG
TAGGAAGTGC ATTGGGACCA GATTGGCATG AAGGCTTGAA TCTTAAAGGA
ACAAAAGAGT TTTCTCATAA GATTACACCTT GAAACTATGT GTGAAGTAAA
GAAAATCTAC CTTCAGGATG AGTTTGAAT TTTGAAGAAA AAGTCCTTCA
CTTTTCCAAG AGACCATAAG TCAGTTACAG CTGATACAGA GATGTGGCCT
GTCTATGTGG AATTGACCAA TGAAAASATA TATGGCTGCG ATTTCAATTGT
CAGTGCTACA GGAGTTACAC CAAATGTAGA ACCTTTTCTC CATGGTAACA
GTTTTGATCT AGGAGAAGAT GGTGGCCTGA AAGTGGATGA TCATATGCAC
ACATCCCTTC CTGATATCTA TGCTGCGGCT GACATCTGTA CTACATCCTG
GCAGCTGAGC CCAGTCTGGC AGCAGATGAG GCTGTGACCC CAGGCTAGAC
AGATGGGATG GTATGCAGCA AAGTGCATGG CTGCAGCGAG TTCAGGAGAC
TCTATTGACA TGGATTTCAG CTTTGAACCTG TTTGCTCATG TGACAAAATT
TTTTAACTAT AAGGTTGTAC TGCTGGGAAA ATACAAATGCA CAGGGCTTAG
GTTTCAGATCA TGAATTAATG CTGAGATGTA CCAAAGGACG AGAATACATC
AAGTCGTCA TGCAAAATGG ACGAATGATG GGAGCTGTCT TAATTGGTGA
AACCAGTTTA GAAGAAACAT TTGAAAACCT AATCTTAAAC CAAATGAATC
TTTCATCATA TGGAGAAGAT CTGCTAGATC CAAATATTGA TATAGAAGAT
TATTTTGACT AA 1630

(SEQ ID NO:113)

FIG. 29

56 ISKILLEEFDV EEQSSTMLGK RFPNIAVIES GVKQLKSEEH CIVTEDGNQH
VYKKLCLCAG AKPHLICEGN PYVLGIRD TD SAQEFQKQLT KAKRIMIIGN
GGIALELVYE IEGCEVIMAI KDKAIGNTFD DAGAAEFLTS KLIAEKSEAK
IAHKRTRYTT EGRNKEARSK SKADNVGSAL GFDWHEGLNL KGTKEFSHKI
HLETMCVKK IYLDDEFRI L KRSPTFFPRD HKSVTADTEM WFPVVELTNE
KIYGCDFIVS ATGVTEVNEP FLHGNSFDLG EDGGLKVDDH MHTSLPDIYA
AGDICTTSWQ LSPVWQMKL WTQARQMGWY AAKCMAAASS GDSIUMDFS F
ELFAHVTKFF NYKVVLGKY NAQGLGSDHE LMLRCKGRE YIKVVMQNGR
MMGAVLIGET DLEETENLI LNQMNLSSYG EDLLDPNIDI EDYFD 500

(SEQ ID NO:114)

FIG. 30

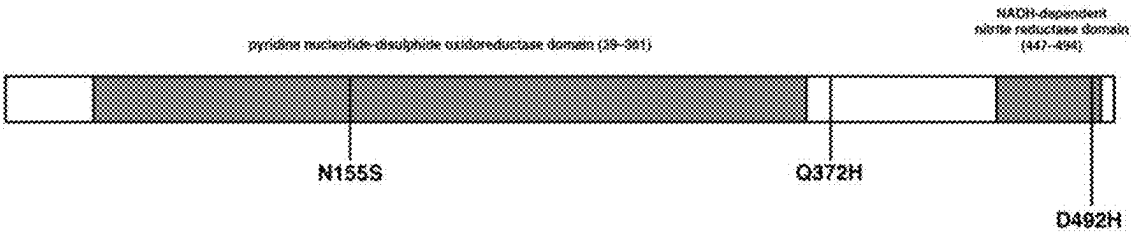
Human
VLLGKYNAQGLGSDHEMLRCTKGREYIKVVMQNGRMMGAVLIGETDLEETFENLILNQMNLSAYGEDLLDPNIDIEDYFD
(SEQ ID NO:115)

Horse
VLLGKYNAQGLGSDHEMLRCTKGQEYVKVVMQNGRMMGAVLIGETDLEETFENLILNQMNLSAYGEDLLDPDIDIEDYFD
(SEQ ID NO:116)

Horse D492H
VLLGKYNAQGLGSDHEMLRCTKGQEYVKVVMQNGRMMGAVLIGETDLEETFENLILNQMNLSAYGEDLLHPDIDIEDYFD
(SEQ ID NO:117)

Human	419	VLLGKYNAQGLGSDHEMLRCTKGREYIKVVMQNGRMMGAVLIGETDLE	469
		*****+*****	
Horse	421	VLLGKYNAQGLGSDHEMLRCTKGQEYVKVVMQNGRMMGAVLIGETDLE	470
Horse D492H	421	VLLGKYNAQGLGSDHEMLRCTKGQEYVKVVMQNGRMMGAVLIGETDLE	470
Human	469	ETFENLILNQMNLSAYGEDLLDPNIDIEDYFD	500
		*****+*****	
Horse	471	ETFENLILNQMNLSAYGEDLLDPDIDIEDYFD	502
Horse D492H	471	ETFENLILNQMNLSAYGEDLLHPDIDIEDYFD	502

FIG. 31



```
(SEQ ID NO:118)   TKGQEYIKVVMQNGRRMMGAVLIGETDLEETFENLILNQMNLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:119)   TKGQEYIVKVVQMONGRRMGAFLIGETDLEETFENLILNQMNLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:120)   TRGQEYVKVVMQNGRRMMGAVLIGETDLEETFENLILNQMDLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:121)   TKGQEYIVKVVQMONGRRMMGAVLIGETDLEETFENLILNQMDLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:122)   TKGQEYIKVVMQNGRRMMGAVLIGETDLEETFENLILNQMNLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:123)   TKGQEYVKVVMQNGRRMMGAVLIGETDLEETFENLILNQMNLSAYGEDLLDPDIDIEDYFD
(SEQ ID NO:124)   TKGQEYIKAVLQNGRRMMGAVLIGETDLEETFENLILNQMNLSAYGEDLLDPNIDIEDYFD
(SEQ ID NO:125)   TKGQEYVKVVMQNGRRMMGAVLIGETDLEETFENLILNQMDLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:126)   TKGREYIKVVMQNGRRMMGAVLIGETDLEETFENLILNQMNLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:127)   TKGQEYIKVVMQNGRRMMGAVLIGETDLEETFENLILNQMDLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:128)   TPQGQEYIKVVMHNGRRMMGAVLIGETDLEETFENLILNQMDLSYGEDLLNPDIIDIEDYFD
(SEQ ID NO:129)   TRGQEYIKVVMQNGRRMMGAVLIGETDLEETFENLILNQMDLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:130)   TRGQEYVKVVMQNGRRMMGAVLIGETDLEETFENLILNQMDLSYGEDLLDPDIDIEDYFD
(SEQ ID NO:131)   TKGQEYVKVVMQNGRRMMGAVLIGETDLEETFENLILNQMDLSPYGERLLDPNIDIEDYFD
(SEQ ID NO:132)   TKGQEYIKVIMQNGRRMMGAVLIGETDLEETFENLILNQMDLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:133)   TKGQEYVKVVMQHGRMMGAVLIGETDLEETFENLILNQDTLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:134)   TKGQEYIKVVMQNGRRMMGAVLIGETDLEETFENLILNQMDLSPYGEDLLDPNIDIEDYFD
(SEQ ID NO:135)   TKGQEYVKAVLQNGRRMMGAVLIGETDLEETFENLILNQMNLSAYGEDLLDPNIDIEDYFD
(SEQ ID NO:136)   TKGQEYVKVVMQNGRRMMGAVLIGETDLEETFENLILNQDTLSYGEDLLDPNIDIEDYFD
(SEQ ID NO:137)   TKGQEYVKVVMQNGRRMGALLIGETDLEETFENLILNQMNLSAYGEDLLDPDIDIEDYFD
(SEQ ID NO:138)   TKGQEYVKVVMQNGRRMMGAILIGETDLEETFENLILNQMNLSAYGEDLLDPNIDIEDYFD
(SEQ ID NO:139)   TKGQEYVKVVMQNGRRMGAILIGETDLEETFENLILNQMNLSAYGEDLLDPSIDIEDYFD
(SEQ ID NO:140)   TKGQEYVKVVMQNGRRMGAVLIGETDLEETFENLILNQMDLSAYGEDLLNPDIIDIEDYFD
(SEQ ID NO:141)   TKGHEYVKVVMQNGRRMMGAVLIGETDLEETFENLILNQMDLSAYGEDLLNPDIIDIEDYFD
(SEQ ID NO:142)   TKGQEYVKVVMQNGRRMVGAVLIGETDLEETFENLILNQMDLSAYGEDLLNPDIIDIEDYFD
(SEQ ID NO:143)   TKGQEYVKVVMQNGRRMGAVLIGETDLEETFENLILNQMDLSAYGEDLLNPDIIDIEDYFD
(SEQ ID NO:144)   TKGHEYVKVVMQNGRRMGAVLIGETDLEETFENLILNPMOLSA YGEDLLNPDIIDIEDYFD
(SEQ ID NO:145)   TKGHEYVKVVMQNGRRMGAVLIGETDLEETFENLILNRMOLSA YGEDLLNPDIIDIEDYFD
(SEQ ID NO:146)   TKGHEYIKVVMQNGRRMGLAVLIGETDLEETFENLILNQMDLSAYGEDLLNPDIIDIEDYFD
(SEQ ID NO:147)   TKGHEYVKVVMQNGRRMGAVLIGETDLEETFENLILNQIDL SA YGEDLLNPDIIDIEDYFD
(SEQ ID NO:148)   TKGQEYVKAVMQNGRRMVGAVLIGETDLEETFENLILNQMDLSAYGEDLLNPDIIDIEDYFD
(SEQ ID NO:149)   TKGQEYVKVVMQSGRRMVGAVLIGETDLEETFENLILNQMDLSAYGEDLLNPVDIIDIEDYFD
(SEQ ID NO:150)   TKGQEYVKVVMQNGRRMGAVLIGETDLEETFENLILNQMDLSRYGEDLLDPNIDIEDYFD
(SEQ ID NO:151)   TKGLEYVKVVMQNGRMLGAVLIGETDLEETFENLILNQMDLSRYGEDLLDPNIDIEDYFD
(SEQ ID NO:152)   TKGQEYVKVVMQNGRMLGAVLIGETDLEETFENLILNQMDLSRYGEDLLDPNIDIEDYFD
(SEQ ID NO:153)   TKGQEYIKVVMQNGRMLGAVLIGETDLEETFENLILNQMDLSRYGENLLDPNIDIEDYFD
(SEQ ID NO:154)   TKGQEYIKVVMQNGRMLGAVLIGETDLEETFENLILNQMDLSRYGEDLLDPNIDIEDYFD
(SEQ ID NO:155)   TKGQEYIKVVMQHGRMLGAVLIGETDLEETFENLILNQMDLSRYGEDLLDPNIDIEDYFD
(SEQ ID NO:156)   TKGQEYIKVVMQNGRMLGAVLIGETDLEETFENLILNQMDLSRYGKDLLDPNIDIEDYFD
(SEQ ID NO:157)   TKGQEYVKVVLQNGRRMGAILIGETDLEETFENLMNLQMDLSAYGEELLPDIIDIEDYFD
(SEQ ID NO:158)   TPQGQEYVKTVMQNGRRMGLVIGETDLEETFENLILNQMDLSAYGEDLLPNIDIEDYFD
(SEQ ID NO:159)   TKGHEYVKAVMQNGRRMGAVLIGETDLEETFENLILNQMDLSYGEELLNPNIIDIEDYFD
(SEQ ID NO:160)   TKGHEYVKVVLSSGRRMLGAVLIGETDLEETFENLILNQMDLT FYGEELLPNIIDIEDYFD
(SEQ ID NO:161)   TKGHEYVKLVMQGGRRMVGALLIGETDLEETFENLILNQDTLSYGERLLDPNIDIEDYFD
(SEQ ID NO:162)   TKGQEYVKVVLTPGRRMVGAVLIGETDLEETFENLILNQMDLSRYGEELLPNIIDIEDYFD
CLUSTAL          *  *  *:  :  *  *  *:*****:*  :  :  :  :  :  :  :  :  :  :  :
D492H                                                    H
```

FIG. 33

23,481,621

TTAGAATTCT GGGGTGGCTG GAGGTGTGTT TGCTCTCTAG CTSCAAAGAT
AAACGTTTGC AAGACCTGTC ATACGAAGAC CTGAGAAGTT TCCCTAAAAT
GTGGTCTCAG GGTACTTTAA GTTTGTTTTA AGAAAAGAGA AGAGTTCGTA
GCTTTTCTTG TTTGAGGCTC TATAAAACCT GTAAGTTTTT GTAGGTGACC
ATTGCGGGCA CCGGCTCTTT CTTGTCCATG AGGCCAGAGT AGCCTGGGGC
ATCGTGCCTC CTGGGGAGGA AGAAGCATTC CCCCCCTGTT GGGGCAGAGA
GTGGGTGCTG ACCGCATGGT CAGGGTCTGT CAGAGGCAGG TGGACGTTCT
GTGGATGGGA TGTGACTGAA GGGACCCGTC TCTTGAGCCT CACATTCATA
TCATCAGGAA TATTTTCTTG AGCAGCTGAT CCTGGGGCGG TGGCTTTTCT
CGCACATGTG GCTCATTTCC CTGACTTGTG TTGTTTTTCA GGTGACTCAG
GGCGGGACAG CCAGCAGAGA GGACCCAAAG GAGAAACCGG AGACCTCGGG
CCCATGGTAC GATGCTCCTT GTTCCCAGAC TCTCCTGTGC TGGAGGCCAG
AGAAACAAAG CTCACACTGG CTCGCCAGGG GCCGAAGGAC CATGGAAGTG
GTCTTTGAGG CTGGGGAGCA TATGCGGTCA GGAACAGAGG GCCCCGTGAA
GCCCCUCTG TGTGCCCATG AAGGACGTTT CCATTGTGGA ACAGAAGCAA
GCCCTACCGA GAGGAGCTTC TTCTGGGCAG AAGTTGTGCT CTTTAGATAT
TTACGAAAGC GCTTGTCAAG CCAGAGGCTG TGGGGGCGGG CACGTACCTA
AGTCTAGCGC ACCTCAGCAC AGAGTACCGA CCTTCCTCTG CCTATCCATC
CCTTTTCTTG TAGTGTCTCG TGCAATATTC CAGAGTTCTT GCTCTTCTTT
GCAGATGAAA ATCTTACACC AATTTTCTTC CCCACGCCAG GGTCTGCCTG

23,480,621

(G/C)
GAGTGTGCT GTCTCGGGAA GTCTTGGAGA ACCAGGGAGG GACGTGAGTG
TCTCTTACAG CTTTGGAGGG AGGTGGCGGG GCCTCTGTGC TGGTATTTAG
CAGGTGCCCT CCAGATCTTC CAGACACTGC CTGGGGCGTT GTTACCCAGC
AACTCTTTGT TTAAGSAATT TAAAGAGATT TCTGAAACCT TTGTCAACCA
AATCTTGATG CTTATGGAAA ACATTCGTCC TTATTGTTTG ATCTGTGCCC
CATCTGTGCE CGTCTCTTCT CCEACCCCECA GTCCCGCTTA CTCCCTTCCC
CAAAACTCAC CGAACAGACA CTTTTTAAAG TATTGTCTTC CCAGCTTTCA
AGTATCTGGC CAGGTTTTAT CCCACTAAAT GTTTAAAAAA ATCCCAACAG
AAACACGAAA ACGGGGGCGG GGGGGGGGGA TGAGGSAATA TGGGGCAGGG
AGGCTGCTGG GGTGTATAT TGTGGTTGC TTTTGAATY GTTTTTCTG
TGCTGCTGTA GCTCAAGCAA GGAAACGAGA CTCAGTTTTT TASTGAATCG
CTTTTGTAAA TCGCTTTCCC AAGGCATTTT AGAAATGCCC TTTGTAGATT
AGCTGATTGC ATGCTTAGTC TTTGAGTCTT TCTGGAAGAG CTGCTAAAGC
CAAGGGGATA AAGATCGTCT TTAGSAAGCC AGTGGATGTC TTCCAGGGCC
CAGGGGCAAG ATGCTCTGGA AGCTCATACT TCCTCTGGGG TGCCTCAGGA
TGCTGGGGGC GGGTGGCCAG GGCAGGTCTC TGCTCAGAAG CTGCCTGGGA
CTATGCAGCC TTGAACGGGG CCUCCTCAGC TGTCTCCCCA GTGGCAGCGA
CACATTAGG GATCAGCAGC TGGACAATCG GGGGGGTGTA TGCTGAGGTG
AGCGCTTGA AAAGAAGTAA CTCTCACCTG TAATCAGATG ATATATCATG
GGCCAAAGTT TGCTTTTAAA ATTCAGAAGA AGACTTAAAA TTTGAAAATG

23,479,621

(SEQ ID NO:163)

FIG. 34

6,356

```
GGACAAAGGG GAGACCGAGG GCCCATCGGC AGCATCGGGC CAAAGGGTGT
TCCCCGAGAA GACCGCTACC GAGGGTACCC CCGTGATGAG GGTGGGGCCG
GTGATCGGGG TCCACCGGGT GTGAACGGCA CTCAGGTTT CCAGGGCTGC
CCCGGGCAGA GAGGAATAAA GGGCTCTCGT GGATTCCCAG GAGAGAAGGG
TGAATTAGGA GAAATCGGAC TTGATGGTCT TGACGGTGAA GATGGAGACA
AAGGATTGUC TGGTCTTCT GGAGAGAAAG GGAATCCTGG AAGGAGGGGT
GACAAAGGAC CTAAAGGAGA CCAAGGGGAA AGAGGAGATG TTGGAATTAG
GGGCGACCCG GGTGACTCAG GCGGGACAG CCAGCAGAGA GGACCCAAAG
GAGAAACCGG AGACCTCGGG CCCATGGGTC TGCCTGGGAG TGATGGTGTG
TCGGGAAGTC CTGGAGAACC AGGGAGGGAC GGTGGCTTTG GCCGAAGGGG
ACCACCAGGA GCTAAGGGCA ACAAGGGCGG TCCTGGCCAG CAGGGCACCG
TGGGAGAGCA GGGGACCAGA GGTGCACAGG GTCCACCTGG TTCCACCGGT
CCTCCAGGGC TGATCGGCGA ACAAGGCATT CCTGGACCTC GGGGAAGCGG
AGGTGCTGTG GCGTCCCTG GAGAACGCGG CCGAACCGGT CCCTTGGGAA
GAAAGGGCGA GCCTGGAGAG CCGGGAGCGA AGGGAGGACT CGGGCCCCGG
GGCCCCCGTG GGGAAACGGG AGATGACGGG CGAGACGGAG TTGGCACTGA
AGGACAAAAA GGCAAAAAAG GAGAAAGAGG ATTCCCTGGA TACCCAGGTC
CAAAGGGTAC CCGTGGTGAG CCAGGGACAG ACGGAACACT AGGACCCAAA
GGCGTCAGAG GCCGAAGGGG AGACTCAGGA CCTCCAGGGG CAGCTGGACA
GAAGGGAGAC CCTGGTTACC CGGGACCATC TGGTCTCAA GGCAACAGAG
GCGACTCG
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7,363

(SEQ ID NO:164)

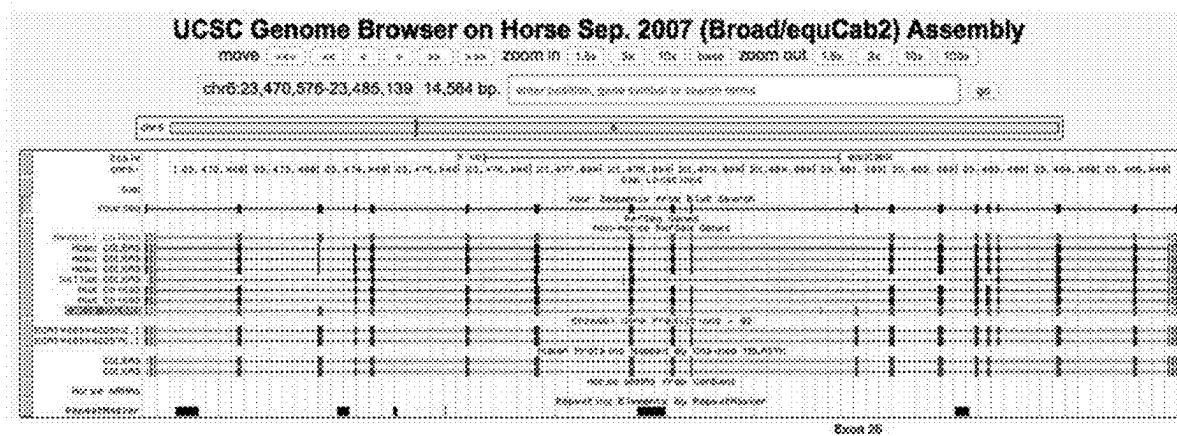
6,356

```
GGACAAAGGG GAGACCGAGG GCCCATCGGC AGCATCGGGC CAAAGGGTGT
TCCCCGAGAA GACCGCTACC GAGGGTACCC CCGTGATGAG GGTGGGGCCG
GTGATCGGGG TCCACCGGGT GTGAACGGCA CTCAGGTTT CCAGGGCTGC
CCCGGGCAGA GAGGAATAAA GGGCTCTCGT GGATTCCCAG GAGAGAAGGG
TGAATTAGGA GAAATCGGAC TTGATGGTCT TGACGGTGAA GATGGAGACA
AAGGATTGCC TGGTCTTCT GGAGAGAAAG GGAATCCTGG AAGGAGGGGT
GACAAAGGAC CTAAAGGAGA CCAAGGGGAA AGAGGAGATG TTGGAATTAG
GGGCGACCCG GGTGACTCAG GCGGGACAG CCAGCAGAGA GGACCCAAAG
GAGAAACCGG AGACCTCGGG CCCATGGGTC TGCCTGGGAG TGATGGTGTG
TCGGGAAGTC CTGGAGAACC AGGGAGGGAC GGTGGCTTTG GCCGAAGGGG
ACCACCAGGA GCTAAGGGCA ACAAGGGCGG TCCTGGCCAG CAGGGCACCG
TGGGAGAGCA GGGGACCAGA GGTGCACAGG GTCCACCTGG TTCCACCGGT
CCTCCAGGGC TGATCGGCGA ACAAGGCATT CCTGGACCTC GGGGAAGCGG
AGGTGCTGTG GCGTCCCTG GAGAACGCGG CCGAACCGGT CCCTTGGGAA
GAAAGGGCGA GCCTGGAGAG CCGGGAGCGA AGGGAGGACT CGGGCCCCGG
GGCCCCCGTG GGGAAACGGG AGATGACGGG CGAGACGGAG TTGGCACTGA
AGGACAAAAA GGCAAAAAAG GAGAAAGAGG ATTCCCTGGA TACCCAGGTC
CAAAGGGTAC CCGTGGTGAG CCAGGGACAG ACGGAACACT AGGACCCAAA
GGCGTCAGAG GCCGAAGGGG AGACTCAGGA CCTCCAGGGG CAGCTGGACA
GAAGGGAGAC CCTGGTTACC CGGGACCATC TGGTCTCAA GGCAACAGAG
GCGACTCG
```

7,363

(SEQ ID NO:165)

FIG. 35



Exon 26

23,480,641 cccacgcccagGGTCTGCCGGTCTGCCTGGGAGTGATGGTGTCTCGGGAAGTCTCTGGAGAACACAGGGAGGGAC
gtgagtgtct 23,480,568

(SEQ ID NO:156)

FIG. 36

2,037 GQRGDRGFIS SIGPKGVFGE DGYRGYPGDE GGPGRGRPPG VNGTQGFQGC
PGQRGINGSR GFPGEKGELG EIGLDGLDGE DGDKGLPGSS GEKGNPGRRG
DKGPKGDQGE RGDVGIRGDF GDSGRDSQQR GPKGETGDLG **PMGLPGSDGV**
SGSPGEPGRD **GGFGRRGPPG** **AKGNKGGPGQ** **QGTVGEQGTR** **GAQGPPGSTG**
PPGLIGEQQI PGPRGSGGAV GVPGERGRTG PLGRKGEPGE PGAKGGLGPR
GPRGETGDDG RDGVGSEGQK GKKGERGFPF YPGPKGTRGE PGTDGTLGPK
GVRGRRGDSG PPGAACQKGD PGYPGPGSLK GNRGDS 2,372

(SEQ ID NO:167)

2,037 GQRGDRGFIS SIGPKGVFGE DGYRGYPGDE GGPGRGRPPG VNGTQGFQGC
PGQRGINGSR GFPGEKGELG EIGLDGLDGE DGDKGLPGSS GEKGNPGRRG
DKGPKGDQGE RGDVGIRGDF GDSGRDSQQR GPKGETGDLG **PMGLPASDGV**
SGSPGEPGRD **GGFGRRGPPG** **AKGNKGGPGQ** **QGTVGEQGTR** **GAQGPPGSTG**
PPGLIGEQQI PGPRGSGGAV GVPGERGRTG PLGRKGEPGE PGAKGGLGPR
GPRGETGDDG RDGVGSEGQK GKKGERGFPF YPGPKGTRGE PGTDGTLGPK
GVRGRRGDSG PPGAACQKGD PGYPGPGSLK GNRGDS 2,372

(SEQ ID NO:168)

FIG. 37

23,481,081 5'-ttcccagac tctcctgtgc
AGACCTCGGG CCCATGGTAC GATGCTCCTT GTTCCCAGAC TCTCCTGTGC
t 3' (SEQ ID NO:171)
TGGAGGCCAG AGAAACAAAG CTCACACTGG CTCGCCAGGG GCCGAAGGAC
CATGGAAGTG GTCTTTGAGG CTGGGGAGCA TATGCCGTCA GGAACAGAGG
GCCCCGTGAA GCCCACCTG TGTGCCATG AAGGACGTTT CCATTGTGGA
ACAGAAGCAA GCCCTACCGA GAGGAGCTTC TTCTGGGCAG AAGTTGTGCT
CTTTAGATAT TTACGAAAGC GCTTGTCAAG CCAAAGGCTG TGGGGGCGGG
CACGTACCTA AGTCTAGCGC ACCTCAGGAC AGAGTACCGA CCTTCCTCTG
CCTATCEATC CCTTTTCCTG TAGTSTCTCG TGCAATATTC CAGAGTTCTT
GCTCTTCTTT GCAGATGAAA ATCTTACACC AATTTTCTTC CCCACGCCAG
23,480,631 GGT CTG CCT GGG AGT GAT GGT GTC TCG GGA AGT CCT GGA
23,480,631 GGT CTG CCT GCG AGT GAT GGT GTC TCG GGA AGT CCT GGA
GAA CCA GGG AGG GAC (SEQ ID NO:169) 23,480,578
GAA CCA GGG AGG GAC (SEQ ID NO:170) 23,480,578
GTGAGTGTCT CTTACAGCTT TGGAGGGAGG TGGCGGGGCT TCTGTGCTGG
TATTTAGCAG GTGCCCTCCA GATCTTCCAG AACTGCTG GGGCGTTGTT
ACCCAGCAAC TCTTTGTTTA AAGAATTTAA AGAGATTTCT GAAACCTTTG
TCACCCAAAT CTTGATGCTT ATGGAAAACA TTGTCCTTA TTSTTTGATC
(SEQ ID NO:172) 3'-caaaactag
TGTGCCCAT CTGTGCCCGT CTCTTCTCCC ACCCCCAGTC CCGCTTACTC 23,480,328
acacggggta ga-5'

FIG. 38

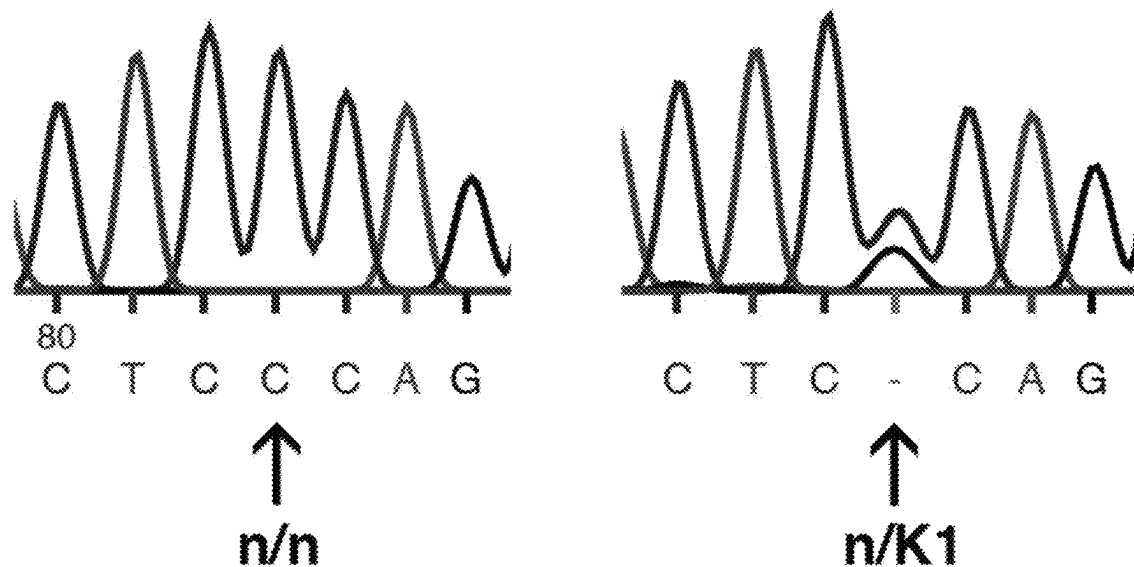


FIG. 39

6,397

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GGGCAGAGGG GAGACCGCGG GCCCATCGGC AGCATCGGGC CAAAGGGTAT
TCCTGAGAGAA GACGECTACC GAGGCTATCC TGGTGATGAG GGTGGACCCG
GTGAGCGTGG TCCGECTGGT GTGAACGGCA CTCAGGTTT CCAGGCGCTG
CCGGGCCAGA GAGGAGTAAA GGCTCTCGG GGATGCCAG GAGAGAAGGG
CGAASTAGGA GAAATYGGAC TGGATGGTCT GGATGGTSA GATGAGACA
AAGGATTGCC TGGTTCTTCT GGAGAGAAAG GGAATCCTGG AAGAAGGGGT
GATAAAGGAC CTCGAGGAGA GAAAGGAGAA AGAGGAGATG TTGGGATTGG
AGGGGACCCG GGTAAACCCG GACAAGACAG CCAGGAGAGA GGACCCAAAG
GAGAAACCGG TGACCTCGGC CCCATGGGTG TCCAGGGAG AGATGGAGTA
CCTGGAGGAC CTGAGAAAC TGGGAAGAAT GGTGGCTTTG GCCGAAGGGG
ACCCCCCGGA GCTAAGGGCA ACAAGGGCGG TCCTGGCCAG CCGGGCTTTG
AGGGAGAGCA GGGGACCAGA GGTGCACAGG GCCCAGCTGG TCCTGCTGGT
CCTCCAGGGE TGATAGGAGA ACAAGGCATT TCTGGACCTE GGGGAAGCGG
AGGTSCCGCT GGTGCTCTCG GAGAACEASG CAGAACCSTG CCATGSGAA
GAAAGGGTGA GCCCGGAGAG CCAGGACCAA AAGGAGGAAT CGGGAACCGG
GGCCCTCGTG GGGAGACGGG AGATGACGGG AGAGACGGAG TTGGCAGTGA
AGGACGCAGA GGCAGAAAAG GAGAAAGAGG ATTGCTGSA TACCCAGGAC
CAAAGGGTAA CCCAGGTGAA CCTGGGCTAA ATGGAACAAC AGGACCCAA
GGCATCAGAG GCCGAAGGGG AANTTCGGGA CCTCCAGGGA TAGTTGACA
GAAGGGAGAC CCTGGCTACC CAGGACCAGC TGGTCCCAAG GGCACAGGG
GCGACTCC
```

7,404

(SEQ ID NO:179)

FIG. 40

2,038 GQRGDRGPIG SIGFEGIPGE DGYRGYFGDE GGPGERGPPG VNQTQGFQGC
PQQBGVKGSR GFFGEKGEVG EIGLDGLDGE DGDKGLPGSS GEKGNPGRRG
DKGPRGENGE RGVVGIRGDF GNPGQDSQER GPKGETGDLG **PMGVPPGRDGV**
PGGFGETGKN GGFGRRGPPG AKGNKGGPGQ PGTEGEQGTG GAQGPAGPAG
PPGLIGEQTG SGPRGSGGAA GAPGERGRTG PLGRKGEFGE PGPKGGIGNR
GPRGETGDDG RDGVGSEGRR GKKGERGFPG YPGPKGNPGE PGLNGTTGPK
GIRGRRGNSG PPGIVGQKGD PGYFGPAGPK GNRGDS 2,373
(SEQ ID NO:174)

FIG. 41

human	2038	GQRGDRGPIGSIGFEGIPGEDGYRGYFGDEGGPGERGPPGVNQTQGFQCCPQQBGVKGSR	2097

horse	2037	GQRGDRGPISSIGPKGVPGEDGYRGYFGDEGGPGERGPPGVNQTQGFQCCPQQBGVKGSR	2096

human	2098	GFPGEKEGEVGIEIGLDGLDGEDGDKGLPGSSGEKGNPGRRGDKGPRGEKGERGDVGIRGDP	2157

horse	2097	GFPGEKGEIGEIGLDGLDGEDGDKGLPGSSGENPGRRGDKGPKGDQGERGDVGIRGDF	2156

human	2158	GNPGQDSQERGPGETGDLGPMGVPPGRDGVPPGGPGETGKNNGGFGRGPPGAKGNKGGPGQ	2217

horse	2157	GDSGRDSQQRGPKGETGDLGPMGLPGSDGVSGSPGERDGGFGRRGPPGAKGNKGGPGQ	2216
G2182A			
		A	
human	2218	PGTEGEQGTGCAQGPAGPAGPPGLIGEQTGSGPRGSGGAACAPGERGRTGPLGRKGEFGE	2277

horse	2217	QCTVGEQGTGCAQGPAGPAGPPGLIGEQTGIPGRGSGGAVGVPERGRTGPLGRKGEFGE	2276

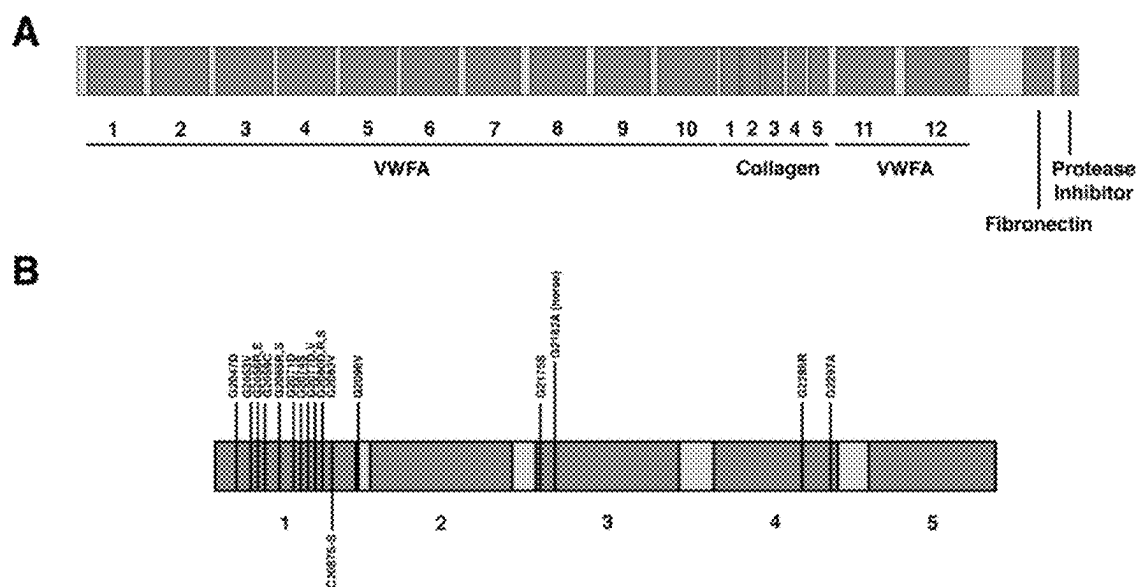
human	2278	PGPKGGIGNRGPRGETGDDGGRDGVGSEGRRGKKGERGFPGYFGPKGNPGEPPGLNGTTGPK	2337

horse	2277	PGAKGGLGPRGPRGETGDDGGRDGVGSEGRRGKKGERGFPGYFGPKGTRGEFGTDGTLGPK	2336

human	2338	GIRGRRGNSGPPGIVGQKGDGPGYFGPAGPKGNRGDS	2373 (SEQ ID NO:175)

horse	2337	GVRGRRGDSGPPGAAGQKGDGPGYFGPSGLKGNRGDS	2372 (SEQ ID NO:176)

FIG. 42



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FIG. 44



METHODS OF DETECTING INHERITED MYOPATHIES IN HORSES

CROSS-REFERENCE TO RELATED APPLICATION

This application is a United States national phase patent application based upon international patent application number PCT/US19/33717 of international filing date May 23, 2019, which claims the benefit U.S. Provisional Patent Application No. 62/717,072 filed Aug. 10, 2018 and U.S. Provisional Patent Application No. 62/737,295, filed Sep. 27, 2018, all three of which applications are incorporated herein by reference in its entirety.

SEQUENCE LISTING

This application contains a Sequence Listing electronically submitted via EFS-Web to the United States Patent and Trademark Office as an ASCII text file entitled "0310_000142WO01_SL" having a size of 223 kilobytes and created on Apr. 30, 2019. The information contained in the Sequence Listing is incorporated by reference herein.

SUMMARY

This disclosure describes, in one aspect, methods for detecting the presence or absence of a set of biomarkers in a horse. Generally, the method includes obtaining a biological sample from a horse that includes a nucleic acid that includes the coding region for three genes: dysferlin (DYSF), pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1), and collagen type VI alpha 3 chain (COL6A3). There are two important biomarkers in DYSF, one important biomarker in PYROXD1, and one important biomarker in COL6A3, to be assayed as follows: (1) Dysferlin (DYSF), determining whether the nucleic acid has the specific substitution of an adenine (A) for a guanine (G) on the forward strand at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 1, or the equivalent substitution in the complement thereof. This base substitution, known as rs1145077095 in dbSNP, corresponding to position 31,306,949 in SEQ ID NO:1, results in a nonconservative amino acid substitution in the dysferlin (DYSF) protein. The amino acid substitution caused by this base substitution is shown in FIG. 4. FIG. 4 shows the partial sequence of an altered dysferlin with tryptophan (W) substituted for arginine (R) at position 253, with SEQ ID NO:10 showing the partial protein sequence encoded by the wild-type or common allele and SEQ ID NO:11 showing the partial protein sequence encoded by the variant. (2) Dysferlin (DYSF), determining whether the nucleic acid has the specific substitution of a thymine (T) for a guanine (G) on the forward strand at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 13, or the equivalent substitution in the complement thereof. This base substitution, known as rs1136366555 in dbSNP, corresponding to position 31,225,630 in SEQ ID NO:56, results in a nonconservative amino acid substitution in the dysferlin (DYSF) protein. The amino acid substitution caused by this base substitution is shown in FIG. 16. FIG. 16 shows the partial sequence of an altered dysferlin with threonine (T) substituted for proline (P) at position 1290, with SEQ ID NO:60 showing the partial protein sequence

encoded by the wild-type or common allele and SEQ ID NO:61 showing the partial protein sequence encoded by the variant. (3) Pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1), determining whether the nucleic acid has the specific substitution of a cytosine (C) for a guanine (G) on the forward strand at chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 22, or the equivalent substitution in the complement thereof. This base substitution, known as rs1136260157 in dbSNP, corresponding to position 47,661,977 in SEQ ID NO:103, results in a non-conservative amino acid substitution in the pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) protein. The amino acid substitution caused by this base substitution is shown in FIG. 25. FIG. 25 shows an altered pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1), with histidine (H) substituted for aspartate (D) at position 492, with SEQ ID NO:107 showing the protein sequence encoded by the wild-type or common allele and SEQ ID NO:108 showing the protein sequence encoded by the variant. (4) Collagen type VI alpha 3 chain (COL6A3), determining whether the nucleic acid has the specific substitution of a guanine (G) for a cytosine (C) on the forward strand at chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 33, or the equivalent substitution in the complement thereof. This base substitution, known as rs1139437410 in dbSNP, corresponding to position 23,480,621 in SEQ ID NO:163, results in a nonconservative amino acid substitution in the collagen type VI alpha 3 chain (COL6A3) protein. The amino acid substitution caused by this base substitution is shown in FIG. 36. FIG. 36 shows an altered collagen type VI alpha 3 chain (COL6A3), with alanine (A) substituted for glycine (G) at position 2182, with SEQ ID NO:167 showing the protein sequence encoded by the wild-type or common allele and SEQ ID NO:168 showing the protein sequence encoded by the variant.

In some embodiments, the method further includes amplifying at least a portion of the DYSF, PYROXD1, or COL6A3 coding regions. In some of these embodiments, all or part of (1) Exon B of the DYSF coding region as identified in FIG. 3 (SEQ ID NO:5) is amplified. This specified exon corresponds to the gene models presented in FIG. 2 in this disclosure; the specific base substitutions detected are presented in FIG. 1, even if alternative gene models or different isoforms result in this exon being named or numbered differently. In another aspect, this disclosure describes a method for detecting the presence or absence of a biomarker in a physiological sample. Generally, the method includes obtaining a physiological sample from a horse that includes a nucleic acid that includes at least a portion of SEQ ID NO:1 that includes nucleotide 31,306,949 of SEQ ID NO:1, determining whether the nucleic acid has an adenine (A) substituted for a guanine (G) at nucleotide 31,306,949 on the forward strand of SEQ ID NO:1; this single-nucleotide polymorphism (SNP) is also known as rs1145077095 in dbSNP. (2) Exon I of the DYSF coding region as identified in FIG. 15 (SEQ ID NO:59) is amplified. This specified exon corresponds to the gene models presented in FIG. 14 in this disclosure; the specific base substitutions detected are presented in FIG. 13, even if alternative gene models or different isoforms result in this exon being named or numbered differently. In another aspect, this disclosure describes a method for detecting the presence or absence of a biomarker in a physiological

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sample. Generally, the method includes obtaining a physiological sample from a horse that includes a nucleic acid that includes at least a portion of SEQ ID NO:56 that includes nucleotide 31,225,630 of SEQ ID NO:56, determining whether the nucleic acid has an thymine (T) substituted for a guanine (G) at nucleotide 31,225,630 of the forward strand of SEQ ID NO:56; this single-nucleotide polymorphism (SNP) is also known as rs1136366555 in dbSNP. (3) Exon 12 of the PYROXD1 coding region, as identified in FIG. 24 (SEQ ID NO:106) is amplified. This specified exon corresponds to the gene models presented in FIG. 23 in this disclosure; the specific base substitutions detected are presented in FIG. 22, even if alternative gene models or different isoforms result in this exon being named or numbered differently. In another aspect, this disclosure describes a method for detecting the presence or absence of a biomarker in a physiological sample. Generally, the method includes obtaining a physiological sample from a horse that includes a nucleic acid that includes at least a portion of SEQ ID NO:103 that includes nucleotide 47,661,977 of SEQ ID NO:103, determining whether the nucleic acid has a cytosine (C) substituted for a guanine (G) at nucleotide 47,661,977 of the forward strand of SEQ ID NO:103; this single nucleotide polymorphism (SNP) is also known as rs1136260157 in dbSNP. (4) Exon 26 of the COL6A3 coding region, as identified in FIG. 35 (SEQ ID NO:166) is amplified. This specified exon corresponds to the gene models presented in FIG. 34 in this disclosure; the specific base substitutions detected are presented in FIG. 33, even if alternative gene models or different isoforms result in this exon being named or numbered differently. In another aspect, this disclosure describes a method for detecting the presence or absence of a biomarker in a physiological sample. Generally, the method includes obtaining a physiological sample from a horse that includes a nucleic acid that includes at least a portion of SEQ ID NO:163 that includes nucleotide 23,480,621 of SEQ ID NO:163, determining whether the nucleic acid has a guanine (G) substituted for a cytosine (C) at nucleotide 23,480,621 of the forward strand of SEQ ID NO:163; this single nucleotide polymorphism (SNP) is also known as rs1139437410 in dbSNP.

In all cases, the nucleotide at the specified position of the forward strand may be inferred by the determination of the nucleotide at the specified position on the reverse (complementary) strand. In some embodiments, the method further includes amplifying at least a portion of the nucleic acid.

In another aspect, this disclosure describes a method for detecting the presence or absence of a biomarker in a physiological sample. Generally, the method includes obtaining a physiological sample from a horse that includes a nucleic acid encoding (1) Dysferlin polypeptide, then determining whether the nucleic acid encodes a dysferlin polypeptide altered as described as follows: a dysferlin polypeptide having, in part, the amino acid sequence of SEQ ID NO:10 (FIG. 4) or a dysferlin polypeptide having a tryptophan (W) substituted for arginine (R) at position 253 as shown in SEQ ID NO:11 (FIG. 4). (2) Dysferlin polypeptide, then determining whether the nucleic acid encodes a dysferlin polypeptide altered as described as follows: a dysferlin polypeptide having, in part, the amino acid sequence of SEQ ID NO:60 (FIG. 16) or a dysferlin polypeptide having a threonine (T) substituted for proline (P) at position 1290 as shown in SEQ ID NO:61 (FIG. 16). (3) Pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) polypeptide, then determining whether the nucleic acid encodes a pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1

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polypeptide altered as described as follows: a pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 polypeptide having, in part, the amino acid sequence of SEQ ID NO:107 (FIG. 25) or a pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 polypeptide having a histidine (H) for an aspartate (D) at position 492 as shown in SEQ ID NO:108 (FIG. 25). (4) Collagen type VI alpha 3 chain (COL6A3) polypeptide, then determining whether the nucleic acid encodes a collagen type VI alpha 3 chain polypeptide altered as described as follows: a collagen type VI alpha 3 chain polypeptide having, in part, the amino acid sequence of SEQ ID NO:167 (FIG. 36) or a collagen type VI alpha 3 chain polypeptide having an alanine (A) for a glycine (G) at position 2182 as shown in SEQ ID NO:168 (FIG. 36).

The above summary is not intended to describe each disclosed embodiment or every implementation. The description that follows more particularly exemplifies illustrative embodiments. In several places throughout the application, guidance is provided through lists of examples, which examples can be used in various combinations. In each instance, the recited list serves only as a representative group and should not be interpreted as an exclusive list.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1. A portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr15:31,306,949 position, the site of a substitution of an adenine (A) for a guanine (G) on the forward strand that results in the substitution of a tryptophan (W) for arginine (R) at amino acid position 253 in dysferlin as shown in FIG. 4 (SEQ ID NO:11). The reverse complement sequence is shown, with the site of a substitution of a thymine (T) for a cytosine (C) as indicated (SEQ ID NO:1). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1145077095 in dbSNP.

FIG. 2. A portion of the normal equine DYSF Coding DNA Sequence (SEQ ID NO:2) and a portion of the mutant DYSF Coding DNA Sequence (SEQ ID NO:3) bearing the C to T mutation at nucleotide position at position 1027 in this figure, corresponding to chr15:31,306,949 as shown in SEQ ID NO:1 (FIG. 1). This sequence is a region of perfect consensus among 21 different experimentally predicted mRNA isoforms. The numbering in FIG. 2 is that of isoform X1 (XM_023618907.1), which for this segment perfectly matches the numbering of isoforms X2 (XM_023618908.1), X3 (XM_023618909.1), X4 (XM_023618910.1), X5 (XM_023618911.1), and X6 (XM_023618912.1). The numbering of the start and end positions for other isoforms is described in detail in the text. In both sequences, the sequence of Exon B as shown in FIG. 3 is indicated in bold. The site of a C to T mutation at nucleotide position 1027, corresponding to 31,306,949 in SEQ ID NO:1 (FIG. 1), and to rs1145077095 in dbSNP, is underlined. The region of sequence comprising Exon B as shown in FIG. 3 is displayed as codons in the correct reading frame for both SEQ ID NO:2 and SEQ ID NO: 3 in FIG. 5.

FIG. 3. A view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the partial DYSF Coding DNA Sequence (SEQ ID NO:2) and the mutant partial DYSF Coding DNA Sequence (SEQ ID NO:3). The DYSF Coding DNA Sequences correspond to Exons A, B, C, D, E, and F as indicated by the regions of sequence similarity of the translated genomic DNA to DYSF protein

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sequences from human (*Homo*), cattle (*Bos*), rat (*Rattus*), mouse (*Mus*), and DYSF protein sequences from the more distantly-related zebrafish (*Danio*) and African claw-toed frog (*Xenopus*). Partial matches to the paralogous protein myoferlin (MYOF) from human (*Homo*), cattle (*Bos*), mouse (*Mus*), rat (*Rattus*), and zebrafish (*Danio*) are also seen. The sequences of Exons A-F with 10 bp of flanking intron sequence and their coordinates in the current horse assembly are displayed below the image from the UCSC Genome Browser. Sequence IDs are Exon A (SEQ ID NO:4), Exon B (SEQ ID NO:5), Exon C (SEQ ID NO:6), Exon D (SEQ ID NO:7), Exon E (SEQ ID NO:8), and Exon F (SEQ ID NO:9).

FIG. 4. Models of part of the normal protein sequence encoded by horse DYSF (XP_023474694.1, presented here as SEQ ID NO:10) corresponding to a translation of SEQ ID NO:2 shown in FIG. 2 and part of the altered protein sequence encoded by horse DYSF with the base substitution at chr15:31,306,949 (based on XP_023474694.1, presented here as SEQ ID NO:11) corresponding to a translation of SEQ ID NO:3 shown in FIG. 2. The portion of the protein encoded by Exon B as shown in FIG. 3 is indicated in bold, while amino acid position 253 affected by the base substitution of an adenine (A) for a guanine (G) on the forward strand at the chr15:31,306,949 position, corresponding to rs1145077095 in dbSNP as shown in FIG. 1, is underlined. The amino acid positions in XP_023474694.1 are indicated at the beginning and end of the sequence.

FIG. 5. Horse DYSF Exon B and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the DYSF-R253W mutation would be most appropriately derived. Genomic coordinates are as in FIG. 1. Exon B from chr15:31,307,036 to chr15:31,306,908 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:12) and the DYSF-R253W allele (SEQ ID NO:13). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to A mutation site at nucleotide position chr15:31,306,949 on the forward strand, corresponding to rs1145077095 in dbSNP as shown in FIG. 1 (C to T in the reverse complement as shown), is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for an arginine (CGG) to one coding for a tryptophan (TGG). Example primers used experimentally to amplify genomic DNA containing the mutation site are shown in lower case [5'-CCCAGATTCTGGCTTCT-3' (SEQ ID NO:14) and 5'-CTCGACAAGTTCTGGGTGT-3' (SEQ ID NO:15)].

FIG. 6. Traces from Sanger DNA sequencing of amplified DYSF genomic DNA using primers shown in FIG. 5 (SEQ ID NO:14 and SEQ ID NO:15). The sequence of the reverse strand is shown. The arrows in the figure indicate nucleotide position chr15:31,306,949, the site of a substitution of a thymine (T) for a cytosine (C) in this position, corresponding to rs1145077095 in dbSNP, that creates the DYSF-R253W variant. The traces show, from left to right, results for a horse homozygous for the wild-type or common allele (N/N), results for a horse heterozygous for the substitution (N/P5), and results for a horse homozygous for the substitution (P5/P5).

FIG. 7. Sequence of the human DYSF coding sequence derived from NM_003494.3 (SEQ ID NO:16). The 5' UTR and 3' UTR have been removed; the sequence begins with the ATG start codon and ends with the TGA stop codon. The numbering of the first and last nucleotides corresponds to

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that of NM_003494.3. The sequence of the human exon corresponding to Exon B in horse as shown in FIG. 3 is indicated in bold.

FIG. 8. Sequence of the human DYSF protein sequence, equivalent to NP_003485.1 (SEQ ID NO:17). The numbering of the first and last amino acids corresponds to that of NP_003485.1. The sequence encoded by the human exon corresponding to Exon B in horse as shown in FIG. 3 is indicated in bold.

FIG. 9. Comparison of that portion of the protein sequence of DYSF encoded by horse Exon B from wild type (XP_023474694.1, shown here as SEQ ID NO:19) and DYSF-R253W (R253W, shown here as SEQ ID NO:20) to the protein sequence of DYSF encoded by human Exon 7 (NP_003485.1, shown here as SEQ ID NO:18). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a space () indicates the nonconservative substitution of an arginine (R) in horse for a glycine (G) in human at human position 247, and a plus sign (+) indicates the conservative substitution of an arginine (R) in horse for a lysine (K) in human at human position 256. No other amino acid substitutions are seen in comparison of the human sequence (SEQ ID NO:18) and the wild-type horse sequence (SEQ ID NO:19). The sequence from horse bearing the DYSF-R253W mutation (SEQ ID NO:20) has a nonconservative substitution of a tryptophan (W) for an arginine (R) that is found in human (SEQ ID NO:18) and the wild-type horse sequence (SEQ ID NO:19). This position, corresponding to position 253 in horse (SEQ ID NO:19), SEQ ID NO:20) and position 251 in human (SEQ ID NO:18) is indicated by bold and underlining.

FIG. 10. Features of the dysferlin protein encoded by the human DYSF gene. (A) The canonical human isoform has 2080 amino acids. The protein contains seven C2 calcium-binding domains designated C2A through C2G, shaded in gray. The secondary structure of the protein sequence encoded by each of the C2 domains consists of eight segments assembling into beta sheet, with two different topologies described. Domains C2C, C2D, C2F, and C2G are designated as topology type I, while domains C2A, C2B, and C2E are designated as topology type II (C. Therrien et al. 2006 J. Neurological Sciences 250: 71-78). The positions of four additional conserved domains (DysfN, DysfC, annexin binding, and transmembrane) are also indicated (C. Therrien et al. 2006 J. Neurological Sciences 250: 71-78), also shaded in gray. (B) topology type II C2 calcium-binding domains C2B, C2A, and C2E are shown expanded, with positions of pathogenic missense alleles listed in TABLE 1 (C2B) and TABLE 2 (C2A and C2E) shown. The DYSF-R253W substitution in the horse C2B domain precisely corresponds to the DYSF-R251W substitution in human.

FIG. 11. Amino acid sequences of proteins encoded by the C2B domain of DYSF, including the position of the equine DYSF-R253W substitution. Species included in the analysis are described in the text. The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated by a period (.), and nonconserved positions are indicated by a blank space (). The last line shows the position of the DYSF-R253W substitution in horse in bold. The position of the DYSF-R253W substitution is indicated in bold in all of the aligned sequences.

FIG. 12. Sequence comparison of the C2A, C2B, and C2E domains of dysferlin encoded by human DYSF to the C2B domain of dysferlin encoded by horse DYSF. (A) The amino acid sequences of two isoforms of the C2A domain of human dysferlin (SEQ ID NO:51 and SEQ ID NO:52), the C2B domain of human dysferlin (SEQ ID NO:53), the C2E domain of human dysferlin (SEQ ID NO:54), and the C2B domain of horse dysferlin (SEQ ID NO:55), are shown. (B) Clustal Omega was used to align these four sequences. The fifth line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated by period (.), and nonconserved positions are indicated by a blank space (). The remaining lines show the position of the horse DYSF-R253W substitution and various pathogenic human substitutions (presented in TABLE 1 and TABLE 2) aligned to the consensus.

FIG. 13. A portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr15:31,225,630 position, the site of a substitution of a thymine (T) for a guanine (G) on the forward strand that results in the substitution of a threonine (T) for proline (P) at amino acid position 1290 in dysferlin as shown in FIG. 16 (SEQ ID NO:61). The reverse complement sequence is shown, with the site of a substitution of an adenine (A) for a cytosine (C) as indicated (SEQ ID NO:56). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1136366555 in dbSNP.

FIG. 14. A portion of the normal equine DYSF Coding DNA Sequence (SEQ ID NO:57) and a portion of the mutant DYSF Coding DNA Sequence (SEQ ID NO:58) bearing the C to A mutation at nucleotide position at position 4174 in this figure, corresponding to chr15:31,225,630 as shown in SEQ ID NO:56 (FIG. 13). [This sequence is a region of perfect consensus among 21 different experimentally predicted mRNA isoforms. The numbering in FIG. 2 is that of isoform X1 (XM_023618907.1), which for this segment perfectly matches the numbering of isoforms X2 (XM_023618908.1), X3 (XM_023618909.1), X4 (XM_023618910.1), X5 (XM_023618911.1), and X6 (XM_023618912.1). The numbering of the start and end positions for other isoforms is described in detail in the text.] In both sequences, the sequence of Exon I as shown in FIG. 15 is indicated in bold. The site of a C to A mutation at nucleotide position 4174, corresponding to 31,225,630 in SEQ ID NO:56 (FIG. 13), and to rs1136366555 in dbSNP, is underlined. The region of sequence comprising Exon I as shown in FIG. 15 is displayed as codons in the correct reading frame for both SEQ ID NO:57 and SEQ ID NO: 58 in FIG. 17.

FIG. 15. A view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the partial DYSF Coding DNA Sequence (SEQ ID NO:57) and the mutant partial DYSF Coding DNA Sequence (SEQ ID NO:58). The DYSF Coding DNA Sequences correspond to Exons G, H, and I as indicated by the regions of sequence similarity of the translated genomic DNA to DYSF protein sequences from human (Homo), orangutan (Pongo), cattle (Bos), mouse (Mus), rat (Rattus) and DYSF protein sequences from the more distantly-related zebrafish (Danio). The sequence of Exon I with 10 bp of flanking intron sequence

and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO: 59).

FIG. 16. Models of part of the normal protein sequence encoded by horse DYSF (XP_023474694.1, presented here as SEQ ID NO:60) corresponding to a translation of SEQ ID NO:57 shown in FIG. 14 and part of the altered protein sequence encoded by horse DYSF with the base substitution at chr15:31,225,630 (based on XP_023474694.1, presented here as SEQ ID NO:61) corresponding to a translation of SEQ ID NO:58 shown in FIG. 14. The portion of the protein encoded by Exon I as shown in FIG. 15 is indicated in bold, while amino acid position 1290 affected by the base substitution of a thymine (T) for a guanine (G) at the chr15:31, 225,630 position on the forward strand, corresponding to rs1136366555 in dbSNP as shown in FIG. 13 (C to A in the reverse complement as shown), is underlined. The amino acid positions in XP_023474694.1 are indicated at the beginning and end of the sequence.

FIG. 17. Horse DYSF Exon I and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the DYSF-P1290T mutation would be most appropriately derived. Genomic coordinates are as in FIG. 13. Exon I from chr15:31,225,648 to chr15: 31,225,619 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:62) and the DYSF-P1290T allele (SEQ ID NO:63). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to T mutation site at nucleotide position chr15:31,225,630 on the forward strand, corresponding to rs1136366555 in dbSNP as shown in FIG. 13 (C to A in the reverse complement as shown), is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for a proline (CCT) to one coding for a threonine (ACT). Example primers used experimentally to amplify genomic DNA containing the mutation site are shown in lower case [5'-GGTTGCAAACCTCCCAACTGT-3' (SEQ ID NO:64) and 5'-GATTTTCAAGCTGCCGAAG-3' (SEQ ID NO:65)].

FIG. 18. Traces from Sanger DNA sequencing of amplified DYSF genomic DNA using primers shown in FIG. 17 (SEQ ID NO:64 and SEQ ID NO:65). The sequence of the forward strand is shown. The arrows in the figure indicate nucleotide position chr15:31,225,630, the site of a substitution of an thymine (T) for a guanine (G) in this position, corresponding to rs1136366555 in dbSNP, that creates the DYSF-P1290T variant. The traces show, from left to right, results for a horse homozygous for the wild-type or common allele (N/N), results for a horse heterozygous for the substitution (N/P6), and results for a horse homozygous for the substitution (P6/P6).

FIG. 19. Comparison of that portion of the protein sequence of DYSF encoded by horse Exon I from wild type (XP_023474694.1, shown here as SEQ ID NO:67) and DYSF-P1290T (P1290T, shown here as SEQ ID NO:68) to the protein sequence of DYSF encoded by human (NP_003485.1, shown here as SEQ ID NO:66). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a space () indicates the nonconservative substitution of a glycine (G) in horse for an arginine (R) in human at human position 1297, and a plus sign (+) indicates the conservative substitution of an alanine (A) in horse for a serine (S) in human at human position 1267, a tyrosine (Y) in horse for a histidine (H) in human position 1285, an aspartic acid (D) in horse for a glutamic acid (E) in human

position 1294, and a glutamic acid (E) in horse for an aspartic acid (D) in human position 1300. No other amino acid substitutions are seen in comparison of the human sequence (SEQ ID NO:66) and the wild-type horse sequence (SEQ ID NO:67). The sequence from horse bearing the DYSF-P1290T mutation (SEQ ID NO:68) has a nonconservative substitution of a threonine (T) for a proline (P) that is found in human (SEQ ID NO:66) and the wild-type horse sequence (SEQ ID NO:67). This position, corresponding to position 1290 in horse (SEQ ID NO:67, SEQ ID NO:68) and position 1288 in human (SEQ ID NO:66) is indicated by bold and underlining.

FIG. 20. Features of the dysferlin protein encoded by the human DYSF gene. (A) The canonical human isoform has 2080 amino acids. The protein contains seven C2 calcium-binding domains designated C2A through C2G, shaded in gray. The positions of four additional conserved domains (DysfN, DysfC, annexin binding, and transmembrane) are also indicated (C.Therrien et al. 2006 *J. Neurological Sciences* 250: 71-78), also shaded in gray. The interdomain region between the C2D and C2E domains, affected by the horse DYSF-P1290T mutation, is indicated in light gray. (B) The interdomain region between the C2D and C2E domains is shown expanded, with positions of pathogenic and potentially pathogenic missense alleles listed in TABLE 3 shown. The horse DYSF-P1290T substitution corresponds to a proline at position 1288 in human, with no pathogenic or potentially pathogenic allele identified at that position in human.

FIG. 21. Amino acid sequences of proteins encoded by the C2D-C2E interdomain region of DYSF, including the position of the equine DYSF-P1290T substitution. Species included in the analysis are described in the text. The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated by a period (.), and nonconserved positions are indicated by a blank space (). The last line shows the position of the DYSF-P1290T substitution in horse in bold. The position of the DYSF-P1290T substitution is indicated in bold in all of the aligned sequences.

FIG. 22. A portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr6:47,661,977 position, the site of a substitution of a cytosine (C) for a guanine (G) on the forward strand that results in the substitution of a histidine (H) for an aspartate (D) at amino acid position 492 in pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) as shown in FIG. 25 (SEQ ID NO:108). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1136260157 in dbSNP.

FIG. 23. A portion of the normal equine PYROXD1 Coding DNA Sequence (SEQ ID NO:104) and a portion of the mutant PYROXD1 Coding DNA Sequence (SEQ ID NO:105) bearing the G to C mutation at nucleotide position 1548 in this figure, corresponding to chr6:47,661,977 as shown in SEQ ID NO:103 (FIG. 22). This sequence is a region of perfect consensus among five different experimentally predicted mRNA isoforms. The numbering in FIG. 23 is that of isoform X1 (XM_001502130.5). The numbering of the start and end positions for other isoforms is described in detail in the text. In both sequences the sequence of Exon 12 as shown in FIG. 24 is indicated in bold. The site of a G to C mutation at nucleotide position 1548, corresponding to

47,661,977 in SEQ ID NO:103 (FIG. 22), and to rs1136260157 in dbSNP, is underlined. The region of sequence comprising Exon 12 as shown in FIG. 24 is displayed as codons in the correct reading frame for both SEQ ID NO:109 and SEQ ID NO:110 in FIG. 26.

FIG. 24. A view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the PYROXD1 Coding DNA Sequence (SEQ ID NO:104) and the mutant PYROXD1 Coding DNA Sequence (SEQ ID NO:105) in a BLAT search. The PYROXD1 Coding DNA Sequences in horse correspond to PYROXD1 coding sequences in other species as indicated by the regions of sequence similarity of the translated genomic DNA to PYROXD1 protein sequences from human (Homo), orangutan (Pongo), cattle (Bos), mouse (Mus), rat (Rattus), African clawed frog (Xenopus), zebrafish (Danio), chicken (Gallus), and fruit fly (Drosophila). Exon 12, which contains the guanine (G) to cytosine (C) variant at chr6:47,661,977 as shown in SEQ ID NO:103 (FIG. 22), is indicated below the image of the browser window. The sequence of horse Exon 12 with 10 nucleotides of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO:106).

FIG. 25. Models of part of the normal protein sequence encoded by horse PYROXD1 (XP_001502180.3, presented here as SEQ ID NO:107) corresponding to a translation of SEQ ID NO:104 shown in FIG. 23 and part of the altered protein sequence encoded by horse PYROXD1 (adapted from XP_001502180.3, presented here as SEQ ID NO:108) with the base substitution at chr6:47,661,977 corresponding to a translation of SEQ ID NO:105 shown in FIG. 23. The portion of the protein encoded by Exon 12 is indicated in bold, while amino acid position 492 affected by the base substitution of a cytosine (C) for a guanine (G) at the chr6:47,661,977 position on the forward strand as shown in FIG. 22, is underlined. The amino acid positions in XP_001502180.3 are indicated at the beginning and end of the sequence.

FIG. 26. Horse PYROXD1 Exon 12 and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the PYROXD1-D492H mutation would be most appropriately derived. Genomic coordinates are as in FIG. 22. Exon 12 from chr6:47,661,764 to chr6:47,662,012 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:109) and the PYROXD1-D492H allele (SEQ ID NO:110). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to C mutation site at nucleotide position chr6:47,661,977, as shown in FIG. 22 is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for an aspartate (GAT) to one coding for a histidine (CAT). Example primers used experimentally to amplify genomic DNA containing the mutation site [5'-CAGAT-TTTCTGCTGGCCATT-3' (SEQ ID NO:111) and 5'-TGGT-CATCATTAATCAGTGCAA-3' (SEQ ID NO:112)] are shown in lower case in the figure.

FIG. 27. Traces from Sanger DNA sequencing of amplified PYROXD1 genomic DNA using primers shown in FIG. 26 (SEQ ID NO:111 and SEQ ID NO:112). The sequence of the forward strand is shown. The arrows in the figure indicate nucleotide position chr6:47,661,977, the site of a substitution of a cytosine (C) for a guanine (G) in this position that creates the PYROXD1-D492H variant. The traces show, from left to right, results for a horse homozy-

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gous for the wild-type or common allele (n/n), results for a horse heterozygous for the substitution (n/P8), and results for a horse homozygous for the substitution (P8/P8).

FIG. 28. Partial sequence of the human PYROXD1 coding sequence derived from NM_024854.4 (SEQ ID NO:113). This sequence is a region of perfect consensus among three different experimentally predicted mRNA isoforms. The numbering of the first and last nucleotides corresponds to that of NM_024854.4. The numbering of the start and end positions for other isoforms is described in the text. The sequence of Exon 12 is indicated in bold. The sequence begins with beginning of the consensus among the three isoforms and ends with the TAA stop codon.

FIG. 29. Partial sequence of the human PYROXD1 protein sequence, showing a translation of SEQ ID NO:113, equivalent to NP_079130.2 (SEQ ID NO:66). The numbering of the first and last amino acids corresponds to that of NP_079130.2.

FIG. 30. Comparison of that portion of the protein sequence of PYROXD1 encoded by horse Exon 12 from wild type (XP_001502180.3, shown here as SEQ ID NO:116) and PYROXD1-D492H (D492H, derived from XP_001502180.3 and shown here as SEQ ID NO:117) to the protein sequence of PYROXD1 encoded by human Exon 12 (NP_079130.2, shown here as SEQ ID NO:115). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a plus sign (+) indicates the following conservative substitutions: a glutamine (Q) for an arginine (R) at human position 444, a valine (V) for an isoleucine (I) at human position 447, an alanine (A) for a serine (S) at human position 483, and an aspartate (D) for an asparagine (N) at human position 492. The sequence from horse bearing the PYROXD1-D492H mutation (SEQ ID NO:117) has a nonconservative substitution of a histidine (H) for an aspartate (D) at horse position 492, corresponding to human position 490. This position is indicated in bold for all three sequences. The wild-type horse sequence (SEQ ID NO:116) matches the human sequence (SEQ ID NO:115) at this position.

FIG. 31. Features of the pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 protein encoded by the human PYROXD1 gene. The human protein has 500 amino acids and two domains: the pyridine nucleotide-disulfide oxidoreductase domain (amino acids 39-361) and the NADH-dependent nitrite reductase domain (447-494) as described in O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105. Positions of two human pathogenic missense alleles (N1555 and Q372H, O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105 DOI: 10.1016/j.ajhg.2016.09.005) are shown, as is the position of the horse D492H mutation described herein.

FIG. 32. Partial amino acid sequences of proteins encoded by PYROXD1, including the position of the equine PYROXD1-D492H substitution. Species included in the analysis are described in the text. The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated by period (.), and nonconserved positions are indicated by a blank space (.). The last line shows the position of the PYROXD1-D492H substitution in horse in bold. The position of the PYROXD1-D492H substitution is indicated in bold in all of the aligned sequences.

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FIG. 33. A portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr6:23,480,621 position, the site of a substitution of a guanine (G) for a cytosine (C) on the forward strand that results in the substitution of an alanine (A) for a glycine (G) at amino acid position 2182 in collagen type VI alpha 3 chain (COL6A3) as shown in FIG. 36 (SEQ ID NO:168). The reverse complement sequence is shown, with the site of a substitution of a cytosine (C) for a guanine (G) on the reverse strand as indicated (SEQ ID NO:163). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1139437410 in dbSNP.

FIG. 34. A portion of the normal equine COL6A3 Coding DNA Sequence (SEQ ID NO:164) and a portion of the mutant COL6A3 Coding DNA Sequence (SEQ ID NO:165) bearing the G to C mutation at nucleotide position 6792 in this figure, corresponding to chr6:23,480,621 as shown in SEQ ID NO:163 (FIG. 33). The numbering in FIG. 34 is that of the COL6A3 coding sequence (CDS) model XM_023642645.1; the sequence presented comprises the coding sequence for the five collagen-like domains in the middle of the protein. See the text for a discussion of the CDS model in NCBI. In both sequences the sequence of the third collagen-like domain, partially encoded by Exon 26, as shown in FIG. 35 is indicated in bold. The site of a G to C mutation at nucleotide position 6792, corresponding to 23,480,621 in SEQ ID NO:163 (FIG. 33), and to rs1139437410 in dbSNP, is underlined. The region of sequence comprising Exon 26 is displayed as codons in the correct reading frame for both SEQ ID NO:164 and SEQ ID NO:165 in FIG. 37.

FIG. 35. A view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the COL6A3 Coding DNA Sequence (SEQ ID NO:164) and the mutant COL6A3 Coding DNA Sequence (SEQ ID NO:165) in a BLAT search. The COL6A3 Coding DNA Sequences in horse correspond to COL6A3 coding sequences in other species as indicated by the regions of sequence similarity of the translated genomic DNA to COL6A3 protein sequences from human (Homo), mouse (Mus), and dog (Canis). Exon 26, which contains the guanine (G) to cytosine (C) variant at chr6:23,480,621 as shown in SEQ ID NO:163 (FIG. 33), is indicated below the image of the browser window. See the text for a discussion of the failure to detect sequence similarity of the translation of Exon 26 to protein sequences from other species. The sequence of horse Exon 26 with 10 nucleotides of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO:166).

FIG. 36. Models of part of the normal protein sequence encoded by horse COL6A3 (XP_023498413.1, presented here as SEQ ID NO:167) corresponding to a translation of SEQ ID NO:164 shown in FIG. 34 and part of the altered protein sequence encoded by horse COL6A3 (adapted from XP_023498413.1, presented here as SEQ ID NO:168) with the base substitution at chr6:23,480,621 corresponding to a translation of SEQ ID NO:165 shown in FIG. 34. The portion of the protein corresponding to all five collagen-like domains is shown; the portion corresponding to the third collagen-like domain, partially encoded by Exon 26, is indicated in bold, while the amino acid at position 2182 affected by the base substitution of a cytosine (C) for a guanine (G) at the chr6:23,480,621 position as shown in

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FIG. 33, is underlined. The amino acid positions in XP_023498413.1 are indicated at the beginning and end of the sequence.

FIG. 37. Horse COL6A3 Exon 26 and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the COL6A3-G2182A mutation would be most appropriately derived. Genomic coordinates are as in FIG. 33. Exon 26 from chr6:23,480,631 to chr6:23,480,578 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:169) and the COL6A3-G2182A allele (SEQ ID NO:170). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the C to G mutation site at nucleotide position chr6:23,480,621 on the forward strand, corresponding to rs1139437410 in dbSNP as shown in FIG. 33 (G to C in the reverse complement as shown), is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for a glycine (GGG) to one coding for an alanine (GCG). Example primers used experimentally to amplify genomic DNA containing the mutation site [5'-AGATGGGGCACA-GATCAAC-3' (SEQ ID NO:172) and 5'-TTCCCA-GACTCTCCTGTGCT-3' (SEQ ID NO:171)] are shown in lower case in the figure.

FIG. 38. Traces from Sanger DNA sequencing of amplified COL6A3 genomic DNA using primers shown in FIG. 37 (SEQ ID NO:171 and SEQ ID NO:172). The sequence of the forward strand is shown. The arrows in the figure indicate nucleotide position chr6:23,480,621, the site of a substitution of a cytosine (C) for a guanine (G) in this position on the reverse strand that creates the COL6A3-G2182A variant. The traces show, from left to right, the sequence of the forward strand for a horse homozygous for the wild-type or common allele (n/n), and results for a horse heterozygous for the substitution (n/K1).

FIG. 39. Partial sequence of the human COL6A3 coding sequence derived from NM_004369.3 (SEQ ID NO:173). This sequence is a region of perfect consensus among multiple different experimentally predicted mRNA isoforms. The numbering of the first and last nucleotides corresponds to that of NM_004369.3. The sequence encoding the third collagen-like domain is indicated in bold.

FIG. 40. Partial sequence of the human COL6A3 protein sequence, showing a translation of SEQ ID NO:173, equivalent to NP_004360.2 (SEQ ID NO:174). The numbering of the first and last amino acids corresponds to that of NP_004360.2. The sequence of the third collagen-like domain is indicated in bold.

FIG. 41. Comparison of the portion of the protein sequence of COL6A3 comprising the five collagen-like repeats from human (NP_004360.2, shown here as SEQ ID NO:175) and horse (XP_023498413.1, shown here as SEQ ID NO:176). The position of the COL6A3-G2182A substitution is shown below the horse sequence. Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a plus sign (+) indicates a conservative substitution, and a space () indicates a nonconservative substitution. The positions of glycine residues that are part of the Gly-X-Y structure of the collagen triple helix are indicated by an asterisk (*) in reverse text (white text on a black background) between the horse and human sequences. All of these glycine residues are conserved between human and horse. The horse G2182A sequence has a nonconservative substitution of an alanine (A) for a glycine (G) at

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position 2182; the wild-type horse sequence matches the human sequence at this position.

FIG. 42. Features of the collagen type VI alpha 3 chain protein encoded by the human COL6A3 gene. (A) The human protein has 3,177 amino acids. It contains five collagen-like domains consisting of Gly-X-Y repeats, labeled as Collagen 1-5 in the figure. Toward the N-terminus of the collagen-like repeats are 10 Von Willebrand Factor-like repeats, labeled as VWFA 1-10 in the figure. Toward the C-terminus of the collagen-like repeats are two additional Von Willebrand Factor-like repeats, labeled as VWFA 11 and 12 in the figure. Near the C-terminus is a fibronectin-like repeat, labeled as Fibronectin in the figure, and a Protease Inhibitor domain (Pancreatic trypsin inhibitor Kunitz domain). (B) The portion of the protein containing the five collagen-like domains extends from position 2038 to 2373 of the human protein. The five collagen-like domains are numbered. Positions of 19 pathogenic human alleles listed in TABLE 4 are indicated above the segment shown, as is the position of the horse G2182A variant. The position of the cysteine involved in the interchain disulfide bond (C2087S-S) is indicated below the segment shown.

FIG. 43. Partial amino acid sequences of proteins encoded by COL6A3, including the position of the equine COL6A3-G2182A substitution. Species included in the analysis are described in the text. The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated by period (.), and nonconserved positions are indicated by a blank space (). The last line shows the position of the COL6A3-G2182A substitution in horse in bold. The position of the COL6A3-G2182A substitution is indicated in bold in all of the aligned sequences.

FIG. 44. An Arabian horse (E682) homozygous for the PYROXD1-D492H substitution (P8/P8), heterozygous for MYOT-S232P (n/P2), and homozygous for the wild-type alleles of FLNC-E753K (P3a), FLNC-A1207T (P3b), MYOZ3-S42L (P4), reported in a prior disclosure. Muscle wasting in the pelvic girdle (hindquarters), shoulder girdle (topline), and proximal limbs is evident. The horse is pregnant in the photo. The horse was reported to be symptomatic with gait abnormalities ("rope walking"), and died of a choking incident. Difficulty in swallowing is observed in human patients with Myofibrillar Myopathy 8 (MFM8) due to mutations in PYROXD1 (O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105 DOI: 10.1016/j.ajhg.2016.09.005).

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

This disclosure describes methods for detecting the presence or absence of a biomarker in horses associated with a defect of muscle integrity that causes a form of inherited exercise intolerance. This disease condition has been previously described as Polysaccharide Storage Myopathy, type 2 (PSSM2). Another previously described form of exercise intolerance in horses is Polysaccharide Storage Myopathy, type 1 (PSSM1), caused by a semidominant allele of glycogen synthase (GYS1-R309H). The term PSSM2 is commonly used to describe horses that show exercise intolerance, a negative test result for the GYS1-R309H variant of Glycogen Synthase 1 that is associated with Polysaccharide Storage Myopathy, type 1 (PSSM1), and abnormal findings on muscle biopsy, including abnormally shaped muscle

fibers, nuclei displaced to the center of muscle fibers rather than the normal position at the edge of fibers, and pools of glycogen granules of normal size in regions of disorganization that give the false appearance of a glycogen storage disease. In one embodiment, the method involves obtaining a physiological sample from a horse and determining whether the biomarker is present in the sample. As used herein, the phrase “physiological sample” refers to a biological sample obtained from a horse that contains nucleic acid. For example, a physiological sample can be a sample collected from an individual horse such as, for example, a cell sample, such as a blood cell, e.g., a lymphocyte, a peripheral blood cell; a sample collected from the spinal cord; a tissue sample such as cardiac tissue or muscle tissue, e.g., cardiac or skeletal muscle; an organ sample, e.g., liver or skin; a hair sample, e.g., a hair sample with roots; and/or a fluid sample, such as blood.

Examples of breeds of affected horse include, but are not limited to, Shires, Clydesdales, Percheron Horses, Belgian Horses, Draft Horses, Quarter Horses, Paint Horses, Warmblood Horses, or related or unrelated breeds. The phrase “related breed” is used herein to refer to breeds that are related to a breed, such as Quarter Horse, Draft Horse, or Warmblood Horse. Such breeds include, but are not limited to stock breeds such as the American Paint horse, the Appaloosa, and the Palomino. The term “Draft Horse” includes many breeds including but not limited to Clydesdale, Belgian, Percheron, and Shire horses. The term “Warmblood” is also a generic term that includes a number of different breeds. “Warmblood” simply distinguishes this type of horse from the “cold bloods” (draft horses) and the “hot bloods” (Thoroughbreds and Arabians). The method described herein also may be performed using a sample obtained from a crossed or mixed breed horse.

The term “biomarker” generally refers herein to a biological indicator, such as a particular molecular feature, that may affect, may be an indicator, and/or be related to diagnosing or predicting an individual’s health. For example, in certain embodiments, the biomarker can refer to (1) a mutation in the equine dyferlin (DYSF) coding region (SEQ ID NO:1), such as a polymorphic allele of *DYSF* that has a substitution of an adenine (A) for a guanine (G) on the forward strand at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 1, (2) a mutation in the equine dyferlin (*DYSF*) coding region (SEQ ID NO:56), a polymorphic allele of *DYSF* that has a substitution of a thymine (T) for a guanine (G) on the forward strand at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 13, (3) a mutation in the equine pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (*PYROXD1*) coding region (SEQ ID NO:103), such as a polymorphic allele of *PYROXD1* that has a substitution of a cytosine (C) for a guanine (G) at chr6:47,661,977 on the forward strand of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 22, or (4) a mutation in the equine collagen type VI alpha 3 chain (*COL6A3*) coding region (SEQ ID NO:163), such as a polymorphic allele of *COL6A3* that has a substitution of a guanine (G) for a cytosine (C) on the forward strand at chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 33. The

specified nucleotide substitution may be inferred by the detection of the complementary base on the reverse strand.

“Oligonucleotide probe” can refer to a nucleic acid segment, such as a primer, that is useful to amplify a sequence in the *DYSF*, *PYROXD1*, or *COL6A3* coding regions that is complementary to, and hybridizes specifically to, a particular nucleotide sequence in *DYSF*, *PYROXD1*, or *COL6A3*, or to a nucleic acid region that flanks *DYSF*, *PYROXD1*, or *COL6A3*.

As used herein, the term “nucleic acid” and “polynucleotide” refers to deoxyribonucleotides or ribonucleotides and polymers thereof in either single-stranded or double-stranded form. Unless specifically limited, the term encompasses nucleic acids containing known analogs of natural nucleotides that have similar binding properties as the reference nucleic acid and are metabolized in a manner similar to naturally occurring nucleotides. Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified variants thereof (e.g., degenerate codon substitutions) and complementary sequences as well as the sequence explicitly indicated. Specifically, degenerate codon substitutions may be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted with mixed-base and/or deoxyinosine residues.

A “nucleic acid fragment” is a portion of a given nucleic acid molecule. Deoxyribonucleic acid (DNA) in the majority of organisms is the genetic material while ribonucleic acid (RNA) is involved in the transfer of information contained within DNA into proteins. The term “nucleotide sequence” refers to DNA or RNA that can be single-stranded or double-stranded, optionally containing synthetic, non-natural, or altered nucleotide bases capable of incorporation into DNA or RNA.

In some embodiments, the method can involve contacting the sample with at least one oligonucleotide probe to form a hybridized nucleic acid and then amplifying the hybridized nucleic acid. “Amplifying” utilizes methods such as the polymerase chain reaction (PCR), ligation amplification (or ligase chain reaction, LCR), strand displacement amplification, nucleic acid sequence-based amplification, and amplification methods based on the use of Q β -replicase. These methods are well known and widely practiced in the art. Reagents and hardware for conducting PCR are commercially available. For example, in certain embodiments, (1) Exon B of the equine dysferlin coding region (also referred to as *DYSF*) as shown in FIG. 3 or portions thereof, (2) Exon I of the equine dysferlin coding region (also referred to as *DYSF*) as shown in FIG. 15 or portions thereof, (3) Exon 12 of the equine pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 coding region (also referred to as *PYROXD1*) as shown in FIG. 24 or portions thereof, or (4) Exon 26 of the equine collagen type VI alpha 3 chain coding region (also referred to as *COL6A3*) as shown in FIG. 35 or portions thereof, may be amplified by PCR. In another embodiment, at least one oligonucleotide probe is immobilized on a solid surface or a semisolid surface.

The methods described herein can be used to detect the presence or absence of a biomarker associated with equine Polysaccharide Storage Myopathy type 2 (PSSM2) in a horse (live or dead) regardless of age (e.g., an embryo, a foal, a neonatal foal, aborted foal, a breeding-age adult, or any horse at any stage of life) or sex (e.g., a mare (dam) or stallion (sire)).

As used herein, the term “presence or absence” refers to affirmatively detecting the presence of a biomarker or

detecting the absence of the biomarker within the experimental limits of the detection methods used to detect the biomarker.

This disclosure further provides a method for detecting and/or diagnosing Polysaccharide Storage Myopathy type 2 (PSSM2), also referred to as Myofibrillar Myopathy (MFM), in a horse, the method involving obtaining a physiological sample from the horse and detecting the presence or absence of biomarkers in the sample, wherein the presence of the biomarkers is indicative of the disease. One embodiment of the method further involves contacting the sample with at least one oligonucleotide probe to form a hybridized nucleic acid and amplifying the hybridized nucleic acid. For example, in four embodiments, (1) Exon B of equine DYSF as shown in FIG. 3, (2) Exon I of equine DYSF as shown in FIG. 15, (3) Exon 12 of equine PYROXD1 as shown in FIG. 24, or (4) Exon 26 of equine COL6A3 as shown in FIG. 35, are amplified using, for example, polymerase chain reaction, strand displacement amplification, ligase chain reaction, amplification methods based on the use of QP-replicase and/or nucleic acid sequence-based amplification. In these embodiments of the method, the biomarkers can include (1) an equine dysferlin (DYSF) coding region having the specific substitution of an adenine (A) for a guanine (G) on the forward strand at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 1, (2) an equine dysferlin (DYSF) coding region having the specific substitution of a thymine (T) for a guanine (G) on the forward strand at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 13, (3) an equine pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) coding region having the specific substitution of a cytosine (C) for a guanine (G) on the forward strand at chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 22, or (4) an equine collagen type VI alpha 3 chain (COL6A3) coding region having the specific substitution of a guanine (G) for a cytosine (C) on the forward strand at chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 33. Biomarkers can also include (1) a coding region that encodes a dysferlin (DYSF) polypeptide (SEQ ID NO:10) having an arginine-to-tryptophan (R to W) substitution at amino acid residue 253 of SEQ ID NO:10, as shown in SEQ ID NO:11, (2) a coding region that encodes a dysferlin (DYSF) polypeptide (SEQ ID NO:60) having an proline-to-threonine (P to T) substitution at amino acid residue 1290 of SEQ ID NO:60, as shown in SEQ ID NO:61, (3) a pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) polypeptide (SEQ ID NO:107) having an aspartate-to-histidine (D to H) substitution at amino acid residue 492 of SEQ ID NO:107, as shown in SEQ ID NO:108, or (4) a collagen type VI alpha 3 chain (COL6A3) polypeptide (SEQ ID NO:167) having an glycine-to-alanine (G to A) substitution at amino acid residue 2182 of SEQ ID NO:167, as shown in SEQ ID NO:168. The method can be used to detect Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), in a horse.

This disclosure further provides a kit that includes a test for diagnosing and/or detecting the presence of equine Polysaccharide Storage Myopathy type 2 (PSSM2), also

known as Myofibrillar Myopathy (MFM), in a horse. The kit generally includes packing material containing, separately packaged, at least one oligonucleotide probe capable of forming hybridized nucleic acids with DYSF, PYROXD1, or COL6A3 and instructions directing the use of the probe in accord with the methods described herein.

Horses affected with Polysaccharide Storage Myopathy type 2 (PSSM2) are either heterozygous or homozygous for the affected DYSF, PYROXD1, or COL6A3 alleles. An “allele” is a variant form of a particular genomic nucleic acid sequence. In the context of the methods described herein, some alleles of the DYSF, PYROXD1, or COL6A3 coding regions cause Polysaccharide Storage Myopathy type 2 (PSSM2) in horses. A “DYSF, PYROXD1, or COL6A3 allele,” refers to a normal allele of the DYSF, PYROXD1, or COL6A3 locus as well as an allele carrying one or more variations that predispose a horse to develop Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM). The coexistence of multiple alleles at a locus is known as “genetic polymorphism.” Any site at which multiple alleles exist as stable components of the population is by definition “polymorphic.” An allele is defined as polymorphic if it is present at a frequency of at least 1% in the population. A “single nucleotide polymorphism (SNP)” is a DNA sequence variation that involves a change in a single nucleotide.

The methods described herein involve the use of isolated or substantially purified nucleic acid molecules. An “isolated” or “purified” nucleic acid molecule is one that, by human intervention, exists apart from its native environment and is therefore not a product of nature. An isolated nucleic acid molecule may exist in a purified form or may exist in a non-native environment. For example, an “isolated” or “purified” nucleic acid molecule, or portion thereof, is substantially free of other cellular material, or culture medium when produced by recombinant techniques, or substantially free of chemical precursors or other chemicals when chemically synthesized. In one embodiment, an “isolated” nucleic acid is free of sequences that naturally flank the nucleic acid (i.e., sequences located at the 5' and 3' ends of the nucleic acid) in the genomic DNA of the organism from which the nucleic acid is derived. For example, in various embodiments, the isolated nucleic acid molecule can contain less than about 5 kb, 4 kb, 3 kb, 2 kb, 1 kb, 0.5 kb, or 0.1 kb of nucleotide sequences that naturally flank the nucleic acid molecule in genomic DNA of the cell from which the nucleic acid is derived. An isolated or purified nucleic acid molecule can be a fragment and/or variant of a reference nucleotide sequence expressly disclosed herein.

A “fragment” or “portion” of a sequence refers to anything less than full-length of the nucleotide sequence encoding—or the amino acid sequence of—a polypeptide. As it relates to a nucleic acid molecule, sequence, or segment when linked to other sequences for expression, a “portion” or a “fragment” refers to a sequence having, for example, at least 80 nucleotides, at least 150 nucleotides, or at least 400 nucleotides. Alternatively, when not employed for expressing—e.g., in the context of a probe or a primer—a “portion” or a “fragment” means, for example, at least 9, at least 12, at least 15, or at least 20 consecutive nucleotides. Alternatively, a fragment or a portion of a nucleotide sequence that is useful as a hybridization probe generally does not encode fragment proteins retaining biological activity. Thus, fragments or portions of a nucleotide sequence may range from at least about 6 nucleotides, about 9, about 12 nucleotides, about 20 nucleotides, about 50 nucleotides, about 100 nucleotides, or more.

A “variant” of a molecule is a sequence that is substantially similar to the sequence of the reference—e.g., native, naturally-occurring, and/or wild-type—molecule. For nucleotide sequences, a variant includes any nucleotide sequence that, because of the degeneracy of the genetic code, encodes the native amino acid sequence of a protein. Naturally occurring allelic variants such as these can be identified with the use of well-known molecular biology techniques, as, for example, with polymerase chain reaction (PCR) and/or hybridization techniques. A variant nucleotide sequence also can include a synthetically-derived nucleotide sequence such as one generated, for example, by using site-directed mutagenesis that encodes the native protein, as well as variant nucleotide sequences that encode a polypeptide having amino acid substitutions. Generally, a nucleotide sequence variant will have at least 40%, at least 50%, at least 60%, at least 70% (e.g., 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%), at least 80% (e.g., 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%), or at least 90% (e.g., 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%) sequence identity to the native (endogenous) nucleotide sequence.

“Synthetic” polynucleotides are those prepared by chemical synthesis.

“Recombinant DNA molecule” is a combination of DNA sequences that are joined together using recombinant DNA technology and procedures that are used to join together DNA sequences as described, for example, in Sambrook and Russell (2001).

“Naturally-occurring,” “native,” or “wild-type” refers to an amino acid sequence or polynucleotide sequence that can be found in nature, without any known mutation, as distinct from being produced artificially or producing a mutated, non-wild-type phenotype. For example, a nucleotide sequence present in an organism (including a virus) that can be isolated from a source in nature and that has not been intentionally modified in the laboratory is naturally occurring. Furthermore, “wild-type” refers to a coding region or organism as found in nature without any known mutation.

A “mutant” dysferlin (DYSF) is a polypeptide or a fragment thereof that is encoded by a DYSF coding region having a mutation, e.g., such as might occur at the DYSF locus. A mutation in one DYSF allele may lead to an alteration in the ability of the encoded polypeptide to interact with calcium ions, with alpha-actinin-2 (encoded by ACTN2), annexin A1 (encoded by ANXA1), annexin A2 (encoded by ANXA2), voltage-dependent L-type calcium channel subunit alpha-1S (encoded by CACNA1S), caveolin 3 (encoded by CAV3), desmin (encoded by DES), diacylglycerol kinase delta (encoded by DGKD), filamin C (encoded by FLNC), myoferlin (encoded by MYOF), myomesin-1 (encoded by MYOM1), gamma-sarcoglycan (encoded by SGCG), myomesin-2 (encoded by MYOM2), myosin-binding protein C, slow-type (encoded by MYBPC1), nebulin (encoded by NEBU), optineurin (encoded by OPTN), beta-parvin (encoded by PARVB), Rho family-interacting cell polarization regular 2 (encoded by RIPOR2), SNARE-associated protein Snapin (encoded by SNAPN), titin (encoded by TTN), tripartite motif-containing protein 72 (encoded by TRIM72), or other proteins that are involved in repair of the sarcolemma of myofibrils, or other proteins that are expressed in skeletal or cardiac muscle that are required for the integrity of myofibrils, leading to alterations in the integrity of myofibrils in a horse heterozygous or homozygous for the allele. Alterations in the interactions of specific proteins can be determined by methods known to the art. Mutations in DYSF may be disease-causing in a horse heterozygous or homozygous for

the mutant DYSF allele, e.g., a horse heterozygous or homozygous for a mutation leading to a mutant DYSF polypeptide such as substitution mutations in Exon B of DYSF as shown in FIG. 3 and FIG. 4, such as that designated herein as DYSF-R253W, or in Exon I of DYSF as shown in FIG. 15 and FIG. 16, such as that designated herein as DYSF-P1290T.

A “mutant” pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) is a polypeptide or a fragment thereof that is encoded by a PYROXD1 coding region having a mutation, e.g., such as might occur at the PYROXD1 locus. A mutation in one PYROXD1 allele may lead to an alteration in the ability of the encoded polypeptide to interact with the cofactors flavin adenine dinucleotide (FAD) or nicotinamide adenine dinucleotide (NAD), with other proteins containing either canonical or non-canonical disulfide (S-S) bonds between cysteine (CVS) residues, such as titin (encoded by TIN), obscurin (encoded by OBSCN), triadin (encoded by TRDN), myosin regulatory light chains (encoded by MYL1, MYL2, MYL3, MYL4, MYL5, MYL6, and others), myosin heavy chains (encoded by MYH1, MYH2, MYH3, MYH4, MYH5, MYH6, and others), the ryanodine receptor (encoded by RYR1), sarcoplasmic reticulum Ca^{2+} -ATPase (encoded by SERCA), nebulin (encoded by NEB), troponin (encoded by TNNT1, TNNT2, and TNNT3), myosin-binding protein C (encoded by MYBPC1), or other proteins that are involved in repair of the sarcolemma of myofibrils, or other proteins that are expressed in skeletal or cardiac muscle that are required for the integrity of myofibrils, leading to alterations in the integrity of myofibrils in a horse heterozygous or homozygous for the allele. Alterations in the interactions of specific proteins can be determined by methods known to the art. It is also possible that a mutation in one or both PYROXD1 alleles may lead to an alteration in the redox state of the cell, which might interfere with signaling processes that are sensitive to the redox state, or to an increased sensitivity to reactive oxygen species such as superoxide, hydrogen peroxide, or other reactive oxygen or reactive nitrogen species. Alterations in the redox state of the cell, as indicated by the ratio of reduced glutathione (GSH) to oxidized glutathione (GSSG), may lead to the irreversible oxidation of sulfhydryl groups in cysteine residues, causing the affected proteins to be ubiquitinated and destroyed. Turnover of particular muscle proteins may alter the stoichiometry of muscle proteins, causing a temporary decrease in muscle function. Oxidative stress is also known to contribute to muscle fatigue by reducing mitochondrial function through the oxidation of key sulfhydryl groups in sarcoplasmic reticulum Ca^{2+} -ATPase (encoded by SERCA) and the ryanodine receptor (encoded by RYR1), which mediate calcium levels. Mutations in PYROXD1 may be disease-causing in a horse heterozygous or homozygous for the mutant PYROXD1 allele, e.g., a horse heterozygous or homozygous for a mutation leading to a mutant PYROXD1 polypeptide such as substitution mutations in Exon 12 of PYROXD1 as shown in FIG. 24 and FIG. 25, such as that designated herein as PYROXD1-D492H.

A “mutant” collagen type VI alpha 3 chain (COL6A3) is a polypeptide or a fragment thereof that is encoded by a COL6A3 coding region having a mutation, e.g., such as might occur at the COL6A3 locus. A mutation in one COL6A3 allele may lead to an alteration in the ability of the encoded polypeptide to interact with the protein encoded by the common or wild-type allele of COL6A3 in a heterozygote, to interact with proteins encoded by COL6A1, COL6A2, COL6A5, or COL6A6, or to assemble into a

correctly-configured collagen triple helix. A mutation in one COL6A3 allele may lead to an alteration in the ability of the encoded polypeptide to interact with the protein products of the COL5A1 gene, which encodes one of the collagen type 5 proteins, with dysferlin (encoded by DYSF), or with other proteins known to interact with collagen. Defects caused by mutations in one or both COL6A3 alleles may interfere with the proper posttranslational modification of collagen, for example by interfering with glycosylation, phosphorylation, the modification of proline residues to hydroxyproline residues, the modification of lysine residues to hydroxylysine residues, or the proper formation of interchain disulfide bonds. Such defects may alter the ability of collagen to interact with other components of the extracellular matrix, or alter the mechanical properties of the extracellular matrix. Alterations in the interactions of specific proteins can be determined by methods known to the art. Mutations in COL6A3 may be disease-causing in a horse heterozygous or homozygous for the mutant COL6A3 allele, e.g., a horse heterozygous or homozygous for a mutation leading to a mutant COL6A3 polypeptide such as substitution mutations in Exon 26 of COL6A3 as shown in FIG. 35 and FIG. 36, such as that designated herein as COL6A3-G2182A.

A "somatic mutation" is a mutation that occurs only in certain tissues, e.g., in liver tissue, and are not inherited in the germline. A "germline" mutation can be found in any of a body's tissues and is inherited. The present COL6A3 mutation is a germline mutation.

"Homology" refers to the percent identity between two polynucleotide sequences or two amino acid sequences. Two sequences are "homologous" to each other when the sequences exhibit at least 70% (e.g., 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%), at least 80% (e.g., 81% 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%), or at least 90% (e.g., 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%) contiguous sequence identity over a defined length of the sequences.

The following terms are used to describe the sequence relationships between two or more nucleic acids or polynucleotides: "reference sequence," "comparison window," "sequence identity," "percentage of sequence identity," and "substantial identity."

As used herein, "reference sequence" refers to a sequence used as a basis for sequence comparison. A reference sequence may be a subset or the entirety of a specified sequence. For example, a reference sequence may be a segment of a full length cDNA or coding region sequence, or the complete cDNA or coding region sequence.

As used herein, "comparison window" refers to a contiguous and specified segment of a polynucleotide sequence, wherein the polynucleotide sequence in the comparison window may reflect one or more additions and/or deletions (i.e., gaps) compared to the reference sequence (which does not exhibit the additions and/or deletions) for optimal alignment of the two sequences. Generally, the comparison window is at least 20 contiguous nucleotides in length, and optionally can be 30, 40, 50, 100, or longer. To avoid a high similarity to a reference sequence due to inclusion of gaps in the polynucleotide sequence, a gap penalty is typically introduced and is subtracted from the number of matches. Methods of alignment of sequences for comparison are well known in the art. Thus, the determination of percent identity between any two sequences can be accomplished using a mathematical algorithm.

Computer implementations of these mathematical algorithms can be used for comparing sequences to determine sequence identity. Such implementations include, but are not

limited to: Clustal Omega (online at EMBL-EBI), COBALT (online at ncbi.nlm.nih.gov), the ALIGN program (Version 2.0), and GAP, BESTFIT, BLAST, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Version 8 (available from the Genetics Computer Group (GCG) Madison, WI, USA). Alignments using these programs can be performed using the default parameters.

Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information (see the World Wide Web at ncbi.nlm.nih.gov). This algorithm involves first identifying high scoring pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as the neighborhood word score threshold. These initial neighborhood word hits act as seeds for initiating searches to find longer HSPs containing them. The word hits are then extended in both directions along each sequence for as far as the cumulative alignment score can be increased. Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when the cumulative alignment score falls off by the quantity X from its maximum achieved value, the cumulative score goes to zero or below due to the accumulation of one or more negative-scoring residue alignments, or the end of either sequence is reached.

In addition to calculating percent sequence identity, the BLAST algorithm also performs a statistical analysis of the similarity between two sequences. One measure of similarity provided by the BLAST algorithm is the smallest sum probability (P(N)), which provides an indication of the probability by which a match between two nucleotide or amino acid sequences would occur by chance. For example, a test nucleic acid sequence is considered similar to a reference sequence if the smallest sum probability in a comparison of the test nucleic acid sequence to the reference nucleic acid sequence is less than about 0.1, less than about 0.01, or even less than about 0.001.

To obtain gapped alignments for comparison purposes, Gapped BLAST (in BLAST 2.0) can be utilized. Alternatively, PSI-BLAST (in BLAST 2.0) can be used to perform an iterated search that detects distant relationships between molecules. When using BLAST, Gapped BLAST, or PSI-BLAST, the default parameters of the respective programs (e.g., BLASTN for nucleotide sequences, BLASTP for proteins) can be used. The BLASTN program (for nucleotide sequences) uses as defaults a wordlength (W) of 11, an expectation (E) of 10, a cutoff of 100, M=5, N=-4, and a comparison of both strands. For amino acid sequences, the BLASTP program uses as defaults a wordlength (W) of 3, an expectation (E) of 10, and the BLOSUM62 scoring matrix. See the World Wide Web at ncbi.nlm.nih.gov. Alignment may also be performed manually by visual inspection. For purposes of the methods described herein, comparison of nucleotide sequences for determination of percent sequence identity to the promoter sequences disclosed herein is preferably made using the BlastN program (version 2.3.0 or later) with its default parameters or any equivalent program. By "equivalent program" is intended any sequence comparison program that, for any two sequences in question, generates an alignment having identical nucleotide or amino acid residue matches and an identical percent sequence

identity when compared to the corresponding alignment generated by a BLAST program.

As used herein, "sequence identity" or "identity" in the context of two nucleic acid or polypeptide sequences refers to a specified percentage of residues in the two sequences that are the same when aligned for maximum correspondence over a specified comparison window, as measured by sequence comparison algorithms or by visual inspection. When percentage of sequence identity is used in reference to a protein, it is recognized that residue positions that are not identical often differ by conservative amino acid substitutions, where amino acid residues are substituted for other amino acid residues with similar chemical properties (e.g., charge or hydrophobicity) and therefore do not change the functional properties of the molecule. When sequences differ in conservative substitutions, the percent sequence identity may be adjusted upwards to correct for the conservative nature of the substitution. Sequences that differ by such conservative substitutions are said to have "sequence similarity" or "similarity." Methods for making this adjustment are well known to those of skill in the art. Typically, this involves scoring a conservative substitution as a partial rather than a full mismatch, thereby increasing the percentage sequence identity. Thus, for example, where an identical amino acid is given a score of 1 and a non-conservative substitution is given a score of zero, a conservative substitution is given a score between zero and 1. The scoring of conservative substitutions is calculated, e.g., as implemented in the program PC/GENE (Intelligenetics, Mountain View, CA).

As used herein, "percentage of sequence identity" refers to the value determined by comparing two optimally aligned sequences over a comparison window, wherein the portion of the polynucleotide sequence in the comparison window may comprise additions or deletions (i.e. gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison, and multiplying the result by 100 to yield the percentage of sequence identity.

The term "substantial identity," in the context of polynucleotide sequences, means that a polynucleotide sequence possesses at least 70% (e.g., 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%), at least 80% (e.g., 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%), or at least 90% (e.g., 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%) sequence identity compared to a reference sequence using one of the alignment programs described using standard parameters. These values can be appropriately adjusted to determine corresponding identity of proteins encoded by two nucleotide sequences by taking into account codon degeneracy, amino acid similarity, reading frame positioning, and the like. Substantial identity of amino acid sequences for these purposes normally means sequence identity of at least 70%, or at least 80%, 90%, or even at least 95%.

Another indication that nucleotide sequences are substantially identical is if two molecules hybridize to each other under stringent conditions (see below). Generally, stringent conditions are selected to be about 5° C. lower than the thermal melting point (T_m) for the specific sequence at a defined ionic strength and pH. However, stringent conditions encompass temperatures in the range of about 1° C. to

about 20° C., depending upon the desired degree of stringency as otherwise qualified herein. Nucleic acids that do not hybridize to each other under stringent conditions are still substantially identical if the polypeptides they encode are substantially identical. This may occur, e.g., when a copy of a nucleic acid is created using the maximum codon degeneracy permitted by the genetic code. One indication that the two nucleic acid sequences are substantially identical is when the polypeptide encoded by the first nucleic acid is immunologically cross reactive with the polypeptide encoded by the second nucleic acid.

The term "substantial identity," in the context of a polypeptide, indicates that a polypeptide possesses a sequence with at least 70% (e.g., 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%), at least 80% (e.g., 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%), or at least 90% (e.g., 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%) amino acid sequence identity to the reference sequence over a specified comparison window. An indication that two polypeptide sequences are substantially identical is that one polypeptide is immunologically reactive with antibodies raised against the second polypeptide.

Thus, a polypeptide is substantially identical to a second polypeptide when, for example, the two polypeptides differ only by a conservative substitution. For sequence comparison, typically one amino acid sequence acts as a reference sequence to which test amino acid sequences are compared. When using a sequence comparison algorithm, test and reference amino acid sequences are input into a computer, subsequence coordinates are designated if necessary, and sequence algorithm program parameters are designated. The sequence comparison algorithm then calculates the percent sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters.

As noted above, another indication that two nucleic acid sequences are substantially identical is that two molecules hybridize to each other under stringent conditions. The phrase "hybridizing specifically to" refers to the binding, duplexing, or hybridizing of a molecule only to a particular nucleotide sequence under stringent conditions when that sequence is present in a complex mixture (e.g., total cellular DNA or RNA). "Bind(s) substantially" refers to complementary hybridization between a probe nucleic acid and a target nucleic acid and embraces minor mismatches that can be accommodated by reducing the stringency of the hybridization media to achieve the desired detection of the target nucleic acid sequence.

"Stringent hybridization conditions" and "stringent hybridization wash conditions" in the context of nucleic acid hybridization experiments such as Southern and Northern hybridizations are sequence dependent, and are different under different environmental parameters. Longer sequences hybridize specifically at higher temperatures. The T_m is the temperature (under defined ionic strength and pH) at which 50% of the target sequence hybridizes to a perfectly matched probe. Specificity is typically the function of post-hybridization washes, the critical factors being the ionic strength and temperature of the final wash solution. For DNA-DNA hybrids, the T_m can be approximated from the equation of Meinkoth and Wahl:

$$T_m = 81.5^\circ \text{C} + 16.6 (\log M) + 0.41 (\% \text{GC}) - 0.61 (\% \text{form}) - 500/L$$

where M is the molarity of monovalent cations, % GC is the percentage of guanosine and cytosine nucleotides in the DNA, % form is the percentage of formamide in the

hybridization solution, and L is the length of the hybrid in base pairs. T_m is reduced by about 1° C. for each 1% of mismatching; thus, T_m , hybridization, and/or wash conditions can be adjusted to hybridize to sequences of the desired identity. For example, if sequences with >90% identity are sought, the T_m can be decreased 10° C. Generally, stringent conditions are selected to be about 5° C. lower than the thermal melting point (T_m) for the specific sequence and its complement at a defined ionic strength and pH. However, severely stringent conditions can utilize a hybridization and/or wash at 1° C., 2° C., 3° C., or 4° C. lower than the thermal melting point (T_m); moderately stringent conditions can utilize a hybridization and/or wash at 6° C., 7° C., 8° C., 9° C., or 10° C. lower than the thermal melting point (T_m); low stringency conditions can utilize a hybridization and/or wash at 11° C., 12° C., 13° C., 14° C., 15° C., or 20° C. lower than the thermal melting point (T_m). Using the equation, hybridization and wash compositions, and desired T , those of ordinary skill will understand that variations in the stringency of hybridization and/or wash solutions are inherently described. If the desired degree of mismatching results in a T of less than 45° C. (aqueous solution) or 32° C. (formamide solution), it is preferred to increase the SSC concentration ($20 \times \text{SSC} = 3.0 \text{ M NaCl}$, $0.3 \text{ M trisodium citrate}$) so that a higher temperature can be used. Generally, highly stringent hybridization and wash conditions are selected to be about 5° C. lower than the thermal melting point (T_m) for the specific sequence at a defined ionic strength and pH.

An example of highly stringent wash conditions is 0.15 M NaCl at 72° C. for about 15 minutes. An example of stringent wash conditions is a $0.2 \times \text{SSC}$ wash at 65° C. for about 15 minutes. Often, a high stringency wash is preceded by a low stringency wash to remove background probe signal. An example medium stringency wash for a duplex of, e.g., more than 100 nucleotides is $4\text{--}6 \times \text{SSC}$ at 40° C. for 15 minutes. For short probes (e.g., about 10 to 50 nucleotides), stringent conditions typically involve salt concentrations of less than about 1.5 M, more preferably about 0.01 M to 1.0 M, Na^+ ion concentration (or other salts) at pH 7.0 to 8.3, and the temperature is typically at least about 30° C. and at least about 60° C. for long probes (e.g., >50 nucleotides). Stringent conditions may also be achieved with the addition of destabilizing agents such as formamide. In general, a signal to noise ratio of $2 \times$ (or higher) than that observed for an unrelated probe in the particular hybridization assay indicates detection of a specific hybridization. Nucleic acids that do not hybridize to each other under stringent conditions are still substantially identical if the proteins that they encode are substantially identical. This occurs, e.g., when a copy of a nucleic acid is created using the maximum codon degeneracy permitted by the genetic code. Very stringent conditions are selected to be equal to the T_m for a particular probe. An example of stringent conditions for hybridization of complementary nucleic acids which have more than 100 complementary residues on a filter in a Southern or Northern blot is 50% formamide, e.g., hybridization in 50% formamide, 1 M NaCl , 1% SDS at 37° C.; and a wash in $0.1 \times \text{SSC}$ at 60° C. to 65° C. Exemplary low stringency conditions include hybridization with a buffer solution of 30 to 35% formamide, 1 M NaCl , 1% SDS (sodium dodecyl sulfate) at 37° C., and a wash in $1 \times$ to $2 \times \text{SSC}$ at 50° C. to 55° C. Exemplary moderate stringency conditions include hybridization in 40% to 45% formamide, 1.0 M NaCl , 1% SDS at 37° C., and a wash in $0.5 \times$ to $1 \times \text{SSC}$ at 55° C. to 60° C.

The term "variant" polypeptide refers to a polypeptide derived from the native protein by deletion (so-called trun-

cation) and/or addition of one or more amino acids to the N-terminal and/or C-terminal end of the native protein, deletion and/or addition of one or more amino acids at one or more sites in the native protein, and/or substitution of one or more amino acids at one or more sites in the native protein. Such variants may result from, for example, genetic polymorphism or human manipulation. Methods for such manipulations are generally known in the art. Variant DYSE, PYROXD1, or COL6A3 polypeptides may be altered in various ways including, for example, being altered to exhibit one or more amino acid substitutions, one or more deletions, one or more truncations, and/or one or more insertions. For example, an amino acid sequence can be prepared by one or more mutations in the DNA encoding the DYSE, PYROXD1, or COL6A3 polypeptides. Guidance regarding appropriate amino acid substitutions that do not affect biological activity of the protein of interest is well known in the art. Conservative substitutions, such as exchanging one amino acid with another having similar properties, are preferred.

Thus, the nucleotide sequences used to practice the methods described herein can include both naturally-occurring sequences or mutant forms. Likewise, the polypeptides referred to herein can include naturally-occurring polypeptides as well as variations and modified forms thereof. Such variants may continue to possess the desired activity. The deletions, insertions, or substitutions of the polypeptide sequence encompassed herein are not expected to produce radical changes in the characteristics of the polypeptide. However, when it is difficult to predict the exact effect of the substitution, deletion, or insertion in advance of doing so, the effect can be evaluated by routine screening assays.

An individual substitution, deletion, or addition that alters, adds, or deletes a single amino acid or a small percentage of amino acids (typically less than 5%, more typically less than 1%) in an encoded sequence are "conservatively modified variations."

"Conservatively modified variations" of a particular nucleic acid sequence refers to those nucleic acid sequences that encode identical or essentially identical amino acid sequences, or where the nucleic acid sequence does not encode an amino acid sequence, to essentially identical sequences. Because of the degeneracy of the genetic code, a large number of functionally identical nucleic acids encode any given polypeptide. For instance, the codons CGT, CGC, CGA, CGG, AGA, and AGG all encode the amino acid arginine. Thus, at every position where an arginine is specified by a codon, the codon can be altered to any of the corresponding codons described without altering the encoded protein. Such nucleic acid variations are "silent variations," which are one species of "conservatively modified variations." Every nucleic acid sequence described herein that encodes a polypeptide also describes every possible silent variation, except where otherwise noted. One of skill will recognize that each codon in a nucleic acid (except ATG, which is ordinarily the only codon for methionine, and TGG, which is ordinarily the only codon for tryptophan) can be modified to yield a functionally identical molecule by standard techniques. Accordingly, each "silent variation" of a nucleic acid that encodes a polypeptide is implicit in each described sequence.

Known methods of PCR include, but are not limited to, methods using paired primers, nested primers, single specific primers, degenerate primers, gene-specific primers, vector-specific primers, partially mismatched primers, and the like.

The terms "heterologous DNA sequence," "exogenous DNA segment," or "heterologous nucleic acid" refer to a sequence that originates from a source foreign to the particular host cell or, if from the same source, is modified from its original form. Thus, a heterologous coding region in a host cell includes a coding region that is endogenous to the particular host cell but has been modified through, for example, the use of single-stranded mutagenesis. The terms also include non-naturally-occurring multiple copies of a naturally occurring DNA sequence. Thus, the terms refer to a DNA segment that is foreign or heterologous to the cell, or homologous to the cell but in a position within the host cell nucleic acid in which the element is not ordinarily found. Exogenous DNA segments, when expressed, yield exogenous polypeptides.

A "homologous" DNA sequence is a DNA sequence that is naturally associated with a host cell into which it is introduced.

"Genome" refers to the complete genetic material of an organism.

"Coding sequence" refers to a DNA or RNA sequence that codes for a specific amino acid sequence and excludes non-coding (e.g., regulatory) nucleotide sequences. For example, a

DNA "coding sequence" or a "sequence encoding" a particular polypeptide is a DNA sequence that is transcribed and translated into a polypeptide in vitro or in vivo when placed under the control of appropriate regulatory elements. The boundaries of the coding sequence are determined by a start codon at the 5'-terminus and a translation stop codon at the 3'-terminus. A coding sequence can include, but is not limited to, prokaryotic sequences, cDNA from eukaryotic mRNA, genomic DNA sequences from eukaryotic (e.g., mammalian) DNA, and/or synthetic DNA sequences. A transcription termination sequence will usually be located 3' to the coding sequence. It may constitute an "uninterrupted coding sequence,"—i.e., lacking an intron, such as in cDNA or it may include one or more introns bounded by appropriate splice junctions. An "intron" is a sequence of RNA that is contained in the primary transcript but that is removed through cleavage and re-ligation of the RNA within the cell to create the mature mRNA that can be translated into a protein.

The terms "open reading frame" and "ORF" refer to the nucleotide sequence between translation initiation and termination codons of a coding sequence. The terms "initiation codon" and "termination codon" refer to a unit of three adjacent nucleotides ("codon") in a coding sequence that specifies initiation and chain termination, respectively, of protein synthesis (mRNA translation).

The term "RNA transcript" refers to the product resulting from RNA polymerase catalyzed transcription of a DNA sequence. When the RNA transcript is a perfect complementary copy of the DNA sequence, it is referred to as a primary transcript or it may be an RNA sequence derived from post transcriptional processing of the primary transcript and is referred to as the mature RNA. "Messenger RNA" (mRNA) refers to the RNA that is without introns and can be translated into protein by the cell. "cDNA" refers to a single- or double-stranded DNA that is complementary to and derived from mRNA.

The term "regulatory sequence" refers to a nucleotide sequence that includes, for example, a promoter, an enhancer, and/or other expression control elements (e.g., polyadenylation signals). Such regulatory sequences are known to those skilled in the art. The design of an expression

vector may depend on such factors as the choice of the host cell to be transfected and/or the amount of fusion protein to be expressed.

The term "DNA control elements" refers collectively to promoters, ribosome binding sites, polyadenylation signals, transcription termination sequences, upstream regulatory domains, enhancers, and the like, that collectively provide for the transcription and translation of a coding sequence in a host cell. Not all of these control sequences need always be present in a recombinant vector so long as the desired coding region is capable of being transcribed and translated.

A control element, such as a promoter, "directs the transcription" of a coding sequence in a cell when RNA polymerase binds to the promoter and transcribes the coding sequence into mRNA, which is then translated into the polypeptide encoded by the coding sequence.

A cell has been "transformed" by exogenous DNA when the exogenous DNA has been introduced inside the cell membrane. Exogenous DNA may or may not be integrated (covalently linked) into chromosomal DNA making up the genome of the cell. In prokaryotes and yeasts, for example, the exogenous DNA may be maintained on an episomal element, such as a plasmid. With respect to other eukaryotic cells, a stably transformed cell is one in which the exogenous DNA has become integrated into the chromosome so that it is inherited by daughter cells through chromosome replication. This stability is demonstrated by the ability of the eukaryotic cell to establish cell lines or clones having a population of daughter cells containing the exogenous DNA.

"Operably linked" refers to the association of nucleic acid sequences on single nucleic acid fragments so that the function of one is affected by the other, e.g., an arrangement of elements wherein the components so described are configured so as to perform their usual function. For example, a regulatory DNA sequence is said to be "operably linked to" a DNA sequence that codes for an RNA or a polypeptide if the two sequences are situated such that the regulatory DNA sequence affects expression of the coding DNA sequence (i.e., that the coding sequence or functional RNA is under the transcriptional control of the promoter). Coding sequences can be operably linked to regulatory sequences in sense or antisense orientation. Control elements operably linked to a coding sequence are capable of effecting the expression of the coding sequence. The control elements need not be contiguous with the coding sequence, so long as they function to direct the expression thereof. Thus, for example, intervening untranslated yet transcribed sequences can be present between a promoter and the coding sequence and the promoter can still be considered "operably linked" to the coding sequence.

"Transcription stop fragment" refers to nucleotide sequences that contain one or more regulatory signals, such as polyadenylation signal sequences, capable of terminating transcription. Examples include the 3' non-regulatory regions of the genes dysferlin (DYSF), pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PY-ROXD1), and collagen type VI alpha 3 chain (COL6A3).

"Translation stop fragment" or "translation stop code" or "stop codon" refers to nucleotide sequences that contain one or more regulatory signals, such as one or more termination codons in all three frames, capable of terminating translation. Insertion of a translation stop fragment adjacent to or near the initiation codon at the 5' end of the coding sequence will result in no translation or improper translation. The change of at least one nucleotide in a nucleic acid sequence can result in an interruption of the coding sequence of the gene, e.g., a premature stop codon. Such sequence changes

can cause a mutation in the polypeptide encoded by the DYSE, PYROXD1, or COL6A3 gene. For example, if the mutation is a nonsense mutation, the mutation results in the generation of a premature stop codon, causing the generation of a truncated DYSE, PYROXD1, or COL6A3 polypeptide.

Nucleic Acids

Nucleotide sequences that are subjected to the methods described herein can be obtained from any prokaryotic or eukaryotic source. For example, they can be obtained from a mammalian, such as equine, cellular source. Alternatively, nucleic acid molecules can be obtained from a library, such as the CHORI-241 Equine BAC library or a similar resource available elsewhere.

As discussed above, the terms “isolated and/or purified” refer to a nucleic acid—e.g. a DNA or RNA molecule—that has been isolated from its natural cellular environment and from association with other components of the cell, such as nucleic acid or polypeptide, so that it can be sequenced, replicated, and/or expressed. For example, an “isolated nucleic acid” may be a DNA molecule that is complementary or hybridizes to a sequence in a coding region of interest—e.g., a nucleic acid sequence encoding an equine collagen type VI alpha 3 chain protein, and remains stably bound under stringent conditions (as defined by methods well known in the art). Thus, an RNA or a DNA is “isolated” in that it is free from at least one contaminating nucleic acid with which it is normally associated in the natural source of the RNA or DNA and in one embodiment of the invention is substantially free of any other mammalian RNA or DNA. The phrase “free from at least one contaminating source nucleic acid with which it is normally associated” includes the case where nucleic acid is reintroduced into the source or natural cell but is in a different chromosomal location or is otherwise flanked by nucleic acid sequences not normally found in the source cell, e.g., in a vector or plasmid.

As used herein, the term “recombinant nucleic acid,” e.g., “recombinant DNA sequence or segment” refers to a nucleic acid, e.g., to DNA that has been derived or isolated from any appropriate cellular source, that may be substantially chemically altered in vitro, so that its sequence is not naturally occurring, or corresponds to naturally occurring sequences that are not positioned as they would be positioned in a genome that has not been transformed with exogenous DNA. An example of preselected DNA “derived” from a source would be a DNA sequence that is identified as a useful fragment within a given organism, and which is then chemically synthesized in essentially pure form. An example of such DNA “isolated” from a source would be a useful DNA sequence that is excised or removed from the source by chemical means, e.g., by the use of restriction endonucleases, so that it can be further manipulated, e.g. amplified, for use in the methods described herein. Thus, recovery or isolation of a given fragment of DNA from a restriction digest can employ separation of the digest on polyacrylamide or agarose gel by electrophoresis, identification of the fragment of interest by comparison of its mobility versus that of marker DNA fragments of known molecular weight, removal of the gel section containing the desired fragment, and separation of the gel from DNA. Therefore, “recombinant DNA” includes completely synthetic DNA sequences, semi-synthetic DNA sequences, DNA sequences isolated from biological sources, and DNA sequences derived from RNA, as well as mixtures thereof.

Nucleic acid molecules having base substitutions (i.e., variants) are prepared by a variety of methods known in the art. These methods include, but are not limited to, isolation

from a natural source (in the case of naturally occurring sequence variants) or preparation by oligonucleotide-mediated (or site-directed) mutagenesis, PCR mutagenesis, and cassette mutagenesis of an earlier prepared variant or non-variant version of the nucleic acid molecule.

Nucleic Acid Amplification Methods

DNA present in a physiological sample may be amplified by any means known to the art. Examples of suitable amplification techniques include, but are not limited to, polymerase chain reaction (including, for RNA amplification, reverse-transcriptase polymerase chain reaction), ligase chain reaction, strand displacement amplification, transcription-based amplification, self-sustained sequence replication (or “3SR”), the Q β -replicase system, nucleic acid sequence-based amplification (or “NASBA”), the repair chain reaction (or “RCR”), and boomerang DNA amplification (or “BDA”).

The bases incorporated into the amplification product may be natural or modified bases (modified before or after amplification), and the bases may be selected to optimize subsequent electrochemical detection steps.

Polymerase chain reaction (PCR) may be performed according to known techniques. In general, PCR involves, first, treating a nucleic acid sample (e.g., in the presence of a heat stable DNA polymerase) with one oligonucleotide primer for each strand of the specific sequence to be detected under hybridizing conditions so that an extension product of each primer is synthesized that is complementary to each nucleic acid strand, with the primers sufficiently complementary to each strand of the specific sequence to hybridize therewith so that the extension product synthesized from each primer, when it is separated from its complement, can serve as a template for synthesis of the extension product of the other primer, and then treating the sample under denaturing conditions to separate the primer extension products from their templates if the sequence or sequences to be detected are present. These steps are cyclically repeated until the desired degree of amplification is obtained. Detection of the amplified sequence may be carried out by adding to the reaction product an oligonucleotide probe capable of hybridizing to the reaction product (e.g., an oligonucleotide probe), the probe carrying a detectable label, and then detecting the label in accordance with known techniques. Where the nucleic acid to be amplified is RNA, amplification may be carried out by initial conversion to DNA by reverse transcriptase in accordance with known techniques.

Strand displacement amplification (SDA) may be performed according to known techniques. For example, SDA may be carried out with a single amplification primer or a pair of amplification primers, with exponential amplification being achieved with the latter. In general, SDA amplification primers comprise, in the 5 to 3 direction, a flanking sequence (the DNA sequence of which is noncritical), a restriction site for the restriction enzyme employed in the reaction, and an oligonucleotide sequence (e.g., an oligonucleotide probe) that hybridizes to the target sequence to be amplified and/or detected. The flanking sequence, which serves to facilitate binding of the restriction enzyme to the recognition site and provides a DNA polymerase priming site after the restriction site has been nicked, is about 15 to 20 nucleotides in length in one embodiment. The restriction site is functional in the SDA reaction: the oligonucleotide probe portion is about 13 to 15 nucleotides in length in one embodiment of the invention.

Ligase chain reaction (LCR) also may be performed according to known techniques. In general, the reaction is carried out with two pairs of oligonucleotide probes: one

pair binds to one strand of the sequence to be detected; the other pair binds to the other strand of the sequence to be detected; each pair together completely overlaps the strand to which it corresponds. The reaction is carried out by, first, denaturing (e.g., separating) the strands of the sequence to be detected, then reacting the strands with the two pairs of oligonucleotide probes in the presence of a heat stable ligase so that each pair of oligonucleotide probes is ligated together, then separating the reaction product, and then cyclically repeating the process until the sequence has been amplified to the desired degree. Detection may then be carried out in like manner as described above with respect to PCR.

In some embodiments, each exon of the DYSF, PYROXD1, or COL6A3 coding regions is amplified by PCR using primers based on the known sequence. The amplified exons are then sequenced using, for example, an automated sequencer. In this manner, the exons of the DYSF, PYROXD1, or COL6A3 coding regions from horses suspected of having Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), in their pedigree are then sequenced until a mutation is found. Examples of such mutations include those in Exon B of the DYSF DNA as shown in FIG. 3 and FIG. 4, in Exon I of the DYSF DNA as shown in FIG. 15 and FIG. 16, in Exon 12 of the PYROXD1 DNA as shown in FIG. 24 and FIG. 25, or in Exon 26 of the COL6A3 DNA as shown in FIG. 35 and FIG. 36. For example, mutations in the DYSF, PYROXD1, or COL6A3 genes include (1) the specific substitution of an adenine (A) for a guanine (G) on the forward strand at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 1, (2) the specific substitution of a thymine (T) for a guanine (G) on the forward strand at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 13, (3) the specific substitution of a cytosine (C) for a guanine (G) on the forward strand at chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 22, and (4) the specific substitution of a guanine (G) for a cytosine (C) on the forward strand at chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 33. Using this technique, additional mutations causing equine Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy, can be identified. Thus, the methods described herein may be used to detect and/or identify an alteration within the wild-type DYSF, PYROXD1, or COL6A3 locus. "Alteration of" a specified locus encompasses all forms of mutations including, for example, a deletion, an insertion, and/or a point mutation in the coding and noncoding regions. A deletion can involve the deletion of all or any portion of the coding region. A point mutation may result in an aberrant stop codon, a frameshift mutation, an amino acid substitution, and/or an alteration in pre-mRNA processing (splicing) that produces a protein with an altered amino acid sequence. Point mutational events may occur in regulatory regions, such as in the promoter of the gene, leading to decreased expression of the mRNA. A point mutation also may interfere with proper RNA processing, leading to decreased expression of the DYSF, PYROXD1, or COL6A3 translation products, decreased mRNA stability, and/or decreased translation efficiency. Polysaccharide Stor-

age Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy, is a disease caused by point mutations (1) at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 1, (2) at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 13, (3) at chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 22, or (4) chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 33. Horses predisposed to or having Polysaccharide Storage Myopathy type 2 (PSSM2) may need only one mutated DYSF, PYROXD1, or COL6A3 allele.

Techniques that are useful in performing the methods described herein include, but are not limited to direct DNA sequencing, PFGE analysis, allele-specific oligonucleotide (ASO), dot blot analysis, and/or denaturing gradient gel electrophoresis.

There are several methods that can be used to detect DNA sequence variation. Direct DNA sequencing, either manual or automated (e.g., fluorescent or semiconductor-based sequencing), can detect sequence variation. Another approach is the single-stranded conformation polymorphism assay (SSCA). This method does not detect all sequence changes, especially if the DNA fragment size is greater than 200 bp, but can be used to detect most DNA sequence variation. SSCA allows for increased throughput compared to direct sequencing for mutation detection on a research basis. The fragments that have shifted mobility on SSCA gels are then sequenced to determine the exact nature of the DNA sequence variation. Other approaches based on the detection of mismatches between the two complementary DNA strands include clamped denaturing gel electrophoresis (CDGE), heteroduplex analysis (HA), and chemical mismatch cleavage (CMC). Once a mutation is known, an allele specific detection approach such as allele specific oligonucleotide (ASO) hybridization can be utilized to rapidly screen large numbers of other samples for that same mutation. Such a technique can utilize probes that are labeled with gold nanoparticles to yield a visual color result.

Detecting point mutations may be accomplished by molecular cloning and then sequencing one or more DYSF, PYROXD1, or COL6A3 alleles. Alternatively, the coding region sequences can be amplified directly from a genomic DNA preparation from equine tissue, using known techniques. The DNA sequence of the amplified sequences can then be determined.

Exemplary methods for a more complete, yet still indirect, test for confirming the presence of a mutant allele include, for example, single stranded conformation analysis (SSCA), denaturing gradient gel electrophoresis (DDGE), an RNase protection assay, allele-specific oligonucleotides (ASOs), the use of a protein that recognizes nucleotide mismatches (e.g., the *E. coli* mutS protein), and allele-specific PCR. For allele-specific PCR, primers are used that hybridize at their 3' ends to a particular DYSF, PYROXD1, or COL6A3 mutation. If the particular mutation is not present, an amplification product is not observed. Allele-specific PCR may also be carried out using quantitative PCR or real-time PCR using a specialized instrument that is capable of detecting and quantifying the appearance of amplification products during each amplification cycle. An Amplification Refractory Mutation System (ARMS) can also be used. Insertions

and deletions of genes can also be detected by cloning, sequencing, and amplification. In addition, restriction fragment length polymorphism (RFLP) probes for the target locus or a surrounding marker locus can be used to score alteration of an allele or an insertion in a polymorphic fragment. Other techniques for detecting insertions or deletions as known in the art can also be used.

In the first three methods (i.e., SSCA, DGGE, and RNase protection assay), a new electrophoretic band appears. SSCA detects a band that migrates differently because the sequence change causes a difference in single-strand, intra-molecular base pairing. RNase protection involves cleaving the mutant polynucleotide into two or more smaller fragments. DGGE detects differences in migration rates of mutant sequences compared to wild-type sequences using a denaturing gradient gel. In an allele-specific oligonucleotide assay, an oligonucleotide is designed that detects a specific sequence, and the assay is performed by detecting the presence or absence of a hybridization signal. In the mutS assay, the protein binds only to sequences that contain a nucleotide mismatch in a heteroduplex between mutant and wild-type sequence.

As used herein, a "nucleotide mismatch" refers to a hybridized nucleic acid duplex in which the two strands are not 100% complementary. Lack of total homology may be due to a deletion, an insertion, an inversion, and/or a substitution. Mismatch detection can be used to detect point mutation in the coding region or its mRNA product. While these techniques are less sensitive than sequencing, they are simpler to perform on a large number of samples. An example of a mismatch cleavage technique is the RNase protection method. In the context of detecting a DYSE-associated, PYROXD1-associated, or COL6A3-associated mismatch, the method involves the use of a labeled riboprobe that is complementary to the horse wild-type DYSE, PYROXD1, or COL6A3 coding region sequence. The riboprobe and either mRNA or DNA isolated from tissue are annealed (i.e., hybridized) and subsequently digested with the enzyme RNase A, which is able to detect some mismatches in a duplex RNA structure. If a mismatch is detected by RNase A, it cleaves at the site of the mismatch. Thus, when the annealed RNA preparation is separated on an electrophoretic gel matrix, if a mismatch has been detected and cleaved by RNase A, an RNA product will be seen that is smaller than the full length duplex RNA for the riboprobe and the mRNA or DNA. The riboprobe need not be the full length of the DYSE, PYROXD1, or COL6A3 mRNA or coding region but can be a segment of either. If the riboprobe includes only a segment of the DYSE, PYROXD1, or COL6A3 mRNA or DNA, it may be desirable to use a number of probes to screen the whole mRNA sequence for mismatches.

In a similar fashion, DNA probes can be used to detect a mismatch through enzymatic and/or chemical cleavage. Alternatively, a mismatch can be detected by shifts in the electrophoretic mobility of mismatched duplexes relative to matched duplexes. With either riboprobes or DNA probes, the cellular mRNA or DNA that might contain a mutation can be amplified using PCR before hybridization.

Nucleic Acid Analysis via Microchip Technology

A DNA sequence of the DYSE, PYROXD1, or COL6A3 coding regions that has been amplified by PCR may be screened using an allele-specific probe. Allele-specific probes are nucleic acid oligomers, each of which contains a region of the DYSE, PYROXD1, or COL6A3 coding region harboring a known mutation. For example, one oligomer may be about 30 nucleotides in length, corresponding to a

portion of the DYSE, PYROXD1, or COL6A3 coding region sequence. Using a battery of such allele-specific probes, a PCR amplification product can be screened to identify the presence of a previously identified mutation in the DYSE, PYROXD1, or COL6A3 coding region. Hybridizing an allele-specific probe with an amplified DYSE, PYROXD1, or COL6A3 sequence can be performed, for example, on a nylon filter. Hybridizing to a particular probe under stringent hybridization conditions indicates the presence of the same mutation in the tissue as in the allele-specific probe.

An alteration of DYSE, PYROXD1, or COL6A3 mRNA expression can be detected by any technique known in the art. Exemplary techniques include, for example, Northern blot analysis, PCR amplification, and/or RNase protection. Decreased mRNA expression indicates an alteration of the wild-type DYSE, PYROXD1, or COL6A3 locus.

Alteration of wild-type DYSE, PYROXD1, or COL6A3 coding region also can be detected by screening for alteration of a wild-type DYSE, PYROXD1, or COL6A3 polypeptide such as, for example, the wild-type DYSE, PYROXD1, or COL6A3 protein or a portion of the wild-type DYSE, PYROXD1, or COL6A3 protein. For example, a monoclonal antibody immunoreactive with wild-type DYSE, PYROXD1, or COL6A3 (or to a specific portion of the DYSE, PYROXD1, or COL6A3 protein) can be used to screen a tissue. Lack of cognate antigen would indicate a mutation. An antibody specific for a product of a mutant allele also can be used to detect a mutation in the DYSE, PYROXD1, or COL6A3 coding region. Such an immunological assay can be performed using conventional methods. Exemplary methods include, for example, Western blot analysis, an immunohistochemical assay, an ELISA assay, and/or any method for detecting an altered DYSE, PYROXD1, or COL6A3 polypeptide. In some embodiments, a functional assay can be used such as, for example, protein binding determination. In addition, an assay can be used that detects DYSE, PYROXD1, or COL6A3 biochemical function. Finding a mutant DYSE, PYROXD1, or COL6A3 polypeptide indicates a mutation at the DYSE, PYROXD1, or COL6A3 locus.

A mutant DYSE, PYROXD1, or COL6A3 coding region or translation product can be detected in a variety of physiological samples collected from a horse. Examples of appropriate samples include a cell sample, such as a blood cell (e.g., a lymphocyte, a peripheral blood cell), a sample collected from the spinal cord, a tissue sample such as cardiac tissue or muscle tissue (e.g. cardiac or skeletal muscle) an organ sample (e.g., liver or skin), a hair sample, especially a hair sample with the hair bulb (roots) attached, and/or a fluid sample (e.g., blood).

The methods described herein are applicable to any equine disease in which DYSE, PYROXD1, or COL6A3 has a role. The method may be particularly useful for, for example, a veterinarian, a Breed Association, and/or individual breeders, so they can decide upon an appropriate course of treatment, and/or to determine if an animal is a suitable candidate as a brood mare or sire.

Oligonucleotide Probes

As described above, the method may be used to detect the presence and/or absence of a polymorphism in equine DNA. In particular, mutations in the DYSE, PYROXD1, or COL6A3 gene include the specific substitution (1) of an adenine (A) for a guanine (G) on the forward strand at chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 1, (2) of a thymine (T) for a guanine (G) on the forward

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strand at chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 13, (3) the specific substitution of a cytosine (C) for a guanine (G) on the forward strand at chr6:47,661, 977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 22, and the specific substitution of a guanine (G) for a cytosine (C) on the forward strand at chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown for the reverse strand in FIG. 33. These substitutions result in (1) an arginine (R) at codon 253 in the dysferlin (DYSF) protein (SEQ ID NO:10) being replaced by a tryptophan (W), as shown in SEQ ID NO: 11, (2) a proline (P) at codon 1290 in the dysferlin (DYSF) protein (SEQ ID NO:57) being replaced by a threonine (T), as shown in SEQ ID NO:58, (3) an aspartate (D) at codon 492 in the pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) protein (SEQ ID NO:104) being replaced by a histidine (H), as shown in SEQ ID NO: 105, and (4) a glycine (G) at codon 2182 in the collagen type VI alpha 3 chain (COL6A3) protein (SEQ ID NO:164) being replaced by an alanine (A), as shown in SEQ ID NO: 165.

A primer pair may be used to determine the nucleotide sequence of a particular DYSF, PYROXD1, or COL6A3 allele using PCR. A pair of single-stranded DNA primers can be annealed to sequences within or surrounding the DYSF, PYROXD1, or COL6A3 coding region in order to prime amplifying DNA synthesis of the DYSF, PYROXD1, or COL6A3 coding region itself. A complete set of primers allows one to synthesize all of the nucleotides of the DYSF, PYROXD1, or COL6A3 coding sequence. In some embodiments, a set of primers can allow synthesis of both intron and exon sequences. In some embodiments, allele-specific primers can be used. Such primers anneal only to particular DYSF, PYROXD1, or COL6A3 mutant alleles, and thus will only amplify product efficiently in the presence of the mutant allele as a template.

The first step of the process involves contacting a physiological sample obtained from a horse, which sample contains nucleic acid, with an oligonucleotide probe to form a hybridized DNA. The oligonucleotide probe can be any probe having from about 4 or 6 bases up to about 80 or 100 bases or more. In one embodiment, the oligonucleotide probe can have between about 10 and about 20 bases.

The primers themselves can be synthesized using conventional techniques and, in some cases, can be made using an automated oligonucleotide synthesizing machine. Given the DYSF, PYROXD1, or COL6A3 genomic sequences as partially set forth in SEQ ID NO:1, SEQ ID NO:56, SEQ ID NO:103, and SEQ ID NO:163, one can design a set of oligonucleotide primers to probe any portion of the DYSF, PYROXD1, or COL6A3 coding sequences. The primers may be designed to hybridize entirely to coding sequence (exons), to noncoding sequence (introns or other noncoding sequences), or to regions spanning the junction of coding and noncoding sequences in genomic DNA.

An oligonucleotide probe may be prepared according to conventional techniques to have any suitable base sequence. Suitable bases for preparing the oligonucleotide probe may be selected from naturally-occurring bases such as adenine, cytosine, guanine, uracil, and thymine. An oligonucleotide probe also can incorporate one or more non-naturally-occurring or "synthetic" nucleotide bases. Exemplary synthetic bases include, for example, 7-deaza-guanine, 8-oxo-

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guanine, 6-mercaptoguanine, N4-acetylcytidine, 5-(carboxyhydroxyethyl)uridine, 2'-O-methylcytidine, 5-(carboxymethylaminomethyl)-2-thiouridine, 5-carboxymethylaminomethyluridine, dihydrouridine, 2'-O-methylpseudouridine, β -D-galactosylqueuosine, 2"-O-methylguanosine, inosine, N6-isopentenyladenosine, 1-methyladenosine, 1-methylpseudouridine, 1-methylguanosine, 1-methylinosine, 2,2-dimethylguanosine, 2-methyladenosine, N2-methylguanosine, 3-methylcytidine, 5-methylcytidine, N6-methyl adenosine, 7-methylguanosine, 5-methylaminomethyluridine, 5-methoxyaminomethyl-2-thiouridine, β -D-mannosylqueuosine, 5-methoxycarbonylmethyluridine, 5-methoxyuridine, 2-methylthio-N6-isopentenyladenosine, N-((9- β -D-ribofuranosyl-2-methylthiopurine-6-yl)carbamoyl)threonine, N-((9- β -D-ribofuranosylpurine-6-yl)N-methyl-carbamoyl)threonine, uridine-5-oxyacetic acid methylester, uridine-5-oxyacetic acid, wybutosine, pseudouridine, queuosine, 2-thiocytidine, 5-methyl-2-thiouridine, 2-thiouridine, 5-methyluridine, N-((9-beta-D-ribofuranosylpurine-6-yl)carbamoyl)threonine, 2'-O-methyl-5-methyluridine, 2'-O-methyluridine, wybutosine, and/or 3-(3-amino-3-carboxypropyl)uridine. Any oligonucleotide backbone may be employed, including DNA, RNA (although RNA may be less preferred than DNA in certain circumstances), modified sugars such as carbocycles, and sugars containing 2 substitutions (e.g., fluoro or methoxy). The oligonucleotides may be oligonucleotides wherein at least one, or all, of the internucleotide bridging phosphate residues is a modified phosphate such as, for example, a methyl phosphate, a methyl phosphonotioate, a phosphoroinorpholidate, a phosphoropiperazidate, and/or a phospholioramidate—for example, every other one of the internucleotide bridging phosphate residues may be modified. The oligonucleotide may be a "peptide nucleic acid" such as described in Nielsen et al., Science, 254, 1497-1500 (1991).

The oligonucleotide probe should possess a sequence at least a portion of which is capable of binding to a known portion of the sequence of the nucleic acid in the physiological sample.

In some embodiments, the nucleic acid in the sample may be contacted with a plurality of oligonucleotide probes having different base sequences (e.g., where there are two or more target nucleic acids in the sample, or where a single target nucleic acid is hybridized to two or more probes in a "sandwich" assay).

The oligonucleotide probes provided herein may be useful for a number of purposes. For example, the oligonucleotide probes can be used to detect PCR amplification products and/or to detect mismatches with the DYSF, PYROXD1, or COL6A3 coding region or mRNA.

Hybridization Methodology

The nucleic acid from the physiological sample may be contacted with the oligonucleotide probe in any conventional manner. For example, the sample nucleic acid may be solubilized in solution and contacted with the oligonucleotide probe by solubilizing the oligonucleotide probe in solution with the sample nucleic acid under conditions that permit hybridization. Suitable hybridization conditions are well known to those skilled in the art. Alternatively, the sample nucleic acid may be solubilized in solution with the oligonucleotide probe immobilized on a solid or semisolid support, whereby the sample nucleic acid may be contacted with the oligonucleotide probe by immersing the solid or semisolid support having the oligonucleotide probe immobilized thereon in the solution containing the sample nucleic acid.

Certain embodiments of the methods described herein relate to mutations in the DYSF, PYROXD1, or COL6A3 coding region or the diagnosis of Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), or the detection of a predisposition for Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), or to the detection of a mutant DYSF, PYROXD1, or COL6A3 allele in a horse.

Mutations in the equine DYSF, PYROXD1, or COL6A3 coding regions (encoding the dysferlin, pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1, and skeletal muscle protein collagen type VI alpha 3 chain) are present in many populations of horses affected by Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM). The differences in the genomic DNA between horses affected by Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM) include (1) point mutations at nucleic acid chr15:31,306,949 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 1, (2) point mutations at nucleic acid chr15:31,225,630 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 13, (3) chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 22, and (4) chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1) as displayed in the UCSC Genome Browser and as shown in FIG. 33.

Scientific Narrative

Dysferlin (DYSF)

A mutation in the equine DYSF coding region (encoding the membrane protein dysferlin) is present in many populations of horses affected by Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM). The difference in the genomic DNA between horses with PSSM2 and control horses include a G-to-A substitution in DYSF Exon B (as defined in FIG. 3) at nucleotide position chr15:31,306,949.

FIG. 1 shows a portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr15:31,306,949 position, the site of a substitution of an adenine (A) for a guanine (G) that results in the substitution of a tryptophan (W) for arginine (R) at amino acid position 253 in dysferlin as shown in FIG. 4 (SEQ ID NO:11). The reverse complement sequence is shown, with the site of a substitution of a thymine (T) for a cytosine (C) as indicated (SEQ ID NO:1). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1145077095 in dbSNP.

There are 21 predicted transcripts of the DYSF coding region in public databases. These models differ somewhat in ways not relevant to the DYSF-R253W mutation. All 21 isoforms share a common segment that includes the chr15:31,306,949 position. This common segment is shown in FIG. 2.

FIG. 2 shows a portion of the normal equine DYSF Coding DNA Sequence (SEQ ID NO:2) and the mutant DYSF Coding DNA Sequence (SEQ ID NO:3) bearing the C to T mutation at nucleotide position 1027 in this figure, corresponding to chr15:31,306,949 as shown in SEQ ID NO:1 (FIG. 1). This sequence is a region of perfect consen-

sus among 21 different experimentally predicted mRNA isoforms. The numbering in FIG. 2 is that of isoform X1 (XM_023618907.1), which for this segment perfectly matches the numbering of isoforms X2 (XM_023618908.1), X3 (XM_023618909.1), X4 (XM_023618910.1), X5 (XM_023618911.1), and X6 (XM_023618912.1). The numbering of the start and end positions for X7 (XM_023618913.1) is 671-1699. The numbering for the start and end positions for X8 (XM_023618914.1) is 763-1791. The numbering for the start and end positions of X9 (XM_023618915.1), X11 (XM_023618917.1), X12 (XM_023618918.1), and X13 (XM_023618919.1) is 670-1698. The numbering for the start and end positions of X10 (XM_023618916.1) is 763-1791. The numbering for the start and end positions of X14 (XM_023618920.1) and X15 (XM_023618921.1) is 1075-2103. The numbering for the start and end positions of X16 (XM_023618922.1) and X18 (XM_023618924.1) is 1074-2102. The numbering for the start and end positions of X17 (XM_023618923.1) and X19 (XM_023618925.1) is 981-2009. The numbering for the start and end positions of X20 (XM_023618926.1) and X21 (XM_023618927.1) is 980-2008. In both sequences, the sequence of Exon B as shown in FIG. 3 is indicated in bold. The site of a C to T mutation site at nucleotide position 1027, corresponding to 31,306,949 in SEQ ID NO:1 (FIG. 1), and to rs1145077095 in dbSNP, is underlined. The region of sequence comprising Exon B as shown in FIG. 3 is displayed as codons in the correct reading frame for both SEQ ID NO:2 and SEQ ID NO: 3 in FIG. 5.

For the sake of simplicity, the amino acid substitution caused by the substitution of an adenine (A) for a guanine (G) at the chr15:31,306,949 position, identified as rs1145077095 in dbSNP, will be referred to as DYSF-R253W, based on SEQ ID NO:11. The amino acid substitution caused by this mutation will remain the same regardless of the numerical position of the affected codon in alternative gene models. The wild-type (e.g., also referred to herein as “normal” or “unaffected”) allele of this coding region may be referred to as R253 and the mutant allele R253W.

FIG. 3 shows a view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the partial DYSF Coding DNA Sequence (SEQ ID NO:2) and the mutant partial DYSF Coding DNA Sequence (SEQ ID NO:3). The DYSF Coding DNA Sequences correspond to Exons A, B, C, D, E, and F as indicated by the regions of sequence similarity of the translated genomic DNA to DYSF protein sequences from human (Homo), cattle (Bos), rat (Rattus), mouse (Mus), and DYSF protein sequences from the more distantly-related zebrafish (Danio) and African claw-toed frog (Xenopus). Partial matches to the paralogous protein myoferlin (MYOF) from human (Homo), cattle (Bos), mouse (Mus), rat (Rattus), and zebrafish (Danio) are also seen. The sequences of Exons A-F with 10 bp of flanking intron sequence and their coordinates in the current horse assembly are displayed below the image from the UCSC Genome Browser. Sequence IDs are Exon A (SEQ ID NO:4), Exon B (SEQ ID NO:5), Exon C (SEQ ID NO:6), Exon D (SEQ ID NO:7), Exon E (SEQ ID NO:8), and Exon F (SEQ ID NO:9).

FIG. 4 shows a model of part of the normal protein sequence encoded by horse DYSF (XP_023474694.1, presented here as SEQ ID NO:10) corresponding to a translation of SEQ ID NO:2 shown in FIG. 2 and part of the altered protein sequence encoded by horse DYSF with the base substitution at chr15:31,306,949 (based on

XP_023474694.1, presented here as SEQ ID NO:11) corresponding to a translation of SEQ ID NO:3 shown in FIG. 2. The portion of the protein encoded by Exon B as shown in FIG. 3 is indicated in bold, while the amino acid position affected by the base substitution of an adenine (A) for a guanine (G) on the forward strand at the chr15:31,306,949 position, corresponding to rs1145077095 in dbSNP as shown in FIG. 1, is underlined. The amino acid positions in XP_023474694.1 are indicated at the beginning and end of the sequence.

FIG. 5 shows horse DYSF Exon B and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the DYSF-R253W mutation would be most appropriately derived. Genomic coordinates are as in FIG. 1. Exon B from chr15:31,307,036 to chr15:31,306,908 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:12) and the DYSF-R253W allele (SEQ ID NO:13). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to A mutation site at nucleotide position chr15:31,306,949, corresponding to rs1145077095 in dbSNP as shown in FIG. 1 (C to T in the reverse complement as shown), is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for an arginine (CGG) to one coding for a tryptophan (TGG). Example primers used experimentally to amplify genomic DNA containing the mutation site are shown in lower case

(SEQ ID NO: 14)
[5' - CCCGAGATTTCTGGCTTTCT - 3'
and
(SEQ ID NO: 15)
5' - CTCGACAAGTTCTGGGGTGT - 3'] .

Genomic DNA obtained from horses can be genotyped by amplifying a region containing a variant in the DYSF gene using Polymerase Chain Reaction (PCR), then sequencing the amplified DNA using Sanger sequencing. The variant allele DYSF-R253W is abbreviated as P5, while the common or wild-type allele is abbreviated as N. The results can be scored as homozygous for the common or wild-type allele (N/N), heterozygous for the nucleotide substitution (N/P5), or homozygous for the nucleotide substitution (P5/P5).

FIG. 6 shows traces from Sanger DNA sequencing of amplified DYSF genomic DNA using primers shown in FIG. 5 (SEQ ID NO:14 and SEQ ID NO:15). The sequence of the reverse strand is shown. The arrows in the figure indicate nucleotide position chr15:31,306,949, the site of a substitution of a thymine (T) for a cytosine (C) in this position, corresponding to rs1145077095 in dbSNP, that creates the DYSF-R253W variant. The traces show, from left to right, results for a horse homozygous for the wild-type or common allele (N/N), results for a horse heterozygous for the substitution (N/P5), and results for a horse homozygous for the substitution (P5/P5).

Dysferlin is a member of a family of genes with a transmembrane domain. The majority of the protein faces the cytoplasm. Dysferlin has seven C2 domains that are implicated in calcium-dependent membrane fusion events. Dysferlin plays an important role in the repair of muscle fibers.

Mice homozygous for targeted mutations in DYSF that are expected to result in a total loss of function display

defects in skeletal muscle. These defects include “dystrophic” muscle that exhibits a progressive muscular weakness. Specific defects in limb grasping are observed. The morphology of muscle fibers in mice homozygous for loss-of-function alleles include centrally located nuclei, increased variability of skeletal muscle fiber size, and skeletal muscle fiber degeneration. Increased serum creatine kinase (CK), a sign of muscle damage, is also seen.

The human ortholog of the equine DYSF gene and the human protein that this gene encodes are richly annotated with experimental data derived from genetic and biochemical studies. It is informative to compare the amino acid substitutions in DYSF found in horses to the information on protein domains and clinically significant variation in the human dysferlin (DYSF) protein. In order to do this, the equine protein models used in this disclosure must be compared to the canonical or reference sequence of the human protein in a public database that captures data from the published literature, such as UniProt.

A large number of pathogenic mutations in human DYSF are known. Many pathogenic alleles are nonsense mutations, that is, the mutation of a codon encoding an amino acid to a termination codon, causing the truncation of the encoded dysferlin at the site of the mutation. Other pathogenic alleles are frameshift mutations, that is, the addition or deletion of bases within coding regions where the number of bases is not an integral multiple of three, therefore altering the reading frame downstream of the mutation. The altered reading frame typically reaches a termination codon, causing truncation of the encoded dysferlin downstream of the mutation following a segment of altered amino acid sequence. There are also pathogenic missense alleles of DYSF, in which a point mutation causes the alteration of a single amino acid in the dysferlin protein sequence. A set of such alleles is presented in TABLE 1. Cited sources in this table are Aoki 2001 (Aoki et al. 2001 Neurology 57:271-278), Nguyen 2005 (Nguyen 2005 Hum Mut DOI: 10.1002/humu.9355), Krahn 2008 (Krahn et al. 2009 Hum Mut 30:E345-375 DOI: 10.1002/humu.20910), LMDD (<http://www.dmd.n1>), and ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>).

Human patients homozygous for missense alleles of DYSF as listed in TABLE 1 may present as cases of Limb-Girdle Muscular Dystrophy 2B or other conditions that overlap this condition clinically, such as Miyoshi Myopathy, Proximal Myopathy, Pseudometabolic Myopathy, or Isolated HyperCKemia (a disorder in which serum levels of creatine kinase or CK, a muscle enzyme, are elevated, indicating damage to the sarcolemma).

TABLE 1

Pathogenic missense alleles of the C2B domain of human DYSF.			
Substitution	Domain	Reference	
S209F	C2B	ClinVar	
Q221H	C2B	ClinVar	
I227T	C2B	Jin 2016	
P233L	C2B	ClinVar	
G234E	C2B	Krahn 2009, ClinVar	
R251W	C2B	ClinVar	
T252M	C2B	Jin 2016, ClinVar	
R253W	C2B	Nguyen 2005, ClinVar	
H255P	C2B	ClinVar	
E264D	C2B	UMD, ClinVar	
L266R	C2B	UMD, ClinVar	
I282T	C2B	UMD	
I284T	C2B	Krahn 2009, ClinVar	

TABLE 1-continued

Pathogenic missense alleles of the C2B domain of human DYSF.		
Substitution	Domain	Reference
T285M	C2B	ClinVar
V286E	C2B	UMD, ClinVar
D288V	C2B	Jin 2016, ClinVar
S289P	C2B	ClinVar
S289F	C2B	ClinVar
G299R	C2B	Wenzel 2006, Spuler 2008, Krahn 2009, ClinVar
G299E	C2B	UMD, ClinVar
G299W	C2B	Spuler 2008, ClinVar, Hofhuis 2017

Cited sources in this table are:

ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>), UMD (<http://www.umd.be/DYSF/>) Davis 2002 (Davis D B et al. 2002 J Biol Chem 277:22883-22888 DOI: 10.1074/jbc.M201858200)

Hofhuis 2017 (Hofhuis J et al. 2017 J Cell Sci 130:841-852 DOI: 10.1242/jcs.198861)

Huang 2007 (Huang Y et al. 2007 FASEB J. 21:732-742. DOI: 10.1096/fj.06-6628com)

Illarioshkin 2000 (Illarioshkin S N et al. 2000 Neurology 55:1931-1933)

Jin 2016 (Jib S-Q et al. 2016 Chin Med J 129:2287-2293 DOI: 10.4103/0366-6999.190671)

Krahn 2009 (Krahn M et al. 2009 Hum Mutat. 30:E345-375 DOI: 10.1002/humu.20910)

Nguyen 2005 (Nguyen K et al. 2005 Hum Mutat. 26:165 DOI: 10.1002/humu.9355),

Spuler 2008 (Spuler S et al. 2008 Ann Neurol. 63:323-328 DOI: 10.1002/ana.21309)

Wenzil 2006 (Wenzil K et al. 2006 Hum Mutat. 27:599-600 DOI: 10.1002/humu.9424)

The C2B domain of dysferlin is highly conserved overall. Comparison to the other two C2 domains with type II topology (C2A and C2E) is helpful in assessing the pathogenicity of missense alleles. TABLE 2 shows pathogenic missense alleles in the other two C2 domains with type II topology.

TABLE 2

Pathogenic missense alleles of the C2A and C2E domains of human DYSF.		
Substitution	Domain	Reference
M1V	C2A	ClinVar
M1T	C2A	ClinVar
I6N	C2A	ClinVar
W46R	C2A	UMD
W52R	C2A	Krahn 2009, ClinVar
V67D	C2A	Illarioshkin 2000, Davis 2002, Huang 2007, ClinVar, Hofhuis 2017
V69G	C2A	ClinVar
T74M	C2A	ClinVar
R1331L	C2E	ClinVar
R1331C	C2E	ClinVar
L1341P	C2E	Wenzel 2006, ClinVar, Hofhuis 2017
R1342Q	C2E	ClinVar
R1342G	C2E	Jin 2016

TABLE 2-continued

Pathogenic missense alleles of the C2A and C2E domains of human DYSF.		
Substitution	Domain	Reference
R1342W	C2E	UMD, ClinVar
C1361R	C2E	UMD, Campanaro 2002
P1400R	C2E	UMD, ClinVar
G1418D	C2E	ClinVar

Cited sources in this table are:

ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>), UMD (<http://www.umd.be/DYSF/>) Davis 2002 (Davis D B et al. 2002 J Biol Chem 277:22883-22888 DOI: 10.1074/jbc.M201858200)

Hofhuis 2017 (Hofhuis J et al. 2017 J Cell Sci 130:841-852 DOI: 10.1242/jcs.198861)

Huang 2007 (Huang Y et al. 2007 FASEB J. 21:732-742. DOI: 10.1096/fj.06-6628com)

Illarioshkin 2000 (Illarioshkin S N et al. 2000 Neurology 55:1931-1933)

Jin 2016 (Jib S-Q et al. 2016 Chin Med J 129:2287-2293 DOI: 10.4103/0366-6999.190671)

Krahn 2009 (Krahn M et al. 2009 Hum Mutat. 30:E345-375 DOI: 10.1002/humu.20910)

Wenzil 2006 (Wenzil K et al. 2006 Hum Mutat. 27:599-600 DOI: 10.1002/humu.9424)

FIG. 7 shows the sequence of the human DYSF coding sequence derived from NM_003494.3 (SEQ ID NO:16). The 5' UTR and 3' UTR have been removed; the sequence begins with the ATG start codon and ends with the TGA stop codon. The numbering of the first and last nucleotides corresponds to that of NM_003494.3. The sequence of the human exon corresponding to Exon B in horse as shown in FIG. 3 is indicated in bold.

FIG. 8 shows the sequence of the human DYSF protein sequence, equivalent to NP_003485.1 (SEQ ID NO:17). The numbering of the first and last amino acids corresponds to that of NP_003485.1. The sequence encoded by the human exon corresponding to Exon B in horse as shown in FIG. 3 is indicated in bold.

Comparison of the equine DYSF-R253W missense allele to the pathogenic human alleles presented in TABLE 1 is informative. FIG. 9 shows a comparison of that portion of the protein sequence of DYSF encoded by horse Exon B as defined in FIG. 3 from wild type (XP_023474694.1, shown here as SEQ ID NO:19) and DYSF-R253W (R253W, shown here as SEQ ID NO:20) to the protein sequence of DYSF encoded by human Exon 7 (NP_003485.1, shown here as SEQ ID NO:18). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a space () indicates the nonconservative substitution of an arginine (R) in horse for a glycine (G) at human position 247, and a plus sign (+) indicates the conservative substitution of an arginine (R) in horse for a lysine (K) at human position 256. No other amino acid substitutions are seen in comparison of the human sequence (SEQ ID NO:18) and the wild-type horse sequence (SEQ ID NO:19). The sequence from horse bearing the DYSF-R253W mutation (SEQ ID NO:20) has a nonconservative substitution of a tryptophan (W) for an arginine (R) that is found in human (SEQ ID NO:18) and the wild-type horse sequence (SEQ ID NO:19). This position, corresponding to position 253 in horse (SEQ ID NO:19, SEQ ID NO:20) and position 251 in human (SEQ ID NO:18) is indicated by bold and underlining.

FIG. 10 shows features of the dysferlin protein encoded by the human DYSF gene. The protein contains seven C2 calcium-binding domains designated C2A through C2G. The secondary structure of the protein sequence encoded by each of the C2 domains consists of eight segments assembling into beta sheet, with two different topologies described. Domains C2C, C2D, C2F, and C2G are designated as topology type I, while domains C2A, C2B, and C2E are designated as topology type II (C.Therrien et al. 2006 J. Neurological Sciences 250: 71-78). The positions of four additional conserved domains (DysfN, DysfC, annexin binding, and transmembrane) are also indicated (C.Therrien et al. 2006 J. Neurological Sciences 250: 71-78). Positions of pathogenic missense alleles listed in TABLE 1 are shown.

The sequence comparison between human and equine dysferlin proteins shows the high degree of sequence conservation between these two species. It also shows that the human equivalent to the equine DYSF-R253W allele would be DYSF-R251W. This allele has been observed in human patients and has been scored as pathogenic (TABLE 1). The extent of sequence conservation between horse and human is high, and it is of value to examine the extent of sequence conservation among a larger number of species.

FIG. 11 shows amino acid sequences of proteins encoded by the C2B domain of DYSF, including the position of the equine DYSF-R253W substitution. The portion of the human DYSF corresponding to the C2B domain, as identified by UniProt annotation (positions 207-302 of NP_003485.1, identical to O75923) was used as a probe in BLASTP searches of the protein sequence database to identify the corresponding region in the orthologous protein in a set of mammals. Extending this approach to more distantly related organisms (birds, reptile, amphibians, and fish) was not successful; either no match was found, or the top match was to a nonorthologous protein (e.g. myoferlin). In the alignment shown in FIG. 11, sequences that are identical over this range of positions in different species are clustered as a single SEQ ID NO.

The 44 species in the alignment shown in FIG. 11 are presented below with their common names, their scientific names, and the protein database ID from which the amino acid sequence is derived. Amino acid sequences identical over this range in different species have been clustered into single SEQ ID NOs for the alignment. The species associated with the SEQ ID NOs shown in the alignment are: SEQ ID NO:21 [Human (*Homo sapiens*, XP_005264642.1), Chimpanzee (*Pan troglodytes*, XP_016804238.1), Bonobo (*Pan paniscus*, XP_003808226.1), Western lowland gorilla (Gorilla gorilla gorilla, XP_004029449.2), Sumatran orang-utan (*Pongo abelii*, XP_024097762.1)], SEQ ID NO:22 [Olive baboon (*Papio anubis*, XP_021780640.1), Rhesus macaque (*Macaca mulatta*, XP_014968111.1), Black snub-nosed monkey (*Rhinopithecus bieti*, XP_017733098.1)], SEQ ID NO:23 [Mouse (*Mus musculus*, XP_006506246.1), Brown rat (*Rattus norvegicus*, XP_006236829.1)], SEQ ID NO:24 [Prairie vole (*Microtus ochrogaster*, XP_005369881.1), Mongolian gerbil (*Meriones unguiculatus*, XP_021497362.1)], SEQ ID NO:25 [Long-tailed chinchilla (*Chinchilla lanigera*, XP_013370151.1), Damora mole-rat (*Fukomys damarensis*, KF021095.1)], SEQ ID NO:26 [David's myotis (*Myotis davidii*, XP_006775358.1), Brandt's bat (*Myotis brandtii*, XP_005867829.1)], SEQ ID NO:27 [Large flying fox (*Pteropus vampyrus*, XP_023389621.1), Alpaca (*Vicugna pacos*, XP_015092494.1), Dromedary (*Camelus dromedaries*, XP_010992957.1), Bactrian camel (*Camelus bactrianus*, XP_010962007.1)], SEQ ID NO:28 [Horse (*Equus caballus*,

XP_023474692.1), Donkey (*Equus asinus*, XP_014692144.1), Przewalski's horse (*Equus przewalskii*, XP_008534159.1)], SEQ ID NO:29 [Cat (*Felis catus*, XP_003984159.2), Leopard (*Panthera pardus*, XP_019297096.1)], SEQ ID NO:30 [Goat (*Capra hircus*, XP_017910544.1), Mouflon (*Ovis aries musimon*, XP_011978930.1)], SEQ ID NO:31 [Panamanian white-throated capuchin (*Cebus capucinus imitator*, XP_017377464.1)], SEQ ID NO:32 [Gray mouse lemur (*Microcebus murinus*, XP_012605526.1)], SEQ ID NO:33 [Philippine tarsier (*Carlito syrichta*, XP_008058745.1)], SEQ ID NO:34 [Northern greater galago (*Otolemur garnettii*, XP_012662868.1)], SEQ ID NO:35 [Coquerel's sifika (*Propithecus coquereli*, XP_012501927.1)], SEQ ID NO:36 [Chinese hamster (*Cricetulus griseus*, ERE66849.1)], SEQ ID NO:37 [Thirteen-lined ground squirrel (*Ictidomys tridecemlineatus*, XP_005322205.1)], SEQ ID NO:38 [Alpine marmot (*Marmota marmota marmota*, XP_015335007.1)], SEQ ID NO:39 [American beaver (*Castor canadensis*, XP_020009964.1)], SEQ ID NO:40 [Ferret (*Mustela putorius furo*, XP_004742226.1)], SEQ ID NO:41 [Great roundleaf bat (*Hipposideros armiger*, XP_019503212.1)], SEQ ID NO:42 [African bush elephant (*Loxodonta africana*, XP_023408298.1)], SEQ ID NO:43 [Wild boar (*Sus scrofa*, XP_013851496.2)], SEQ ID NO:44 [Cattle (*Bos taurus*, NP_001095960.1)], SEQ ID NO:45 [Minke whale (*Balaenoptera acutorostrata scammoni*, XP_007175590.1)], SEQ ID NO:46 [Giant panda (*Ailuropoda melanoleuca*, XP_019650380.1)], SEQ ID NO:47 [Killer whale (*Orcinus orca*, XP_004277106.1)], SEQ ID NO:48 [Koala (*Phascogale cinereus*, XP_020856837.1)], SEQ ID NO:49 [Tasmanian devil (*Sarcophilus harrisii*, XP_023351344.1)], SEQ ID NO:50 [Platypus (*Ornithorhynchus anatinus*, XP_007665017.1)].

The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated by period (.), and nonconserved positions are indicated by a blank space (). The last line shows the position of the DYSF-R253W substitution in horse in bold. The position of the DYSF-R253W substitution is indicated in bold in all of the sequences.

The overall sequence conservation of the C2B domain of dysferlin is quite high across mammals. The amino acid alteration seen in the equine DYSF-R253W substitution is not seen in any of the 44 species; all have an arginine (R) at this position. The comparison of dysferlin C2B sequences presented here refutes the hypothesis that the DYSF-R253W substitution is selectively neutral, and supports the claim that the DYSF-R253W mutation found in horses with PSSM2 is pathogenic.

FIG. 12 shows comparison of the C2A, C2B, and C2E domains of dysferlin encoded by human DYSF to the C2B domain of dysferlin encoded by horse DYSF. (A) The amino acid sequences of two isoforms of the C2A domain of human dysferlin (SEQ ID NO:51 and SEQ ID NO:52), the C2B domain of human dysferlin (SEQ ID NO:53), the C2E domain of human dysferlin (SEQ ID NO:54), and the C2B domain of horse dysferlin (SEQ ID NO:55), are shown. (B) Clustal Omega was used to align these four sequences. The fifth line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated by

cated by period (.), and nonconserved positions are indicated by a blank space (). The remaining lines show the position of the horse DYSF-R253W substitution and various pathogenic human substitutions (presented in TABLE 1 and TABLE 2) aligned to the consensus.

The comparison of C2A, C2B, and C2E domains reveals a much more limited number of highly conserved positions. The position affected by the equine DYSF-R253W mutation is highly conserved, with only arginine (R) and glutamine (Q) seen in this position. In general, there is a good correspondence between positions that are conserved and the positions of substitutions identified in pathogenic human DYSF alleles. This analysis also refutes the hypothesis that the DYSF-R253W substitution is selectively neutral, and supports the claim that the DYSF-R253W mutation found in horses with Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM) is pathogenic.

FIG. 13 shows portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr15:31,225,630 position, the site of a substitution of an thymine (T) for a guanine (G) that results in the substitution of a threonine (T) for proline (P) at amino acid position 1290 in dysferlin as shown in FIG. 16 (SEQ ID NO:61). The reverse complement sequence is shown, with the site of a substitution of an adenine (A) for a cytosine (C) as indicated (SEQ ID NO:56). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1136366555 in dbSNP

FIG. 14 shows a portion of the normal equine DYSF Coding DNA Sequence (SEQ ID NO:57) and the mutant DYSF Coding DNA Sequence (SEQ ID NO:58) bearing the C to A mutation at nucleotide position 4174 in this figure, corresponding to chr15:31,225,630 as shown in SEQ ID NO:56 (FIG. 13). This sequence is a region of perfect consensus among 21 different experimentally predicted mRNA isoforms. The numbering in FIG. 14 is that of isoform X1 (XM_023618907.1), which for this segment perfectly matches the numbering of isoforms X2 (XM_023618908.1), X4 (XM_023618910.1), and X5 (XM_023618911.1). The numbering of the start and end positions for X3 (XM_023618909.1) and X6 (XM_023618912.1) is 3685-4143. The numbering of the start and end positions for X7 (XM_023618913.1) is 3634-4092. The numbering of the start and end positions for X8 (XM_023618914.1) is 3726-4184. The numbering of the start and end positions for X9 (XM_023618915.1) and X12 (XM_023618918.1) is 3633-4091. The numbering of the start and end positions for X10 (XM_023618916.1) is 3684-4142. The numbering of the start and end positions for X11 (XM_023618917.1) and X13 (XM_023618919.1) is 3591-4049. The numbering of the start and end positions for X14 (XM_023618920.1) is 4038-4496. The numbering of the start and end positions for X15 (XM_023618921.1) is 3996-4454. The numbering of the start and end positions for X16 (XM_023618922.1) is 4037-4495. The numbering of the start and end positions for X17 (XM_023618923.1) is 3944-4402. The numbering of the start and end positions for X18 (XM_023618924.1) is 3995-4453. The numbering of the start and end positions for X19 (XM_023618925.1) is 3902-4360. The numbering of the start and end positions for X20 (XM_023618926.1) is 3943-4401. The numbering of the start and end positions for X21 (XM_023618927.1) is 3901-4359. In both sequences, the sequence of Exon I as shown in FIG. 15 is indicated in bold. The site of a C to A mutation site at nucleotide position 4174, corresponding to chr15:31,

225,630 in SEQ ID NO:56 (FIG. 13), and to rs1136366555 in dbSNP, is underlined. The region of sequence comprising Exon I as shown in FIG. 15 is displayed as codons in the correct reading frame for both SEQ ID NO:57 and SEQ ID NO:58 in FIG. 17.

FIG. 15 shows a view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the partial DYSF Coding DNA Sequence (SEQ ID NO:57) and the mutant partial DYSF Coding DNA Sequence (SEQ ID NO:58). The DYSF Coding DNA Sequences correspond to Exons G, H, and I as indicated by the regions of sequence similarity of the translated genomic DNA to DYSF protein sequences from human (Homo), orangutan (Pongo), cattle (Bos), mouse (Mus), rat (Rattus) and DYSF protein sequences from the more distantly-related zebrafish (Danio). The sequence of Exon I with 10 bp of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO: 59).

FIG. 16 shows models of part of the normal protein sequence encoded by horse DYSF (XP_023474694.1, presented here as SEQ ID NO:60) corresponding to a translation of SEQ ID NO:57 shown in FIG. 14 and part of the altered protein sequence encoded by horse DYSF with the base substitution at chr15:31,225,630 (based on XP_023474694.1, presented here as SEQ ID NO:61) corresponding to a translation of SEQ ID NO:58 shown in FIG. 14. The portion of the protein encoded by Exon I as shown in FIG. 15 is indicated in bold, while the amino acid position affected by the base substitution of an adenine (A) for a cytosine (C) at the chr15:31,225,630 position, corresponding to rs1136366555 in dbSNP as shown in FIG. 13, is underlined. The amino acid positions in XP_023474694.1 are indicated at the beginning and end of the sequence.

FIG. 17 shows horse DYSF Exon I and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the DYSF-P1290T mutation would be most appropriately derived. Genomic coordinates are as in FIG. 13. Exon I from chr15:31,225,648 to chr15:31,225,619 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:62) and the DYSF-P1290T allele (SEQ ID NO:63). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to T mutation site at nucleotide position chr15:31,225,630, corresponding to rs1136366555 in dbSNP as shown in FIG. 13 (C to A in the reverse complement as shown), is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for a proline (CCT) to one coding for a threonine (ACT). Example primers used experimentally to amplify genomic DNA containing the mutation site are shown in lower case [5'-GGTTGCAAACTCC-CAACTGT-3' (SEQ ID NO:64) and 5'-GATTTTT-CAAGCTGCCGAAG-3' (SEQ ID NO:65)].

Genomic DNA obtained from horses can be genotyped by amplifying a region containing a variant in the DYSF gene using Polymerase Chain Reaction (PCR), then sequencing the amplified DNA using Sanger sequencing. The variant allele DYSF-P1290T is abbreviated as P6, while the common or wild-type allele is abbreviated as N. The results can be scored as homozygous for the common or wild-type allele (N/N), heterozygous for the nucleotide substitution (N/P6), or homozygous for the nucleotide substitution (P6/P6).

FIG. 18 shows traces from Sanger DNA sequencing of amplified DYSF genomic DNA using primers shown in FIG. 17 (SEQ ID NO:64 and SEQ ID NO:65). The sequence of the forward strand is shown. The arrows in the figure indicate nucleotide position chr15:31,225,630, the site of a substitution of a thymine (T) for a guanine (G) in this position, corresponding to rs1136366555 in dbSNP, that creates the DYSF-P1290T variant. The traces show, from left to right, results for a horse homozygous for the wild-type or common allele (N/N), results for a horse heterozygous for the substitution (N/P6), and results for a horse homozygous for the substitution (P6/P6).

The sequence of the human DYSF coding sequence is presented in FIG. 7, while the sequence of the human DYSF protein sequence is presented in FIG. 8, as discussed above.

Comparison of the equine DYSF-P1290T missense allele to the pathogenic human alleles presented in TABLE 3 is informative. FIG. 19 shows a comparison of that portion of the protein sequence of DYSF encoded by horse Exon I from wild type (XP_023474694.1, shown here as SEQ ID NO:67) and DYSF-P1290T (P1290T, shown here as SEQ ID NO:68) to the protein sequence of DYSF encoded by human (NP_003485.1, shown here as SEQ ID NO:66). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a space () indicates the nonconservative substitution of an glycine (G) in horse for an arginine (R) in human at human position 1297, and a plus sign (+) indicates the conservative substitution of an alanine (A) in horse for a serine (S) in human at human position 1267, a tyrosine (Y) in horse for a histidine (H) in human position 1285, an aspartic acid (D) in horse for a glutamic acid (E) in human position 1294, and a glutamic acid (E) in horse for an aspartic acid (D) in human position 1300. No other amino acid substitutions are seen in comparison of the human sequence (SEQ ID NO:66) and the wild-type horse sequence (SEQ ID NO:67). The sequence from horse bearing the DYSF-P1290T mutation (SEQ ID NO:68) has a nonconservative substitution of a threonine (T) for a proline (P) that is found in human (SEQ ID NO:66) and the wild-type horse sequence (SEQ ID NO:67). This position, corresponding to position 1290 in horse (SEQ ID NO:67, SEQ ID NO:68) and position 1288 in human (SEQ ID NO:66) is indicated by bold and underlining.

Pathogenic and potentially pathogenic missense alleles are spread across the whole of dysferlin. Besides the pathogenic missense alleles described in the C2 calcium-binding domains C2B, C2A, and C2E described earlier and shown in TABLE 1 and TABLE 2, there are pathogenic and potentially pathogenic missense alleles in the interdomain region between C2D and C2E, as shown in TABLE 3.

TABLE 3

Pathogenic missense alleles of the C2D - C2E interdomain region of human DYSF.		
Substitution	Domain	Reference
P1247L	C2D-C2E Interdomain	ClinVar
R1254W	C2D-C2E Interdomain	ClinVar
T1261M	C2D-C2E Interdomain	ClinVar
G1268E	C2D-C2E Interdomain	ClinVar
L1270R	C2D-C2E Interdomain	ClinVar
L1270P	C2D-C2E Interdomain	ClinVar
L1276P	C2D-C2E Interdomain	Jin 2016
S1302P	C2D-C2E Interdomain	ClinVar
T1305K	C2D-C2E Interdomain	ClinVar
V1321F	C2D-C2E Interdomain	ClinVar

TABLE 3-continued

Pathogenic missense alleles of the C2D - C2E interdomain region of human DYSF.		
Substitution	Domain	Reference
R1331L	C2D-C2E Interdomain	UMD, Hofhuis 2017
E1335G	C2D-C2E Interdomain	UMD

Cited sources in this table are:

ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>), UMD (<http://www.umd.be/DYSF/>) Hofhuis 2017 (Hofhuis J et al. 2017 J Cell Sci 130:841-852 DOI: 10.1242/jcs.198861) Jin 2016 (Jib S-Q et al. 2016 Chin Med J 129:2287-2293 DOI: 10.4103/0366-6999.190671)

FIG. 20 shows features of the dysferlin protein encoded by the human DYSF gene. (A) The protein contains seven C2 calcium-binding domains designated C2A through C2G, shaded in gray. The positions of four additional conserved domains (DysfN, DysfC, annexin binding, and transmembrane) are also indicated (C.Therrien et al. 2006 J. Neurological Sciences 250: 71-78), also shaded in gray. The interdomain region between the C2D and C2E domains, affected by the horse DYSF-P1290T mutation, is indicated in light gray. (B) The interdomain region between the C2D and C2E domains is shown expanded, with positions of pathogenic and potentially pathogenic missense alleles listed in TABLE 3 shown. The horse DYSF-P1290T substitution corresponds to a proline at position 1288 in human, with no pathogenic or potentially pathogenic alleles identified at that position in human.

The extent of sequence conservation between horse and human is high, and it is of value to examine the extent of sequence conservation among a larger number of species, in order to assess the likelihood that the horse DYSF-P1290T allele is pathogenic.

FIG. 21 shows amino acid sequences of proteins encoded by part of the C2D-C2E interdomain region of DYSF, including the position of the equine DYSF-P1290T substitution. Part of the portion of the human DYSF corresponding to the C2D-C2E interdomain region, as identified by UniProt annotation (positions 1261-1320 of NP_003485.1, identical to 075923) was used as a probe in BLASTP searches of the protein sequence database to identify the corresponding region in the orthologous protein in a set of mammals. Extending this approach to more distantly related organisms (birds, reptile, amphibians, and fish) was not successful; either no match was found, or the top match was to a nonorthologous protein (e.g. myoferlin). In the alignment shown in FIG. 11, sequences that are identical over this range of positions in different species are clustered as a single SEQ ID NO.

The 46 species in the alignment shown in FIG. 21 are presented below with their common names, their scientific names, and the protein database ID from which the amino acid sequence is derived. Amino acid sequences identical over this range in different species have been clustered into single SEQ ID NOs for the alignment. The species associated with the SEQ ID NOs shown in the alignment are: SEQ ID NO:69 [Human (*Homo sapiens*, XP_520786.2), Chimpanzee (*Pan troglodytes*, XP_016804234.1), Bonobo (*Pan paniscus*, XP_003808233.1), Western lowland gorilla (*Gorilla gorilla gorilla*, XP_004029453.2)], SEQ ID NO:70 [Sumatran orangutan (*Pongo abelii*, XP_024097764.1), Panamanian white-throated capuchin (*Cebus capucinus imitator*, XP_017377462.1)], SEQ ID NO:71 [Mouse (*Mus musculus*, XP_006506242.1), Rat (*Rattus norvegicus*,

XP_006236834.1)], SEQ ID NO:72 [David's myotis (*Myotis davidii*, XP_006775352.1), Great roundleaf bat (*Hipposideros armiger*, XP_019503204.1)], SEQ ID NO:73 [Dromedary (*Camelus dromedarius*, XP_010992957.1), Bactrian camel (*Camelus bactrianus*, XP_010962003.1), Alpaca (*Vicugna pacos*, XP_015092494.1)], SEQ ID NO:74 [Cat (*Felis catus*, XP_003984159.2), Leopard (*Panthera pardus*, XP_019297033.1)], SEQ ID NO:75 [Cattle (*Bos taurus*, NP_001095960.1), Goat (*Capra hircus*, XP_017910552.1), Mouflon (*Ovis aries musimon*, XP_011978926.1)], SEQ ID NO:76 [Wild boar (*Sus scrofa*, XP_013851496.2), Sunda pangolin (*Manis javanica*, XP_017524151.1)], SEQ ID NO:77 Olive baboon (*Papio anubis*, XP_021780643.1), SEQ ID NO:78 Rhesus macaque (*Macaca mulatta*, XP_014968113.1), SEQ ID NO:79 Black snub-nosed monkey (*Rhinopithecus bieti*, XP_017733087.1), SEQ ID NO:80 Gray mouse lemur (*Microcebus murinus*, XP_012605531.1), SEQ ID NO:81 Philippine tarsier (*Carlito syrichta*, XP_008058745.1), SEQ ID NO:82 Northern greater galago (*Otolemur garnettii*, XP_012662869.1), SEQ ID NO:83 Coquerel's sifaka (*Propithecus coquereli*, XP_012501928.1), SEQ ID NO:84 Chinese hamster (*Cricetulus griseus*, XP_027284348.1), SEQ ID NO:85 Prairie vole (*Microtus ochrogaster*, XP_005369881.1), SEQ ID NO:86 Mongolian gerbil (*Meriones unguiculatus*, XP_021497376.1), SEQ ID NO:87 Thirteen-lined ground squirrel (*Ictidomys tridecemlineatus*, XP_005322200.1), SEQ ID NO:88 Long-tailed chinchilla (*Chinchilla lanigera*, XP_005385648.1), SEQ ID NO:89 Alpine marmot (*Marmota marmota marmota*, XP_015335021.1), SEQ ID NO:90 American beaver (*Castor canadensis*, XP_020009964.1), SEQ ID NO:91 Ferret (*Mustela putorius furo*, XP_004742233.1), SEQ ID NO:92 Damora mole-rat (*Fukomys damarensis*, XP_010607925.1), SEQ ID NO:93 Brandt's bat (*Myotis brandtii*, XP_005867829.1), SEQ ID NO:94 Large flying fox (*Pteropus vampyrus*, XP_023389621.1), SEQ ID NO:95 Horse (*Equus caballus*, XP_023474694.1), SEQ ID NO:96 Donkey (*Equus asinus*, XP_014692152.1), SEQ ID NO:97 African bush elephant (*Loxodonta africana*, XP_023408298.1), SEQ ID NO:98 Minke whale (*Balaenoptera acutorostrata scammoni*, XP_007175594.1), SEQ ID NO:99 Giant panda (*Ailuropoda melanoleuca*, XP_019650381.1), SEQ ID NO:100 Killer whale (*Orcinus orca*, XP_004277100.1), SEQ ID NO:101 Koala (*Phascogaleus cinereus*, XP_020856831.1), SEQ ID NO:102 Platypus (*Ornithorhynchus anatinus*, XP_007665017.1)

The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated by period (.), and nonconserved positions are indicated by a blank space (). The last line shows the position of the DYSF-P1290T substitution in horse in bold. The position of the DYSF-P1290T substitution is indicated in bold in all of the sequences.

The overall sequence conservation of the C2D-C2E interdomain region of dysferlin is quite high across mammals. The amino acid alteration seen in the equine DYSF-P1290T substitution is not seen in any of the 44 species; all have a proline (P) at this position. The comparison of dysferlin C2D-C2E interdomain sequences presented here refutes the hypothesis that the DYSF-P1290T substitution is selectively neutral, and supports the claim that the DYSF-P1290T mutation found in horses with PSSM2 is pathogenic.

Pyridine Nucleotide-Disulfide Oxidoreductase Domain-Containing Protein 1 (PYROXD1)

A mutation in the equine PYROXD1 coding region (encoding the muscle protein pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1) is present in many populations of horses affected by Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM). The difference in the genomic DNA between horses with PSSM2, also known as MFM, and control horses include a G-to-C substitution in PYROXD1 Exon 12 (as defined in FIG. 15) at nucleotide position chr6:47,661,977 of the current horse genome assembly (EquCab2, GCA_000002305.1), as shown in FIG. 13).

FIG. 22 shows a portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr6:47,661,977 position, the site of a substitution of a cytosine (C) for a guanine (G) that results in the substitution of a histidine (H) for an aspartate (D) at amino acid position 492 in pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) as shown in FIG. 16 (SEQ ID NO:60). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1136260157 in dbSNP.

There are five predicted transcripts of the PYROXD1 coding region in public databases. These models differ somewhat in ways not relevant to the PYROXD1-D492H mutation. All five isoforms share a common segment that includes the chr6:47,661,977 position. This common segment is shown in FIG. 23.

FIG. 23 shows a portion of the normal equine PYROXD1 Coding DNA Sequence (SEQ ID NO:104) and a portion of the mutant PYROXD1 Coding DNA Sequence (SEQ ID NO:105) bearing the G to C mutation at nucleotide position 1548 in this figure, corresponding to chr6:47,661,977 as shown in SEQ ID NO:103 (FIG. 22). This sequence is a region of perfect consensus among five different experimentally predicted mRNA isoforms. The numbering in FIG. 23 is that of isoform X1 (XM_001502130.5), with the segment from 238 to 1583 shown. The numbering of the start and end positions for X2 (XM_023643216.1) is 148-1493. The numbering of the start and end positions for X3 (XM_023643217.1) is 187-1532. The numbering of the start and end positions for X4 (XM_023643218.1) is 151-1496. The numbering of the start and end positions for X5 (XM_023643219.1) is 191-1536. In both sequences the sequence of Exon 12 as shown in FIG. 24 is indicated in bold. The site of a G to C mutation at nucleotide position 1548, corresponding to 47,661,977 in SEQ ID NO:103 (FIG. 22), and to rs1136260157 in dbSNP, is underlined. The region of sequence comprising Exon 12 as shown in FIG. 24 is displayed as codons in the correct reading frame for both SEQ ID NO:104 and SEQ ID NO:1057 in FIG. 26.

For the sake of simplicity, the amino acid substitution caused by the substitution of an cytosine (C) for a guanine (G) at the chr6:47,661,977 position, identified as rs1136260157 in dbSNP, will be referred to as PYROXD1-D492H, based on SEQ ID NO:108. The amino acid substitution caused by this mutation will remain the same regardless of the numerical position of the affected codon in alternative gene models. The wild-type (e.g., also referred to herein as "normal" or "unaffected") allele of this coding region may be referred to as D492 and the mutant allele D492H.

FIG. 24 shows a view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the

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PYROXD1 Coding DNA Sequence (SEQ ID NO:104) and the mutant PYROXD1 Coding DNA Sequence (SEQ ID NO:105) in a BLAT search. The PYROXD1 Coding DNA Sequences in horse correspond to PYROXD1 coding sequences in other species as indicated by the regions of sequence similarity of the translated genomic DNA to PYROXD1 protein sequences from human (Homo), orangutan (Pongo), cattle (Bos), mouse (Mus), rat (Rattus), African clawed frog (Xenopus), zebrafish (Danio), chicken (Gallus), and fruit fly (Drosophila). Exon 12, which contains the guanine (G) to cytosine (C) variant at chr6:47,661,977 as shown in SEQ ID NO:55 (FIG. 13), is indicated below the image of the browser window. The sequence of horse Exon 12 with 10 nucleotides of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO:106).

Conceptual translation of the nucleic acid sequences shown in FIG. 23 (SEQ ID NO:104 and SEQ ID NO:105) yields the amino acid sequences for the proteins shown in FIG. 25. FIG. 25 shows models of part of the normal protein sequence encoded by horse PYROXD1 (XP_001502180.3, presented here as SEQ ID NO:107) corresponding to a translation of SEQ ID NO:104 shown in FIG. 23 and part of the altered protein sequence encoded by mutant horse PYROXD1 (adapted from XP_001502180.3, presented here as SEQ ID NO:108) with the base substitution at chr6:47,661,977 corresponding to a translation of SEQ ID NO:105 shown in FIG. 23. The portion of the protein encoded by Exon 12 is indicated in bold, while the amino acid at position 492 affected by the base substitution of a cytosine (C) for a guanine (G) at the chr6:47,661,977 position as shown in FIG. 23, is underlined. The amino acid positions in XP_001502180.3 are indicated at the beginning and end of the sequence.

FIG. 26 shows horse PYROXD1 Exon 12 and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the PYROXD1-D492H mutation would be most appropriately derived. Genomic coordinates are as in FIG. 22. Exon 12 from chr6:47,661,764 to chr6:47,662,012 is shown broken into codons in the correct reading frame for the wild-type allele (SEQ ID NO:104) and the PYROXD1-D492H allele (SEQ ID NO:105). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to C mutation site at nucleotide position chr6:47,661,977, as shown in FIG. 22 is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for an aspartate (GAT) to one coding for a histidine (CAT). Example primers used experimentally to amplify genomic DNA containing the mutation site [5'-CAGATTTTCTGCTGGCCATT-3' (SEQ ID NO:111) and 5'-TGGTCATCATTAATCAGTGCAA-3' (SEQ ID NO:112)] are shown in lower case in the figure.

Genomic DNA obtained from horses can be genotyped by amplifying a region containing a variant in the PYROXD1 gene using Polymerase Chain Reaction (PCR), then sequencing the amplified DNA using Sanger sequencing. The variant allele PYROXD1-D492H is abbreviated as P8, while the common or wild-type allele is abbreviated as n. The results can be scored as homozygous for the common or wild-type allele (n/n), heterozygous for the nucleotide substitution (n/P8), or homozygous for the nucleotide substitution (P8/P8).

FIG. 27 shows traces from Sanger DNA sequencing of amplified PYROXD1 genomic DNA using primers shown in

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FIG. 26 (SEQ ID NO:111 and SEQ ID NO:112). The arrows in the figure indicate nucleotide position chr6:47,661,977, the site of a substitution of a cytosine (C) for a guanine (G) in this position that creates the PYROXD1-D492H variant. The traces show, from left to right, results for a horse homozygous for the wild-type or common allele (n/n), results for a horse heterozygous for the substitution (n/P8), and results for a horse homozygous for the substitution (P8/P8).

The human ortholog of the equine PYROXD1 gene encodes pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1, one member of a family of nuclear-cytoplasmic pyridine nucleotide-disulfide reductases (PNDR). These are flavoproteins, bound to flavin adenine dinucleotide (FAD), that catalyze pyridine-nucleotide-dependent (nicotinamide adenine dinucleotide or NAD) reduction of thiol residues in other proteins. PYROXD1 differs from five other human class I PNDRs, dihydrolipoamide dehydrogenase (DLD), glutathione reductase (GSR), thioredoxin 1, 2, and 3 (TXNRD1, TXNRD2, and TXNRD3), in not having a consensus redox active site in the oxidoreductase domain, and in not having a conserved C-terminal dimerization domain found in all other class I PNDRs; instead, it bears a highly conserved C-terminal nitrile reductase domain. Recessive mutations causing a partial loss of function of PYROXD1 are associated with early-onset myopathy in humans (O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105 DOI: 10.1016/j.ajhg.2016.09.005).

Human patients with various mutations in PYROXD1 show progressive weakness, reduction in muscle bulk, and some experience difficulty walking in the second decade of life. Muscle biopsies show central nuclei and disorganized inclusions with positive staining for the Z disc proteins desmin (encoded by DES) and myotilin (encoded by MYOT). Electron microscopy shows sarcomeric disorganization including disorganized Z discs and nemaline rods (a finding associated with several different types of Nemaline Myopathy). The myopathy associated with mutations in PYROXD1 in humans therefore has features associated with several different types of inherited myopathies in humans (Centronuclear Myopathy, Myofibrillar Myopathy, and Nemaline Myopathy).

In mouse, the Knockout Mouse Project (KOMP), a high-throughput gene knockout project, generated a loss-of-function allele of PYROXD1. Mice homozygous for the knockout allele die as embryos prior to organogenesis. Heterozygotes were smaller than normal and had abnormal vertebrae morphology. The muscle phenotype of heterozygotes has not been studied in detail, but it is evident from these results that a reduction in the level of pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 has phenotypic consequences, that is, the gene is haploinsufficient, and the activity of the pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 polypeptide is limiting.

In zebrafish, injection of morpholinos (a synthetic molecule containing DNA bases attached to a backbone of methylenemorpholine rings linked through phosphorodiamidate groups) targeting expression of ryroxd1 (the zebrafish ortholog of horse and human PYROXD1) causes reduction of the Ryroxd1 protein. Treated embryos show fragmentation of muscle fibers, loss of Z disc structure, and the formation of nemaline bodies. Treated embryos exhibit impaired swimming performance. These defects can be partially reversed through the injection of human PYROXD1 mRNA. These experiments show that

PYROXD1 function is necessary for both normal muscle structure and function in zebrafish (O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105).

In *Drosophila*, two different lines ubiquitously expressing RNA interference constructs for CG10721 (the *Drosophila* homolog of horse and human PYROXD1) were developed (Saha et al. 2018 Physiol Genomics doi:10.1152). Flies bearing these constructs failed to complete development, failing to emerge from their pupal cases, demonstrating the importance of PYROXD1 and its homologs in development.

It is therefore informative to compare the horse PYROXD1 gene and the protein that it encodes with the human ortholog. FIG. 28 shows the partial sequence of the human PYROXD1 coding sequence derived from NM_024854.4 (SEQ ID NO:113). This sequence is a region of perfect consensus among three different experimentally predicted mRNA isoforms. The numbering of the first and last nucleotides corresponds to that of NM_024854.4, with the segment from 619 to 1630 shown. The numbering of the start and end positions for transcript variant 2 (NM_001350912.1) is 1032-2043. The numbering of the start and end positions for transcript variant 3 (NM_001350913.1) is 545-1556. The sequence of Exon 12 is indicated in bold. The 3' UTR has been removed; the sequence begins with beginning of the consensus among the three isoforms and ends with the TAA stop codon.

FIG. 29 shows the partial sequence of the human PYROXD1 protein sequence, showing a translation of SEQ ID NO:113, equivalent to NP_079130.2 (SEQ ID NO:114). The numbering of the first and last amino acids corresponds to that of NP_079130.2.

FIG. 30 shows a comparison of that portion of the protein sequence of PYROXD1 encoded by horse Exon 12 from wild type (XP_001502180.3, shown here as SEQ ID NO:116) and PYROXD1-D492H (D492H, derived from XP_001502180.3 and shown here as SEQ ID NO:117) to the protein sequence of PYROXD1 encoded by human Exon 12 (NP_079130.2, shown here as SEQ ID NO:115). Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a plus sign (+) indicates the following conservative substitutions: a glutamine (Q) for an arginine (R) at human position 444, a valine (V) for an isoleucine (I) at human position 447, an alanine (A) for a serine (S) at human position 483, and an aspartate (D) for an asparagine (N) at human position 492. The sequence from horse bearing the PYROXD1-D492H mutation (SEQ ID NO:117) has a nonconservative substitution of a histidine (H) for an aspartate (D) at horse position 492, corresponding to human position 490. This position is indicated in bold for all three sequences. The wild-type horse sequence (SEQ ID NO:116) matches the human sequence (SEQ ID NO:115) at this position.

FIG. 31 shows features of the pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 protein encoded by the human PYROXD1 gene. The human protein has 500 amino acids and two domains: the pyridine nucleotide-disulfide oxidoreductase domain (amino acids 39-361) and the NADH-dependent nitrite reductase domain (447-494) as described in O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105. Positions of two human pathogenic missense alleles (N1555 and Q372H, O'Grady et al. 2016 Am J Hum Genet. 99:1086-1105) are shown, as is the position of the horse D492H mutation described herein.

The sequence comparison between the proteins encoded by horse and human PYROXD1 shows the high degree of sequence conservation between these two species. The

extent of sequence conservation is so high that it is difficult to evaluate whether this comparison offers any evidence that the equine PYROXD1-D492H allele would be expected to be pathogenic.

FIG. 32 shows partial amino acid sequences of proteins encoded by PYROXD1, including the position of the equine PYROXD1-D492H substitution. A portion of the human PYROXD1 protein sequence was used as a probe to identify the corresponding region in the orthologous protein across a wide range of species, including mammals, birds, reptiles, amphibians, and fish. In the alignment shown in FIG. 32, sequences that are identical over this range of positions are clustered as a single SEQ ID NO.

The 92 species in the alignment shown in FIG. 32 are presented below with their common names, their scientific names, and the protein database ID from which the amino acid sequence is derived. Amino acid sequences identical over this range in different species have been clustered into single SEQ ID NOs for the alignment. The species associated with the SEQ ID NOs shown in the alignment are: SEQ ID NO:118 [Chimpanzee (*Pan troglodytes*, XP_520786.2), Bonobo (*Pan paniscus*, XP_003828944.1), Western lowland gorilla (*Gorilla gorilla gorilla*, XP_018894407.1), Olive baboon (*Papio anubis*, XP_009178608.1), Rhesus macaque (*Macaca mulatta*, EHH20571.1), Panamanian white-throated capuchin (*Cebus capucinus imitator*, XP_017378442.1), Philippine tarsier (*Carlito syrichta*, XP_008064020.1), Coquerel's sifaka (*Propithecus coquereli*, XP_012496437.1)], SEQ ID NO:119 [Sumatran orang-utan (*Pongo abelii*, NP_001127199.1), Gray mouse lemur (*Microcebus murinus*, XP_012640956.1), Large flying fox (*Pteropus vampyrus*, XP_011379886.1), Great roundleaf bat (*Hipposideros armiger*, XP_019485916.1), African bush elephant (*Loxodonta africana*, XP_003405731.1)], SEQ ID NO:120 [Mouse (*Mus musculus*, XP_017177048.1), Mongolian gerbil (*Meriones unguiculatus*, XP_021514531.1), Sunda pangolin (*Manis javanica*, XP_017517789.1)], SEQ ID NO:121 [Black snub-nosed monkey (*Rhinopithecus bieti*, XP_017744103.1), Thirteen-lined ground squirrel (*Ictidomys tridecemlineatus*, XP_021584516.1), Alpine marmot (*Marmota marmota marmota*, XP_015353707.1)], SEQ ID NO:122 [David's myotis (*Myotis davidii*, ELK23147.1), Brandt's bat (*Myotis brandtii*, XP_014397831.1)], SEQ ID NO:123 [Horse (*Equus caballus*, XP_005611026.1), Donkey (*Equus asinus*, XP_014687887.1), Przewalski's horse (*Equus przewalskii*, XP_008533269.1)], SEQ ID NO:124 [Dromedary (*Camelus dromedarius*, XP_010987132.1), Bactrian camel (*Camelus bactrianus*, XP_010944895.1), Wild boar (*Sus scrofa*, XP_020947912.1), Cattle (*Bos taurus*, XP_005206989.1), Goat (*Capra hircus*, XP_017903901.1), Sheep (*Ovis aries*, XP_012030611.1), Mouflon (*Ovis aries musimon*, XP_012018062.1), Central European red deer (*Cervus elaphus hippelaphus*, OWK03976.1), Minke whale (*Balaenoptera acutorostrata scammoni*, XP_007195125.1), Killer whale (*Orcinus orca*, XP_004270982.1)], SEQ ID NO:125 [Cat (*Felis catus*, XP_006933619.1), Leopard (*Panthera pardus*, XP_019307710.1)], SEQ ID NO:126 Human (*Homo sapiens*, NP_079130.2), SEQ ID NO:127 Northern greater galago (*Otolemur garnettii*, XP_003792495.1), SEQ ID NO:128 Rat (*Rattus norvegicus*, EDM01504.1), SEQ ID NO:129 Chinese hamster (*Cricetulus griseus*, ERE65977.1), SEQ ID NO:130 Prairie vole (*Microtus ochrogaster*, XP_005364695.1), SEQ ID NO:131 Long-tailed chinchilla (*Chinchilla lanigera*, XP_005379114.1), SEQ ID NO:132 American beaver (*Castor canadensis*, XP_020016064.1), SEQ ID NO:133 Ferret (*Mustela putorius furo*,

XP_004755775.1), SEQ ID NO:134 Damora mole-rat (*Fukomys damarensis*, XP_010613965.1), SEQ ID NO:135 Alpaca (*Vicugna pacos*, XP_015091099.1), SEQ ID NO:136 Giant panda (*Ailuropoda melanoleuca*, XP_002926616.1), SEQ ID NO:137 Koala (*Phascolarctos cinereus*, XP_020849955.1), SEQ ID NO:138 Tasmanian devil (*Sarcophilus harrisii*, XP_003771207.1), SEQ ID NO:139 Gray short-tailed opossum (*Monodelphis domestica*, XP_001365452.2), SEQ ID NO:140 [Bald eagle (*Haliaeetus leucocephalus*, XP_010564402.1), Golden eagle (*Aquila chrysaetos canadensis*, XP_011574851.1), Chicken (*Gallus gallus*, NP_001264205.1), Downy woodpecker (*Picoides pubescens*, XP_009904475.1), Turkey (*Meleagris gallopavo*, XP_003202569.1), Speckled mousebird (*Colius striatus*, XP_010208275.1), Emperor penguin (*Aptenodytes forsteri*, XP_009277280.1), Rock dove (*Columba livia*, XP_021151605.1), Band-tailed pigeon (*Patagioenas fasciata monilis*, OPJ70652.1), Adélie penguin (*Pygoscelis adeliae*, KFW73339.1), Ruff (*Calidris pugnax*, XP_014815036.1), Japanese quail (*Coturnix japonica*, XP_015723825.1), Crested ibis (*Nipponia nippon*, XP_009463403.1), Anna's hummingbird (*Calypte anna*, XP_008500751.1), Little egret (*Egretta garzetta*, KFP11017.1), Chimney swift (*Chaetura pelagica*, KFU93618.1), Scaled quail (*Callipepla squamata*, OXB63337.1), Common cuckoo (*Cuculus canorus*, XP_009558366.1), Peregrine falcon (*Falco peregrinus*, XP_013156768.1)], SEQ ID NO:141 [Zebra finch (*Taeniopygia guttata*, XP_002200247.1), Society finch (*Lonchura striata domestica*, XP_021407730.1)], SEQ ID NO:142 Northern carmine bee-eater (*Merops nubicus*, KFK27091.1), SEQ ID NO:143 North Island brown kiwi (*Apteryx australis mantelli*, XP_013802597.1), SEQ ID NO:144 American crow (*Corvus brachyrhynchos*, KF060564.1), SEQ ID NO:145 Hooded crow (*Corvus cornix cornix*, XP_010402574.2), SEQ ID NO:146 Atlantic canary (*Serinus canaria*, XP_009093423.1), SEQ ID NO:147 Common starling (*Sturnus vulgaris*, XP_014735476.1), SEQ ID NO:148 Blue-crowned manakin (*Lepidothrix coronata*, XP_017660339.1), SEQ ID NO:149 Turquoise-fronted amazon (*Amazona aestiva*, KQK78799.1), SEQ ID NO:150 Schlegel's Japanese gecko (*Gekko japonicus*, XP_015277871.1), SEQ ID NO:151 Green anole (*Anolis carolinensis*, XP_003220806.1), SEQ ID NO:152 Burmese python (*Python bivittatus*, XP_007420993.1), SEQ ID NO:153 Central bearded dragon (*Pogona vitticeps*, XP_020669436.1), SEQ ID NO:154 Brown spotted pit viper (*Protobothrops mucrosquamatus*, XP_015682645.1), SEQ ID NO:155 King cobra (*Ophiophagus hannah*, ETE62148.1), SEQ ID NO:156 Common garter snake (*Thamnophis sirtalis*, XP_013928623.1), SEQ ID NO:157 West Indian Ocean coelacanth (*Latimeria chalumnae*, XP_005998922.1), SEQ ID NO:158 Whale shark (*Rhincodon typus*, XP_020373590.1), SEQ ID NO:159 Australian ghostshark (*Callorhynchus milii*, XP_007884060.1), SEQ ID NO:160 Atlantic salmon (*Salmo salar*, XP_014008456.1), SEQ ID NO:161 African clawed frog (*Xenopus laevis*, XP_018107310.1), SEQ ID NO:162 Common carp (*Cyprinus carpio*, KTF74061.1)

The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated by period (.), and nonconserved positions are indicated by a blank space (). The last line shows the position of the PYROXD1-D492H substitution in horse in

bold. The position of the PYROXD1-D492H substitution is indicated in bold in all of the aligned sequences.

The overall sequence conservation of this portion of the PYROXD1 protein is quite high. The amino acid alteration seen in the equine PYROXD1-D492H is not seen in any of the 92 species; all have an aspartate (D) or asparagine (N) at this position. The comparison of PYROXD1 sequences presented here refutes the hypothesis that the PYROXD1-D492H substitution is selectively neutral, and supports the claim that the PYROXD1-D492H mutation found in horses with Polysaccharide Storage Myopathy type 2 (PSSM2), also known as Myofibrillar Myopathy (MFM), is pathogenic.

Collagen Type VI Alpha 3 Chain (COL6A3)

A mutation in the equine COL6A3 coding region (encoding the muscle protein collagen type VI alpha 3 chain) is present in many populations of horses affected by Polysaccharide Storage Myopathy type 2 (PSSM2). The difference in the genomic DNA between horses with PSSM2 and control horses include a G-to-C substitution in COL6A3 Exon 26 (as defined in FIG. 33) at nucleotide position chr6:23,480,621 of the current horse genome assembly (EquCab2, GCA_000002305.1), as shown in FIG. 33.

FIG. 33 shows portion of the current horse genome assembly (EquCab2, GCA_000002305.1) with coordinates as displayed in the UCSC Genome Browser centered on the chr6:23,480,621 position, the site of a substitution of a guanine (G) for a cytosine (C) on the forward strand that results in the substitution of an alanine (A) for a glycine (G) at amino acid position 2182 in collagen type VI alpha 3 chain (COL6A3) as shown in FIG. 36 (SEQ ID NO:168). The reverse complement sequence is shown, with the site of a substitution of a cytosine (C) for a guanine (G) on the reverse strand as indicated (SEQ ID NO:163). The single nucleotide polymorphism (SNP) defined by this base substitution is identified as rs1139437410 in dbSNP.

The NCBI database contains five models for the mature horse mRNA containing the coding sequence (CDS) for COL6A3: XM_023642645.1, XM_023642646.1, XM_014740385.2, XM_023642647.1, and XM_023642648.1. There is a complete consensus among the five models for the portion of the coding sequence that encodes the five collagen-like repeats in the middle of the COL6A3 protein (see FIG. 34). For XM_023642645.1 (transcript variant X1), the coordinates are 6356-7363. The other four isoforms have coordinates as follows: XM_023642646.1 (5751-6758), XM_014740385.2 (5733-6740), XM_023642647.1 (5130-6137), and XM_023642648.1 (4530-5537).

FIG. 34 shows a portion of the normal equine COL6A3 Coding DNA Sequence (SEQ ID NO:164) and a portion of the mutant COL6A3 Coding DNA Sequence (SEQ ID NO:165) bearing the G to C mutation at nucleotide position 6792 in this figure, corresponding to chr6:23,480,621 as shown in SEQ ID NO:163 (FIG. 33). The numbering in FIG. 34 is that of the COL6A3 coding sequence (CDS) model XM_023642645.1; the sequence presented comprises the coding sequence for the five collagen-like domains in the middle of the protein. In both sequences the sequence of the third collagen-like domain, partially encoded by Exon 26, as shown in FIG. 35 is indicated in bold. The site of a G to C mutation at nucleotide position 6792, corresponding to 23,480,621 in SEQ ID NO:163 (FIG. 33), and to rs1139437410 in dbSNP, is underlined. The region of sequence comprising Exon 26 is displayed as codons in the correct reading frame for both SEQ ID NO:164 and SEQ ID NO:165 in FIG. 37.

FIG. 35 shows a view of the current horse genome assembly (EquCab2, GCA_000002305.1) in the UCSC Genome Browser with exon sequences that match the COL6A3 Coding DNA Sequence (SEQ ID NO:164) and the mutant COL6A3 Coding DNA Sequence (SEQ ID NO:165) in a BLAT search. The COL6A3 Coding DNA Sequences in horse correspond to COL6A3 coding sequences in other species as indicated by the regions of sequence similarity of the translated genomic DNA to COL6A3 protein sequences from human (*Homo*), mouse (*Mus*), and dog (*Canis*). Exon 26, which contains the guanine (G) to cytosine (C) variant at chr6:23,480,621 on the reverse strand as shown in SEQ ID NO:163 (FIG. 33), is indicated below the image of the browser window. The sequence of horse Exon 26 with 10 nucleotides of flanking intron sequence and its coordinates in the current horse assembly is displayed below the image from the UCSC Genome Browser (SEQ ID NO:166).

The translation of horse COL6A3 CDS does not always produce a significant match to the protein sequences of COL6A3 from other species, as shown in the UCSC Genome Browser window. This is particularly apparent for Exon 26. There are two reasons for this. First, the protein sequence of the collagen-like repeat portion of the protein, as discussed below, is a repeating sequence of Gly-X-Y. With every third residue a glycine, the sequence is of low complexity. Second, Exon 26 is very short. A short sequence with low complexity will not always produce a significant match to orthologous proteins, even if the proteins are truly orthologous.

Conceptual translation of the nucleic acid sequences shown in FIG. 34 (SEQ ID NO:164 and SEQ ID NO:165) yields the amino acid sequences for the proteins shown in FIG. 36. FIG. 36 shows models of part of the normal protein sequence encoded by horse COL6A3 (XP_023498413.1, presented here as SEQ ID NO:167) corresponding to a translation of SEQ ID NO:164 shown in FIG. 34 and part of the altered protein sequence encoded by horse COL6A3 (adapted from XP_023498413.1, presented here as SEQ ID NO:168) with the base substitution at chr6:23,480,621 corresponding to a translation of SEQ ID NO:165 shown in FIG. 34. The portion of the protein corresponding to all five collagen-like domains is shown; the portion corresponding to the third collagen-like domain, partially encoded by Exon 26, is indicated in bold, while the amino acid at position 2182 affected by the base substitution of a cytosine (C) for a guanine (G) at the chr6:23,480,621 position as shown in FIG. 33, is underlined. The amino acid positions in XP_023498413.1 are indicated at the beginning and end of the sequence.

For the sake of simplicity, the amino acid substitution caused by the substitution of an cytosine (C) for a guanine (G) at the chr6:23,480,621 position on the reverse strand as shown in FIG. 33, identified as rs1139437410 in dbSNP, will be referred to as COL6A3-G2182A, based on SEQ ID NO:168. The amino acid substitution caused by this mutation will remain the same regardless of the numerical position of the affected codon in alternative gene models. The wild-type (e.g., also referred to herein as "normal" or "unaffected") allele of this coding region may be referred to as G2182 and the mutant allele G2182A.

FIG. 37 shows horse COL6A3 Exon 26 and flanking genomic DNA sequence from which PCR primers to amplify genomic DNA containing the site of the COL6A3-G2182A mutation would be most appropriately derived. Genomic coordinates are as in FIG. 24. Exon 26 from chr6:23,480,631 to chr6:23,480,578 is shown broken into codons in the correct reading frame for the wild-type allele

(SEQ ID NO:169) and the COL6A3-G2182A allele (SEQ ID NO:170). Only the reference sequence from the assembly is shown for the flanking sequences. The codon affected by the G to C mutation site at nucleotide position chr6:23,480,621 on the reverse strand, as shown in FIG. 33, is shown in bold, with the position of the base substitution indicated by underlining. The base substitution changes the bold three base codon from one coding for a glycine (GGG) to one coding for an alanine (GCG). Example primers used experimentally to amplify genomic DNA containing the mutation site [5'-AGATGGGGCACAGATCAAAC-3' (SEQ ID NO:172) and 5'-TTCCAGACTCTCTGTGCT-3' (SEQ ID NO:171)] are shown in lower case in the figure.

Genomic DNA obtained from horses can be genotyped by amplifying a region containing a variant in the COL6A3 gene using Polymerase Chain Reaction (PCR), then sequencing the amplified DNA using Sanger sequencing. The variant allele COL6A3-G2182A is abbreviated as K1, while the common or wild-type allele is abbreviated as n. The results can be scored as homozygous for the common or wild-type allele (n/n), heterozygous for the nucleotide substitution (n/K1), or homozygous for the nucleotide substitution (K1/K1).

FIG. 38 shows traces from Sanger DNA sequencing of amplified COL6A3 genomic DNA using primers shown in FIG. 37 (SEQ ID NO:171 and SEQ ID NO:172). The arrows in the figure indicate nucleotide position chr6:23,480,621, the site of a substitution of a cytosine (C) for a guanine (G) in this position on the reverse strand that creates the COL6A3-G2182A variant. The traces show, from left to right, the sequence of the forward strand for a horse homozygous for the wild-type or common allele (n/n), and results for a horse heterozygous for the substitution (n/K1).

Collagen is the main structural protein of the extracellular matrix. Collagens are synthesized with N- and C-terminal extensions that are cleaved off proteolytically upon export from the cell. The N- and C-terminal extensions assist in localization within the extracellular matrix. The core of the collagen protein is the triple helical domain, consisting of Gly-X-Y repeats. The collagen triple helix consists of a right-handed intertwining of three left-handed helices. The glycine residues are critical, because every third residue occupies a position at the center of the triple helix, where there is no room for any other R group besides the hydrogen found in glycine. The collagen triple helix is stiffened by the post-translation modification of proline and lysine residues to hydroxyproline and hydroxylysine. As collagen fibers assemble, each triple helix is joined to its neighbors. In the fully assembled array, each triple helix overlaps with a neighboring triple helix with an offset of about one quarter of its length. The array is stabilized by covalent links through interchain disulfide bonds and post-translationally modified hydroxylysine.

The human ortholog of the equine COL6A3 gene encodes collagen type VI alpha 3 chain, one of a set of collagen type VI genes. The others are COL6A1, COL6A2, COL6A5, and COL6A6 (COL6A4 is a pseudogene). Collagens play important roles in maintaining extracellular matrix structure and function. Members of the collagen VI family, like COL6A3, form distinct networks of microfibrils in connective tissue and interact with other extracellular matrix components. Mutations in the human genes COL6A1, COL6A2, and COL6A3 are associated with Bethlem Myopathy, Ullrich Congenital Muscular Dystrophy, and dystonia. Mutations may be either dominant or recessive, with the age of onset and the severity of the condition varying with the exact mutation.

In mouse, an in-frame intragenic deletion of six Gly-X-Y repeats of COL6A3 (Col6a^{2m.2.1Chu}) is a dominant mutation causing abnormal muscle fiber morphology and myopathy. Targeted mutation of COL6A1 is also a dominant mutation causing myopathy. Electron microscopy of tendons from homozygous mice shows a reduced diameter of collagen fibrils (Pan et al. 2013 J Biol Chem, 288(20):14329-14331).

It is therefore informative to compare the horse COL6A3 gene and the protein that it encodes with the human ortholog. FIG. 39 shows the partial sequence of the human COL6A3 coding sequence derived from NM_004369.3 (SEQ ID NO:173). This sequence is a region of perfect consensus among multiple different experimentally predicted mRNA isoforms. The numbering of the first and last nucleotides corresponds to that of NM_004369.3. The sequence encoding the third collagen-like domain is indicated in bold.

FIG. 40 shows a partial sequence of the human COL6A3 protein, showing a translation of SEQ ID NO:175, equivalent to NP_004360.2 (SEQ ID NO:176). The numbering of the first and last amino acids corresponds to that of NP_004360.2. The sequence of the third collagen-like domain is indicated in bold.

FIG. 41 shows a comparison of the portion of the protein sequence of COL6A3 comprising the five collagen-like repeats from human (NP_079130.2, shown here as SEQ ID NO:175) and horse (XP_001502180.3, shown here as SEQ ID NO:176). The position of the COL6A3-G2182A substitution is shown below the horse sequence. Between the sequences of the horse and human proteins in the alignment, an asterisk (*) indicates an identical amino acid in that position, while a plus sign (+) indicates a conservative substitution, and a space () indicates a nonconservative substitution. The positions of glycine residues that are part of the Gly-X-Y structure of the collagen triple helix are indicated by an asterisk (*) in reverse text (white text on a black background) between the horse and human sequences. All of these glycine residues are conserved between human and horse. The horse G2182A sequence has a nonconservative substitution of an alanine (A) for a glycine (G) at position 2182; the wild-type horse sequence matches the human sequence at this position.

FIG. 42 shows features of the collagen type VI alpha 3 chain protein encoded by the human COL6A3 gene. A) The human protein has 3,162 amino acids. It contains five collagen-like domains consisting of Gly-X-Y repeats, labeled as Collagen 1-5 in the figure. Toward the N-terminus of the collagen-like repeats are 10 Von Willebrand Factor-like repeats, labeled as VWFA 1-10 in the figure. Toward the C-terminus of the collagen-like repeats are two additional Von Willebrand Factor-like repeats, labeled as VWFA 11 and 12 in the figure. Near the C-terminus is a fibronectin-like repeat, labeled as Fibronectin in the figure, and a Protease Inhibitor domain (Pancreatic trypsin inhibitor Kunitz domain). B) The portion of the protein containing the five collagen-like domains extends from position 2038 to 2373 of the human protein. The five collagen-like domains are numbered. Positions of 19 pathogenic human alleles listed in TABLE 2 below are indicated above the segment shown, as is the position of the horse G2182A variant. The position of the cysteine involved in the interchain disulfide bond (C2087S-S) is indicated below the segment shown.

TABLE 4 shows nineteen missense alleles of human COL6A3 shown in FIG. 42. Although pathogenic missense alleles of human COL6A3 outside of the collagen-like repeats in the middle of the protein have been described, we focus on mutations in this region in humans because of the

relevance to the COL6A3-G2182A allele in horse. All nineteen mutations are substitutions of other amino acid residues for the glycine residues that are part of the Gly-X-Y repeat structure of the collagen-like region. Human missense alleles of COL6A3 are associated with Bethlem Myopathy and with Ullrich Congenital Muscular Dystrophy. Bethlem Myopathy has also been referred to as Limb-Girdle Muscular Dystrophy D5 (LGMDD5). Both dominant and recessive forms of Bethlem Myopathy, caused by mutations in COL6A1, COL6A2, and COL6A3, are known. Symptoms include progressive muscular weakness with muscle atrophy of the trunk and limbs and joint contractures. Missense alleles affecting the glycine residues of the collagen repeats account for almost one-third of all pathogenic alleles. The symptoms of Ullrich Congenital Muscular Dystrophy overlap those of Bethlem Myopathy. Ullrich Congenital Muscular Dystrophy is caused by mutations in COL6A1, COL6A2, and COL6A3.

TABLE 4

Pathogenic missense alleles of human COL6A3 and associated diseases.			
Substitution	Domain	Disease	Reference
G2047D	Collagen 1	Bethlem Myopathy	Lampe 2005
G2053V	Collagen 1	Not provided	ClinVar
G2056E	Collagen 1	Not provided	ClinVar
G2056R	Collagen 1	Bethlem Myopathy	Pepe 1999
G2059C	Collagen 1	Bethlem Myopathy	ClinVar
G2065S	Collagen 1	Bethlem Myopathy/ Ulrich congenital muscular dystrophy	ClinVar
G2055R	Collagen 1	Bethlem Myopathy/ Ulrich congenital muscular dystrophy	ClinVar
G2071D	Collagen 1	Bethlem Myopathy/ Ulrich congenital muscular dystrophy	ClinVar
G2074S	Collagen 1	Bethlem Myopathy/ Ulrich congenital muscular dystrophy	ClinVar
G2077D	Collagen 1	Bethlem Myopathy/ Ulrich congenital muscular dystrophy	ClinVar
G2077V	Collagen 1	Bethlem Myopathy	ClinVar
G2080S	Collagen 1	Ulrich congenital muscular dystrophy	ClinVar
G2080R	Collagen 1	Not provided	ClinVar
G2080D	Collagen 1	Bethlem Myopathy/ Ulrich congenital muscular dystrophy	ClinVar, Lampe 2005
G2083V	Collagen 1	Not provided	ClinVar
G2098V	Interdomain	Bethlem Myopathy/ Ulrich congenital muscular dystrophy	ClinVar
G2174S	Collagen 3	Bethlem Myopathy	ClinVar
G2267S	Collagen 4	Bethlem Myopathy	ClinVar
G2285R	Collagen 4	Bethlem Myopathy	ClinVar
G2297A	Collagen 4	Bethlem Myopathy/ Ulrich congenital muscular dystrophy	ClinVar

Cited sources in this table are:

ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>)

Lampe 2005 (Lampe A K et al. 2005 J Med Genet 42:108-120 DOI: 10.1136/jmg.2004.023754)

Pepe 1999 (Pepe G et al. 1999 Neuromuscul Disord 9:264-271)

The glycine substitutions seen in human pathogenic alleles of COL6A3 are clustered in the amino-terminal portion of the collagen-like domain, as shown in FIG. 42 and TABLE 4. This may represent a functional domain within the triple helix (Butterfield et al. Hum Mutat. 2013 34(11):

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1558-1567). The most severe cases are caused by glycine substitutions in Gly-X-Y triplets 10 to 15. Glycine substitutions in this region are reported as dominant; the only cases of recessive glycine substitutions are located outside of this region in the C-terminal portion of the triple helix. This generalization applies to glycine substitutions in COL6A1, COL6A2, and COL6A3.

The position of the COL6A3-G2182A mutation in horse is shown in FIG. 42. COL6A3-G2182A affects Gly-X-Y triplet 48, as shown in FIG. 41. The position of the COL6A3-G2182A mutation in horse is closest to the human COL6A3-G2174S allele. The phenotype produced by human pathogenic glycine substitution alleles depends not only on the position of the mutation, but also on the amino acid substitution. There is a published destabilization scale for glycine substitutions in the triple helix (Ala<Ser<Cys<Arg<Val<Glu<Asp<Trp) (Persikov et al. Hum Mutat. 2004 24(4):330-337). The horse COL6A3-G2182A allele is therefore the least damaging substitution in a moderately damaging position in the protein, consistent with the moderate phenotype of affected horses described below.

The sequence comparison between the proteins encoded by horse and human COL6A3 shows the high degree of sequence conservation between these two species. The extent of sequence conservation is so high that it is difficult to evaluate whether this comparison offers any evidence that the equine COL6A3-G2182A allele would be expected to be pathogenic.

FIG. 43 shows partial amino acid sequences of proteins encoded by COL6A3, including the position of the equine COL6A3-G2182A substitution. A portion of the human COL6A3 protein sequence was used as a probe to identify the corresponding region in the orthologous protein across a wide range of species, including mammals, birds, reptiles, amphibians, and fish. In the alignment shown in FIG. 43, sequences that are identical over this range of positions are clustered as a single SEQ ID NO.

The 89 species in the alignment shown in FIG. 43 are presented below with their common names, their scientific names, and the protein database ID from which the amino acid sequence is derived. Amino acid sequences identical over this range in different species have been clustered into single SEQ ID NOs for the alignment. The species associated with the SEQ ID NOs shown in the alignment are:

SEQ ID NO:177 [Human (*Homo sapiens*, NP_004360.2), Chimpanzee (*Pan troglodytes*, XP_001153544.3), Bonobo (*Pan paniscus*, XP_003813155.1), Western lowland gorilla (*Gorilla gorilla gorilla*, XP_018878003.1), Sumatran orangutan (*Pongo abelii*, XP_009236555.2)], SEQ ID NO:178 [Thirteen-lined ground squirrel (*Ictidomys tridecemlineatus*, XP_013214606.1), Alpine marmot (*Marmota marmota marmota*, XP_015337108.1)], SEQ ID NO:179 [David's myotis (*Myotis davidii*, XP_015417293.1), Brandt's bat (*Myotis brandtii*, XP_014393090.1)], SEQ ID NO:180 [Horse (*Equus caballus*, XP_023498413.1), Donkey (*Equus asinus*, XP_014698264.1), Przewalski's horse (*Equus przewalskii*, XP_008528449.1)], SEQ ID NO:181 [Dromedary (*Camelus dromedarius*, XP_010994640.1), Bactrian camel (*Camelus bactrianus*, XP_010946449.1), Alpaca (*Vicugna pacos*, XP_015098979.1)], SEQ ID NO:182 Ferret [(*Mustela putorius furo*, XP_012907287.1), Cat (*Felis catus*, XP_019694498.2)], SEQ ID NO:183 [Cattle (*Bos taurus*, XP_024846030.1), Goat (*Capra hircus*, XP_017896230.1), Sheep (*Ovis aries*, XP_014947129.1), Mouflon (*Ovis aries musimon*, XP_011989955.1)], SEQ ID NO:184 [Leopard (*Panthera pardus*, XP_019301005.1),

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Giant panda (*Ailuropoda melanoleuca*, XP_019662500.1)], SEQ ID NO:185 Olive baboon (*Papio anubis*, XP_021779879.1), SEQ ID NO:186 Rhesus macaque (*Macaca mulatta*, XP_014966890.1), SEQ ID NO:187 Panamanian white-throated capuchin (*Cebus capucinus imitator*, XP_017393233.1), SEQ ID NO:188 Black snub-nosed monkey (*Rhinopithecus bieti*, XP_017745835.1), SEQ ID NO:189 Gray mouse lemur (*Microcebus murinus*, XP_012634739.2), SEQ ID NO:190 Philippine tarsier (*Carlito syrichta*, XP_008057564.1), SEQ ID NO:191 Northern greater galago (*Otolemur garnettii*, XP_003798279.2), SEQ ID NO:192 Coquerel's sifika (*Propithecus coquereli*, XP_012504274.1), SEQ ID NO:193 Mouse (*Mus musculus*, NP_001229937.1), SEQ ID NO:194 Rat (*Rattus norvegicus*, XP_008756297.1), SEQ ID NO:195 Chinese hamster (*Cricetulus griseus*, XP_016824341.1), SEQ ID NO:196 Prairie vole (*Microtus ochrogaster*, XP_005361761.1), SEQ ID NO:197 Mongolian gerbil (*Meriones unguiculatus*, XP_021513097.1), SEQ ID NO:198 Long-tailed chinchilla (*Chinchilla lanigera*, XP_005411290.1), SEQ ID NO:199 American beaver (*Castor canadensis*, XP_020043158.1), SEQ ID NO:200 Damora mole-rat (*Fukomys damarensis*, XP_010624430.1), SEQ ID NO:201 Large flying fox (*Pteropus vampyrus*, XP_023380958.1), SEQ ID NO:202 Sunda pangolin (*Manis javanica*, XP_017531172.1), SEQ ID NO:203 Great roundleaf bat (*Hipposideros armiger*, XP_019485542.1), SEQ ID NO:204 African bush elephant (*Loxodonta africana*, XP_010595897.1), SEQ ID NO:205 Wild boar (*Sus scrofa*, XP_020931400.1), SEQ ID NO:206 Minke whale (*Balaenoptera acutorostrata scammoni*, XP_007184961.1), SEQ ID NO:207 Killer whale (*Orcinus orca*, XP_004262601.1), SEQ ID NO:208 Koala (*Phascogaleus cinereus*, XP_020827927.1), SEQ ID NO:209 Tasmanian devil (*Sarcophilus harrisii*, XP_023359576.1), SEQ ID NO:210 Platypus (*Ornithorhynchus anatinus*, XP_007666682.1), SEQ ID NO:211 [American crow (*Corvus brachyrhynchos*, XP_017602594.1), Hooded crow (*Corvus cornix cornix*, XP_019146502.1)], SEQ ID NO:212 Bald eagle (*Haliaeetus leucocephalus*, XP_010576193.1), SEQ ID NO:213 Golden eagle (*Aquila chrysaetos canadensis*, XP_011584187.1), SEQ ID NO:214 Chicken (*Gallus gallus*, NP_990865.2), SEQ ID NO:215 Downy woodpecker (*Picoides pubescens*, XP_009899132.1), SEQ ID NO:216 Turkey (*Meleagris gallopavo*, XP_010711547.1), SEQ ID NO:217 Northern carmine bee-eater (*Merops nubicus*, XP_008936095.1), SEQ ID NO:218 Speckled mousebird (*Colius striatus*, XP_010208169.1), SEQ ID NO:219 North Island brown kiwi (*Apteryx australis mantelli*, XP_013798018.1), SEQ ID NO:220 Emperor penguin (*Aptenodytes forsteri*, XP_019329168.1), SEQ ID NO:221 Rock dove (*Columba livia*, XP_021142382.1), SEQ ID NO:222 Band-tailed pigeon (*Patagioenas fasciata monilis*, OPJ80253.1), SEQ ID NO:223 Adélie penguin (*Pygoscelis adeliae*, XP_009327902.1), SEQ ID NO:224 Ruff (*Calidris pugnax*, XP_014807202.1), SEQ ID NO:225 Japanese quail (*Coturnix japonica*, XP_015723569.1), SEQ ID NO:226 Crested ibis (*Nipponia nippon*, XP_009460310.1), SEQ ID NO:227 Anna's hummingbird (*Calypte anna*, XP_008493236.1), SEQ ID NO:228 Little egret (*Egretta garzetta*, XP_009633772.1), SEQ ID NO:229 Atlantic canary (*Serinus canaria*, XP_009086444.1), SEQ ID NO:230 Common starling (*Sturnus vulgaris*, XP_014737483.1), SEQ ID NO:231 Zebra finch (*Taeniopygia guttata*, XP_004175063.1), SEQ ID NO:232 Society finch (*Lonchura striata domestica*, XP_021391774.1), SEQ ID NO:233 Chimney swift (*Chaetura pelagica*, XP_010003753.1), SEQ ID NO:234 Scaled quail (*Calli-*

pepla squamata, OXB61690.1), SEQ ID NO:235 Common cuckoo (*Cuculus canorus*, XP_009568416.1), SEQ ID NO:236 Blue-crowned manakin (*Lepidothrix coronata*, XP_017691139.1), SEQ ID NO:237 Peregrine falcon (*Falco peregrinus*, XP_005244480.1), SEQ ID NO:238 Budgerigar (*Melopsittacus undulatus*, XP_012985362.1), SEQ ID NO:239 Schlegel's Japanese gecko (*Gekko japonicus*, XP_015273310.1), SEQ ID NO:240 Green anole (*Anolis carolinensis*, XP_008104462.1), SEQ ID NO:241 Burmese python (*Python bivittatus*, XP_025027494.1), SEQ ID NO:242 Brown spotted pit viper (*Protobothrops mucrosquamatus*, XP_015676224.1), SEQ ID NO:243 King cobra (*Ophiophagus hannah*, ETE68401.1), SEQ ID NO:244 Common garter snake (*Thamnophis sirtalis*, XP_013925229.1), SEQ ID NO:245 West Indian Ocean coelacanth (*Latimeria chalumnae*, XP_014344739.1), SEQ ID NO:246 Whale shark (*Rhincodon typus*, XP_020391663.1), SEQ ID NO:247 Australian ghostshark (*Callorhynchus milii*, XP_007904071.1), SEQ ID NO:248 Atlantic salmon (*Salmo salar*, XP_014028364.1), SEQ ID NO:249 African clawed frog (*Xenopus laevis*, KTG32498.1), SEQ ID NO:250 Common carp (*Cyprinus carpio*, KTG32498.1).

The next to the last line (labeled CLUSTAL) shows the consensus sequence, where positions with fully conserved amino acids are represented by an asterisk (*), positions with strongly conserved amino acids are indicated by a colon (:), positions with weakly conserved amino acids are indicated by a period (.), and nonconserved positions are indicated by a blank space (). The last line shows the position of the COL6A3-G2182A substitution in horse. The position of the COL6A3-G2182A substitution is indicated in bold in all of the aligned sequences.

The strongly conserved positions in the aligned sequences are all glycine (G) residues that are part of the Gly-X-Y repeat structure of the collagen triple helix. The sequences to the N-terminal position of the strongly conserved glycine residues also show a Gly-X-Y repeat structure that is evident from the examination of individual sequences, but a variable number of amino acids between the Gly-X-Y repeats showing sequence conservation and the Gly-X-Y repeats at the N-terminal position prevents this from showing in the multiple alignment. Separation of Gly-X-Y repeats by sequences not conforming to the Gly-X-Y pattern is common among collagens.

The overall sequence conservation of this portion of the COL6A3 protein is quite high. The amino acid alteration seen in the equine COL6A3-G2182A is not seen in any of the 89 species; all have a glycine (G) at this position. The comparison of COL6A3 sequences presented here refutes the hypothesis that the COL6A3-G2182A substitution is selectively neutral, and supports the claim that the COL6A3-G2182A mutation found in horses with PSSM2 is pathogenic.

Phenotypic Effects of the Genetic Variants Associated with PSSM2

This disclosure, together with a prior filing, identifies eight variants associated with PSSM2: MYOT-S232P (P2), FLNC-E753K (P3a), FLNC-A1207T (P3b), MYOZ3-S42L (P4), DYSF-R285W (P5), PYROXD1-D492H (P8), and COL6A3-G2182A (K1). These eight variants correspond to single-base substitutions having the coordinates given as follows on the forward strand in the public horse genome assembly (EquCab 2): chr14:38519183 A/G (P2), chr4:83736244 G/A (P3a), chr4:83738769 G/A (P3b), chr14:27399222 G/A (P4), chr15:31306949 G/A (P5), chr15:5:31,225,630 T/G (P6), chr6:47,661,977 G/C (P8), and chr6:23,

480,621 C/G (K1). The P3a and P3b variants are a haplotype, that is, among hundreds of horses tested, horses with the P3a variant also have the P3b variant and vice versa, whether homozygous or heterozygous. Only two types of chromosomes are seen: those with both P3a and P3b, or those with the wild-type or common alleles of both variants. The haplotype with both variants is therefore abbreviated as P3.

One of the earliest symptoms of PSSM2 is a change in behavior apparently associated with pain. Owners note a difference in temperament, with horses reacting badly to being ridden or even saddled. Common behaviors include biting at the flanks or even at the rider or trainer, and bucking, rearing, and other displays of resistance that trainers often blame on lack of discipline from the owner.

Another early symptom is stifle problems. The stifle is the largest joint in the horse's body, equivalent to the human knee, but in contrast to the human knee, the equine stifle is held at an angle when the horse is standing still. Stifle problems commonly result from injury or arthritis, degenerative joint disease, or injury. In stifle problems resulting from Polysaccharide Storage Myopathy type 2 (PSSM2), there will be no radiographic findings. Stifle problems are one example of shifting lameness. A horse with Polysaccharide Storage Myopathy type 2 (PSSM2) will exhibit lameness that appears first in one limb, then another. There will be no radiographic findings.

Changes in gait are often apparent. These include stiffness in the hindquarters and limited range of motion of the hind legs ("short-gaited"). At canter, disunited canter ("cross-firing") and "bunny hopping" (bringing both hind legs forward at the same time) are seen. "Rope walking" (placing one foot directly in front of the other along the centerline as if walking a tightrope) is sometimes seen in all four legs or in the rear legs only.

Other gait changes resulting from weakness in the hind limbs are described by horse owners as "heavy on the forehand, not able to come from behind." This means that the horse's gait is altered in such a way that it appears to be pulling itself forward with its front hooves instead of pushing from the rear. Farriers note this as a pattern of wear in the front hooves for unshod horses.

There is no evidence of cardiomyopathy.

Phenotypic Effects of the DYSF-R285W and DYSF-P1290T Variants

The DYSF-R253W variant (hereafter abbreviated as P5) was discovered by analysis of whole genome sequencing data from a Shire draft horse (E016) diagnosed via muscle biopsy with Polysaccharide Storage Myopathy type 2 (PSSM2). This horse was homozygous for the P5 allele (P5/P5) and homozygous wild type for six other variants with PSSM2: MYOT-S232P (P2), FLNC-E753K (P3a), FLNC-A1207T (P3b), MYOZ3-S42L (P4), PYROXD1-D492H (P8), and COL6A3-G2182A (K1). The P6 allele of DYSF (DYSF-P1290T) was not present.

We analyzed additional horses using PCR amplification and Sanger sequencing to identify P5/P5 homozygotes. A Shire horse (E799) scored as P5/P5 and homozygous for the wild-type alleles of P2, P3, P4, P5, P8, and K1 was identified at 11 years of age. This horse had been retired from regular work the prior year following an episode of lameness, muscle fasciculation, and profuse sweating during work.

Other individual horses of the Shire breed and the closely related Clydesdale breed that were determined to be normal, that is, not affected by PSSM2, were found to be heterozygous for the P5 allele (N/P5) or homozygous for the wild-type allele (N/N). Therefore, the P5 variant appears to be

recessive, as is typical for missense alleles of the human DYSF gene shown in TABLE 1, TABLE 2, and TABLE 3.

The DYSF-P1290T variant (hereafter abbreviated as P6) was discovered by analysis of whole genome sequencing data from a number of horses diagnosed via muscle biopsy with Polysaccharide Storage Myopathy type 2 (PSSM2). Each of these horses had other variants, and all were found to be heterozygous of P6 (N/P6). Because mutations in DYSF analyzed in human patients and mice have all been recessive, we analyzed additional horses using PCR amplification and Sanger sequencing to identify P6/P6 homozygotes.

An Icelandic horse (E885) scored as P6/P6 and homozygous for the wild-type alleles of P2, P3, P4, P5, P8, and K1 was identified at 25 years of age. The horse had been retired from work for at least five years. The owner reported that this horse had gait abnormalities, described by the owner as “tripping.” The owner refused an exercise challenge, expected to induce a rise in serum creatine kinase (CK); serum CK measured in this horse in the absence of an exercise challenge was in the normal range, as were the serum CK levels in eight other Icelandics tested at the same time in the same location. All eight of the other Icelandics were also homozygous for the wild-type alleles of P2, P3, P4, P8, and K1; their genotypes with respect to DYSF were N/N (4), N/P6 (3), and N/P5 (1). None of these other eight Icelandics was reported to be symptomatic.

A Thoroughbred (E456) scored as P6/P6, n/P3 and homozygous for the wild-type alleles of P2, P4, P8, and K1 was identified. This horse was also homozygous for the wild-type allele of P5. This Thoroughbred was reported by the owner to have shown frequent tie-ups at the track, was only raced once, and was described by an experienced trainer as “not quite right.” The horse died of colic at four years of age.

Two compound heterozygotes scored as P5/P6 and homozygous for the wild-type alleles of P2, P3, P4, P8, and K1 were identified. Both the P5/P6 Percheron (E797) and the P5/P6 Clydesdale (E823) were described by the owners as symptomatic, as were two other P5/P5 Clydesdales (E818 and E820) with the owner of the P5/P6 Clydesdale (E823).

Recombination between the P5 and P6 base substitution can produce a chromosome with both of the mutations in cis. Because both whole-genome sequencing on the Illumina platform and Sanger sequencing of DNA amplified by PCR do not yield phase information, such a chromosome could only be definitively identified in a horse scored as homozygous for P5 and heterozygous for P6, or in a horse scored as homozygous for P6 and heterozygous for P5. No such result has been seen in any sample.

The phenotype of human patients with the various alleles listed in TABLE 1, TABLE 2, and TABLE 3 suggests that horses homozygous for the DYSF-R253W allele (P5) would be expected to show increased serum creatine kinase (CK) and aspartate aminotransferase (AST), especially upon an exercise challenge. Affected horses are also expected to show abnormalities on muscle biopsy, including necrosis and regeneration, which may lead to adipose and fibrotic infiltration of the endomysial space. We do not expect heterozygous horses to be affected.

The novel DYSF-P1290T (P6) allele does not precisely replicate any known human mutation. Nevertheless, the phenotype observed in P6/P6 homozygotes and in P5/P6 compound heterozygotes suggests that these horses would also be expected to show increased serum creatine kinase (CK) and aspartate aminotransferase (AST), especially upon

an exercise challenge, as well as defects observable by muscle biopsy, as for horses homozygous for the DYSF-R253W (P5) allele.

Phenotypic Effects of the PYROXD1-D492H Variant

The PYROXD1-D492H variant (hereafter abbreviated as P8) was discovered by analysis of whole genome sequencing data from Icelandic, Arabian, Thoroughbred, and Quarter Horses. Some of these horses were diagnosed via muscle biopsy with Polysaccharide Storage Myopathy type 2 (PSSM2) or Myofibrillar Myopathy (MFM). These horses were heterozygous for the P8 allele (n/P8); some were homozygous wild-type for seven other variants associated with PSSM2: MYOT-S232P (P2), FLNC-E753K (P3a), FLNC-A1207T (P3b), MYOZ3-S42L (P4), DYSF-R285W (P5), DYSF-P1290T (P6) and COL6A3-G2182A (K1).

An Icelandic horse (E013) free of the P2, P3, P4, P5, P6, and K1 variants was diagnosed as having PSSM2 by muscle biopsy. The horse was heterozygous for P8 (n/P8). The symptoms observed in this horse correspond to the general description of the symptoms of PSSM2, and are similar to symptoms produced by homozygosity or heterozygosity for the P2, P3, P4, P5, and K1 variants in various combinations.

An Arabian horse (E361) scored as n/P8 and homozygous for the wild-type alleles of P2, P3, P4, P5, P6, and K1 was diagnosed as having Recurrent Exertional Rhabdomyolysis (RER) from a severe episode of exercise intolerance during which serum creatine kinase (CK) and aspartate aminotransferase (AST) rose to levels above 20,000 units; the baseline is below 500. Myoglobinuria (dark brown urine resulting from the presence of myoglobin) was also observed.

An Arabian horse (E049) scored as n/P8 and homozygous for the wild-type alleles of P2, P3, P4, P5, P6, and K1 was highly symptomatic and was euthanized at 12 years of age. A necropsy showed gluteal muscles affected but normal muscles in the lower limbs, neck, and inner thighs. In the most affected regions, muscle cells were replaced by fat cells. Up to 50% of the muscle cells were replaced by fat in some regions.

A horse owner who had submitted samples for a number of Arabians received results for P8 several months after submission. One of her Arabian mares (E682) was n/P2 and P8/P8 and homozygous for the wild-type alleles of P3, P4 and not yet scored for P5, P6, and K1. The owner reported that this mare had never been able to keep weight on her, and “looked like a living skeleton no matter what diet was fed.” This horse is shown in FIG. 44. Muscle wasting in the pelvic girdle (hindquarters), shoulder girdle (topline), and proximal limbs is evident. The horse is pregnant in the photo. The horse was reported to be symptomatic with gait abnormalities (“rope walking”), and died of a choking incident between the time of submission of the sample and the test results. Difficulty in swallowing is observed in human patients with Myofibrillar Myopathy 8 (MFM8) due to mutations in PYROXD1 (O’Grady et al. 2016 Am J Hum Genet. 99:1086-1105 DOI: 10.1016/j.ajhg.2016.09.005).

The phenotype of human patients with the missense alleles described here suggests that horses heterozygous or homozygous for PYROXD1-D492H (P8) would be expected to show an array of myopathic changes resembling Centronuclear Myopathy, Myofibrillar Myopathy, and Nemaline Myopathy in muscle biopsies. Specifically, we expect to see ectopic aggregates of desmin-positive material outside of the Z disc, Z disc streaming, and other changes seen in Myofibrillar Myopathy.

There is a high incidence of the P8 variant among Arabians (TABLE 7). Arabian horses diagnosed with Myofibrillar Myopathy (MFM) by muscle biopsy and symptoms

of exercise intolerance were analyzed in a recent published study (Valberg et al. 2018 *Physiol Genomics* doi: 10.1152). Samples were taken pre- and post-exercise and analyzed by RNA-Seq and iTRAQ (proteomics). Differential expression was seen for genes involved in pathways for structure morphogenesis, fiber organization, tissue development, and cell differentiation. Proteomic analysis showed lower levels of the antioxidant protein peroxiredoxin 6 in resting muscle; the authors proposed that altered cysteine metabolism and a deficiency of cysteine-containing antioxidants produced oxidative stress during aerobic exercise. Irreversible oxidation of cysteine residues in proteins that are structural components of the contractile apparatus, for example desmin, might induce proteolytic degradation of these proteins, causing Z disc fragmentation and streaming seen in muscle biopsies. While the genotype of the affected horses in this study is unknown, the defect in PYROXD1 described in this disclosure, and the phenotype of PYROXD1 mutants in humans, suggests that the affected horses in this study are n/P8 or P8/P8.

Three Appaloosas kept at the same farm under the same conditions were coincidentally tested for vitamin E levels around the same time that they were genotyped with respect to P8. All three horses were clear for P2, P3, P4, P5, and K1. Two are n/n for P8; one is n/P8.

Vitamin E levels (reference range is 200-1000 ug/dL)

Horse 1 (n/n)	255 ug/dL
Horse 2 (n/n)	383 ug/dL
Horse 3 (n/P8)	164 ug/dL

Vitamin E is an antioxidant that is consumed when it scavenges superoxide radicals in lipid membranes. Increased levels of oxidative stress would be expected to deplete vitamin E.

Prospects for treatment: If the myopathy caused by the defect in PYROXD1 described in this disclosure results from oxidative stress, it is possible that treatments with antioxidants would be effective at preventing symptoms or the progression of the disease. For example, dietary supplementation with vitamin E might be effective. Also, the combination of methylsulfonylmethane (MSM) and vitamin C has been shown to be effective at combatting oxidative stress caused by sustained exercise in horses, leading to an increase in glutathione synthesis and in levels of glutathione transferase (Marñón G et al. 2008 *Acta Vet Scand* 50:45 DOI: 10.1186/1751-0147-50-45; Williams CA 2016 *J Anim Sci* 94:4067-4075 DOI: 10.2527/jas.2015-9988). N-acetyl cysteine is a safe, low-cost compound that increases the level of glutathione, and has been shown to protect against muscle damage in mice subjected to oxidative stress through a mutation in CASQ1, the gene encoding calsequestrin (Paolini C et al. *Skelet Muscle* 5:10 DOI: 10.1186/s13395-015-0035-9).

Phenotypic Effects of the COL6A3-G2182A Variant

The COL6A3-G2182A variant (hereafter abbreviated as K1) was discovered by analysis of whole genome sequencing data from Icelandic, Arabian, Thoroughbred, and Quarter Horses. Some of these horses were diagnosed via muscle biopsy with Polysaccharide Storage Myopathy type 2 (PSSM2). These horses were heterozygous for the K1 allele; some were homozygous wild-type for six other variants previously associated with PSSM2: MYOT-S232P (P2), FLNC-E753K (P3a), FLNC-A1207T (P3b), MYOZ3-S42L (P4), DYSF-R285W (P5), DYSF-P1290T (P6), and PYROXD1-D492H (P8).

A Paint horse (E008) scored as n/K1 and homozygous for the wild-type alleles of P2, P3, P4, P5, P6, and P8 was diagnosed as having PSSM2 by muscle biopsy. The symptoms observed in this horse correspond to the general description of the symptoms of PSSM2, and are similar to symptoms produced by homozygosity or heterozygosity for the P2, P3, P4, P5, and P8 variants in various combinations.

The phenotype of human patients with missense alleles of COL6A3 and of mice with the targeted allele suggests that muscle biopsies from horses bearing the COL6A3-G2182A mutation (K1) will show changes in the connective tissue layer that ensheaths the muscle fiber. The size of the connective tissue layer is expected to increase. In late stages of the disease, replacement of some of this layer with adipose or scar tissue is expected, without evidence of necrosis. This condition has been called skeletal muscle endomysial fibrosis. It has not yet been directly observed in muscle biopsies from n/K1 or K1/K1 horses.

Incidence of Genetic Variants by Breed

The incidence of the previously described genetic variants (P2, P3, and P4) and the genetic variants described in this disclosure (P5, P6, P8, and K1) varies among breeds. It is useful to have a measurement of the allele frequency of different variants among breeds, as this can facilitate a further examination of the phenotype produced by a specific variant by identifying a breed in which it occurs at high incidence, perhaps with other genetic variants at a low incidence.

TABLE 5 shows the observed incidence of P2, P3, P4, P5, P6, P8, and K1 in Quarter Horse-related breeds. For purposes of this tabulation, a "Quarter Horse-related" breed is a Quarter Horse, Appendix Quarter Horse (the result of a cross of a Quarter Horse to a Thoroughbred), Paint Horse, Appaloosa, or a horse resulting from a cross of any two of these types to each other. This sample cannot be considered unselected, as the majority of horse owners volunteering their horses for study did so because their horses had symptoms of some kind. When we attempted to recruit breeding herds, the majority of horse owners did not volunteer; it is possible that those volunteering did so because they observed symptoms of exercise intolerance in their herd.

Quarter Horse-related breeds are of relatively recent origin, with gene flow from other breeds. For example, the American Quarter Horse Association allows the registration of Appendix Quarter Horses that have a Thoroughbred parent. In contrast, Thoroughbreds have a closed breeding book, meaning that a horse can only be registered as a Thoroughbred if both parents are Thoroughbreds. All seven genetic variants are found in Quarter Horse-related breeds. The incidence ranges from an allele frequency of 0.204 for P2 to an allele frequency of 0.007 for K1. Quarter Horse-related breeds are therefore a good choice for the further study of the interaction of all seven genetic variants. It should be noted that P2 (chr14:38519183 A/G) and P4 (chr14:27399222 G/A) are located approximately 11.2 Megabases (Mb) apart on chromosome 14 and are expected to display genetic linkage. Two different genotypes confirm that the chromosome having both P2 and P4 in cis exists in horse populations: P2/P2 n/P4 and n/P2 P4/P4 individuals have been observed in Quarter Horse-related breeds. No horse that is P2/P2 P4/P4, or more properly written, P2 P4/P2 P4, has been observed to date in Quarter Horse-related breeds or in any other breed in a sample of nearly 1,000 horses tested for both variants.

TABLE 6 shows the observed incidence of P2, P3, P4, P5, P6, P8, and K1 in Thoroughbreds. Because as noted above,

Thoroughbreds have a closed breeding book, it is possible that Thoroughbreds may be entirely free of one or more of these genetic variants. In the sample of modest size tested to date, P5 and K1 have not yet been observed in Thoroughbreds, while the observed frequency of other variants ranges from 0.028 for P8 to 0.193 for P6. The Thoroughbreds included in this sample are less highly selected for symptomatic horses than is the Quarter Horse-related sample because the method of recruitment of horse owners was different. However, some horse owners were aware of the nature of the research and likely volunteered symptomatic horses. The results show that Thoroughbreds are well suited for the further study of the P6 genetic variant, as it occurs at a reasonably high frequency in the absence of P5, the other

allele of DYSF. TABLE 7 shows the observed incidence of P2, P3, P4, P5, P6, P8, and K1 in Arabians. So far, P3 has not yet been observed in Arabians, nor has P5 or P6. P2, P4, and K1 are observed. P8 is observed at an allele frequency of 0.218. As noted previously in this disclosure, it is possible that the PYROXD1 variant P8 is responsible for the defect in the redox state of cysteine-containing antioxidants seen in Arabians diagnosed with Myofibrillar Myopathy (MFM) in a recent published study (Valberg et al. 2018 Physiol Genomics doi: 10.1152). This shows that Arabians are ideal for the further study of the effects of the PYROXD1 variant P8.

TABLE 8 shows the observed incidence of P2, P3, P4, P5, P6, P8, and K1 in draft breeds, defined for this sample as Shires, Clydesdales, and Percherons, or horses derived from crosses among these breeds. There are many other draft breeds for which we have obtained samples of very modest size, but we initially concentrated on these three breeds because the observed incidence of P5 was very high. The observed frequency of P5 in this modest sample is 0.543. The observed frequency of P6 in this modest sample is 0.136. These frequencies are high enough that compound heterozygotes (P5/P6) can be expected; indeed, two symptomatic compound heterozygotes have been identified. The observed frequency of P4, P8, and K1 in this small sample is zero; P2 and P3 are present. This shows that Shires, Clydesdales, and Percherons are ideal for the further study of the effects of the DYSF variants P5 and P6.

TABLE 5

Incidence of genetic variants in Quarter Horse-related breeds					
P2	n/n	n/P2	P2/P2	Total	P2 allele frequency
	390	167	38	595	0.204
P3	n/n	n/P3	P3/P3	Total	P3 allele frequency
	503	92	4	599	0.083
P4	n/n	n/P4	P4/P4	Total	P4 allele frequency
	587	81	4	587	0.076
P5	n/n	n/P5	P5/P5	Total	P5 allele frequency
	35	3	0	37	0.041
P6	n/n	n/P6	P6/P6	Total	P6 allele frequency
	64	8	0	72	0.056
P8	n/n	n/P8	P8/P8	Total	P8 allele frequency
	79	7	0	86	0.041
K1	n/n	n/K1	K1/K1	Total	K1 allele frequency
	145	2	0	147	0.007

TABLE 6

Incidence of genetic variants in Thoroughbreds					
P2	n/n	n/P2	P2/P2	Total	P2 allele frequency
	149	51	8	208	0.161

TABLE 6-continued

Incidence of genetic variants in Thoroughbreds					
P3	n/n	n/P3	P3/P3	Total	P3 allele frequency
	170	42	2	214	0.107
P4	n/n	n/P4	P4/P4	Total	P4 allele frequency
	177	34	1	212	0.085
P5	n/n	n/P5	P5/P5	Total	P5 allele frequency
	102	0	0	102	0.0
P6	n/n	n/P6	P6/P6	Total	P6 allele frequency
	69	33	4	106	0.193
P8	n/n	n/P8	P8/P8	Total	P8 allele frequency
	116	7	0	123	0.028
K1	n/n	n/K1	K1/K1	Total	K1 allele frequency
	121	0	0	121	0.0

TABLE 7

Incidence of genetic variants in Arabians					
P2	n/n	n/P2	P2/P2	Total	P2 allele frequency
	43	17	4	64	0.195
P3	n/n	n/P3	P3/P3	Total	P3 allele frequency
	64	0	0	64	0.0
P4	n/n	n/P4	P4/P4	Total	P4 allele frequency
	58	8	1	65	0.077
P5	n/n	n/P5	P5/P5	Total	P5 allele frequency
	22	0	0	22	0.0
P6	n/n	n/P6	P6/P6	Total	P6 allele frequency
	16	0	0	16	0.0
P8	n/n	n/P8	P8/P8	Total	P8 allele frequency
	38	21	3	62	0.218
K1	n/n	n/K1	K1/K1	Total	K1 allele frequency
	18	3	0	21	0.071

TABLE 8

Incidence of genetic variants in Draft breeds					
P2	n/n	n/P2	P2/P2	Total	P2 allele frequency
	15	5	5	25	0.300
P3	n/n	n/P3	P3/P3	Total	P3 allele frequency
	24	1	0	25	0.020
P4	n/n	n/P4	P4/P4	Total	P4 allele frequency
	23	0	0	23	0.0
P5	n/n	n/P5	P5/P5	Total	P5 allele frequency
	6	9	8	23	0.543
P6	n/n	n/P6	P6/P6	Total	P6 allele frequency
	16	6	0	22	0.136
P8	n/n	n/P8	P8/P8	Total	P8 allele frequency
	22	0	0	22	0.0
K1	n/n	n/K1	K1/K1	Total	K1 allele frequency
	23	0	0	23	0.0

In the preceding description and following claims, the term “and/or” means one or all of the listed elements or a combination of any two or more of the listed elements; the terms “comprises,” “comprising,” and variations thereof are to be construed as open ended—i.e., additional elements or steps are optional and may or may not be present; unless otherwise specified, “a,” “an,” “the,” and “at least one” are used interchangeably and mean one or more than one; and the recitations of numerical ranges by endpoints include all numbers subsumed within that range (e.g., 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, 5, etc.).

In the preceding description, particular embodiments may be described in isolation for clarity. Unless otherwise expressly specified that the features of a particular embodiment are incompatible with the features of another embodiment, certain embodiments can include a combination of compatible features described herein in connection with one or more embodiments.

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For any method disclosed herein that includes discrete steps, the steps may be conducted in any feasible order. And, as appropriate, any combination of two or more steps may be conducted simultaneously.

This is illustrated by the following examples. It is to be understood that the particular examples, materials, amounts, and procedures are to be interpreted broadly in accordance with the scope and spirit of the invention as set forth herein.

EXAMPLES

Example 1

Method of Detecting DNA Mutations Associated with Equine Polysaccharide Storage Myopathy Type 2 (PSSM2), Also Known as Myofibrillar Myopathy (MFM)

The complete DNA sequences of the horse DYSF, PYROXD1, and COL6A3 coding regions were obtained from the current version of the public horse genome assembly (EquCab2).

Using the DYSF, PYROXD1, and COL6A3 sequences, PCR primers are developed that can amplify the site of genomic DNA containing the DYSF-R253W, DYSF-P1290T, PYROXD1-D492H, and COL6A3-G2182A mutations. For example, a PCR primer pair that has been successfully and reliably used to amplify the region including DYSF-R253W from isolated horse DNA samples lies in the region around Exon B (FIG. 3). These sequences are

(SEQ ID NO: 14)
5' - CCCGAGATTTCTGGCTTTCT - 3'
and

(SEQ ID NO: 15)
5' - CTCGACAAGTTCTGGGGTGT - 3'.

A PCR primer pair that has been successfully and reliably used to amplify the region including DYSF-P1290T from isolated horse DNA samples lies in the region around Exon I (FIG. 3). These sequences are

(SEQ ID NO: 64)
5' - GGTGCAAACTCCCAACTGT - 3'
and

(SEQ ID NO: 65)
5' - GATTTTTCAGCTGCCGAAG - 3'.

A PCR primer pair that has been successfully and reliably used to amplify the region including PYROXD1-D492H from isolated horse DNA samples lies in the region around Exon 12 (FIG. 15). These sequences are

(SEQ ID NO: 111)
5' - CAGATTTTCTGCTGGCCATT - 3'
and

(SEQ ID NO: 112)
5' - TGGTCATCATTAAATCAGTGCAA - 3'.

A PCR primer pair that has been successfully and reliably used to amplify the region including COL6A3-G2182A from isolated horse DNA samples lies in the region around Exon 12 (FIG. 26). These sequences are 5'-AG-ATGGGGCACAGATCAAAC-3' (SEQ ID NO:172) and

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5'-TTCCCAGACTCTCCTGTGCT-3' (SEQ ID NO:171). Many other primer pairs are also possible.

Using the above PCR primers to amplify the region, the genotype of any horse (G/G, G/A, or A/A) for the DNA sequence of the forward strand at chr15:31,306,949, and R/R, R/W, or W/W for the amino acid sequence of the DYSF-R253W variant can be obtained. In this method, the amplified DNA may be cloned and then sequenced or sequenced directly without cloning. Alternatively, the appearance of amplified product in the presence of primers specific to the wild type or mutant allele may be monitored in real time using a qPCR instrument designed for this purpose. Many other methods of detecting the nucleotides at the positions of the horse DYSF sequence are possible.

Using the above PCR primers to amplify the region, the genotype of any horse (G/G, G/T, or T/T) for the DNA sequence of the forward strand at chr15:31,306,949, and P/P, P/T, or T/T for the amino acid sequence of the DYSF-P1290T variant can be obtained. In this method, the amplified DNA may be cloned and then sequenced or sequenced directly without cloning. Alternatively, the appearance of amplified product in the presence of primers specific to the wild type or mutant allele may be monitored in real time using a qPCR instrument designed for this purpose. Many other methods of detecting the nucleotides at the positions of the horse DYSF sequence are possible.

Using the above PCR primers to amplify the region, the genotype of any horse (G/G, G/C, or C/C) for the DNA sequence of the forward strand at chr6:47,661,977, and D/D, D/H, or H/H for the amino acid sequence of the PYROXD1-D492H variant can be obtained. In this method, the amplified DNA may be cloned and then sequenced or sequenced directly without cloning. Alternatively, the appearance of amplified product in the presence of primers specific to the wild type or mutant allele may be monitored in real time using a qPCR instrument designed for this purpose. Many other methods of detecting the nucleotides at the positions of the horse PYROXD1 sequence are possible.

Using the above PCR primers to amplify the region, the genotype of any horse (G/G, G/C, or C/C) for the DNA sequence of the forward strand at chr6:23,480,621, and G/G, G/A, or A/A for the amino acid sequence of the COL6A3-G2182A variant can be obtained. In this method, the amplified DNA may be cloned and then sequenced or sequenced directly without cloning. Alternatively, the appearance of amplified product in the presence of primers specific to the wild type or mutant allele may be monitored in real time using a qPCR instrument designed for this purpose. Many other methods of detecting the nucleotides at the positions of the horse COL6A3 sequence are possible.

DNA testing now provides veterinarians and veterinary pathologists with a means to more accurately determine if a horse with the clinical signs of Polysaccharide Storage Myopathy type 2 (PSSM2) has the heritable and common form of the disease that can be specifically attributed to the DYSF-R253W, DYSF-P1290T, PYROXD1-D492H, or COL6A3-G2182A coding region mutation. All that is needed is a tissue sample containing the individual's DNA (typically hair root or blood) and appropriate PCR and sequence analysis technology to detect the distinct nucleotide changes. Such PCR primers are based in (1) DYSF Exon B (as shown in FIG. 3) and the flanking intron sequences, as shown in FIG. 3, or in other DNA sequences of this gene, (2) DYSF Exon I (as shown in FIG. 15) and the flanking intron sequences, as shown in FIG. 15, or in other DNA sequences of this gene, (3) PYROXD1 Exon 12 (as shown in FIG. 24) and the flanking intron sequences, as

shown in FIG. 24, or in other DNA sequences of this gene, or (4) COL6A3 Exon 26 (as shown in FIG. 35) and the flanking intron sequences, as shown in FIG. 35, or in other DNA sequences of this gene.

Also, DNA testing provides owners and breeders with a means to determine if any horse can be expected to produce offspring with these forms of Polysaccharide Storage Myopathy type 2 (PSSM2). Abbreviating the DYSF-R253W allele as P5, a P5/P5 horse would produce a carrier foal 100% of the time, while an N/P5 horse would produce a carrier foal 50% of the time when mated to an N/N horse. Mating of an N/P5 horse to an N/P5 horse would produce an affected foal 25% of the time and a carrier foal 50% of the time. Abbreviating the DYSF-P1290T allele as P6, a P6/P6 horse would produce a carrier foal 100% of the time, while an N/P6 horse would produce a carrier foal 50% of the time when mated to an N/N horse. Mating of an N/P6 horse to an N/P6 horse would produce an affected foal 25% of the time and a carrier foal 50% of the time. Abbreviating the PYROXD1-D492H allele as P8, a P8/P8 horse would produce a heterozygous foal 100% of the time when mated to an n/n horse, while an n/P8 horse would produce a heterozygous foal 50% of the time when mated to an n/n horse. Mating of an n/P8 horse to an n/P8 horse would produce an affected P8/P8 foal 25% of the time, and an affected n/P8 foal 50% of the time. Abbreviating the COL6A3-G2182A allele as K1, a K1/K1 horse would produce a heterozygous foal 100% of the time when mated to an n/n horse, while an n/K1 horse would produce a heterozygous foal 50% of the time when mated to an n/n horse. Mating of an n/K1 horse to an n/K1 horse would produce an affected K1/K1 foal 25% of the time, and an affected n/K1 foal 50% of the time. Breeding programs could incorporate this information in the selection of parents that could eventually reduce and even eliminate these forms of Polysaccharide Storage Myopathy type 2 (PSSM2) in their herds.

The complete disclosure of all patents, patent applications, and publications, and electronically available material

(including, for instance, nucleotide sequence submissions in, e.g., GenBank and RefSeq, and amino acid sequence submissions in, e.g., SwissProt, PIR, PRF, PDB, and translations from annotated coding regions in GenBank and RefSeq) cited herein are incorporated by reference in their entirety. In the event that any inconsistency exists between the disclosure of the present application and the disclosure(s) of any document incorporated herein by reference, the disclosure of the present application shall govern. The foregoing detailed description and examples have been given for clarity of understanding only. No unnecessary limitations are to be understood therefrom. The invention is not limited to the exact details shown and described, for variations obvious to one skilled in the art will be included within the invention defined by the claims.

Unless otherwise indicated, all numbers expressing quantities of components, molecular weights, and so forth used in the specification and claims are to be understood as being modified in all instances by the term "about." Accordingly, unless otherwise indicated to the contrary, the numerical parameters set forth in the specification and claims are approximations that may vary depending upon the desired properties sought to be obtained. At the very least, and not as an attempt to limit the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques.

Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. All numerical values, however, inherently contain a range necessarily resulting from the standard deviation found in their respective testing measurements.

All headings are for the convenience of the reader and should not be used to limit the meaning of the text that follows the heading, unless so specified.

SEQUENCE LISTING

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Glu Ala Val Phe Ile Thr Val Val Asp Ser Cys Ser Leu Arg Thr Asp	130	135	140
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Asp Lys Glu Asp Ile Glu Ser Asn Leu Leu Arg Pro Thr Gly Met Ala	210	215	220
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Phe Asp Ser Asn Lys Lys Asn Leu Val Asp Pro Phe Val Glu Val Ser	260	265	270
Phe Ala Gly Lys Met Leu Cys Ser Lys Ile Leu Glu Lys Met Ala Asn	275	280	285
Pro Gln Trp Asn Gln Ser Ile Thr Leu Pro Val Met Phe Pro Ser Met	290	295	300
Ser Glu Lys Met Arg Ile Arg Val Ile Asp Trp Asp Arg Leu Thr His	305	310	315
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Gly Lys Arg Lys Arg Ser Thr Pro Ala Pro Arg Lys Leu Leu Ser Asp	50	55	60	
Lys Pro Gln Asp Phe Gln Ile Arg Val Gln Val Ile Glu Gly Arg Gln	65	70	75	80
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 Glu Ala Val Phe Ile Thr Val Val Asp Ser Cys Ser Leu Arg Thr Asp
 130 135 140
 Ala Leu Ile Gly Glu Phe Arg Met Asp Val Gly Thr Ile Tyr Arg Glu
 145 150 155 160
 Pro Arg His Ala Tyr Leu Arg Lys Trp Leu Leu Leu Ser Asp Pro Asp
 165 170 175
 Asp Phe Ser Ala Gly Pro Lys Gly Tyr Leu Lys Ala Ser Leu Cys Val
 180 185 190
 Leu Gly Pro Gly Asp Glu Ala Pro Leu Glu Arg Lys Asp Pro Ser Glu
 195 200 205
 Asp Lys Glu Asp Ile Glu Ser Asn Leu Leu Arg Pro Thr Gly Met Ala
 210 215 220
 Leu Arg Gly Ala His Phe Cys Leu Lys Val Phe Arg Ala Glu Asp Leu
 225 230 235 240
 Pro Gln Met Asp Asp Ala Val Val Asp Ser Val Lys Gln Ile Phe Gly
 245 250 255
 Phe Asp Ser Asn Lys Lys Asn Leu Val Asp Pro Phe Val Glu Val Ser
 260 265 270
 Phe Ala Gly Lys Met Leu Cys Ser Lys Ile Leu Glu Lys Met Ala Asn
 275 280 285
 Pro Gln Trp Asn Gln Ser Ile Thr Leu Pro Val Met Phe Pro Ser Met
 290 295 300
 Ser Glu Lys Met Arg Ile Arg Val Ile Asp Trp Asp Arg Leu Thr His
 305 310 315 320
 Asn Asp Ile Val Ala Thr Thr Tyr Leu Asn Met Ser Lys Ile Ser Ala
 325 330 335
 Pro Gly Gly Glu Ile Ala
 340

<210> SEQ ID NO 12
 <211> LENGTH: 129
 <212> TYPE: DNA
 <213> ORGANISM: Equus caballus

<400> SEQUENCE: 12

```

atcagggtcc aggtgatcga ggggcgccag ctgccagggg tgaacatcaa gcctgtggtc    60
aaggtcaccg cggccaggca gaccaagcgg actcggattc acaggggaaa cagcccgtc    120
ttcaacgag                                     129
  
```

<210> SEQ ID NO 13
 <211> LENGTH: 129
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 13

```

atcagggtcc aggtgatcga ggggcgccag ctgccagggg tgaacatcaa gcctgtggtc    60
aaggtcaccg cggccaggca gaccaagtgg actcggattc acaggggaaa cagcccgtc    120
ttcaacgag                                     129
  
```

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<210> SEQ ID NO 14
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        primer

<400> SEQUENCE: 14

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<210> SEQ ID NO 15
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        primer

<400> SEQUENCE: 15

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<210> SEQ ID NO 16
<211> LENGTH: 6243
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 16

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agcgtgaacc ctgtatggaa tgagggtattt gaatgggacc tcaagggcat cccctggac      180
cagggctctg agcttcatgt ggtgggtcaaa gaccatgaga cgatggggag gaacaggttc      240
ctgggggaag ccaaggtccc actccgagag gtcctcgcca cccctagtct gtccgccagc      300
ttcaatgccc ccctgctgga caccaagaag cagccacag gggcctcgct ggtcctgcag      360
gtgtcctaca caccgctgcc tggagctgtg cccctgttcc cgccccctac tcctctggag      420
ccctccccga ctctgcctga cctggatgta gtggcagaca caggaggaga ggaagacaca      480
gaggaccagg gactcactgg agatgaggcg gagccattcc tggatcaaag cggaggccccg      540
ggggctccca ccacccaag gaaactacct tcacgtcttc cgccccacta ccccgggatc      600
aaaagaaagc gaagtgcgcc tacatctaga aagctgctgt cagacaaacc gcaggatttc      660
cagatcaggg tccagggtgat cgaggggcgc cagctgccgg ggggtgaacat caagcctgtg      720
gtcaagggtta ccgctgcagg gcagaccaag cggacgcgga tccacaaggg aaacagccca      780
ctcttcaatg agactctttt cttcaacttg tttgactctc ctggggagct gtttgatgag      840
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ctgctgctct cagacctga tgacttctct gctggggcca gaggetacct gaaaacaagc     1020
ctttgtgtgc tggggcctgg ggacgaagcg cctctggaga gaaaagacc ctctgaagac     1080
aaggaggaca ttgaaagcaa cctgctccgg ccacaggcg tagccctgcg aggagccac      1140
ttctgcctga aggtcttcgg ggccgaggac ttgccgcaga tggacgatgc cgtgatggac     1200
aacgtgaaac agatcttttg cttcgagagt aacaagaaga acttggtgga cccctttgtg     1260
gaggtcagct ttgcggggaa aatgctgtgc agcaagatct tggagaagac ggccaaccct     1320
cagtggaacc agaacatcac actgcctgcc atgtttccct ccatgtgcga aaaaatgagg     1380

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attcgatatca tagactggga ccgcctgact cacaatgaca tcgtgggtac cacctacctg	1440
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aagccttcga aagcctcaga cttggatgac tacctgggct tcctcccccac ttttggggccc	1560
tgctacatca acctctatgg cagtcccaga gagttcacag gcttcccaga cccctacaca	1620
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gagaccaagc tgggtggagca cagtgaacag aaggtggagg accttccctgc ggatgacatc	1740
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ctggcgggca agctggaaat gaccttgag attgtagcag agagtgagca tgaggagcgg	6000
cctgctggcc agggccggga tgagcccaac atgaacctc agcttgagga cccaaggcgc	6060
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eggcggtttcc ggtgggccaat caccctcttc atcaccctct taccctctgct gctgttctcg 6180
gccatcttca tctacgcctt cccgaactat gctgccatga agctggtgaa gcccttcagc 6240
tga 6243

```

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<210> SEQ ID NO 17
<211> LENGTH: 2080
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 17

```

```

Met Leu Arg Val Phe Ile Leu Tyr Ala Glu Asn Val His Thr Pro Asp
1      5      10      15
Thr Asp Ile Ser Asp Ala Tyr Cys Ser Ala Val Phe Ala Gly Val Lys
20     25     30
Lys Arg Thr Lys Val Ile Lys Asn Ser Val Asn Pro Val Trp Asn Glu
35     40     45
Gly Phe Glu Trp Asp Leu Lys Gly Ile Pro Leu Asp Gln Gly Ser Glu
50     55     60
Leu His Val Val Val Lys Asp His Glu Thr Met Gly Arg Asn Arg Phe
65     70     75     80
Leu Gly Glu Ala Lys Val Pro Leu Arg Glu Val Leu Ala Thr Pro Ser
85     90     95
Leu Ser Ala Ser Phe Asn Ala Pro Leu Leu Asp Thr Lys Lys Gln Pro
100    105    110
Thr Gly Ala Ser Leu Val Leu Gln Val Ser Tyr Thr Pro Leu Pro Gly
115    120    125
Ala Val Pro Leu Phe Pro Pro Pro Thr Pro Leu Glu Pro Ser Pro Thr
130    135    140
Leu Pro Asp Leu Asp Val Val Ala Asp Thr Gly Gly Glu Glu Asp Thr
145    150    155    160
Glu Asp Gln Gly Leu Thr Gly Asp Glu Ala Glu Pro Phe Leu Asp Gln
165    170    175
Ser Gly Gly Pro Gly Ala Pro Thr Thr Pro Arg Lys Leu Pro Ser Arg
180    185    190
Pro Pro Pro His Tyr Pro Gly Ile Lys Arg Lys Arg Ser Ala Pro Thr
195    200    205
Ser Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile Arg Val
210    215    220
Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys Pro Val
225    230    235    240
Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile His Lys
245    250    255
Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Leu Phe Asp
260    265    270
Ser Pro Gly Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val Val Asp
275    280    285
Ser Arg Ser Leu Arg Thr Asp Ala Leu Leu Gly Glu Phe Arg Met Asp
290    295    300
Val Gly Thr Ile Tyr Arg Glu Pro Arg His Ala Tyr Leu Arg Lys Trp
305    310    315    320
Leu Leu Leu Ser Asp Pro Asp Asp Phe Ser Ala Gly Ala Arg Gly Tyr
325    330    335
Leu Lys Thr Ser Leu Cys Val Leu Gly Pro Gly Asp Glu Ala Pro Leu

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340							345					350				
Glu	Arg	Lys	Asp	Pro	Ser	Glu	Asp	Lys	Glu	Asp	Ile	Glu	Ser	Asn	Leu	
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Leu	Arg	Pro	Thr	Gly	Val	Ala	Leu	Arg	Gly	Ala	His	Phe	Cys	Leu	Lys	
		370				375						380				
Val	Phe	Arg	Ala	Glu	Asp	Leu	Pro	Gln	Met	Asp	Asp	Ala	Val	Met	Asp	
385						390				395				400		
Asn	Val	Lys	Gln	Ile	Phe	Gly	Phe	Glu	Ser	Asn	Lys	Lys	Asn	Leu	Val	
				405				410						415		
Asp	Pro	Phe	Val	Glu	Val	Ser	Phe	Ala	Gly	Lys	Met	Leu	Cys	Ser	Lys	
		420						425				430				
Ile	Leu	Glu	Lys	Thr	Ala	Asn	Pro	Gln	Trp	Asn	Gln	Asn	Ile	Thr	Leu	
		435				440						445				
Pro	Ala	Met	Phe	Pro	Ser	Met	Cys	Glu	Lys	Met	Arg	Ile	Arg	Ile	Ile	
		450				455						460				
Asp	Trp	Asp	Arg	Leu	Thr	His	Asn	Asp	Ile	Val	Ala	Thr	Thr	Tyr	Leu	
465						470				475				480		
Ser	Met	Ser	Lys	Ile	Ser	Ala	Pro	Gly	Gly	Glu	Ile	Glu	Glu	Glu	Pro	
				485				490						495		
Ala	Gly	Ala	Val	Lys	Pro	Ser	Lys	Ala	Ser	Asp	Leu	Asp	Asp	Tyr	Leu	
				500				505						510		
Gly	Phe	Leu	Pro	Thr	Phe	Gly	Pro	Cys	Tyr	Ile	Asn	Leu	Tyr	Gly	Ser	
		515				520						525				
Pro	Arg	Glu	Phe	Thr	Gly	Phe	Pro	Asp	Pro	Tyr	Thr	Glu	Leu	Asn	Thr	
		530				535						540				
Gly	Lys	Gly	Glu	Gly	Val	Ala	Tyr	Arg	Gly	Arg	Leu	Leu	Leu	Ser	Leu	
545						550				555				560		
Glu	Thr	Lys	Leu	Val	Glu	His	Ser	Glu	Gln	Lys	Val	Glu	Asp	Leu	Pro	
				565				570						575		
Ala	Asp	Asp	Ile	Leu	Arg	Val	Glu	Lys	Tyr	Leu	Arg	Arg	Arg	Lys	Tyr	
				580				585						590		
Ser	Leu	Phe	Ala	Ala	Phe	Tyr	Ser	Ala	Thr	Met	Leu	Gln	Asp	Val	Asp	
		595				600						605				
Asp	Ala	Ile	Gln	Phe	Glu	Val	Ser	Ile	Gly	Asn	Tyr	Gly	Asn	Lys	Phe	
		610				615						620				
Asp	Met	Thr	Cys	Leu	Pro	Leu	Ala	Ser	Thr	Thr	Gln	Tyr	Ser	Arg	Ala	
625						630				635				640		
Val	Phe	Asp	Gly	Cys	His	Tyr	Tyr	Tyr	Leu	Pro	Trp	Gly	Asn	Val	Lys	
				645				650						655		
Pro	Val	Val	Val	Leu	Ser	Ser	Tyr	Trp	Glu	Asp	Ile	Ser	His	Arg	Ile	
		660						665						670		
Glu	Thr	Gln	Asn	Gln	Leu	Leu	Gly	Ile	Ala	Asp	Arg	Leu	Glu	Ala	Gly	
		675				680						685				
Leu	Glu	Gln	Val	His	Leu	Ala	Leu	Lys	Ala	Gln	Cys	Ser	Thr	Glu	Asp	
		690				695						700				
Val	Asp	Ser	Leu	Val	Ala	Gln	Leu	Thr	Asp	Glu	Leu	Ile	Ala	Gly	Cys	
705						710				715				720		
Ser	Gln	Pro	Leu	Gly	Asp	Ile	His	Glu	Thr	Pro	Ser	Ala	Thr	His	Leu	
				725				730						735		
Asp	Gln	Tyr	Leu	Tyr	Gln	Leu	Arg	Thr	His	His	Leu	Ser	Gln	Ile	Thr	
		740						745						750		
Glu	Ala	Ala	Leu	Ala	Leu	Lys	Leu	Gly	His	Ser	Glu	Leu	Pro	Ala	Ala	
		755				760						765				

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Leu Glu Gln Ala Glu Asp Trp Leu Leu Arg Leu Arg Ala Leu Ala Glu
 770 775 780
 Glu Pro Gln Asn Ser Leu Pro Asp Ile Val Ile Trp Met Leu Gln Gly
 785 790 795 800
 Asp Lys Arg Val Ala Tyr Gln Arg Val Pro Ala His Gln Val Leu Phe
 805 810 815
 Ser Arg Arg Gly Ala Asn Tyr Cys Gly Lys Asn Cys Gly Lys Leu Gln
 820 825 830
 Thr Ile Phe Leu Lys Tyr Pro Met Glu Lys Val Pro Gly Ala Arg Met
 835 840 845
 Pro Val Gln Ile Arg Val Lys Leu Trp Phe Gly Leu Ser Val Asp Glu
 850 855 860
 Lys Glu Phe Asn Gln Phe Ala Glu Gly Lys Leu Ser Val Phe Ala Glu
 865 870 875 880
 Thr Tyr Glu Asn Glu Thr Lys Leu Ala Leu Val Gly Asn Trp Gly Thr
 885 890 895
 Thr Gly Leu Thr Tyr Pro Lys Phe Ser Asp Val Thr Gly Lys Ile Lys
 900 905 910
 Leu Pro Lys Asp Ser Phe Arg Pro Ser Ala Gly Trp Thr Trp Ala Gly
 915 920 925
 Asp Trp Phe Val Cys Pro Glu Lys Thr Leu Leu His Asp Met Asp Ala
 930 935 940
 Gly His Leu Ser Phe Val Glu Glu Val Phe Glu Asn Gln Thr Arg Leu
 945 950 955 960
 Pro Gly Gly Gln Trp Ile Tyr Met Ser Asp Asn Tyr Thr Asp Val Asn
 965 970 975
 Gly Glu Lys Val Leu Pro Lys Asp Asp Ile Glu Cys Pro Leu Gly Trp
 980 985 990
 Lys Trp Glu Asp Glu Glu Trp Ser Thr Asp Leu Asn Arg Ala Val Asp
 995 1000 1005
 Glu Gln Gly Trp Glu Tyr Ser Ile Thr Ile Pro Pro Glu Arg Lys
 1010 1015 1020
 Pro Lys His Trp Val Pro Ala Glu Lys Met Tyr Tyr Thr His Arg
 1025 1030 1035
 Arg Arg Arg Trp Val Arg Leu Arg Arg Arg Asp Leu Ser Gln Met
 1040 1045 1050
 Glu Ala Leu Lys Arg His Arg Gln Ala Glu Ala Glu Gly Glu Gly
 1055 1060 1065
 Trp Glu Tyr Ala Ser Leu Phe Gly Trp Lys Phe His Leu Glu Tyr
 1070 1075 1080
 Arg Lys Thr Asp Ala Phe Arg Arg Arg Arg Trp Arg Arg Arg Met
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 Glu Pro Leu Glu Lys Thr Gly Pro Ala Ala Val Phe Ala Leu Glu
 1100 1105 1110
 Gly Ala Leu Gly Gly Val Met Asp Asp Lys Ser Glu Asp Ser Met
 1115 1120 1125
 Ser Val Ser Thr Leu Ser Phe Gly Val Asn Arg Pro Thr Ile Ser
 1130 1135 1140
 Cys Ile Phe Asp Tyr Gly Asn Arg Tyr His Leu Arg Cys Tyr Met
 1145 1150 1155
 Tyr Gln Ala Arg Asp Leu Ala Ala Met Asp Lys Asp Ser Phe Ser
 1160 1165 1170

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Asp	Pro	Tyr	Ala	Ile	Val	Ser	Phe	Leu	His	Gln	Ser	Gln	Lys	Thr
1175						1180					1185			
Val	Val	Val	Lys	Asn	Thr	Leu	Asn	Pro	Thr	Trp	Asp	Gln	Thr	Leu
1190						1195					1200			
Ile	Phe	Tyr	Glu	Ile	Glu	Ile	Phe	Gly	Glu	Pro	Ala	Thr	Val	Ala
1205						1210					1215			
Glu	Gln	Pro	Pro	Ser	Ile	Val	Val	Glu	Leu	Tyr	Asp	His	Asp	Thr
1220						1225					1230			
Tyr	Gly	Ala	Asp	Glu	Phe	Met	Gly	Arg	Cys	Ile	Cys	Gln	Pro	Ser
1235						1240					1245			
Leu	Glu	Arg	Met	Pro	Arg	Leu	Ala	Trp	Phe	Pro	Leu	Thr	Arg	Gly
1250						1255					1260			
Ser	Gln	Pro	Ser	Gly	Glu	Leu	Leu	Ala	Ser	Phe	Glu	Leu	Ile	Gln
1265						1270					1275			
Arg	Glu	Lys	Pro	Ala	Ile	His	His	Ile	Pro	Gly	Phe	Glu	Val	Gln
1280						1285					1290			
Glu	Thr	Ser	Arg	Ile	Leu	Asp	Glu	Ser	Glu	Asp	Thr	Asp	Leu	Pro
1295						1300					1305			
Tyr	Pro	Pro	Pro	Gln	Arg	Glu	Ala	Asn	Ile	Tyr	Met	Val	Pro	Gln
1310						1315					1320			
Asn	Ile	Lys	Pro	Ala	Leu	Gln	Arg	Thr	Ala	Ile	Glu	Ile	Leu	Ala
1325						1330					1335			
Trp	Gly	Leu	Arg	Asn	Met	Lys	Ser	Tyr	Gln	Leu	Ala	Asn	Ile	Ser
1340						1345					1350			
Ser	Pro	Ser	Leu	Val	Val	Glu	Cys	Gly	Gly	Gln	Thr	Val	Gln	Ser
1355						1360					1365			
Cys	Val	Ile	Arg	Asn	Leu	Arg	Lys	Asn	Pro	Asn	Phe	Asp	Ile	Cys
1370						1375					1380			
Thr	Leu	Phe	Met	Glu	Val	Met	Leu	Pro	Arg	Glu	Glu	Leu	Tyr	Cys
1385						1390					1395			
Pro	Pro	Ile	Thr	Val	Lys	Val	Ile	Asp	Asn	Arg	Gln	Phe	Gly	Arg
1400						1405					1410			
Arg	Pro	Val	Val	Gly	Gln	Cys	Thr	Ile	Arg	Ser	Leu	Glu	Ser	Phe
1415						1420					1425			
Leu	Cys	Asp	Pro	Tyr	Ser	Ala	Glu	Ser	Pro	Ser	Pro	Gln	Gly	Gly
1430						1435					1440			
Pro	Asp	Asp	Val	Ser	Leu	Leu	Ser	Pro	Gly	Glu	Asp	Val	Leu	Ile
1445						1450					1455			
Asp	Ile	Asp	Asp	Lys	Glu	Pro	Leu	Ile	Pro	Ile	Gln	Glu	Glu	Glu
1460						1465					1470			
Phe	Ile	Asp	Trp	Trp	Ser	Lys	Phe	Phe	Ala	Ser	Ile	Gly	Glu	Arg
1475						1480					1485			
Glu	Lys	Cys	Gly	Ser	Tyr	Leu	Glu	Lys	Asp	Phe	Asp	Thr	Leu	Lys
1490						1495					1500			
Val	Tyr	Asp	Thr	Gln	Leu	Glu	Asn	Val	Glu	Ala	Phe	Glu	Gly	Leu
1505						1510					1515			
Ser	Asp	Phe	Cys	Asn	Thr	Phe	Lys	Leu	Tyr	Arg	Gly	Lys	Thr	Gln
1520						1525					1530			
Glu	Glu	Thr	Glu	Asp	Pro	Ser	Val	Ile	Gly	Glu	Phe	Lys	Gly	Leu
1535						1540					1545			
Phe	Lys	Ile	Tyr	Pro	Leu	Pro	Glu	Asp	Pro	Ala	Ile	Pro	Met	Pro
1550						1555					1560			
Pro	Arg	Gln	Phe	His	Gln	Leu	Ala	Ala	Gln	Gly	Pro	Gln	Glu	Cys

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1565	1570	1575
Leu Val Arg Ile Tyr Ile Val Arg Ala Phe Gly Leu Gln Pro Lys		
1580	1585	1590
Asp Pro Asn Gly Lys Cys Asp Pro Tyr Ile Lys Ile Ser Ile Gly		
1595	1600	1605
Lys Lys Ser Val Ser Asp Gln Asp Asn Tyr Ile Pro Cys Thr Leu		
1610	1615	1620
Glu Pro Val Phe Gly Lys Met Phe Glu Leu Thr Cys Thr Leu Pro		
1625	1630	1635
Leu Glu Lys Asp Leu Lys Ile Thr Leu Tyr Asp Tyr Asp Leu Leu		
1640	1645	1650
Ser Lys Asp Glu Lys Ile Gly Glu Thr Val Val Asp Leu Glu Asn		
1655	1660	1665
Arg Leu Leu Ser Lys Phe Gly Ala Arg Cys Gly Leu Pro Gln Thr		
1670	1675	1680
Tyr Cys Val Ser Gly Pro Asn Gln Trp Arg Asp Gln Leu Arg Pro		
1685	1690	1695
Ser Gln Leu Leu His Leu Phe Cys Gln Gln His Arg Val Lys Ala		
1700	1705	1710
Pro Val Tyr Arg Thr Asp Arg Val Met Phe Gln Asp Lys Glu Tyr		
1715	1720	1725
Ser Ile Glu Glu Ile Glu Ala Gly Arg Ile Pro Asn Pro His Leu		
1730	1735	1740
Gly Pro Val Glu Glu Arg Leu Ala Leu His Val Leu Gln Gln Gln		
1745	1750	1755
Gly Leu Val Pro Glu His Val Glu Ser Arg Pro Leu Tyr Ser Pro		
1760	1765	1770
Leu Gln Pro Asp Ile Glu Gln Gly Lys Leu Gln Met Trp Val Asp		
1775	1780	1785
Leu Phe Pro Lys Ala Leu Gly Arg Pro Gly Pro Pro Phe Asn Ile		
1790	1795	1800
Thr Pro Arg Arg Ala Arg Arg Phe Phe Leu Arg Cys Ile Ile Trp		
1805	1810	1815
Asn Thr Arg Asp Val Ile Leu Asp Asp Leu Ser Leu Thr Gly Glu		
1820	1825	1830
Lys Met Ser Asp Ile Tyr Val Lys Gly Trp Met Ile Gly Phe Glu		
1835	1840	1845
Glu His Lys Gln Lys Thr Asp Val His Tyr Arg Ser Leu Gly Gly		
1850	1855	1860
Glu Gly Asn Phe Asn Trp Arg Phe Ile Phe Pro Phe Asp Tyr Leu		
1865	1870	1875
Pro Ala Glu Gln Val Cys Thr Ile Ala Lys Lys Asp Ala Phe Trp		
1880	1885	1890
Arg Leu Asp Lys Thr Glu Ser Lys Ile Pro Ala Arg Val Val Phe		
1895	1900	1905
Gln Ile Trp Asp Asn Asp Lys Phe Ser Phe Asp Asp Phe Leu Gly		
1910	1915	1920
Ser Leu Gln Leu Asp Leu Asn Arg Met Pro Lys Pro Ala Lys Thr		
1925	1930	1935
Ala Lys Lys Cys Ser Leu Asp Gln Leu Asp Asp Ala Phe His Pro		
1940	1945	1950
Glu Trp Phe Val Ser Leu Phe Glu Gln Lys Thr Val Lys Gly Trp		
1955	1960	1965

-continued

Trp Pro Cys Val Ala Glu Glu Gly Glu Lys Lys Ile Leu Ala Gly
 1970 1975 1980
 Lys Leu Glu Met Thr Leu Glu Ile Val Ala Glu Ser Glu His Glu
 1985 1990 1995
 Glu Arg Pro Ala Gly Gln Gly Arg Asp Glu Pro Asn Met Asn Pro
 2000 2005 2010
 Lys Leu Glu Asp Pro Arg Arg Pro Asp Thr Ser Phe Leu Trp Phe
 2015 2020 2025
 Thr Ser Pro Tyr Lys Thr Met Lys Phe Ile Leu Trp Arg Arg Phe
 2030 2035 2040
 Arg Trp Ala Ile Ile Leu Phe Ile Ile Leu Phe Ile Leu Leu Leu
 2045 2050 2055
 Phe Leu Ala Ile Phe Ile Tyr Ala Phe Pro Asn Tyr Ala Ala Met
 2060 2065 2070
 Lys Leu Val Lys Pro Phe Ser
 2075 2080

<210> SEQ ID NO 18
 <211> LENGTH: 45
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 18

Phe Gln Ile Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val
 1 5 10 15
 Asn Ile Lys Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg
 20 25 30
 Thr Arg Ile His Lys Gly Asn Ser Pro Leu Phe Asn Glu
 35 40 45

<210> SEQ ID NO 19
 <211> LENGTH: 45
 <212> TYPE: PRT
 <213> ORGANISM: Equus caballus

<400> SEQUENCE: 19

Phe Gln Ile Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val
 1 5 10 15
 Asn Ile Lys Pro Val Val Lys Val Thr Ala Ala Arg Gln Thr Lys Arg
 20 25 30
 Thr Arg Ile His Arg Gly Asn Ser Pro Leu Phe Asn Glu
 35 40 45

<210> SEQ ID NO 20
 <211> LENGTH: 45
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
 polypeptide

<400> SEQUENCE: 20

Phe Gln Ile Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val
 1 5 10 15
 Asn Ile Lys Pro Val Val Lys Val Thr Ala Ala Arg Gln Thr Lys Trp
 20 25 30
 Thr Arg Ile His Arg Gly Asn Ser Pro Leu Phe Asn Glu
 35 40 45

-continued

<210> SEQ ID NO 21
 <211> LENGTH: 96
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 DYSF sequence

<400> SEQUENCE: 21

Pro	Thr	Ser	Arg	Lys	Leu	Leu	Ser	Asp	Lys	Pro	Gln	Asp	Phe	Gln	Ile
1				5					10					15	
Arg	Val	Gln	Val	Ile	Glu	Gly	Arg	Gln	Leu	Pro	Gly	Val	Asn	Ile	Lys
		20					25						30		
Pro	Val	Val	Lys	Val	Thr	Ala	Ala	Gly	Gln	Thr	Lys	Arg	Thr	Arg	Ile
		35				40						45			
His	Lys	Gly	Asn	Ser	Pro	Leu	Phe	Asn	Glu	Thr	Leu	Phe	Phe	Asn	Leu
	50					55					60				
Phe	Asp	Ser	Pro	Gly	Glu	Leu	Phe	Asp	Glu	Pro	Ile	Phe	Ile	Thr	Val
65				70					75					80	
Val	Asp	Ser	Arg	Ser	Leu	Arg	Thr	Asp	Ala	Leu	Leu	Gly	Glu	Phe	Arg
			85						90					95	

<210> SEQ ID NO 22
 <211> LENGTH: 96
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 DYSF sequence

<400> SEQUENCE: 22

Pro	Thr	Ser	Arg	Lys	Leu	Leu	Ser	Asp	Lys	Pro	Gln	Asp	Phe	Gln	Ile
1				5					10					15	
Arg	Val	Gln	Val	Ile	Glu	Gly	Arg	Gln	Leu	Pro	Gly	Val	Asn	Ile	Lys
		20					25						30		
Pro	Val	Val	Lys	Val	Thr	Ala	Ala	Gly	Gln	Thr	Lys	Arg	Thr	Arg	Ile
		35				40						45			
His	Lys	Gly	Asn	Ser	Pro	Leu	Phe	Asn	Glu	Thr	Leu	Phe	Phe	Asn	Leu
	50					55					60				
Phe	Asp	Cys	Pro	Ala	Glu	Leu	Phe	Asp	Glu	Pro	Ile	Phe	Ile	Thr	Val
65				70					75					80	
Val	Asp	Ser	Arg	Ser	Leu	Arg	Thr	Asp	Ala	Leu	Leu	Gly	Glu	Phe	Arg
			85						90					95	

<210> SEQ ID NO 23
 <211> LENGTH: 93
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 DYSF sequence

<400> SEQUENCE: 23

Arg	Lys	Leu	Leu	Ser	Asp	Lys	Pro	Gln	Asp	Phe	Gln	Ile	Arg	Val	Gln
1				5					10					15	
Val	Ile	Glu	Gly	Arg	Gln	Leu	Pro	Gly	Val	Asn	Ile	Lys	Pro	Val	Val
		20					25					30			
Lys	Val	Thr	Ala	Ala	Gly	Gln	Thr	Lys	Arg	Thr	Arg	Ile	Gln	Lys	Gly
		35				40						45			
Asn	Ser	Pro	Leu	Phe	Asn	Glu	Thr	Leu	Phe	Phe	Asn	Val	Phe	Asp	Ser

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50	55	60
Pro Leu Glu Leu Phe Asp	Glu Pro Ile Phe Ile	Thr Val Val Asp Ser
65	70	75 80

Arg Ser Leu Arg Thr Asp	Ala Leu Leu Gly	Glu Phe Arg
85	90	

<210> SEQ ID NO 24

<211> LENGTH: 93

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:

DYSF sequence

<400> SEQUENCE: 24

Arg Lys Leu Leu Ser Asp	Lys Pro Gln Asp Phe	Gln Ile Arg Val Gln
1	5	10 15

Val Ile Glu Gly Arg Gln	Leu Pro Gly Val Asn	Ile Lys Pro Val Val
20	25	30

Lys Val Thr Ala Ala Gly	Gln Thr Lys Arg Thr	Arg Ile Gln Lys Gly
35	40	45

Asn Ser Pro Leu Phe Asn	Glu Thr Leu Phe Phe	Asn Val Phe Asp Ser
50	55	60

Pro Ser Glu Leu Phe Asp	Glu Pro Ile Phe Ile	Thr Val Val Asp Ser
65	70	75 80

Arg Ser Leu Arg Thr Asp	Ala Leu Leu Gly	Glu Phe Arg
85	90	

<210> SEQ ID NO 25

<211> LENGTH: 96

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:

DYSF sequence

<400> SEQUENCE: 25

Pro Pro Pro Arg Lys Leu	Leu Ser Asp Lys Pro	Gln Asp Phe Gln Ile
1	5	10 15

Arg Val Gln Val Ile Glu	Gly Arg Gln Leu Pro	Gly Val Asn Ile Lys
20	25	30

Pro Val Val Lys Val Thr	Ala Ala Gly Gln Thr	Lys Arg Thr Arg Ile
35	40	45

His Lys Gly Asn Ser Pro	Leu Phe Asn Glu Thr	Leu Phe Phe Asn Leu
50	55	60

Phe Asp Ser Pro Ser Glu	Leu Phe Asp Glu Pro	Ile Phe Ile Thr Val
65	70	75 80

Val Asp Ser Arg Ser Leu	Arg Thr Asp Ala Leu	Ile Gly Glu Phe Arg
85	90	95

<210> SEQ ID NO 26

<211> LENGTH: 96

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:

DYSF sequence

<400> SEQUENCE: 26

Pro Ala Pro Arg Lys Val	Leu Ser Asp Lys Pro	Gln Asp Phe Gln Ile
1	5	10 15

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Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
      20                      25                      30

Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Arg Arg Thr Arg Ile
      35                      40                      45

His Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
      50                      55                      60

Phe Asp Ser Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
      65                      70                      75                      80

Val Asp Ser Arg Ser Arg Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
      85                      90                      95

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<210> SEQ ID NO 27
<211> LENGTH: 96
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
      DYSF sequence

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<400> SEQUENCE: 27

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Pro Thr Pro Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
  1                      5                      10                      15

Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
      20                      25                      30

Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
      35                      40                      45

His Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
      50                      55                      60

Phe Asp Ser Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
      65                      70                      75                      80

Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
      85                      90                      95

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<210> SEQ ID NO 28
<211> LENGTH: 96
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
      DYSF sequence

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<400> SEQUENCE: 28

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Pro Ala Pro Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
  1                      5                      10                      15

Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
      20                      25                      30

Pro Val Val Lys Val Thr Ala Ala Arg Gln Thr Lys Arg Thr Arg Ile
      35                      40                      45

His Arg Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
      50                      55                      60

Phe Asp Ser Pro Ser Glu Leu Phe Asp Glu Ala Val Phe Ile Thr Val
      65                      70                      75                      80

Val Asp Ser Cys Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
      85                      90                      95

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<210> SEQ ID NO 29
<211> LENGTH: 96
<212> TYPE: PRT
<213> ORGANISM: Unknown

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-continued

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:
DYSF sequence

<400> SEQUENCE: 29

Pro Thr Ser Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
1 5 10 15Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
20 25 30Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
35 40 45Gln Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
50 55 60Phe Asp Ser Pro Ser Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
65 70 75 80Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
85 90 95

<210> SEQ ID NO 30

<211> LENGTH: 96

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:
DYSF sequence

<400> SEQUENCE: 30

Pro Ala Pro Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
1 5 10 15Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
20 25 30Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
35 40 45His Lys Gly Asn Ser Pro Phe Phe Asn Glu Thr Leu Phe Phe Asn Val
50 55 60Phe Asp Ser Pro Ala Glu Leu Phe Asn Glu Pro Ile Phe Ile Thr Val
65 70 75 80Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
85 90 95

<210> SEQ ID NO 31

<211> LENGTH: 96

<212> TYPE: PRT

<213> ORGANISM: Cebus capucinus imitator

<400> SEQUENCE: 31

Pro Thr Pro Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
1 5 10 15Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
20 25 30Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
35 40 45His Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Leu
50 55 60Phe Asp Ser Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
65 70 75 80Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Leu Gly Glu Phe Arg
85 90 95

-continued

<210> SEQ ID NO 32
 <211> LENGTH: 96
 <212> TYPE: PRT
 <213> ORGANISM: Microcebus murinus

<400> SEQUENCE: 32

```

Pro Pro Ser Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
1          5          10          15
Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
          20          25          30
Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
          35          40          45
His Lys Gly Asn Ser Pro Leu Phe Ser Glu Thr Leu Phe Phe Asn Leu
50          55          60
Phe Glu Ser Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
65          70          75          80
Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Leu Gly Glu Phe Arg
          85          90          95

```

<210> SEQ ID NO 33
 <211> LENGTH: 96
 <212> TYPE: PRT
 <213> ORGANISM: Carlito syrichta

<400> SEQUENCE: 33

```

Pro Thr Ser Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
1          5          10          15
Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
          20          25          30
Pro Val Val Lys Val Thr Ala Ala Gly His Thr Lys Arg Thr Arg Ile
          35          40          45
Gln Lys Gly Asn Asn Pro Leu Phe Ser Glu Thr Leu Phe Phe Asn Leu
50          55          60
Phe Asp Ser Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
65          70          75          80
Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Leu Gly Glu Phe Arg
          85          90          95

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<210> SEQ ID NO 34
 <211> LENGTH: 93
 <212> TYPE: PRT
 <213> ORGANISM: Otolemur garnettii

<400> SEQUENCE: 34

```

Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile Arg Val Gln
1          5          10          15
Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys Pro Val Val
          20          25          30
Lys Val Thr Ala Ala Gly His Thr Lys Arg Thr Arg Ile His Lys Gly
          35          40          45
Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Leu Val Glu Ser
50          55          60
Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val Val Asp Ser
65          70          75          80
Arg Ser Leu Arg Thr Asp Ala Leu Leu Gly Glu Phe Arg
          85          90

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<210> SEQ ID NO 35
 <211> LENGTH: 96
 <212> TYPE: PRT
 <213> ORGANISM: *Propithecus coquereli*

<400> SEQUENCE: 35

Pro Thr Ser Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
 1 5 10 15
 Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
 20 25 30
 Pro Val Val Lys Val Thr Ala Ala Gly His Thr Lys Arg Thr Arg Ile
 35 40 45
 Gln Lys Gly Asn Asn Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Leu
 50 55 60
 Phe Asp Ser Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
 65 70 75 80
 Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Leu Gly Glu Phe Arg
 85 90 95

<210> SEQ ID NO 36
 <211> LENGTH: 93
 <212> TYPE: PRT
 <213> ORGANISM: *Cricetulus griseus*

<400> SEQUENCE: 36

Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile Arg Val Gln
 1 5 10 15
 Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys Pro Val Val
 20 25 30
 Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile Gln Lys Gly
 35 40 45
 Asn Ser Pro Val Phe Asn Glu Thr Leu Phe Phe Asn Val Phe Asp Ser
 50 55 60
 Pro Ser Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val Val Asp Ser
 65 70 75 80
 Arg Ser Leu Arg Thr Asp Ala Leu Leu Gly Glu Phe Arg
 85 90

<210> SEQ ID NO 37
 <211> LENGTH: 96
 <212> TYPE: PRT
 <213> ORGANISM: *Ictidomys tridecemlineatus*

<400> SEQUENCE: 37

Pro Pro Ser Arg Lys Pro Leu Ser Asp Lys Gln Gln Asp Phe Gln Ile
 1 5 10 15
 Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Arg
 20 25 30
 Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
 35 40 45
 His Lys Gly Asn Ser Pro Phe Phe Asn Glu Thr Leu Phe Phe Asn Val
 50 55 60
 Phe Asp Ser Pro Leu Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
 65 70 75 80
 Val Asp Ser Cys Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
 85 90 95

<210> SEQ ID NO 38

-continued

<211> LENGTH: 96
 <212> TYPE: PRT
 <213> ORGANISM: Marmota marmota marmota

<400> SEQUENCE: 38

```

Pro Pro Ser Arg Lys Pro Leu Ser Asp Lys Gln Gln Asp Phe Gln Ile
1          5          10          15
Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Arg
          20          25          30
Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
          35          40          45
His Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
          50          55          60
Phe Asp Ser Pro Leu Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
65          70          75          80
Val Asp Ser Cys Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
          85          90          95

```

<210> SEQ ID NO 39
 <211> LENGTH: 96
 <212> TYPE: PRT
 <213> ORGANISM: Castor canadensis

<400> SEQUENCE: 39

```

Pro Thr Ser Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
1          5          10          15
Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Arg
          20          25          30
Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
          35          40          45
His Lys Gly Asn Ser Pro Val Phe Asn Glu Thr Leu Phe Phe Asn Met
          50          55          60
Phe Glu Ser Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
65          70          75          80
Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
          85          90          95

```

<210> SEQ ID NO 40
 <211> LENGTH: 96
 <212> TYPE: PRT
 <213> ORGANISM: Mustela putorius furo

<400> SEQUENCE: 40

```

Pro Thr Pro Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
1          5          10          15
Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
          20          25          30
Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
          35          40          45
His Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
          50          55          60
Phe Asp Ser Pro Ser Glu Leu Phe Glu Glu Pro Ile Phe Ile Thr Val
65          70          75          80
Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
          85          90          95

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<210> SEQ ID NO 41
 <211> LENGTH: 96

-continued

<212> TYPE: PRT

<213> ORGANISM: *Hipposideros armiger*

<400> SEQUENCE: 41

```

Pro Lys Ala Arg Thr Leu Leu Ser Glu Lys Pro Gln Asp Phe Gln Ile
1           5           10           15
Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Leu Asn Ile Lys
           20           25           30
Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
           35           40           45
His Lys Gly Asn Ser Pro Leu Phe Ser Glu Thr Leu Phe Phe Asn Val
50           55           60
Phe Asp Ser Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
65           70           75           80
Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
           85           90           95

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<210> SEQ ID NO 42

<211> LENGTH: 90

<212> TYPE: PRT

<213> ORGANISM: *Loxodonta africana*

<400> SEQUENCE: 42

```

Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile Arg Val Gln Val Ile Glu
1           5           10           15
Gly Arg Gln Leu Pro Gly Val Asn Ile Lys Pro Val Val Lys Val Thr
           20           25           30
Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile His Lys Gly Asn Ser Pro
           35           40           45
Leu Phe Asn Glu Thr Leu Phe Phe Asn Val Cys Glu Ser Pro Ala Gln
50           55           60
Leu Phe Asp Glu Pro Ile Phe Ile Thr Val Val Asp Ser Arg Ser Leu
65           70           75           80
Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
           85           90

```

<210> SEQ ID NO 43

<211> LENGTH: 96

<212> TYPE: PRT

<213> ORGANISM: *Sus scrofa*

<400> SEQUENCE: 43

```

Pro Ala Pro Arg Lys Pro Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
1           5           10           15
Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
           20           25           30
Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
           35           40           45
His Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
50           55           60
Phe Asp Ser Pro Ala Glu Leu Phe Asn Glu Pro Ile Phe Ile Thr Val
65           70           75           80
Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
           85           90           95

```

<210> SEQ ID NO 44

<211> LENGTH: 96

<212> TYPE: PRT

-continued

<213> ORGANISM: Bos taurus

<400> SEQUENCE: 44

Pro Ala Pro Ser Lys Pro Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
 1 5 10 15

Arg Val Gln Val Ile Lys Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
 20 25 30

Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
 35 40 45

His Lys Gly Asn Ser Pro Val Phe Asn Glu Thr Leu Phe Phe Asn Val
 50 55 60

Phe Asp Ser Pro Ala Glu Leu Phe Asn Glu Pro Ile Phe Ile Thr Val
 65 70 75 80

Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
 85 90 95

<210> SEQ ID NO 45

<211> LENGTH: 96

<212> TYPE: PRT

<213> ORGANISM: Balaenoptera acutorostrata scammoni

<400> SEQUENCE: 45

Pro Ala Pro Arg Lys Leu Leu Ser Asp Lys Ser Gln Asp Phe Gln Ile
 1 5 10 15

Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
 20 25 30

Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
 35 40 45

His Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
 50 55 60

Phe Asp Ser Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
 65 70 75 80

Val Asp Ser Cys Ser Leu Arg Thr Asp Ala Phe Leu Gly Glu Phe Arg
 85 90 95

<210> SEQ ID NO 46

<211> LENGTH: 96

<212> TYPE: PRT

<213> ORGANISM: Ailuropoda melanoleuca

<400> SEQUENCE: 46

Pro Thr Pro Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
 1 5 10 15

Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
 20 25 30

Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
 35 40 45

His Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
 50 55 60

Phe Asp Ser Pro Ser Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
 65 70 75 80

Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
 85 90 95

<210> SEQ ID NO 47

<211> LENGTH: 96

<212> TYPE: PRT

<213> ORGANISM: Orcinus orca

-continued

<400> SEQUENCE: 47

```

Pro Ala Pro Arg Lys Pro Leu Ser Asp Lys Ser Gln Asp Phe Gln Ile
1           5           10           15
Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
20           25           30
Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
35           40           45
His Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
50           55           60
Phe Asp Ser Pro Ala Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
65           70           75           80
Val Asp Ser Cys Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
85           90           95

```

<210> SEQ ID NO 48

<211> LENGTH: 95

<212> TYPE: PRT

<213> ORGANISM: Phascolarctos cinereus

<400> SEQUENCE: 48

```

Ser Ser Lys Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Val Arg
1           5           10           15
Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Leu Asn Ile Arg Pro
20           25           30
Val Val Lys Val Met Ala Cys Gly Gln Thr Lys Arg Thr Arg Ile Arg
35           40           45
Lys Gly Asn Gly Pro Val Phe Asp Glu Thr Leu Phe Phe Asn Val Phe
50           55           60
Asp Ser Pro Val Glu Leu Phe Asp Glu Val Ile Cys Ile Thr Val Val
65           70           75           80
Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Leu Gly Glu Phe Arg
85           90           95

```

<210> SEQ ID NO 49

<211> LENGTH: 89

<212> TYPE: PRT

<213> ORGANISM: Sarcophilus harrisii

<400> SEQUENCE: 49

```

Ser Asp Leu Ser Leu Cys Leu Gln Ile Arg Val Gln Val Val Glu Gly
1           5           10           15
Arg Gln Leu Pro Gly Val Asn Ile Arg Pro Val Val Lys Val Thr Ala
20           25           30
Ala Gly Gln Thr Lys Arg Thr Arg Ile Arg Lys Gly Asn Ser Pro Val
35           40           45
Phe Asp Glu Thr Leu Phe Phe Asn Val Phe Asp Ser Pro Leu Glu Leu
50           55           60
Phe Asp Glu Pro Ile Phe Ile Thr Val Val Asp Ser Arg Ser Leu Arg
65           70           75           80
Thr Asp Ala Leu Ile Gly Glu Phe Arg
85

```

<210> SEQ ID NO 50

<211> LENGTH: 78

<212> TYPE: PRT

<213> ORGANISM: Ornithorhynchus anatinus

-continued

<400> SEQUENCE: 50

```

Lys Ile Arg Val Arg Val Ile Glu Gly Arg Gln Leu Pro Gly Ala Ser
1           5           10           15
Ile Arg Pro Val Val Lys Val Thr Ala Ala Gly Lys Thr Lys Arg Thr
20           25           30
Arg Ile Cys Lys Gly Asn Ser Pro Phe Phe Asp Glu Thr Leu Phe Phe
35           40           45
Asn Val Phe Glu Ser Pro Val Glu Leu Phe Glu Asp Pro Ile Phe Ile
50           55           60
Thr Val Arg Thr Asn Ala Ser Cys Arg Ser Pro Ser Pro Leu
65           70           75

```

<210> SEQ ID NO 51

<211> LENGTH: 85

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 51

```

Met Leu Arg Val Phe Ile Leu Tyr Ala Glu Asn Val His Thr Pro Asp
1           5           10           15
Thr Asp Ile Ser Asp Ala Tyr Cys Ser Ala Val Phe Ala Gly Val Lys
20           25           30
Lys Arg Thr Lys Val Ile Lys Asn Ser Val Asn Pro Val Trp Asn Glu
35           40           45
Gly Phe Glu Trp Asp Leu Lys Gly Ile Pro Leu Asp Gln Gly Ser Glu
50           55           60
Leu His Val Val Val Lys Asp His Glu Thr Met Gly Arg Asn Arg Phe
65           70           75           80
Leu Gly Glu Ala Lys
85

```

<210> SEQ ID NO 52

<211> LENGTH: 86

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 52

```

Met Leu Cys Cys Leu Leu Val Arg Ala Ser Asn Leu Pro Ser Ala Lys
1           5           10           15
Lys Asp Arg Arg Ser Asp Pro Val Ala Ser Leu Thr Phe Arg Gly Val
20           25           30
Lys Lys Arg Thr Lys Val Ile Lys Asn Ser Val Asn Pro Val Trp Asn
35           40           45
Glu Gly Phe Glu Trp Asp Leu Lys Gly Ile Pro Leu Asp Gln Gly Ser
50           55           60
Glu Leu His Val Val Val Lys Asp His Glu Thr Met Gly Arg Asn Arg
65           70           75           80
Phe Leu Gly Glu Ala Lys
85

```

<210> SEQ ID NO 53

<211> LENGTH: 96

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 53

```

Pro Thr Ser Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
1           5           10           15

```

-continued

```

Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
      20                      25                      30

Pro Val Val Lys Val Thr Ala Ala Gly Gln Thr Lys Arg Thr Arg Ile
      35                      40                      45

His Lys Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Leu
      50                      55                      60

Phe Asp Ser Pro Gly Glu Leu Phe Asp Glu Pro Ile Phe Ile Thr Val
      65                      70                      75                      80

Val Asp Ser Arg Ser Leu Arg Thr Asp Ala Leu Leu Gly Glu Phe Arg
      85                      90                      95

```

```

<210> SEQ ID NO 54
<211> LENGTH: 88
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

```

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<400> SEQUENCE: 54

```

```

Ile Leu Ala Trp Gly Leu Arg Asn Met Lys Ser Tyr Gln Leu Ala Asn
1      5                      10                      15

Ile Ser Ser Pro Ser Leu Val Val Glu Cys Gly Gly Gln Thr Val Gln
      20                      25                      30

Ser Cys Val Ile Arg Asn Leu Arg Lys Asn Pro Asn Phe Asp Ile Cys
      35                      40                      45

Thr Leu Phe Met Glu Val Met Leu Pro Arg Glu Glu Leu Tyr Cys Pro
      50                      55                      60

Pro Ile Thr Val Lys Val Ile Asp Asn Arg Gln Phe Gly Arg Arg Pro
      65                      70                      75                      80

Val Val Gly Gln Cys Thr Ile Arg
      85

```

```

<210> SEQ ID NO 55
<211> LENGTH: 96
<212> TYPE: PRT
<213> ORGANISM: Equus caballus

```

```

<400> SEQUENCE: 55

```

```

Pro Ala Pro Arg Lys Leu Leu Ser Asp Lys Pro Gln Asp Phe Gln Ile
1      5                      10                      15

Arg Val Gln Val Ile Glu Gly Arg Gln Leu Pro Gly Val Asn Ile Lys
      20                      25                      30

Pro Val Val Lys Val Thr Ala Ala Arg Gln Thr Lys Arg Thr Arg Ile
      35                      40                      45

His Arg Gly Asn Ser Pro Leu Phe Asn Glu Thr Leu Phe Phe Asn Val
      50                      55                      60

Phe Asp Ser Pro Ser Glu Leu Phe Asp Glu Ala Val Phe Ile Thr Val
      65                      70                      75                      80

Val Asp Ser Cys Ser Leu Arg Thr Asp Ala Leu Ile Gly Glu Phe Arg
      85                      90                      95

```

```

<210> SEQ ID NO 56
<211> LENGTH: 2001
<212> TYPE: DNA
<213> ORGANISM: Equus caballus

```

```

<400> SEQUENCE: 56

```

```

gactcttttt tctccacgtc gggctctaggg tgcagatgag tttatgggac gctgcacctg      60
tcagccgact ctggaacgga tgccccggct gacctgggtc cactgacta ggggcagcca      120

```

-continued

```

gccggcgggc gagctgctgg cctcttttga gctcatccag agagagaagg tgaggctggt 180
ccccatcctg atccaggaag cccagacagg gagtacatgg taggagaggg tagggggcac 240
aggcagggat gccccgaagg aggggagggg ctgccgatac atccttctcc ccccaccagg 300
cctgctcagt ggggagcact gaaccgccac ctctggaatt caggcttctc gctgagactg 360
gttttgcccc tcttttctga accccaagt aggtactgt ctaaaccatt gggtagaatg 420
ttggaatctg ggctgtgctg gaatgtctga gtctctggc cattggatgt atttttccct 480
gggtcatatc ctgacgtggt ttctgagagg aggacctccg tgatctggtc ggggagactt 540
aattgtgatg gtggcaccca gggacaggta aggtcaggac ctccggtgta gtgttggtc 600
tgccatggcc tggctgtgcg accctgggcc aggcacttcc ctgctgtgaa ataggggggt 660
tgggttagaa ccatatcttc agctttgtc ttagagtgat gtgaacccca ggtgtaagct 720
ggatggaagt caggctactc tgggtcaagt gtctggaggg ggcacacagt atgctgcaac 780
ctggcttccc ctggcacctg gctgccaaact cccgggcacc tccaaggagt ccctaggttg 840
caaaactcca actgtgatca ctgacatatt ccatgagtgt cattttgtgg ctgaagggtg 900
gtttgctcgg cccggacaga aaggagggtg atggggcctt gggtgacagg cactgaccaa 960
agctcttttt tctaccccg agccggccat ctaccacatt mctgggtttg aggtaagtct 1020
tgctcttccc tcttcttctt caaactcatg gcccgctct gtgtgtttgc agccccctat 1080
gagctaggaa gggcagtcag gtgtatacca ccaccggccc gccttcaggg atgggggtgg 1140
tgagggccgg ggtccctggc cagctcaggc ctttcccgcc cttcttccct gagactcaga 1200
ctgtagggac ctcatggtca tccagccac gggctctcag ctgggctgca tgttagaatc 1260
ccttcggcag cttgaaaaat caccagtgtc caggcccccac cccagaccag atggaacaca 1320
gtctctgggg atgtggccca ggcattggta tcttccagag cttcccaggg gattttaatg 1380
tgctgctagg gctgagcacc atcgacttag cccagtgggt ctcagtgtga tatcacctgg 1440
gatgttgta gaaatgccga ttcttggtc ctgcttcaca cctactgaag caaactctga 1500
ggggcagtc agcaacctgc aaattaacaa gcctccaggg gattctgatg cccactcgag 1560
tttcagaagc tctggcttag accatcttcc ccaactcatt gttccccca tgaccggttt 1620
ttctctctc actagtgatg ggggctctgt tcccttctcc ctgctctcct tccttctcct 1680
gggggcgatg tagaaaatgc agacccttc ctggtaacca ggctaatttg attcaggaga 1740
tactgttctc agtcaggttc agctgcatct tggaggagcg gtcacccggt ctgtcgttgg 1800
atgccttccc ccttacaagg cagtcacgc aatctgcaga cagtttttac catgcaggcc 1860
ttcttcacc cctgtctctg ctccaccccc ggcacccctc agccccctct ctctcctgct 1920
ccctctctg tccagcccaa gctcccgctc tcagggtctg cagccctgg ggatccagat 1980
tctttctccc tggtcttcca a 2001

```

<210> SEQ ID NO 57

<211> LENGTH: 459

<212> TYPE: DNA

<213> ORGANISM: Equus caballus

<400> SEQUENCE: 57

```

agaccacga tttcctgcat cttcgactac gggaaaccgt accatctacg ctgtacatg 60
taccaggccc gggacctgcc cgccatggac aaggactctt tttctgatcc ctacgcaatc 120
gtctccttcc tgcaccagag ccagaagaca gtggtggtga agaacacct gaacccacc 180

```

tgggaccaga	cactcatctt	ctatgagatc	gagatctttg	gtgagccgcc	cagcatcgcc	240
gagcagccgc	ccagcatcgt	ggtggagcta	tacgaccatg	acacctacgg	tgcatgatgag	300
tttatgggac	gctgcatctg	tcagccgact	ctggaacgga	tgccccggct	gacctggttc	360
ccactgacta	ggggcgagcca	gccggcgggc	gagctgctgg	cctcttttga	gctcatccag	420
agagagaagc	cggccatcta	ccacattcct	ggttttgag			459

```
<210> SEQ ID NO 58
<211> LENGTH: 459
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide
```

<400> SEQUENCE: 58

agaccacga	tttctgcat	cttcgactac	gggaaccgct	accatctacg	ctgctacatg	60
taccaggccc	gggacctgcc	cgccatggac	aaggactctt	tttctgatcc	ctacgcaatc	120
gtctccttcc	tgcaccagag	ccagaagaca	gtggtggtga	agaacaccct	gaacccacc	180
tgggaccaga	cactcatctt	ctatgagatc	gagatctttg	gtgagccgcc	cagcatcgcc	240
gagcagccgc	ccagcatcgt	ggtggagcta	tacgaccatg	acacctacgg	tgcagatgag	300
tttatgggac	gctgcatctg	tcagccgact	ctggaacgga	tgccccggct	gacctgggtc	360
ccactgacta	ggggcgaccca	gccggccgggc	gagctgctgg	cctcttttga	gctcatccag	420
aqaqagaagc	cqcccatcta	ccacattact	qgttttqaq			450

```
<210> SEQ ID NO 59
<211> LENGTH: 50
<212> TYPE: DNA
<213> ORGANISM: Equus caballus
```

<400> SEQUENCE: 59

taccccgag cggccatct accacattcc tggttttgag gtaagtcttg 50

```
<210> SEQ ID NO 60
<211> LENGTH: 153
<212> TYPE: PRT
<213> ORGANISM: Equus caballus
```

<400> SEQUENCE: 60

Arg 1	Pro	Thr	Ile 5	Ser	Cys	Ile	Phe	Asp	Tyr 10	Gly	Asn	Arg	Tyr	His 15	Leu
Arg	Cys	Tyr	Met 20	Tyr	Gln	Ala	Arg	Asp 25	Leu	Pro	Ala	Met	Asp 30	Lys	Asp
Ser	Phe	Ser 35	Asp	Pro	Tyr	Ala	Ile 40	Val	Ser	Phe	Leu	His 45	Gln	Ser	Gln
Lys 50	Thr	Val	Val	Val	Lys	Asn 55	Thr	Leu	Asn	Pro	Thr 60	Trp	Asp	Gln	Thr
Leu 65	Ile	Phe	Tyr	Glu	Ile 70	Glu	Ile	Phe	Gly	Glu 75	Pro	Pro	Ser	Ile 80	Ala
Glu	Gln	Pro	Pro	Ser 85	Ile	Val	Val	Glu	Leu 90	Tyr	Asp	His	Asp	Thr 95	Tyr
Gly	Ala	Asp 100	Glu	Phe	Met	Gly	Arg	Cys 105	Ile	Cys	Gln	Pro	Thr 110	Leu	Glu
Arg	Met 115	Pro	Arg	Leu	Thr	Trp	Phe 120	Pro	Leu	Thr	Arg	Gly 125	Ser	Gln	Pro

-continued

Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu Ile Gln Arg Glu Lys Pro
130 135 140

Ala Ile Tyr His Ile Pro Gly Phe Glu
145 150

<210> SEQ ID NO 61
<211> LENGTH: 153
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 61

Arg Pro Thr Ile Ser Cys Ile Phe Asp Tyr Gly Asn Arg Tyr His Leu
1 5 10 15

Arg Cys Tyr Met Tyr Gln Ala Arg Asp Leu Pro Ala Met Asp Lys Asp
20 25 30

Ser Phe Ser Asp Pro Tyr Ala Ile Val Ser Phe Leu His Gln Ser Gln
35 40 45

Lys Thr Val Val Val Lys Asn Thr Leu Asn Pro Thr Trp Asp Gln Thr
50 55 60

Leu Ile Phe Tyr Glu Ile Glu Ile Phe Gly Glu Pro Pro Ser Ile Ala
65 70 75 80

Glu Gln Pro Pro Ser Ile Val Val Glu Leu Tyr Asp His Asp Thr Tyr
85 90 95

Gly Ala Asp Glu Phe Met Gly Arg Cys Ile Cys Gln Pro Thr Leu Glu
100 105 110

Arg Met Pro Arg Leu Thr Trp Phe Pro Leu Thr Arg Gly Ser Gln Pro
115 120 125

Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu Ile Gln Arg Glu Lys Pro
130 135 140

Ala Ile Tyr His Ile Thr Gly Phe Glu
145 150

<210> SEQ ID NO 62
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Equus caballus

<400> SEQUENCE: 62

ccggccatct accacattcc tggttttgag 30

<210> SEQ ID NO 63
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 63

ccggccatct accacattac tggttttgag 30

<210> SEQ ID NO 64
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
primer

-continued

<400> SEQUENCE: 64

ggttgcaaac tcccaactgt

20

<210> SEQ ID NO 65

<211> LENGTH: 20

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic primer

<400> SEQUENCE: 65

gatttttcaa gctgccgaag

20

<210> SEQ ID NO 66

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 66

Thr	Arg	Gly	Ser	Gln	Pro	Ser	Gly	Glu	Leu	Leu	Ala	Ser	Phe	Glu	Leu
1				5					10					15	

Ile	Gln	Arg	Glu	Lys	Pro	Ala	Ile	His	His	Ile	Pro	Gly	Phe	Glu	Val
			20					25					30		

Gln	Glu	Thr	Ser	Arg	Ile	Leu	Asp	Glu	Ser	Glu	Asp	Thr	Asp	Leu	Pro
		35					40					45			

Tyr	Pro	Pro	Pro	Gln	Arg	Glu	Ala	Asn	Ile	Tyr	Met
	50					55					60

<210> SEQ ID NO 67

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Equus caballus

<400> SEQUENCE: 67

Thr	Arg	Gly	Ser	Gln	Pro	Ala	Gly	Glu	Leu	Leu	Ala	Ser	Phe	Glu	Leu
1				5					10					15	

Ile	Gln	Arg	Glu	Lys	Pro	Ala	Ile	Tyr	His	Ile	Pro	Gly	Phe	Glu	Val
			20					25					30		

Gln	Asp	Thr	Ser	Gly	Ile	Leu	Glu	Glu	Ser	Glu	Asp	Thr	Asp	Leu	Pro
		35					40					45			

Tyr	Pro	Pro	Pro	Gln	Arg	Glu	Ala	Asn	Ile	Tyr	Met
	50					55					60

<210> SEQ ID NO 68

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 68

Thr	Arg	Gly	Ser	Gln	Pro	Ala	Gly	Glu	Leu	Leu	Ala	Ser	Phe	Glu	Leu
1				5					10					15	

Ile	Gln	Arg	Glu	Lys	Pro	Ala	Ile	Tyr	His	Ile	Thr	Gly	Phe	Glu	Val
			20					25					30		

Gln	Asp	Thr	Ser	Gly	Ile	Leu	Glu	Glu	Ser	Glu	Asp	Thr	Asp	Leu	Pro
		35					40					45			

Tyr	Pro	Pro	Pro	Gln	Arg	Glu	Ala	Asn	Ile	Tyr	Met
	50					55					60

-continued

<210> SEQ ID NO 69
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
DYSF sequence

<400> SEQUENCE: 69

Thr Arg Gly Ser Gln Pro Ser Gly Glu Leu Leu Ala Ser Phe Glu Leu
1 5 10 15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
20 25 30

Gln Glu Thr Ser Arg Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro
35 40 45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50 55 60

<210> SEQ ID NO 70
<211> LENGTH: 59
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
DYSF sequence

<400> SEQUENCE: 70

Arg Gly Ser Gln Pro Ser Gly Glu Leu Leu Ala Ser Phe Glu Leu Ile
1 5 10 15

Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val Gln
20 25 30

Glu Thr Ser Arg Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro Tyr
35 40 45

Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50 55

<210> SEQ ID NO 71
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
DYSF sequence

<400> SEQUENCE: 71

Thr Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ala Phe Glu Leu
1 5 10 15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Met
20 25 30

His Glu Thr Ser Arg Ile Leu Asp Glu Thr Glu Asp Thr Asp Leu Pro
35 40 45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50 55 60

<210> SEQ ID NO 72
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
DYSF sequence

-continued

<400> SEQUENCE: 72

```

Thr Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
1           5           10           15
Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
20          25          30
Gln Asp Thr Ser Arg Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro
35          40          45
Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50          55          60

```

<210> SEQ ID NO 73

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:
DYSF sequence

<400> SEQUENCE: 73

```

Thr Arg Gly Asn Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
1           5           10           15
Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
20          25          30
Gln Asp Thr Ala Gly Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro
35          40          45
Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50          55          60

```

<210> SEQ ID NO 74

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:
DYSF sequence

<400> SEQUENCE: 74

```

Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu Ile
1           5           10           15
Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val Gln
20          25          30
Asp Thr Ser Gly Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro Tyr
35          40          45
Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50          55

```

<210> SEQ ID NO 75

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Unknown

<220> FEATURE:

<223> OTHER INFORMATION: Description of Unknown:
DYSF sequence

<400> SEQUENCE: 75

```

Thr Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
1           5           10           15
Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
20          25          30
Gln Asp Thr Thr Gly Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro
35          40          45

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-continued

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50 55 60

<210> SEQ ID NO 76
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 DYSF sequence

<400> SEQUENCE: 76

Thr Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
1 5 10 15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
20 25 30

Gln Asp Thr Ser Gly Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro
35 40 45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50 55 60

<210> SEQ ID NO 77
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Papio anubis

<400> SEQUENCE: 77

Thr Lys Gly Ser Gln Pro Thr Gly Glu Leu Leu Ala Ser Phe Glu Leu
1 5 10 15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
20 25 30

Gln Glu Thr Ser Arg Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro
35 40 45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50 55 60

<210> SEQ ID NO 78
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Macaca mulatta

<400> SEQUENCE: 78

Thr Arg Gly Ser Gln Pro Thr Gly Glu Leu Leu Ala Ser Phe Glu Leu
1 5 10 15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
20 25 30

Gln Glu Thr Ser Arg Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro
35 40 45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50 55 60

<210> SEQ ID NO 79
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Rhinopithecus bieti

<400> SEQUENCE: 79

Thr Arg Gly Ser Gln Pro Met Gly Glu Leu Leu Ala Ser Phe Glu Leu
1 5 10 15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val

20								25				30			
Gln	Glu	Thr	Ser	Arg	Ile	Leu	Asp	Glu	Ser	Glu	Asp	Thr	Asp	Leu	Pro
35				40				45							
Tyr	Pro	Pro	Pro	Gln	Arg	Glu	Ala	Asn	Ile	Tyr	Met				
50						55				60					

<400> SEQUENCE: 80

Arg	Gly	Ser	Gln	Pro	Ser	Gly	Glu	Leu	Leu	Ala	Ser	Phe	Glu	Leu	Ile
1				5					10					15	
Gln	Arg	Glu	Lys	Pro	Ala	Ile	Tyr	His	Ile	Pro	Gly	Phe	Glu	Val	Gln
			20					25					30		
Asp	Thr	Ser	Arg	Ile	Leu	Asp	Glu	Ser	Glu	Asp	Thr	Asp	Leu	Pro	Tyr
		35					40					45			
Pro	Pro	Pro	Gln	Arg	Glu	Ala	Asn	Val	Tyr	Met					
	50					55									

<400> SEQUENCE: 81

Arg	Gly	Gly	Gln	Pro	Ser	Gly	Glu	Leu	Leu	Ala	Ala	Phe	Glu	Leu	Ile
1				5					10					15	
Gln	Arg	Glu	Lys	Pro	Ala	Ile	His	His	Ile	Pro	Gly	Phe	Glu	Val	Gln
			20					25					30		
Asp	Ala	Ser	Arg	Ile	Leu	Asp	Glu	Ala	Glu	Asp	Thr	Asp	Leu	Pro	Tyr
		35					40					45			
Pro	Pro	Pro	Gln	Arg	Glu	Ala	Asn	Val	His	Met					
	50					55									

<400> SEQUENCE: 82

Arg	Gly	Ser	Gln	Pro	Ser	Gly	Glu	Leu	Leu	Ala	Ser	Phe	Glu	Leu	Ile
1				5					10					15	
Gln	Arg	Glu	Lys	Pro	Ala	Ile	His	His	Ile	Pro	Gly	Phe	Glu	Val	Gln
			20					25					30		
Asp	Thr	Ala	Arg	Ile	Leu	Asp	Glu	Ser	Glu	Asp	Thr	Asp	Leu	Pro	Tyr
		35					40					45			
Pro	Pro	Pro	Gln	Arg	Glu	Ala	Asn	Ile	Tyr	Val					
	50					55									

<400> SEQUENCE: 83

Arg Gly Ser Gln Ala Ser Gly Glu Leu Leu Ala Ser Phe Glu Leu Ile
1 5 10 15

-continued

Gln Arg Glu Lys Pro Ala Ile Tyr His Ile Pro Gly Phe Glu Val Gln
 20 25 30

Asp Thr Ser Arg Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro Tyr
 35 40 45

Pro Pro Pro Gln Arg Glu Ala Asn Val Tyr Met
 50 55

<210> SEQ ID NO 84
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Cricetulus griseus*

<400> SEQUENCE: 84

Thr Arg Gly Gly Gln Pro Ala Gly Glu Leu Leu Ala Ala Phe Glu Leu
 1 5 10 15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Met
 20 25 30

His Glu Thr Ser Arg Ile Leu Asp Glu Thr Glu Asp Thr Asp Leu Pro
 35 40 45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
 50 55 60

<210> SEQ ID NO 85
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Microtus ochrogaster*

<400> SEQUENCE: 85

Thr Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ala Phe Glu Leu
 1 5 10 15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Met
 20 25 30

His Glu Thr Ser Ser Ile Leu Glu Glu Thr Glu Asp Thr Asp Leu Pro
 35 40 45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
 50 55 60

<210> SEQ ID NO 86
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Meriones unguiculatus*

<400> SEQUENCE: 86

Thr Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ala Phe Glu Leu
 1 5 10 15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
 20 25 30

Gln Glu Thr Ser Arg Thr Leu Asp Glu Thr Glu Asp Thr Asp Leu Pro
 35 40 45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
 50 55 60

<210> SEQ ID NO 87
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Ictidomys tridecemlineatus*

<400> SEQUENCE: 87

Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu Ile
 1 5 10 15

-continued

Gln Arg Glu Lys Pro Ala Ile Tyr His Ile Pro Gly Phe Glu Val Gln
 20 25 30

Asp Thr Ser Arg Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro Tyr
 35 40 45

Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Val
 50 55

<210> SEQ ID NO 88
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Chinchilla lanigera

<400> SEQUENCE: 88

Thr Arg Gly Asn Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
 1 5 10 15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
 20 25 30

Gln Asp Thr Ser Arg Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro
 35 40 45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
 50 55 60

<210> SEQ ID NO 89
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Marmota marmota marmota

<400> SEQUENCE: 89

Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu Ile
 1 5 10 15

Gln Arg Glu Lys Pro Ala Ile Tyr His Ile Pro Gly Phe Glu Val Gln
 20 25 30

Asp Thr Ser Arg Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro Tyr
 35 40 45

Pro Pro Pro Gln Arg Glu Ala Asn Val Tyr Val
 50 55

<210> SEQ ID NO 90
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Castor canadensis

<400> SEQUENCE: 90

Arg Ser Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu Ile
 1 5 10 15

Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val Gln
 20 25 30

Asp Thr Ser Arg Ile Leu Asp Glu Ser Glu Asp Ala Asp Leu Pro Tyr
 35 40 45

Pro Pro Pro Gln Arg Glu Val Asn Ile Tyr Met
 50 55

<210> SEQ ID NO 91
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Mustela putorius furo

<400> SEQUENCE: 91

Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu Ile

-continued

1	5	10	15
Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val His	20	25	30
Asp Thr Ser Gly Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro Tyr	35	40	45
Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met	50	55	

<210> SEQ ID NO 92
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Fukomys damarensis

<400> SEQUENCE: 92

Thr Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu	1	5	10	15
Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val	20	25	30	
Gln Asp Thr Ser Arg Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro	35	40	45	
Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met	50	55	60	

<210> SEQ ID NO 93
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Myotis brandtii

<400> SEQUENCE: 93

Thr Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu	1	5	10	15
Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val	20	25	30	
Gln Asp Ala Ser Arg Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro	35	40	45	
Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met	50	55	60	

<210> SEQ ID NO 94
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Pteropus vampyrus

<400> SEQUENCE: 94

Thr Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu	1	5	10	15
Ile Gln Arg Glu Lys Pro Ala Ile Tyr His Ile Pro Gly Phe Glu Val	20	25	30	
Gln Asp Thr Ser Arg Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro	35	40	45	
Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Ile	50	55	60	

<210> SEQ ID NO 95
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Equus caballus

<400> SEQUENCE: 95

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Thr Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
1           5           10           15

Ile Gln Arg Glu Lys Pro Ala Ile Tyr His Ile Pro Gly Phe Glu Val
          20           25           30

Gln Asp Thr Ser Gly Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro
          35           40           45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50           55           60

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<210> SEQ ID NO 96
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Equus asinus

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<400> SEQUENCE: 96

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Thr Arg Gly Ser Gln Leu Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
1           5           10           15

Ile Gln Arg Glu Lys Pro Ala Ile Tyr His Leu Pro Gly Phe Glu Val
          20           25           30

Gln Asp Thr Ser Gly Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro
          35           40           45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50           55           60

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<210> SEQ ID NO 97
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Loxodonta africana

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<400> SEQUENCE: 97

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Thr Arg Gly Ser Gln Ala Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
1           5           10           15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
          20           25           30

Gln Asp Thr Ser Arg Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro
          35           40           45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50           55           60

```

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<210> SEQ ID NO 98
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Balaenoptera acutorostrata scammoni

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<400> SEQUENCE: 98

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```

Thr Arg Gly Asn Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
1           5           10           15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
          20           25           30

Gln Asp Thr Ser Gly Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro
          35           40           45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
50           55           60

```

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<210> SEQ ID NO 99
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Ailuropoda melanoleuca

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<400> SEQUENCE: 99

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-continued

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Val Arg Gly Ser Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
1           5           10           15

Ile Gln Arg Glu Lys Pro Ala Ile His His Ile Pro Gly Phe Glu Val
           20           25           30

Gln Asp Thr Ser Gly Ile Leu Asp Glu Ser Glu Asp Thr Asp Leu Pro
           35           40           45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
           50           55           60

```

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<210> SEQ ID NO 100
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Orcinus orca

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<400> SEQUENCE: 100

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Thr Arg Gly Asn Gln Pro Ala Gly Glu Leu Leu Ala Ser Phe Glu Leu
1           5           10           15

Ile Gln Arg Glu Lys Pro Ala Ile Tyr His Ile Pro Gly Phe Glu Val
           20           25           30

Gln Asp Thr Ser Gly Ile Leu Glu Glu Ser Glu Asp Thr Asp Leu Pro
           35           40           45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Met
           50           55           60

```

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<210> SEQ ID NO 101
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Phascolarctos cinereus

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<400> SEQUENCE: 101

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Val Arg Gly Ser Arg Pro Ala Gly Glu Leu Leu Ala Ala Phe Glu Leu
1           5           10           15

Ile Met Arg Glu Lys Pro Ala Thr His His Ile Pro Gly Phe Glu Pro
           20           25           30

Glu Glu Ile Ser Gly Val Ala Asp Glu Ile Gly Asp Thr Asp Leu Pro
           35           40           45

Tyr Pro Pro Pro Gln Arg Glu Ala Asn Ile Tyr Val
           50           55           60

```

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<210> SEQ ID NO 102
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Ornithorhynchus anatinus

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<400> SEQUENCE: 102

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```

Thr Lys Lys Asn Lys Pro Ala Gly Asp Leu Leu Ala Ala Phe Glu Leu
1           5           10           15

Ile Gln Arg Glu Lys Pro Ser Thr His His Ser Asp Pro Gly Phe Asp
           20           25           30

Pro Ala Val Arg Thr Ser Arg Pro Leu Gly Lys Ala Gly Asp Ser Asp
           35           40           45

Leu Pro Tyr Pro Pro Pro Gln Arg Glu Pro Asn Val
           50           55           60

```

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<210> SEQ ID NO 103
<211> LENGTH: 2001
<212> TYPE: DNA
<213> ORGANISM: Equus caballus

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-continued

<400> SEQUENCE: 103

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cttcagatt tgcctcctgt tactgcattt ctaaatttta cttgagattc attctttcct    60
tgtgagggca actatcagct atcatactca agaatgaagc aagtgccttc ccacatgtat    120
tatcttaatc ctctactttg aggcaagtat tttctccctt atttactgat aagtagaaac    180
acacacatac aaacacgtgc acaaggttca aggagagtaa ataatttgcc catgaaagat    240
ctaactcagt tccttattaa atctttaaat catgactcca ggctaagtga gtctacatcc    300
tacatctaca tatttgtata tgccttatat tttttttttt tttttgagga agattgtgag    360
ctgacgtcta ttgccaatct tcctcttttt gcttaaggaa gacctgagct aacatctgtg    420
ccatcttctc gtattttgta tgtgggccac tggcacagca ttacttgatg accggtgtgt    480
aggctctgtc ctgggatccg aaccccaggc caccaaagcg gagcgtgtga acttaaccac    540
tacgccacca ggccggcccc tccttagata tttttaatgt tcatatagat tatgtggtat    600
tatcatctgt aggataaatg tatgccattc aaatgttttc tttggaactc tctccagtga    660
cacttttagc tgtaaaattt tgtgaacaga ttttctgctg gccattctaa ataacacaaa    720
ataactatat attgtgtttc ctttttatac tcaactagaat aacttcttta tcgcctttta    780
tatataggtt gtattgtctg gaaaatacaa tgcacagggc ttaggttcag atcatgaatt    840
aatgctgaga tgtaccaaag gacaagaata cgtcaaagtc gtcatgcaa atgggcgaat    900
gatgggagct gtcttaattg gtgaaaccga tttagaagaa acatttgaaa acttgatttt    960
aaaccagatg aatctttcag catcaggaga agatctgctg satccagata ttgacataga   1020
agattatttt gactaaaaag gtcattccaa gaaccacata aagttccaaa taagacaaaa   1080
aagtcacaca tcaataaagt aaatgattgc actgatttaa tgatgaccac attgaagtta   1140
aaagtacaga agtgataatg atttcagtgg aaaaatatta aaaaataaat tctaaagata   1200
aaatcaattc aagtaactta tttacagatt tttttcttaa cacaaaattg accaattatg   1260
taagaattct caagttattc atttctgtgt tttaaacgta cacacattca tttgtgattt   1320
tagctttgga gcacatttag ctaggctgtt atctgctcag ccacacaactt ggtcttggtt   1380
agtagaccag agccattctt tgattggaaa acgtcaagaa gcttgtaatt ttattttact   1440
tagagatgcc gcaatatctt ctgctaata ttatttgtat tgatacctta ttcctttttt   1500
gtttgaattc tacagctatt attttaacct gaaatcattc actttctgtc atgagcctgt   1560
gacttttcag tacaataaca ctaccatcaa aagagagtta tgtgaaaga aaaaatatag   1620
ctaggtatat caccgcataa agagcttaag acagttatgt gaacttactc ttttaaggag   1680
tttacataaa aaaagattta ttttccttg tttcatactg taaatttata caattcatgc   1740
tctcaaattt tcaaaagatc tcaaattctc aagtcctcaa ttctttagag ataagttctt   1800
aaaatatagc tccatttgta taaacatta atgaaattgt cttaaaaaatt aaactttgat   1860
taactacaaa aataatgggtg taaacatgca taaactgttc tgttcttgga aaatttagat   1920
acactcatgc atcatcaatt tttcttttct tagccctgtg gttcttagcc tcagattgct   1980
gaagcatgtt tgcagaattc t                                     2001

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<210> SEQ ID NO 104

<211> LENGTH: 1346

<212> TYPE: DNA

<213> ORGANISM: Equus caballus

<400> SEQUENCE: 104

```

aggtttccag agtattggaa gaatttgatg ttgaagaaca gccaaatacc atgttagaaa    60

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-continued

atcgctttcc caacattaag gttatagaat ctggagtaaa gcaactgaag agtgaagaac	120
actgcatttt aacagaagat ggcaatcagc atgtatataa gaaactctgt ctgtgtgctg	180
gagctaaacc aaagttgata tgtgaaggaa atccttatgt attaggaatc cgtgacacag	240
acagtgtctca ggaatttcag aaacagctta ctaaggctaa aagaataatg atcataggca	300
atggtgggat cgcaacttgaa ttagtgtatg aaattgaagg ctgtgaagtc atttgggtca	360
ttaaagacaa agctattggg aatacgttct tcgatgcagg agcagctgaa ttcttgactt	420
caaagctcat cgctgaaaaa ccagagggtta aaattgcaca tagaagaacc agatatacaa	480
ctgaaggaag gaaaaaggaa gcacgaacca aaggatgatgc tgctaagtga ggagtgccc	540
tgggacctga ctggcatgaa ggcttgaatc ttaaaggaa aaaagagttt tctcataaga	600
ttcacattga aactatgtgt gaagtaaaga aaatctacct tcaggaagaa tttagaattt	660
ccaagaaaaa gtccttgact ttccaagag accataataa tcagtcagtt tcaactgata	720
aagagatatg gcctgtatat gtggaattga ccaatgaaaa gatatatggc tgcgatttca	780
ttgtcagtcg tacaggagtt acaccaaata tagaaccttt cctctgtggc aacaattttg	840
atctaggaga agatgggtggc ctgaaagtga ataatcatat gcacacgtcc ctctctgaca	900
tctatgctgc aggtgacatc tgcaactgcct cctgggaacc cagcccagtg tggcagcaga	960
tgaggctgtg gacgcaggct agacagatgg gatggtatgc agccaagtgc atggctgcag	1020
ctactttagg agactccatt gacatggatt tcagcttcga actgtttgct catgtaacaa	1080
aattttttaa ctataaggtt gtattgctgg gaaaatacaa tgcacagggc ttaggttcag	1140
atcatgaatt aatgctgaga tgtaccaaag gacaagaata cgtcaaagtc gtcagtcaaa	1200
atgggcgaat gatgggagct gtcttaattg gtgaaaccga tttagaagaa acatttgaaa	1260
acttgatttt aaaccagatg aatctttcag catatggaga agatctgctg gatccagata	1320
ttgacataga agattatttt gactaa	1346

<210> SEQ ID NO 105

<211> LENGTH: 1346

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 105

aggtttccag agtattggaa gaatttgatg ttgaagaaca gccaaatacc atgttagaaa	60
atcgctttcc caacattaag gttatagaat ctggagtaaa gcaactgaag agtgaagaac	120
actgcatttt aacagaagat ggcaatcagc atgtatataa gaaactctgt ctgtgtgctg	180
gagctaaacc aaagttgata tgtgaaggaa atccttatgt attaggaatc cgtgacacag	240
acagtgtctca ggaatttcag aaacagctta ctaaggctaa aagaataatg atcataggca	300
atggtgggat cgcaacttgaa ttagtgtatg aaattgaagg ctgtgaagtc atttgggtca	360
ttaaagacaa agctattggg aatacgttct tcgatgcagg agcagctgaa ttcttgactt	420
caaagctcat cgctgaaaaa ccagagggtta aaattgcaca tagaagaacc agatatacaa	480
ctgaaggaag gaaaaaggaa gcacgaacca aaggatgatgc tgctaagtga ggagtgccc	540
tgggacctga ctggcatgaa ggcttgaatc ttaaaggaa aaaagagttt tctcataaga	600
ttcacattga aactatgtgt gaagtaaaga aaatctacct tcaggaagaa tttagaattt	660
ccaagaaaaa gtccttgact ttccaagag accataataa tcagtcagtt tcaactgata	720

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aagagatatg gcctgtatat gtggaattga ccaatgaaaa gatatatggc tgcgatttca    780
ttgtcagtgc tacaggaggt acaccaaata tagaaccttt cctctgtggc aacaattttg    840
atctaggaga agatgggtggc ctgaaagtga ataatcatat gcacacgtcc ctctctgaca    900
tctatgctgc aggtgacatc tgcactgcct cctgggaacc cagcccagtg tggcagcaga    960
tgaggctgtg gacgcaggct agacagatgg gatggtatgc agccaagtgc atggctgcag   1020
ctactttagg agactccatt gacatggatt tcagcttcga actgtttgct catgtaacaa   1080
aatTTTTTaa ctataagggtt gtattgctgg gaaaatacaa tgcacagggc ttaggttcag   1140
atcatgaatt aatgctgaga tgtaccaaag gacaagaata cgtcaaagtc gtcatgcaaa   1200
atgggcgaat gatgggagct gtcttaattg gtgaaaccga tttagaagaa acatttgaaa   1260
acttgatttt aaaccagatg aatctttcag catatggaga agatctgctg catccagata   1320
ttgacataga agattatttt gactaa                                     1346

```

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<210> SEQ ID NO 106
<211> LENGTH: 269
<212> TYPE: DNA
<213> ORGANISM: Equus caballus

```

```

<400> SEQUENCE: 106

```

```

ttatatatag gttgtattgc tgggaaaata caatgcacag ggcttaggtt cagatcatga    60
attaatgctg agatgtacca aaggacaaga atacgtcaaa gtcgtcatgc aaaatgggcg   120
aatgatggga gctgtcttaa ttggtgaaac cgatttagaa gaaacatttg aaaacttgat   180
tttaaaccag atgaatcttt cagcatacgg agaagatctg ctggatccag atattgacat   240
agaagattat tttgactaaa aaggtcatt                                     269

```

```

<210> SEQ ID NO 107
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Equus caballus

```

```

<400> SEQUENCE: 107

```

```

Val Ser Arg Val Leu Glu Glu Phe Asp Val Glu Glu Gln Pro Asn Thr
 1             5             10             15
Met Leu Glu Asn Arg Phe Pro Asn Ile Lys Val Ile Glu Ser Gly Val
          20             25             30
Lys Gln Leu Lys Ser Glu Glu His Cys Ile Leu Thr Glu Asp Gly Asn
          35             40             45
Gln His Val Tyr Lys Lys Leu Cys Leu Cys Ala Gly Ala Lys Pro Lys
          50             55             60
Leu Ile Cys Glu Gly Asn Pro Tyr Val Leu Gly Ile Arg Asp Thr Asp
        65             70             75             80
Ser Ala Gln Glu Phe Gln Lys Gln Leu Thr Lys Ala Lys Arg Ile Met
          85             90             95
Ile Ile Gly Asn Gly Gly Ile Ala Leu Glu Leu Val Tyr Glu Ile Glu
          100            105            110
Gly Cys Glu Val Ile Trp Val Ile Lys Asp Lys Ala Ile Gly Asn Thr
          115            120            125
Phe Phe Asp Ala Gly Ala Ala Glu Phe Leu Thr Ser Lys Leu Ile Ala
          130            135            140
Glu Lys Pro Glu Gly Lys Ile Ala His Arg Arg Thr Arg Tyr Thr Thr
          145            150            155            160

```

-continued

Glu Gly Arg Lys Lys Glu Ala Arg Thr Lys Gly Asp Ala Ala Asn Val
 165 170 175
 Gly Ser Ala Leu Gly Pro Asp Trp His Glu Gly Leu Asn Leu Lys Gly
 180 185 190
 Thr Lys Glu Phe Ser His Lys Ile His Ile Glu Thr Met Cys Glu Val
 195 200 205
 Lys Lys Ile Tyr Leu Gln Glu Glu Phe Arg Ile Ser Lys Lys Lys Ser
 210 215 220
 Leu Thr Phe Pro Arg Asp His Asn Asn Gln Ser Val Ser Thr Asp Lys
 225 230 235 240
 Glu Ile Trp Pro Val Tyr Val Glu Leu Thr Asn Glu Lys Ile Tyr Gly
 245 250 255
 Cys Asp Phe Ile Val Ser Ala Thr Gly Val Thr Pro Asn Ile Glu Pro
 260 265 270
 Phe Leu Cys Gly Asn Asn Phe Asp Leu Gly Glu Asp Gly Gly Leu Lys
 275 280 285
 Val Asn Asn His Met His Thr Ser Leu Pro Asp Ile Tyr Ala Ala Gly
 290 295 300
 Asp Ile Cys Thr Ala Ser Trp Glu Pro Ser Pro Val Trp Gln Gln Met
 305 310 315 320
 Arg Leu Trp Thr Gln Ala Arg Gln Met Gly Trp Tyr Ala Ala Lys Cys
 325 330 335
 Met Ala Ala Ala Thr Leu Gly Asp Ser Ile Asp Met Asp Phe Ser Phe
 340 345 350
 Glu Leu Phe Ala His Val Thr Lys Phe Phe Asn Tyr Lys Val Val Leu
 355 360 365
 Leu Gly Lys Tyr Asn Ala Gln Gly Leu Gly Ser Asp His Glu Leu Met
 370 375 380
 Leu Arg Cys Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn
 385 390 395 400
 Gly Arg Met Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu
 405 410 415
 Thr Phe Glu Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Ala Tyr Gly
 420 425 430
 Glu Asp Leu Leu Asp Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
 435 440 445

<210> SEQ ID NO 108

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Equus caballus

<400> SEQUENCE: 108

Val Ser Arg Val Leu Glu Glu Phe Asp Val Glu Glu Gln Pro Asn Thr
 1 5 10 15
 Met Leu Glu Asn Arg Phe Pro Asn Ile Lys Val Ile Glu Ser Gly Val
 20 25 30
 Lys Gln Leu Lys Ser Glu Glu His Cys Ile Leu Thr Glu Asp Gly Asn
 35 40 45
 Gln His Val Tyr Lys Lys Leu Cys Leu Cys Ala Gly Ala Lys Pro Lys
 50 55 60
 Leu Ile Cys Glu Gly Asn Pro Tyr Val Leu Gly Ile Arg Asp Thr Asp
 65 70 75 80
 Ser Ala Gln Glu Phe Gln Lys Gln Leu Thr Lys Ala Lys Arg Ile Met
 85 90 95

-continued

Ile Ile Gly Asn Gly Gly Ile Ala Leu Glu Leu Val Tyr Glu Ile Glu
 100 105 110
 Gly Cys Glu Val Ile Trp Val Ile Lys Asp Lys Ala Ile Gly Asn Thr
 115 120 125
 Phe Phe Asp Ala Gly Ala Ala Glu Phe Leu Thr Ser Lys Leu Ile Ala
 130 135 140
 Glu Lys Pro Glu Gly Lys Ile Ala His Arg Arg Thr Arg Tyr Thr Thr
 145 150 155 160
 Glu Gly Arg Lys Lys Glu Ala Arg Thr Lys Gly Asp Ala Ala Asn Val
 165 170 175
 Gly Ser Ala Leu Gly Pro Asp Trp His Glu Gly Leu Asn Leu Lys Gly
 180 185 190
 Thr Lys Glu Phe Ser His Lys Ile His Ile Glu Thr Met Cys Glu Val
 195 200 205
 Lys Lys Ile Tyr Leu Gln Glu Glu Phe Arg Ile Ser Lys Lys Lys Ser
 210 215 220
 Leu Thr Phe Pro Arg Asp His Asn Asn Gln Ser Val Ser Thr Asp Lys
 225 230 235 240
 Glu Ile Trp Pro Val Tyr Val Glu Leu Thr Asn Glu Lys Ile Tyr Gly
 245 250 255
 Cys Asp Phe Ile Val Ser Ala Thr Gly Val Thr Pro Asn Ile Glu Pro
 260 265 270
 Phe Leu Cys Gly Asn Asn Phe Asp Leu Gly Glu Asp Gly Gly Leu Lys
 275 280 285
 Val Asn Asn His Met His Thr Ser Leu Pro Asp Ile Tyr Ala Ala Gly
 290 295 300
 Asp Ile Cys Thr Ala Ser Trp Glu Pro Ser Pro Val Trp Gln Gln Met
 305 310 315 320
 Arg Leu Trp Thr Gln Ala Arg Gln Met Gly Trp Tyr Ala Ala Lys Cys
 325 330 335
 Met Ala Ala Ala Thr Leu Gly Asp Ser Ile Asp Met Asp Phe Ser Phe
 340 345 350
 Glu Leu Phe Ala His Val Thr Lys Phe Phe Asn Tyr Lys Val Val Leu
 355 360 365
 Leu Gly Lys Tyr Asn Ala Gln Gly Leu Gly Ser Asp His Glu Leu Met
 370 375 380
 Leu Arg Cys Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn
 385 390 395 400
 Gly Arg Met Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu
 405 410 415
 Thr Phe Glu Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Ala Tyr Gly
 420 425 430
 Glu Asp Leu Leu His Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
 435 440 445

<210> SEQ ID NO 109

<211> LENGTH: 249

<212> TYPE: DNA

<213> ORGANISM: Equus caballus

<400> SEQUENCE: 109

gttgtattgc tgggaaaata caatgcacag ggcttaggtt cagatcatga attaatgctg 60

agatgtacca aaggacaaga atacgtcaaa gtcgtcatgc aaaatgggag aatgatggga 120

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gctgtcttaa ttggtgaaac cgatttagaa gaaacatttg aaaacttgat tttaaaccag 180
atgaatcttt cagcatatgg agaagatctg ctggatccag atattgacat agaagattat 240
tttgactaa 249

```

```

<210> SEQ ID NO 110
<211> LENGTH: 249
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        polynucleotide

```

```

<400> SEQUENCE: 110

```

```

gttgatttgc tgggaaaata caatgcacag ggcttaggtt cagatcatga attaatgctg 60
agatgtacca aaggacaaga atacgtcaaa gtcgtcatgc aaaatgggag aatgatggga 120
gctgtcttaa ttggtgaaac cgatttagaa gaaacatttg aaaacttgat tttaaaccag 180
atgaatcttt cagcatatgg agaagatctg ctgcatccag atattgacat agaagattat 240
tttgactaa 249

```

```

<210> SEQ ID NO 111
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        primer

```

```

<400> SEQUENCE: 111

```

```

cagattttct gctggccatt 20

```

```

<210> SEQ ID NO 112
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
                        primer

```

```

<400> SEQUENCE: 112

```

```

tggatcatcat taaatcagtg caa 23

```

```

<210> SEQ ID NO 113
<211> LENGTH: 1012
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

```

```

<400> SEQUENCE: 113

```

```

tgaaattgaa ggctgtgaag tgatttgggc cattaaagat aaagctatag ggaatacttt 60
cttcgatgca ggagcagctg aattcttgac ttcaaagctc attgctgaaa aatcagaggc 120
taaaattgca cataaaagaa ccagatatac aactgaagga aggaaaaagg aagctagaag 180
caaatctaaa gcagataatg taggaagtgc attgggacca gattggcatg aaggcttgaa 240
tcttaaagga acaaaagagt tttctcataa gattcacctt gaaactatgt gtgaagtaaa 300
gaaaatctac cttcaggatg agtttagaat ttgaagaaa aagtccttca cttttccaag 360
agaccataag tcagttacag ctgatacaga gatgtggcct gtctatgtgg aattgaccaa 420
tgaaaagata tatggctgag atttcattgt cagtgtctaca ggagttacac caaatgtaga 480
accttttctc catggtaaca gttttgatct aggagaagat ggtggcctga aagtggatga 540

```

-continued

```

tcatatgcac acatcccttc ctgatatcta tgetgccggt gacatctgta ctacatcctg    600
gcagctgagc ccagctctggc agcagatgag gctgtggacc caggctagac agatgggatg    660
gtatgcagca aagtgcattg ctgcagcgag ttcaggagac tctattgaca tggatttcag    720
ctttgaactg tttgctcatg tgacaaaatt ttttaactat aaggttgtag tgctgggaaa    780
atacaatgca cagggtcttag gttcagatca tgaattaatg ctgagatgta ccaaaggacg    840
agaatacatc aaagtcgtca tgcaaaatgg acgaatgatg ggagctgtct taattggtga    900
aaccgattta gaagaaacat ttgaaaacct aatcttaaac caaatgaatc tttcatcata    960
tggagaagat ctgctagatc caaatattga tatagaagat tattttgact aa          1012

```

<210> SEQ ID NO 114

<211> LENGTH: 445

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 114

```

Ile Ser Lys Ile Leu Glu Glu Phe Asp Val Glu Glu Gln Ser Ser Thr
1           5           10           15
Met Leu Gly Lys Arg Phe Pro Asn Ile Lys Val Ile Glu Ser Gly Val
20          25          30
Lys Gln Leu Lys Ser Glu Glu His Cys Ile Val Thr Glu Asp Gly Asn
35          40          45
Gln His Val Tyr Lys Lys Leu Cys Leu Cys Ala Gly Ala Lys Pro Lys
50          55          60
Leu Ile Cys Glu Gly Asn Pro Tyr Val Leu Gly Ile Arg Asp Thr Asp
65          70          75          80
Ser Ala Gln Glu Phe Gln Lys Gln Leu Thr Lys Ala Lys Arg Ile Met
85          90          95
Ile Ile Gly Asn Gly Gly Ile Ala Leu Glu Leu Val Tyr Glu Ile Glu
100         105         110
Gly Cys Glu Val Ile Trp Ala Ile Lys Asp Lys Ala Ile Gly Asn Thr
115         120         125
Phe Phe Asp Ala Gly Ala Ala Glu Phe Leu Thr Ser Lys Leu Ile Ala
130         135         140
Glu Lys Ser Glu Ala Lys Ile Ala His Lys Arg Thr Arg Tyr Thr Thr
145         150         155         160
Glu Gly Arg Lys Lys Glu Ala Arg Ser Lys Ser Lys Ala Asp Asn Val
165         170         175
Gly Ser Ala Leu Gly Pro Asp Trp His Glu Gly Leu Asn Leu Lys Gly
180         185         190
Thr Lys Glu Phe Ser His Lys Ile His Leu Glu Thr Met Cys Glu Val
195         200         205
Lys Lys Ile Tyr Leu Gln Asp Glu Phe Arg Ile Leu Lys Lys Lys Ser
210         215         220
Phe Thr Phe Pro Arg Asp His Lys Ser Val Thr Ala Asp Thr Glu Met
225         230         235         240
Trp Pro Val Tyr Val Glu Leu Thr Asn Glu Lys Ile Tyr Gly Cys Asp
245         250         255
Phe Ile Val Ser Ala Thr Gly Val Thr Pro Asn Val Glu Pro Phe Leu
260         265         270
His Gly Asn Ser Phe Asp Leu Gly Glu Asp Gly Gly Leu Lys Val Asp
275         280         285
Asp His Met His Thr Ser Leu Pro Asp Ile Tyr Ala Ala Gly Asp Ile

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290	295	300
Cys Thr Thr Ser Trp Gln Leu Ser Pro Val Trp Gln Gln Met Arg Leu		
305	310	315 320
Trp Thr Gln Ala Arg Gln Met Gly Trp Tyr Ala Ala Lys Cys Met Ala		
	325	330 335
Ala Ala Ser Ser Gly Asp Ser Ile Asp Met Asp Phe Ser Phe Glu Leu		
	340	345 350
Phe Ala His Val Thr Lys Phe Phe Asn Tyr Lys Val Val Leu Leu Gly		
	355	360 365
Lys Tyr Asn Ala Gln Gly Leu Gly Ser Asp His Glu Leu Met Leu Arg		
	370	375 380
Cys Thr Lys Gly Arg Glu Tyr Ile Lys Val Val Met Gln Asn Gly Arg		
385	390	395 400
Met Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe		
	405	410 415
Glu Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Ser Tyr Gly Glu Asp		
	420	425 430
Leu Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp		
	435	440 445

<210> SEQ ID NO 115
 <211> LENGTH: 82
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 115

Val Val Leu Leu Gly Lys Tyr Asn Ala Gln Gly Leu Gly Ser Asp His
1 5 10 15
Glu Leu Met Leu Arg Cys Thr Lys Gly Arg Glu Tyr Ile Lys Val Val
20 25 30
Met Gln Asn Gly Arg Met Met Gly Ala Val Leu Ile Gly Glu Thr Asp
35 40 45
Leu Glu Glu Thr Phe Glu Asn Leu Ile Leu Asn Gln Met Asn Leu Ser
50 55 60
Ser Tyr Gly Glu Asp Leu Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr
65 70 75 80
Phe Asp

<210> SEQ ID NO 116
 <211> LENGTH: 82
 <212> TYPE: PRT
 <213> ORGANISM: Equus caballus

<400> SEQUENCE: 116

Val Val Leu Leu Gly Lys Tyr Asn Ala Gln Gly Leu Gly Ser Asp His
1 5 10 15
Glu Leu Met Leu Arg Cys Thr Lys Gly Gln Glu Tyr Val Lys Val Val
20 25 30
Met Gln Asn Gly Arg Met Met Gly Ala Val Leu Ile Gly Glu Thr Asp
35 40 45
Leu Glu Glu Thr Phe Glu Asn Leu Ile Leu Asn Gln Met Asn Leu Ser
50 55 60
Ala Tyr Gly Glu Asp Leu Leu Asp Pro Asp Ile Asp Ile Glu Asp Tyr
65 70 75 80
Phe Asp

-continued

<210> SEQ ID NO 117
 <211> LENGTH: 82
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 117

Val	Val	Leu	Leu	Gly	Lys	Tyr	Asn	Ala	Gln	Gly	Leu	Gly	Ser	Asp	His
1			5						10				15		
Glu	Leu	Met	Leu	Arg	Cys	Thr	Lys	Gly	Gln	Glu	Tyr	Val	Lys	Val	Val
		20						25					30		
Met	Gln	Asn	Gly	Arg	Met	Met	Gly	Ala	Val	Leu	Ile	Gly	Glu	Thr	Asp
		35					40					45			
Leu	Glu	Glu	Thr	Phe	Glu	Asn	Leu	Ile	Leu	Asn	Gln	Met	Asn	Leu	Ser
	50					55					60				
Ala	Tyr	Gly	Glu	Asp	Leu	Leu	His	Pro	Asp	Ile	Asp	Ile	Glu	Asp	Tyr
65					70					75				80	

Phe Asp

<210> SEQ ID NO 118
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown: PYROXD1 sequence

<400> SEQUENCE: 118

Thr	Lys	Gly	Gln	Glu	Tyr	Ile	Lys	Val	Val	Met	Gln	Asn	Gly	Arg	Met
1				5					10					15	
Met	Gly	Ala	Val	Leu	Ile	Gly	Glu	Thr	Asp	Leu	Glu	Glu	Thr	Phe	Glu
		20						25					30		
Asn	Leu	Ile	Leu	Asn	Gln	Met	Asn	Leu	Ser	Ser	Tyr	Gly	Glu	Asp	Leu
		35					40					45			
Leu	Asp	Pro	Asn	Ile	Asp	Ile	Glu	Asp	Tyr	Phe	Asp				
	50					55					60				

<210> SEQ ID NO 119
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown: PYROXD1 sequence

<400> SEQUENCE: 119

Thr	Lys	Gly	Gln	Glu	Tyr	Val	Lys	Val	Val	Met	Gln	Asn	Gly	Arg	Met
1				5					10					15	
Met	Gly	Ala	Val	Leu	Ile	Gly	Glu	Thr	Asp	Leu	Glu	Glu	Thr	Phe	Glu
		20						25					30		
Asn	Leu	Ile	Leu	Asn	Gln	Met	Asn	Leu	Ser	Ser	Tyr	Gly	Glu	Asp	Leu
		35					40					45			
Leu	Asp	Pro	Asn	Ile	Asp	Ile	Glu	Asp	Tyr	Phe	Asp				
	50					55					60				

<210> SEQ ID NO 120
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Unknown

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<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
    PYROXD1 sequence

<400> SEQUENCE: 120

Thr Arg Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1          5          10          15

Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
          20          25          30

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ser Tyr Gly Glu Asp Leu
          35          40          45

Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
          50          55          60

<210> SEQ ID NO 121
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
    PYROXD1 sequence

<400> SEQUENCE: 121

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1          5          10          15

Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
          20          25          30

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ser Tyr Gly Glu Asp Leu
          35          40          45

Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
          50          55          60

<210> SEQ ID NO 122
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
    PYROXD1 sequence

<400> SEQUENCE: 122

Thr Lys Gly Gln Glu Tyr Ile Lys Val Val Met Gln Asn Gly Arg Met
1          5          10          15

Met Gly Ala Val Leu Ile Gly Glu Thr Asn Leu Glu Glu Thr Phe Glu
          20          25          30

Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Ser Tyr Gly Glu Asp Leu
          35          40          45

Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
          50          55          60

<210> SEQ ID NO 123
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
    PYROXD1 sequence

<400> SEQUENCE: 123

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1          5          10          15

Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu

```

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20	25	30
Asn Leu Ile Leu Asn Gln Met	Asn Leu Ser Ala Tyr Gly Glu Asp Leu	
35	40	45
Leu Asp Pro Asp Ile Asp Ile	Glu Asp Tyr Phe Asp	
50	55	60

<210> SEQ ID NO 124
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 PYROXD1 sequence

<400> SEQUENCE: 124

Thr Lys Gly Gln Glu Tyr Ile Lys Ala Val Leu Gln Asn Gly Arg Met
1 5 10 15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30
Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Ala Tyr Gly Glu Asp Leu
35 40 45
Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 125
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 PYROXD1 sequence

<400> SEQUENCE: 125

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1 5 10 15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ser Tyr Gly Glu Asp Leu
35 40 45
Leu Asp Pro Asn Ile Asp Leu Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 126
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 126

Thr Lys Gly Arg Glu Tyr Ile Lys Val Val Met Gln Asn Gly Arg Met
1 5 10 15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30
Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Ser Tyr Gly Glu Asp Leu
35 40 45
Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 127
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Otolemur garnettii

-continued

<400> SEQUENCE: 127

```

Thr Lys Gly Gln Glu Tyr Ile Lys Val Val Met Gln Asn Gly Arg Met
1           5           10           15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
          20           25           30
Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Gln Tyr Gly Glu Asp Leu
          35           40           45
Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
          50           55           60

```

<210> SEQ ID NO 128

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 128

```

Thr Arg Gly Gln Glu Tyr Ile Lys Val Val Met His Asn Gly Arg Met
1           5           10           15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
          20           25           30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ser Tyr Gly Glu Asp Leu
          35           40           45
Leu Asn Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
          50           55           60

```

<210> SEQ ID NO 129

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Cricetulus griseus

<400> SEQUENCE: 129

```

Thr Arg Gly Gln Glu Tyr Ile Lys Val Val Met Gln Asn Gly Arg Met
1           5           10           15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
          20           25           30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ser Tyr Gly Glu Asp Leu
          35           40           45
Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
          50           55           60

```

<210> SEQ ID NO 130

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Microtus ochrogaster

<400> SEQUENCE: 130

```

Thr Arg Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1           5           10           15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
          20           25           30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ser Tyr Gly Glu Asp Leu
          35           40           45
Leu Asp Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
          50           55           60

```

<210> SEQ ID NO 131

<211> LENGTH: 60

<212> TYPE: PRT

-continued

<213> ORGANISM: Chinchilla lanigera

<400> SEQUENCE: 131

Thr	Lys	Gly	Gln	Glu	Tyr	Val	Lys	Val	Val	Met	Gln	Asn	Gly	Arg	Met
1				5					10					15	

Met	Gly	Ala	Val	Leu	Ile	Gly	Glu	Thr	Asp	Leu	Glu	Glu	Thr	Phe	Glu
			20					25					30		

Asn	Leu	Ile	Leu	Asn	Gln	Met	Asp	Leu	Ser	Pro	Tyr	Gly	Glu	Glu	Leu
		35				40						45			

Leu	Asp	Pro	Asn	Ile	Asp	Ile	Glu	Asp	Tyr	Phe	Asp
	50				55					60	

<210> SEQ ID NO 132

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Castor canadensis

<400> SEQUENCE: 132

Thr	Lys	Gly	Gln	Glu	Tyr	Ile	Lys	Val	Ile	Met	Gln	Asn	Gly	Arg	Met
1				5					10					15	

Met	Gly	Ala	Val	Leu	Ile	Gly	Glu	Thr	Asp	Leu	Glu	Glu	Thr	Phe	Glu
			20					25					30		

Asn	Leu	Ile	Leu	Asn	Gln	Met	Asp	Leu	Ser	Ser	Tyr	Gly	Glu	Asp	Leu
		35				40						45			

Leu	Asp	Pro	Asn	Ile	Asp	Ile	Glu	Asp	Tyr	Phe	Asp
	50				55					60	

<210> SEQ ID NO 133

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Mustela putorius furo

<400> SEQUENCE: 133

Thr	Lys	Gly	Gln	Glu	Tyr	Val	Lys	Val	Val	Met	Gln	His	Gly	Arg	Met
1				5					10					15	

Met	Gly	Ala	Val	Leu	Ile	Gly	Glu	Thr	Asp	Leu	Glu	Glu	Thr	Phe	Glu
			20					25					30		

Asn	Leu	Ile	Leu	Asn	Gln	Thr	Asp	Leu	Ser	Ser	Tyr	Gly	Glu	Asp	Leu
		35				40						45			

Leu	Asp	Pro	Asn	Ile	Asp	Leu	Glu	Asp	Tyr	Phe	Asp
	50				55					60	

<210> SEQ ID NO 134

<211> LENGTH: 60

<212> TYPE: PRT

<213> ORGANISM: Fukomys damarensis

<400> SEQUENCE: 134

Thr	Lys	Gly	Gln	Glu	Tyr	Ile	Lys	Val	Val	Met	Gln	Asn	Gly	His	Met
1				5					10					15	

Met	Gly	Ala	Val	Leu	Ile	Gly	Glu	Thr	Asp	Leu	Glu	Glu	Thr	Phe	Glu
			20					25					30		

Asn	Leu	Ile	Leu	Asn	Gln	Met	Asp	Leu	Ser	Pro	Tyr	Gly	Glu	Asp	Leu
		35				40						45			

Leu	Asp	Pro	Asn	Ile	Asp	Ile	Glu	Asp	Tyr	Phe	Asp
	50				55					60	

<210> SEQ ID NO 135

<211> LENGTH: 60

-continued

<212> TYPE: PRT
 <213> ORGANISM: *Vicugna pacos*

<400> SEQUENCE: 135

```

Thr Lys Gly Gln Glu Tyr Val Lys Ala Val Leu Gln Asn Gly Arg Met
1          5          10          15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
          20          25          30
Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Ala Tyr Gly Glu Asp Leu
          35          40          45
Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
          50          55          60

```

<210> SEQ ID NO 136
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Ailuropoda melanoleuca*

<400> SEQUENCE: 136

```

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1          5          10          15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
          20          25          30
Asn Leu Ile Leu Asn Gln Thr Asp Leu Ser Ser Tyr Gly Glu Asp Leu
          35          40          45
Leu Asp Pro Asn Ile Asp Leu Glu Asp Tyr Phe Asp
          50          55          60

```

<210> SEQ ID NO 137
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Phascolarctos cinereus*

<400> SEQUENCE: 137

```

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1          5          10          15
Met Gly Ala Ile Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
          20          25          30
Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Ala Tyr Gly Glu Asp Leu
          35          40          45
Leu Asp Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
          50          55          60

```

<210> SEQ ID NO 138
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Sarcophilus harrisii*

<400> SEQUENCE: 138

```

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1          5          10          15
Met Gly Ala Ile Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
          20          25          30
Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Ala Tyr Gly Glu Asp Leu
          35          40          45
Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
          50          55          60

```

<210> SEQ ID NO 139

-continued

<211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Monodelphis domestica

<400> SEQUENCE: 139

```

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1           5           10           15
Met Gly Ala Ile Leu Ile Gly Glu Thr Asn Leu Glu Glu Thr Phe Glu
                20           25           30
Asn Leu Ile Leu Asn Gln Met Asn Leu Ser Ala Tyr Gly Glu Asp Leu
                35           40           45
Leu Asp Pro Ser Ile Asp Ile Glu Asp Tyr Phe Asp
    50           55           60

```

<210> SEQ ID NO 140
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 PYROXD1 sequence

<400> SEQUENCE: 140

```

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1           5           10           15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
                20           25           30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ala Tyr Gly Glu Asp Leu
                35           40           45
Leu Asn Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
    50           55           60

```

<210> SEQ ID NO 141
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 PYROXD1 sequence

<400> SEQUENCE: 141

```

Thr Lys Gly His Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1           5           10           15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
                20           25           30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ala Tyr Gly Glu Asp Leu
                35           40           45
Leu Asn Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
    50           55           60

```

<210> SEQ ID NO 142
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Merops nubicus

<400> SEQUENCE: 142

```

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1           5           10           15
Val Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
                20           25           30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ala Tyr Gly Glu Asp Leu

```

-continued

35	40	45
Leu Asn Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp		
50	55	60

<210> SEQ ID NO 143
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Apteryx australis mantelli
 <400> SEQUENCE: 143

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1 5 10 15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ala Tyr Gly Glu Asp Leu
35 40 45
Leu Asn Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 144
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Corvus brachyrhynchos
 <400> SEQUENCE: 144

Thr Lys Gly His Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1 5 10 15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30
Asn Leu Ile Leu Asn Arg Met Asp Leu Ser Ala Tyr Gly Glu Asp Leu
35 40 45
Leu Asn Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 145
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Corvus cornix cornix
 <400> SEQUENCE: 145

Thr Lys Gly His Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1 5 10 15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30
Asn Leu Ile Leu Asn Arg Met Asp Leu Ser Ala Tyr Gly Glu Asp Leu
35 40 45
Leu Asn Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 146
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Serinus canaria
 <400> SEQUENCE: 146

Thr Lys Gly His Glu Tyr Ile Lys Val Val Met Gln Asn Gly Arg Met
1 5 10 15
Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30

-continued

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ala Tyr Gly Glu Asp Leu
35 40 45

Leu Asn Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 147
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: *Sturnus vulgaris*

<400> SEQUENCE: 147

Thr Lys Gly His Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1 5 10 15

Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30

Asn Leu Ile Leu Asn Gln Ile Asp Leu Ser Ala Tyr Gly Glu Asp Leu
35 40 45

Leu Asn Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 148
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: *Lepidothrix coronata*

<400> SEQUENCE: 148

Thr Lys Gly Gln Glu Tyr Val Lys Ala Val Met Gln Asn Gly Arg Met
1 5 10 15

Val Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ala Tyr Gly Glu Asp Leu
35 40 45

Leu Asn Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 149
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: *Amazona aestiva*

<400> SEQUENCE: 149

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Ser Gly Arg Met
1 5 10 15

Val Gly Ala Val Leu Ile Gly Asp Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ala Tyr Gly Glu Asp Leu
35 40 45

Leu Asn Pro Asp Val Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 150
<211> LENGTH: 60
<212> TYPE: PRT
<213> ORGANISM: *Gekko japonicus*

<400> SEQUENCE: 150

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
1 5 10 15

Met Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30

-continued

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Arg Tyr Gly Glu Asp Leu
 35 40 45

Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
 50 55 60

<210> SEQ ID NO 151
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Anolis carolinensis*

<400> SEQUENCE: 151

Thr Lys Gly Leu Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
 1 5 10 15

Leu Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
 20 25 30

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Arg Tyr Gly Glu Asp Leu
 35 40 45

Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
 50 55 60

<210> SEQ ID NO 152
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Python bivittatus*

<400> SEQUENCE: 152

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Met Gln Asn Gly Arg Met
 1 5 10 15

Leu Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
 20 25 30

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Arg Tyr Gly Glu Asp Leu
 35 40 45

Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
 50 55 60

<210> SEQ ID NO 153
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Pogona vitticeps*

<400> SEQUENCE: 153

Thr Lys Gly Gln Glu Tyr Ile Lys Val Val Met Gln Asn Gly Arg Met
 1 5 10 15

Leu Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
 20 25 30

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Arg Tyr Gly Glu Asn Leu
 35 40 45

Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
 50 55 60

<210> SEQ ID NO 154
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Protobothrops mucrosquamatus*

<400> SEQUENCE: 154

Thr Lys Gly Gln Glu Tyr Ile Lys Val Val Met Gln Asn Gly Arg Met
 1 5 10 15

Leu Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu

-continued

20	25	30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Arg Tyr Gly Glu Asp Leu		
35	40	45
Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp		
50	55	60

<210> SEQ ID NO 155
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Ophiophagus hannah*

<400> SEQUENCE: 155

Thr Lys Gly Gln Glu Tyr Ile Lys Val Val Met Gln His Gly Arg Met
1 5 10 15
Leu Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Arg Tyr Gly Glu Asp Leu
35 40 45
Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 156
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Thamnophis sirtalis*

<400> SEQUENCE: 156

Thr Lys Gly Gln Glu Tyr Ile Lys Val Val Met Gln Asn Gly Arg Met
1 5 10 15
Leu Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30
Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Arg Tyr Gly Lys Asp Leu
35 40 45
Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 157
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Latimeria chalumnae*

<400> SEQUENCE: 157

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Leu Gln Asn Gly Arg Met
1 5 10 15
Met Gly Ala Ile Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30
Asn Leu Met Leu Asn Gln Met Asp Leu Ser Ala Tyr Gly Glu Glu Leu
35 40 45
Leu Asn Pro Asp Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 158
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: *Rhincodon typus*

<400> SEQUENCE: 158

Thr Thr Gly Gln Glu Tyr Val Lys Thr Val Met Gln Asn Gly Arg Met
1 5 10 15

-continued

Lys Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
 20 25 30

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ala Tyr Gly Glu Asp Leu
 35 40 45

Leu Asn Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
 50 55 60

<210> SEQ ID NO 159
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Callorhinchus milii

<400> SEQUENCE: 159

Thr Lys Gly His Glu Tyr Val Lys Ala Val Met Gln Asn Gly Arg Met
 1 5 10 15

Met Gly Ala Val Leu Ile Gly Glu Thr Asn Leu Glu Glu Thr Phe Glu
 20 25 30

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Ser Tyr Gly Glu Glu Leu
 35 40 45

Leu Asn Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
 50 55 60

<210> SEQ ID NO 160
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Salmo salar

<400> SEQUENCE: 160

Thr Lys Gly His Glu Tyr Val Lys Val Val Leu Ser Gly Gly Arg Met
 1 5 10 15

Leu Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
 20 25 30

Asn Leu Ile Leu Asn Gln Met Asp Leu Thr Pro Tyr Gly Glu Glu Leu
 35 40 45

Leu Asn Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
 50 55 60

<210> SEQ ID NO 161
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Xenopus laevis

<400> SEQUENCE: 161

Thr Arg Gly His Glu Tyr Val Lys Leu Val Met Gln Gly Gly Arg Met
 1 5 10 15

Val Gly Ala Leu Leu Ile Gly Asp Thr Asp Leu Glu Glu Thr Phe Glu
 20 25 30

Asn Leu Ile Leu Asn Gln Thr Asp Leu Ser Ser Tyr Gly Glu Arg Leu
 35 40 45

Leu Asp Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
 50 55 60

<210> SEQ ID NO 162
 <211> LENGTH: 60
 <212> TYPE: PRT
 <213> ORGANISM: Cyprinus carpio

<400> SEQUENCE: 162

Thr Lys Gly Gln Glu Tyr Val Lys Val Val Leu Thr Gly Gly Arg Met
 1 5 10 15

-continued

Val Gly Ala Val Leu Ile Gly Glu Thr Asp Leu Glu Glu Thr Phe Glu
20 25 30

Asn Leu Ile Leu Asn Gln Met Asp Leu Ser Arg Tyr Gly Glu Glu Leu
35 40 45

Leu Asn Pro Asn Ile Asp Ile Glu Asp Tyr Phe Asp
50 55 60

<210> SEQ ID NO 163

<211> LENGTH: 2001

<212> TYPE: DNA

<213> ORGANISM: Equus caballus

<400> SEQUENCE: 163

```

ttagaattct ggggtggctg gaggtgtgtt tgctctctag ctgcaaagat aaacgtttgc      60
aagacctgtc atacgaagac ctgagaagtt tccctaaaat gtggtctcag ggtactttaa      120
gtttgtttta agaaaagaga agagttcgta gcttttcttg tttcaggctc tataaaacct      180
gtaagttttt gtaggtgacc attgcgggga ccggcctctt cttgtccatg aggccagagt      240
agcctggggc atcgtgtctc ctggggagga agaagcattg cccctctgtt ggggcagaga      300
gtgggtgctg accgcatggt cagggctctgt cagaggcagg tggacgttct gtggatggga      360
tgtgactgaa gggaccctgc tcttgagcct cacattcata tcatcacgaa tatttttctg      420
agcacctgat cctgcggcgg tggtcttcct cgcacatgtg gctcatttcc ctgacttgtg      480
ttgttttcag ggtgactcag ggcgggacag ccagcagaga ggacccaaag gagaaaccgg      540
agacctcggg cccatggtac gatgctcctt gttcccagac tctcctgtgc tggaggccag      600
agaaacaaag ctcacactgg ctgcgcaggg gccgaaggac catggaagtg gtctttgagg      660
ctggggagca tatcgggtca ggaacagagg gcccctgtaa gcccaccctg tgtgcccatg      720
aaggacgttt ccattgtgga acagaagcaa gccctaccga gaggagcttc ttctgggcag      780
aagttgtgct ctttagatat ttacgaaagc gcttgtcaag ccaaaggctg tgggggcggg      840
cacgtacctc agtctagcgc acctcacgac agagtaccga ccttctctct cctatccatc      900
ccttttctct tagtgtctcg tgcaatattc cagagttcct gctcttcttt gcagatgaaa      960
atcttacacc aattttcttc cccacgccag ggtctgcctg sgagtgatgg tgtctcggga     1020
agtctcggag aaccaggagg ggacgtgagt gtctcttaca gctttggagg gaggtggcgg     1080
ggcctctgtg ctggtattta gcaggtgcc tccagatctt ccagacactg cctggggcgt     1140
tgttaccacg caactctttg tttaaagaat ttaaagagat ttctgaaacc ttgtcaccc     1200
aaatcttgat gcttatggaa aacattcgtc cttattgttt gatctgtgcc ccatctgtgc     1260
ccgtctcttc tcccaccccc agtcccgctt actccctcc caaaaactca ccgaacagac     1320
actttttaa gttattgtct cccagctttc aagtatctgg ccagggtttta tcccactaaa     1380
tgtttaaaaa aatcccaaca gaaacacgaa aacgggggcy gggggggggg atgagggaa     1440
atggggcagg gaggtgctg gggttgtata tttgtggttg cttttagaat tgttttcgt     1500
gtgctgctgt agctcaagca aggaaacgag actcagtttt ctagtgaatc gctttttgaa     1560
atcgctttcc caaggcattt cagaaatgcc ctttgtagat tagctgattg catgcctagt     1620
ctttgagtc ttctggaaga gctgctaaag ccaaagggat aaagatcgtc tttaggaagc     1680
cagtggaatg cttccagggc ccaggggcaa gatggtctgg aagctcatac atcctctggg     1740
gtgcgtcagg atgctggggg cgggtgccc gggcaggtct ctgctcagaa gctgctggg     1800
actatgcagc cttgaacggg gcccctcag ctgtctcccc agtggcagcg acacatttag     1860

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ggatcagcag ctggacaatc ggggggggtgt atgctgaggt gagcgctctg aaaagaagta 1920
actctcacct gtaatcagat gatatatcat gggccaaagt ttgcttttaa aattcagaag 1980
aagacttaaa atttgaaaat g 2001

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<210> SEQ ID NO 164
<211> LENGTH: 1008
<212> TYPE: DNA
<213> ORGANISM: Equus caballus

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<400> SEQUENCE: 164

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```

ggacaaaggg gagaccgagg gcccatcggc agcatcgggc caaaggggtgt tcccggagaa 60
gacggctacc gaggttacc cggtgatgag ggtgggccc gtgatcggg tccaccgggt 120
gtgaacggca ctcaagggtt ccagggtctgc cccgggcaga gaggaataaa gggctctcgt 180
ggattccag gagagaaggg tgaattagga gaaatcggac ttgatggtct tgacggtgaa 240
gatggagaca aaggattgcc tgggtcttct ggagagaaa ggaatcctgg aaggagggt 300
gacaaaggac ctaaaggaga ccaaggggaa agaggagatg ttggaattag gggcgacccg 360
ggtgactcag ggcgggacag ccagcagaga ggacccaaag gagaaccgg agacctcggg 420
cccatgggtc tgcttgggag tgatggtgtc tcgggaagtc ctggagaacc agggagggac 480
ggtggctttg gccgaagggg accaccagga gctaagggca acaagggcgg tcttgccag 540
cagggcaccg tgggagagca ggggaccaga ggtgcacagg gtccacctgg tccaccggt 600
cctccagggc tgatcggcga acaaggcatt cctggacctc ggggaagcgg aggtgctgtg 660
ggcgctccctg gagaacgcgg ccgaaccggt ccttgggaa gaaagggcga gcctggagag 720
ccgggagcga agggaggact cgggccccgg gggccccgtg gggaaacggg agatgacggg 780
cgagacggag ttggcagtga aggacaaaaa ggcaaaaaag gagaagagg attccctgga 840
taccagggtc caaaggttac ccgtggtgag ccagggacag acggaacact aggacccaaa 900
ggcgtcagag gccgaagggg agactcagga cctccagggg cagctggaca gaagggagac 960
cctggttacc cgggaccatc tgggtctcaa ggcaacagag gcgactcg 1008

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<210> SEQ ID NO 165
<211> LENGTH: 1008
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

```

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<400> SEQUENCE: 165

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```

ggacaaaggg gagaccgagg gcccatcggc agcatcgggc caaaggggtgt tcccggagaa 60
gacggctacc gaggttacc cggtgatgag ggtgggccc gtgatcggg tccaccgggt 120
gtgaacggca ctcaagggtt ccagggtctgc cccgggcaga gaggaataaa gggctctcgt 180
ggattccag gagagaaggg tgaattagga gaaatcggac ttgatggtct tgacggtgaa 240
gatggagaca aaggattgcc tgggtcttct ggagagaaa ggaatcctgg aaggagggt 300
gacaaaggac ctaaaggaga ccaaggggaa agaggagatg ttggaattag gggcgacccg 360
ggtgactcag ggcgggacag ccagcagaga ggacccaaag gagaaccgg agacctcggg 420
cccatgggtc tgcttggcag tgatggtgtc tcgggaagtc ctggagaacc agggagggac 480
ggtggctttg gccgaagggg accaccagga gctaagggca acaagggcgg tcttgccag 540
cagggcaccg tgggagagca ggggaccaga ggtgcacagg gtccacctgg tccaccggt 600

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cctccagggc tgatcggcga acaaggcatt cctggacctc ggggaagcgg aggtgctgtg      660
ggcgtccttg gagaacgcgg ccgaaccggt cccttgggaa gaaagggcga gcctggagag      720
ccgggagcga agggaggact cgggcccccg ggccccctg gggaaacggg agatgacggg      780
cgagacggag ttggcagtga aggacaaaaa ggcaaaaaag gaaaaagagg attccctgga      840
taccacagtc caaagggtag cctgtgtgag ccagggacag acggaacact aggacccaaa      900
ggcgtcagag gccgaagggg agactcagga cctccagggg cagctggaca gaagggagac      960
cctggttacc cgggaccatc tggctcctaaa ggcaacagag gcgactcg                    1008

```

```

<210> SEQ ID NO 166
<211> LENGTH: 82
<212> TYPE: DNA
<213> ORGANISM: Equus caballus

```

```

<400> SEQUENCE: 166

```

```

cccacgccag ggtctgccgg tctgcctggg agtgatggtg tctcgggaag tcctggagaa      60
ccagggaggg acgtgagtgt ct                                              82

```

```

<210> SEQ ID NO 167
<211> LENGTH: 336
<212> TYPE: PRT
<213> ORGANISM: Equus caballus

```

```

<400> SEQUENCE: 167

```

```

Gly Gln Arg Gly Asp Arg Gly Pro Ile Gly Ser Ile Gly Pro Lys Gly
1         5         10        15
Val Pro Gly Glu Asp Gly Tyr Arg Gly Tyr Pro Gly Asp Glu Gly Gly
20        25        30
Pro Gly Asp Arg Gly Pro Pro Gly Val Asn Gly Thr Gln Gly Phe Gln
35        40        45
Gly Cys Pro Gly Gln Arg Gly Ile Lys Gly Ser Arg Gly Phe Pro Gly
50        55        60
Glu Lys Gly Glu Leu Gly Glu Ile Gly Leu Asp Gly Leu Asp Gly Glu
65        70        75        80
Asp Gly Asp Lys Gly Leu Pro Gly Ser Ser Gly Glu Lys Gly Asn Pro
85        90        95
Gly Arg Arg Gly Asp Lys Gly Pro Lys Gly Asp Gln Gly Glu Arg Gly
100       105       110
Asp Val Gly Ile Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln
115       120       125
Gln Arg Gly Pro Lys Gly Glu Thr Gly Asp Leu Gly Pro Met Gly Leu
130       135       140
Pro Gly Ser Asp Gly Val Ser Gly Ser Pro Gly Glu Pro Gly Arg Asp
145       150       155       160
Gly Gly Phe Gly Arg Arg Gly Pro Pro Gly Ala Lys Gly Asn Lys Gly
165       170       175
Gly Pro Gly Gln Gln Gly Thr Val Gly Glu Gln Gly Thr Arg Gly Ala
180       185       190
Gln Gly Pro Pro Gly Ser Thr Gly Pro Pro Gly Leu Ile Gly Glu Gln
195       200       205
Gly Ile Pro Gly Pro Arg Gly Ser Gly Gly Ala Val Gly Val Pro Gly
210       215       220
Glu Arg Gly Arg Thr Gly Pro Leu Gly Arg Lys Gly Glu Pro Gly Glu
225       230       235       240

```

Pro	Gly	Ala	Lys	Gly	Gly	Leu	Gly	Pro	Arg	Gly	Pro	Arg	Gly	Glu	Thr
				245					250					255	
Gly	Asp	Asp	Gly	Arg	Asp	Gly	Val	Gly	Ser	Glu	Gly	Gln	Lys	Gly	Lys
			260					265					270		
Lys	Gly	Glu	Arg	Gly	Phe	Pro	Gly	Tyr	Pro	Gly	Pro	Lys	Gly	Thr	Arg
			275				280					285			
Gly	Glu	Pro	Gly	Thr	Asp	Gly	Thr	Leu	Gly	Pro	Lys	Gly	Val	Arg	Gly
						295					300				
Arg	Arg	Gly	Asp	Ser	Gly	Pro	Pro	Gly	Ala	Ala	Gly	Gln	Lys	Gly	Asp
305					310					315					320
Pro	Gly	Tyr	Pro	Gly	Pro	Ser	Gly	Leu	Lys	Gly	Asn	Arg	Gly	Asp	Ser
				325					330					335	

```
<210> SEQ ID NO 168
<211> LENGTH: 336
<212> TYPE: PRT
<213> ORGANISM: Equus caballus
```

<400> SEQUENCE: 168

Gly 1	Gln	Arg	Gly 5	Asp	Arg	Gly	Pro	Ile 10	Gly	Ser	Ile	Gly	Pro	Lys 15	Gly
Val	Pro	Gly	Glu 20	Asp	Gly	Tyr	Arg	Gly 25	Tyr	Pro	Gly	Asp	Glu 30	Gly	Gly
Pro	Gly	Asp	Arg 35	Gly	Pro	Pro	Gly 40	Val	Asn	Gly	Thr	Gln 45	Gly	Phe	Gln
Gly	Cys 50	Pro	Gly	Gln	Arg	Gly 55	Ile	Lys	Gly	Ser	Arg 60	Gly	Phe	Pro	Gly
Glu 65	Lys	Gly	Glu	Leu 70	Gly	Glu	Ile	Gly	Leu 75	Asp	Gly	Leu	Asp	Gly	Glu 80
Asp	Gly	Asp	Lys 85	Gly	Leu	Pro	Gly	Ser	Ser 90	Gly	Glu	Lys	Gly	Asn 95	Pro
Gly	Arg	Arg	Gly 100	Asp	Lys	Gly	Pro	Lys 105	Gly	Asp	Gln	Gly	Glu 110	Arg	Gly
Asp	Val 115	Gly	Ile	Arg	Gly	Asp	Pro 120	Gly	Asp	Ser	Gly	Arg 125	Asp	Ser	Gln
Gln 130	Arg	Gly	Pro	Lys	Gly	Glu 135	Thr	Gly	Asp	Leu 140	Gly	Pro	Met	Gly	Leu
Pro 145	Ala	Ser	Asp	Gly 150	Val	Ser	Gly	Ser	Pro	Gly 155	Glu	Pro	Gly	Arg	Asp 160
Gly	Gly	Phe	Gly	Arg 165	Arg	Gly	Pro	Pro	Gly 170	Ala	Lys	Gly	Asn 175	Lys	Gly
Gly	Pro	Gly	Gln 180	Gln	Gly	Thr	Val	Gly 185	Glu	Gln	Gly	Thr	Arg 190	Gly	Ala
Gln	Gly 195	Pro	Pro	Gly	Ser	Thr	Gly 200	Pro	Pro	Gly	Leu	Ile 205	Gly	Glu	Gln
Gly 210	Ile	Pro	Gly	Pro	Arg	Gly 215	Ser	Gly	Gly	Ala	Val 220	Gly	Val	Pro	Gly
Glu 225	Arg	Gly	Arg	Thr 230	Gly	Pro	Leu	Gly	Arg	Lys 235	Gly	Glu	Pro	Gly	Glu 240
Pro	Gly	Ala	Lys 245	Gly	Gly	Leu	Gly	Pro	Arg 250	Gly	Pro	Arg	Gly	Glu 255	Thr
Gly	Asp	Asp	Gly 260	Arg	Asp	Gly	Val	Gly 265	Ser	Glu	Gly	Gln	Lys	Gly	Lys
Lys	Gly	Glu	Arg	Gly	Phe	Pro	Gly	Tyr	Pro	Gly	Pro	Lys	Gly	Thr	Arg

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275	280	285	
Gly Glu Pro Gly Thr Asp	Gly Thr Leu Gly Pro	Lys Gly Val Arg Gly	
290	295	300	
Arg Arg Gly Asp Ser	Gly Pro Pro Gly Ala Ala	Gly Gln Lys Gly Asp	
305	310	315	320
Pro Gly Tyr Pro Gly Pro Ser	Gly Leu Lys Gly Asn Arg Gly Asp Ser		
	325	330	335
<210> SEQ ID NO 169			
<211> LENGTH: 54			
<212> TYPE: DNA			
<213> ORGANISM: Equus caballus			
<400> SEQUENCE: 169			
ggctcgcctg ggagtgatgg tgtctcgga agtcctggag aaccagggag ggac			54
<210> SEQ ID NO 170			
<211> LENGTH: 54			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic oligonucleotide			
<400> SEQUENCE: 170			
ggctcgcctg cgagtgatgg tgtctcgga agtcctggag aaccagggag ggac			54
<210> SEQ ID NO 171			
<211> LENGTH: 20			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic primer			
<400> SEQUENCE: 171			
ttcccagact ctctgtgct			20
<210> SEQ ID NO 172			
<211> LENGTH: 20			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic primer			
<400> SEQUENCE: 172			
agatggggca cagatcaaac			20
<210> SEQ ID NO 173			
<211> LENGTH: 1008			
<212> TYPE: DNA			
<213> ORGANISM: Homo sapiens			
<400> SEQUENCE: 173			
gggcagaggg gagaccgagg gcccatcggc agcatcgggc caaagggat tcttgagaa			60
gacggctacc gaggtatcc tggatgatgag ggtggaccgg gtgagcgtgg tccgcctggt			120
gtgaacggca ctcaaggttt ccagggctgc ccgggccaga gaggagtaaa gggctctcgg			180
ggattccag gagagaagg cgaagtagga gaaattggac tggatggtct ggaatggtgaa			240
gatggagaca aaggattgcc tggttcttct ggagagaaa ggaatcctgg aagaaggggt			300
gataaaggac ctcgaggaga gaaaggagaa agaggagatg ttgggattcg aggggacccg			360

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ggtaaccag gacaagacag ccaggagaga ggacccaaag gagaaaccgg tgacctggc 420
cccatgggtg tcccaggag agatggagta cctggaggac ctggagaaac tgggaagaat 480
ggtggctttg gccgaagggg acccccggga gctaagggca acaagggcgg tcttgccag 540
ccgggctttg agggagagca ggggaccaga ggtgcacagg gccagctgg tctgctggt 600
cctccagggc tgataggaga acaaggcatt tctggacctc ggggaagcgg aggtgccgct 660
ggtgctcctg gagaacgagg cagaaccggc cactgggaa gaaagggta gcccgagag 720
ccaggaccaa aaggaggaat cggaaccgg gccctcgtg gggagacggg agatgacggg 780
agagacggag ttggcagtga aggacgcaga ggcaaaaaag gagaaagagg attccctgga 840
taccaggac caaagggtaa ccaggtgaa cctgggctaa atggaacaac aggacccaaa 900
ggcatcagag gccgaagggg aaattcggga cctccaggga tagttggaca gaaggagag 960
cctggctacc caggaccagc tggccaag ggcaacagg gcgactcc 1008

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<210> SEQ ID NO 174

<211> LENGTH: 336

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 174

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Gly Gln Arg Gly Asp Arg Gly Pro Ile Gly Ser Ile Gly Pro Lys Gly
1           5           10          15
Ile Pro Gly Glu Asp Gly Tyr Arg Gly Tyr Pro Gly Asp Glu Gly Gly
20          25          30
Pro Gly Glu Arg Gly Pro Pro Gly Val Asn Gly Thr Gln Gly Phe Gln
35          40          45
Gly Cys Pro Gly Gln Arg Gly Val Lys Gly Ser Arg Gly Phe Pro Gly
50          55          60
Glu Lys Gly Glu Val Gly Glu Ile Gly Leu Asp Gly Leu Asp Gly Glu
65          70          75          80
Asp Gly Asp Lys Gly Leu Pro Gly Ser Ser Gly Glu Lys Gly Asn Pro
85          90          95
Gly Arg Arg Gly Asp Lys Gly Pro Arg Gly Glu Lys Gly Glu Arg Gly
100         105         110
Asp Val Gly Ile Arg Gly Asp Pro Gly Asn Pro Gly Gln Asp Ser Gln
115        120        125
Glu Arg Gly Pro Lys Gly Glu Thr Gly Asp Leu Gly Pro Met Gly Val
130        135        140
Pro Gly Arg Asp Gly Val Pro Gly Gly Pro Gly Glu Thr Gly Lys Asn
145        150        155        160
Gly Gly Phe Gly Arg Arg Gly Pro Pro Gly Ala Lys Gly Asn Lys Gly
165        170        175
Gly Pro Gly Gln Pro Gly Phe Glu Gly Glu Gln Gly Thr Arg Gly Ala
180        185        190
Gln Gly Pro Ala Gly Pro Ala Gly Pro Pro Gly Leu Ile Gly Glu Gln
195        200        205
Gly Ile Ser Gly Pro Arg Gly Ser Gly Gly Ala Ala Gly Ala Pro Gly
210        215        220
Glu Arg Gly Arg Thr Gly Pro Leu Gly Arg Lys Gly Glu Pro Gly Glu
225        230        235        240
Pro Gly Pro Lys Gly Gly Ile Gly Asn Arg Gly Pro Arg Gly Glu Thr
245        250        255

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Gly Asp Asp Gly Arg Asp Gly Val Gly Ser Glu Gly Arg Arg Gly Lys
 260 265 270

Lys Gly Glu Arg Gly Phe Pro Gly Tyr Pro Gly Pro Lys Gly Asn Pro
 275 280 285

Gly Glu Pro Gly Leu Asn Gly Thr Thr Gly Pro Lys Gly Ile Arg Gly
 290 295 300

Arg Arg Gly Asn Ser Gly Pro Pro Gly Ile Val Gly Gln Lys Gly Asp
 305 310 315 320

Pro Gly Tyr Pro Gly Pro Ala Gly Pro Lys Gly Asn Arg Gly Asp Ser
 325 330 335

<210> SEQ ID NO 175

<211> LENGTH: 336

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 175

Gly Gln Arg Gly Asp Arg Gly Pro Ile Gly Ser Ile Gly Pro Lys Gly
 1 5 10 15

Ile Pro Gly Glu Asp Gly Tyr Arg Gly Tyr Pro Gly Asp Glu Gly Gly
 20 25 30

Pro Gly Glu Arg Gly Pro Pro Gly Val Asn Gly Thr Gln Gly Phe Gln
 35 40 45

Gly Cys Pro Gly Gln Arg Gly Val Lys Gly Ser Arg Gly Phe Pro Gly
 50 55 60

Glu Lys Gly Glu Val Gly Glu Ile Gly Leu Asp Gly Leu Asp Gly Glu
 65 70 75 80

Asp Gly Asp Lys Gly Leu Pro Gly Ser Ser Gly Glu Lys Gly Asn Pro
 85 90 95

Gly Arg Arg Gly Asp Lys Gly Pro Arg Gly Glu Lys Gly Glu Arg Gly
 100 105 110

Asp Val Gly Ile Arg Gly Asp Pro Gly Asn Pro Gly Gln Asp Ser Gln
 115 120 125

Glu Arg Gly Pro Lys Gly Glu Thr Gly Asp Leu Gly Pro Met Gly Val
 130 135 140

Pro Gly Arg Asp Gly Val Pro Gly Gly Pro Gly Glu Thr Gly Lys Asn
 145 150 155 160

Gly Gly Phe Gly Arg Arg Gly Pro Pro Gly Ala Lys Gly Asn Lys Gly
 165 170 175

Gly Pro Gly Gln Pro Gly Phe Glu Gly Glu Gln Gly Thr Arg Gly Ala
 180 185 190

Gln Gly Pro Ala Gly Pro Ala Gly Pro Pro Gly Leu Ile Gly Glu Gln
 195 200 205

Gly Ile Ser Gly Pro Arg Gly Ser Gly Gly Ala Ala Gly Ala Pro Gly
 210 215 220

Glu Arg Gly Arg Thr Gly Pro Leu Gly Arg Lys Gly Glu Pro Gly Glu
 225 230 235 240

Pro Gly Pro Lys Gly Gly Ile Gly Asn Arg Gly Pro Arg Gly Glu Thr
 245 250 255

Gly Asp Asp Gly Arg Asp Gly Val Gly Ser Glu Gly Arg Arg Gly Lys
 260 265 270

Lys Gly Glu Arg Gly Phe Pro Gly Tyr Pro Gly Pro Lys Gly Asn Pro
 275 280 285

Gly Glu Pro Gly Leu Asn Gly Thr Thr Gly Pro Lys Gly Ile Arg Gly
 290 295 300

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Arg Arg Gly Asn Ser Gly Pro Pro Gly Ile Val Gly Gln Lys Gly Asp
305 310 315 320

Pro Gly Tyr Pro Gly Pro Ala Gly Pro Lys Gly Asn Arg Gly Asp Ser
325 330 335

<210> SEQ ID NO 176

<211> LENGTH: 336

<212> TYPE: PRT

<213> ORGANISM: Equus caballus

<400> SEQUENCE: 176

Gly Gln Arg Gly Asp Arg Gly Pro Ile Gly Ser Ile Gly Pro Lys Gly
1 5 10 15

Val Pro Gly Glu Asp Gly Tyr Arg Gly Tyr Pro Gly Asp Glu Gly Gly
20 25 30

Pro Gly Asp Arg Gly Pro Pro Gly Val Asn Gly Thr Gln Gly Phe Gln
35 40 45

Gly Cys Pro Gly Gln Arg Gly Ile Lys Gly Ser Arg Gly Phe Pro Gly
50 55 60

Glu Lys Gly Glu Leu Gly Glu Ile Gly Leu Asp Gly Leu Asp Gly Glu
65 70 75 80

Asp Gly Asp Lys Gly Leu Pro Gly Ser Ser Gly Glu Lys Gly Asn Pro
85 90 95

Gly Arg Arg Gly Asp Lys Gly Pro Lys Gly Asp Gln Gly Glu Arg Gly
100 105 110

Asp Val Gly Ile Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln
115 120 125

Gln Arg Gly Pro Lys Gly Glu Thr Gly Asp Leu Gly Pro Met Gly Leu
130 135 140

Pro Gly Ser Asp Gly Val Ser Gly Ser Pro Gly Glu Pro Gly Arg Asp
145 150 155 160

Gly Gly Phe Gly Arg Arg Gly Pro Pro Gly Ala Lys Gly Asn Lys Gly
165 170 175

Gly Pro Gly Gln Gln Gly Thr Val Gly Glu Gln Gly Thr Arg Gly Ala
180 185 190

Gln Gly Pro Pro Gly Ser Thr Gly Pro Pro Gly Leu Ile Gly Glu Gln
195 200 205

Gly Ile Pro Gly Pro Arg Gly Ser Gly Gly Ala Val Gly Val Pro Gly
210 215 220

Glu Arg Gly Arg Thr Gly Pro Leu Gly Arg Lys Gly Glu Pro Gly Glu
225 230 235 240

Pro Gly Ala Lys Gly Gly Leu Gly Pro Arg Gly Pro Arg Gly Glu Thr
245 250 255

Gly Asp Asp Gly Arg Asp Gly Val Gly Ser Glu Gly Gln Lys Gly Lys
260 265 270

Lys Gly Glu Arg Gly Phe Pro Gly Tyr Pro Gly Pro Lys Gly Thr Arg
275 280 285

Gly Glu Pro Gly Thr Asp Gly Thr Leu Gly Pro Lys Gly Val Arg Gly
290 295 300

Arg Arg Gly Asp Ser Gly Pro Pro Gly Ala Ala Gly Gln Lys Gly Asp
305 310 315 320

Pro Gly Tyr Pro Gly Pro Ser Gly Leu Lys Gly Asn Arg Gly Asp Ser
325 330 335

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<210> SEQ ID NO 177
<211> LENGTH: 59
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
        COL6A3 sequence

<400> SEQUENCE: 177

Arg Gly Asp Pro Gly Asn Pro Gly Gln Asp Ser Gln Glu Arg Gly Pro
1             5             10             15

Lys Gly Glu Thr Gly Asp Leu Gly Pro Met Gly Val Pro Gly Arg Asp
                20             25             30

Gly Val Pro Gly Gly Pro Gly Glu Thr Gly Lys Asn Gly Gly Phe Gly
            35             40             45

Arg Arg Gly Pro Pro Gly Ala Lys Gly Asn Lys
50             55

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<210> SEQ ID NO 178
<211> LENGTH: 59
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
        COL6A3 sequence

<400> SEQUENCE: 178

Arg Gly Asp Pro Gly Glu Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
1             5             10             15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
                20             25             30

Gly Val Ser Gly Ser Pro Gly Glu Thr Gly Lys Asp Gly Gly Phe Gly
            35             40             45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Arg
50             55

```

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<210> SEQ ID NO 179
<211> LENGTH: 59
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
        COL6A3 sequence

<400> SEQUENCE: 179

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
1             5             10             15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
                20             25             30

Gly Val Pro Gly Arg Pro Gly Asp Pro Gly Lys Asp Gly Gly Tyr Gly
            35             40             45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Arg
50             55

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<210> SEQ ID NO 180
<211> LENGTH: 59
<212> TYPE: PRT
<213> ORGANISM: Unknown
<220> FEATURE:
<223> OTHER INFORMATION: Description of Unknown:
        COL6A3 sequence

<400> SEQUENCE: 180

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-continued

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15
 Lys Gly Glu Thr Gly Asp Leu Gly Pro Met Gly Leu Pro Gly Ser Asp
 20 25 30
 Gly Val Ser Gly Ser Pro Gly Glu Pro Gly Arg Asp Gly Gly Phe Gly
 35 40 45
 Arg Arg Gly Pro Pro Gly Ala Lys Gly Asn Lys
 50 55

<210> SEQ ID NO 181
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 COL6A3 sequence

<400> SEQUENCE: 181

Arg Gly Asp Pro Gly Asp Ser Gly Gln Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15
 Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
 20 25 30
 Gly Val Ser Gly Ser Pro Gly Glu Pro Gly Lys Ser Gly Gly Phe Gly
 35 40 45
 Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys
 50 55

<210> SEQ ID NO 182
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 COL6A3 sequence

<400> SEQUENCE: 182

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15
 Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
 20 25 30
 Gly Val Ser Gly Arg Pro Gly Glu Thr Gly Lys Asp Gly Gly Phe Gly
 35 40 45
 Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys
 50 55

<210> SEQ ID NO 183
 <211> LENGTH: 56
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 COL6A3 sequence

<400> SEQUENCE: 183

Arg Gly Asp Pro Gly Asn Ser Gly Gln Asp Ser Gln Gln Arg Gly Ala
 1 5 10 15
 Lys Gly Glu Thr Gly Asp Ile Gly Pro Val Gly Leu Pro Gly Arg Asp
 20 25 30
 Gly Val Ser Gly Gly Pro Gly Glu Pro Gly Lys Ser Gly Gly Phe Gly
 35 40 45
 Arg Arg Gly Pro Ala Gly Ala Lys

-continued

50 55

<210> SEQ ID NO 184
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 COL6A3 sequence

<400> SEQUENCE: 184

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
 20 25 30

Gly Val Ser Gly Ser Pro Gly Glu Thr Gly Lys Asp Gly Gly Phe Gly
 35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys
 50 55

<210> SEQ ID NO 185
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Papio anubis

<400> SEQUENCE: 185

Arg Gly Asp Pro Gly Asn Pro Gly Gln Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Thr Gly Asp Leu Gly Pro Met Gly Ala Pro Gly Thr Asp
 20 25 30

Gly Val Pro Gly Gly Pro Gly Glu Thr Gly Lys Asn Gly Gly Phe Gly
 35 40 45

Arg Arg Gly Pro Pro Gly Ala Lys Gly Asn Lys
 50 55

<210> SEQ ID NO 186
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Macaca mulatta

<400> SEQUENCE: 186

Arg Gly Asp Pro Gly Asn Pro Gly Gln Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Thr Gly Asp Leu Gly Pro Met Gly Val Pro Gly Arg Asp
 20 25 30

Gly Val Pro Gly Gly Pro Gly Glu Thr Gly Lys Asn Gly Gly Phe Gly
 35 40 45

Arg Arg Gly Pro Pro Gly Ala Lys Gly Asn Lys
 50 55

<210> SEQ ID NO 187
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Cebus capucinus imitator

<400> SEQUENCE: 187

Arg Gly Asp Pro Gly Asn Pro Gly Gln Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Val Gly Val Pro Gly Arg Asp
 20 25 30

-continued

Gly Val Pro Gly Gly Pro Gly Glu Thr Gly Lys Asn Gly Gly Phe Gly
35 40 45

Arg Arg Gly Pro Pro Gly Val Lys Gly Asn Lys
50 55

<210> SEQ ID NO 188
<211> LENGTH: 59
<212> TYPE: PRT
<213> ORGANISM: *Rhinopithecus bieti*

<400> SEQUENCE: 188

Arg Gly Asp Pro Gly Asn Pro Gly Gln Asp Ser Gln Gln Arg Gly Pro
1 5 10 15

Lys Gly Glu Ile Gly Asp Leu Gly Pro Met Gly Val Pro Gly Arg Asp
20 25 30

Gly Val Pro Gly Gly Pro Gly Glu Thr Gly Lys Asn Gly Gly Phe Gly
35 40 45

Arg Arg Gly Pro Pro Gly Ala Lys Gly Asn Lys
50 55

<210> SEQ ID NO 189
<211> LENGTH: 59
<212> TYPE: PRT
<213> ORGANISM: *Microcebus murinus*

<400> SEQUENCE: 189

Arg Gly Asp Pro Gly Glu Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
20 25 30

Gly Val Ser Gly Ser Pro Gly Glu Asn Gly Arg Asn Gly Gly Phe Gly
35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys
50 55

<210> SEQ ID NO 190
<211> LENGTH: 59
<212> TYPE: PRT
<213> ORGANISM: *Carlito syrichta*

<400> SEQUENCE: 190

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
20 25 30

Gly Val Pro Gly Arg Pro Gly Glu Thr Gly Lys Asn Gly Gly Phe Gly
35 40 45

Arg Arg Gly Ser Ala Gly Ala Lys Gly Asn Lys
50 55

<210> SEQ ID NO 191
<211> LENGTH: 59
<212> TYPE: PRT
<213> ORGANISM: *Otolemur garnettii*

<400> SEQUENCE: 191

Arg Gly Asp Pro Gly Glu Ser Gly Gln Asp Ser Gln Gln Arg Gly Pro
1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
20 25 30

-continued

Gly Val Ser Gly Ser Pro Gly Glu Thr Gly Lys Asn Gly Gly Phe Gly
 35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys
 50 55

<210> SEQ ID NO 192
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Propithecus coquereli

<400> SEQUENCE: 192

Arg Gly Asp Pro Gly Glu Ser Gly Gln Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
 20 25 30

Gly Val Ser Gly Ser Pro Gly Glu Asn Gly Lys Asn Gly Gly Phe Gly
 35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys
 50 55

<210> SEQ ID NO 193
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 193

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
 20 25 30

Gly Ile Pro Gly Ser Pro Gly Asp Pro Gly Lys Asp Gly Gly Ser Gly
 35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Arg
 50 55

<210> SEQ ID NO 194
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 194

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
 20 25 30

Gly Ile Pro Gly Ser Pro Gly Asp Pro Gly Lys Asp Gly Gly Phe Gly
 35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Arg
 50 55

<210> SEQ ID NO 195
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Cricetulus griseus

<400> SEQUENCE: 195

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp

-continued

20	25	30
Gly Ile Pro Gly Arg Pro Gly Asp Pro Gly Lys Asp Gly Gly Ser Gly		
35	40	45

Arg Arg Gly Pro Ala Gly Lys Gly Asn Arg
50 55

<210> SEQ ID NO 196
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Microtus ochrogaster*

<400> SEQUENCE: 196

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
20 25 30

Gly Val Pro Gly Arg Pro Gly Asp Pro Gly Lys Asp Gly Gly Phe Gly
35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Arg
50 55

<210> SEQ ID NO 197
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Meriones unguiculatus*

<400> SEQUENCE: 197

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
20 25 30

Gly Ile Ser Gly Arg Pro Gly Asp Pro Gly Lys Asp Gly Gly Ser Gly
35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys
50 55

<210> SEQ ID NO 198
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Chinchilla lanigera*

<400> SEQUENCE: 198

Arg Gly Asp Pro Gly Glu Ser Gly Gln Asp Ser Gln Gln Arg Gly Pro
1 5 10 15

Lys Gly Glu Thr Gly Asp Leu Gly Pro Met Gly Leu Pro Gly Arg Asp
20 25 30

Gly Val Ser Gly Ser Pro Gly Glu Thr Gly Lys Asp Gly Gly Phe Gly
35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Arg
50 55

<210> SEQ ID NO 199
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Castor canadensis*

<400> SEQUENCE: 199

Arg Gly Asp Pro Gly Asp Ser Gly Gln Asp Ser Gln Gln Arg Gly Pro
1 5 10 15

-continued

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Val Pro Gly Arg Asp
 20 25 30

Gly Ala Ser Gly Ser Pro Gly Glu Thr Gly Lys Asp Gly Gly Phe Gly
 35 40 45

Arg Arg Gly Pro Ala Gly Val Lys Gly Asn Arg
 50 55

<210> SEQ ID NO 200
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Fukomys damarensis

<400> SEQUENCE: 200

Arg Gly Asp Pro Gly Asp Ser Gly Gln Asp Asn Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
 20 25 30

Gly Val Pro Gly Ser Pro Gly Gln Ala Gly Lys Asn Gly Gly Phe Gly
 35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Arg
 50 55

<210> SEQ ID NO 201
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Pteropus vampyrus

<400> SEQUENCE: 201

Arg Gly Val Lys Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Ile Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
 20 25 30

Gly Val Ser Gly Ser Pro Gly Glu Pro Gly Lys Asp Gly Gly Phe Gly
 35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys
 50 55

<210> SEQ ID NO 202
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Manis javanica

<400> SEQUENCE: 202

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Ser Gln Gln Arg Gly Pro
 1 5 10 15

Lys Gly Glu Thr Gly Asp Leu Gly Pro Met Gly Leu Pro Gly Ser Asp
 20 25 30

Gly Leu Pro Gly Ser Ser Gly Glu Pro Gly Lys Asn Gly Gly Phe Gly
 35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Arg
 50 55

<210> SEQ ID NO 203
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Hipposideros armiger

<400> SEQUENCE: 203

Arg Gly Asp Pro Gly Glu Pro Gly Arg Asp Asn Gln Gln Arg Gly Pro
 1 5 10 15

-continued

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
20 25 30

Gly Ala Pro Gly Arg Pro Gly Glu Pro Gly Glu Ser Gly Gly Phe Gly
35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys
50 55

<210> SEQ ID NO 204

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Loxodonta africana*

<400> SEQUENCE: 204

Arg Gly Asp Pro Gly Asp Ser Gly Arg Asp Gly Gln Gln Pro Gly Pro
1 5 10 15

Lys Gly Glu Thr Gly Glu Ile Gly Pro Met Gly Leu Pro Gly Arg Asp
20 25 30

Gly Val Ser Gly Gly Pro Gly Glu Pro Gly Lys Asp Gly Ser Phe Gly
35 40 45

Arg Arg Gly Pro Ala Gly Asn Lys Gly Ser Lys
50 55

<210> SEQ ID NO 205

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Sus scrofa*

<400> SEQUENCE: 205

Arg Gly Asp Pro Gly Asp Thr Gly Arg Asp Ser Gln Gln Arg Gly Pro
1 5 10 15

Lys Gly Glu Ser Gly Asp Leu Gly Pro Met Gly Leu Pro Gly Arg Asp
20 25 30

Gly Val Ser Gly Ser Pro Gly Glu Pro Gly Lys Ser Gly Gly Phe Gly
35 40 45

Arg Arg Gly Pro Ala Gly Leu Lys Gly Asn Lys
50 55

<210> SEQ ID NO 206

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Balaenoptera acutorostrata scammoni*

<400> SEQUENCE: 206

Arg Gly Asp Pro Gly Asn Ser Gly Gln Asp Ser Gln Gln Arg Gly Pro
1 5 10 15

Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Ala Asp
20 25 30

Gly Val Ser Gly Gly Pro Gly Glu Pro Gly Lys Ser Gly Gly Phe Gly
35 40 45

Arg Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys
50 55

<210> SEQ ID NO 207

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Orcinus orca*

<400> SEQUENCE: 207

Arg Gly Asp Pro Gly Asn Ser Gly Gln Asp Ser Gln Gln Arg Gly Pro

-continued

1	5	10	15
Lys Gly Glu Thr Gly Asp Ile Gly Pro Met Gly Leu Pro Gly Thr Asp			
	20	25	30
Gly Val Ser Gly Ala Pro Gly Glu Pro Gly Lys Ser Gly Gly Phe Gly			
	35	40	45
Gln Arg Gly Pro Ala Gly Ala Lys Gly Asn Lys			
	50	55	

<210> SEQ ID NO 208
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Phascolarctos cinereus

<400> SEQUENCE: 208

Arg Gly Asp Pro Gly Glu Ile Gly Leu Asp Ser Gln Gln Gln Gly Ala			
1	5	10	15
Glu Gly Glu Lys Gly Glu Ile Gly Pro Met Gly Met Pro Gly Leu Asp			
	20	25	30
Gly Thr Pro Gly Gly Pro Gly Val Val Gly Lys Asp Gly Gly Pro Gly			
	35	40	45
Arg Arg Gly Pro Pro Gly Val Lys Gly Asn Lys			
	50	55	

<210> SEQ ID NO 209
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Sarcophilus harrisii

<400> SEQUENCE: 209

Arg Gly Asp Pro Gly Glu Ile Gly Lys Asp Asn Gln Gln Gln Gly Pro			
1	5	10	15
Arg Gly Glu Lys Gly Glu Ile Gly Pro Met Gly Val Pro Gly Leu Asp			
	20	25	30
Gly Arg Ala Gly Gly Pro Gly Gly Leu Gly Lys Asp Gly Gly Pro Gly			
	35	40	45
Arg Arg Gly Pro Pro Gly Ala Lys Gly Asn Lys			
	50	55	

<210> SEQ ID NO 210
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Ornithorhynchus anatinus

<400> SEQUENCE: 210

Arg Gly Asp Pro Gly Asp Ser Gly Leu Asp Ser Arg Gln Arg Gly Pro			
1	5	10	15
Lys Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Leu Pro Gly Asn Asp			
	20	25	30
Gly Ala Pro Gly Ser Pro Gly Gly Asn Gly Lys Glu Gly Gly Phe Gly			
	35	40	45
Arg Arg Gly Thr Val Gly Val Lys Gly Ser Lys			
	50	55	

<210> SEQ ID NO 211
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Unknown
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of Unknown:
 COL6A3 sequence

-continued

<400> SEQUENCE: 211

```

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Asp Pro Gly Pro Ala
          20           25           30
Gly Leu Glu Gly Gln Asp Gly Gly Val Gly Arg Lys Gly Met Ala Gly
          35           40           45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
          50           55

```

<210> SEQ ID NO 212

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Haliaeetus leucocephalus*

<400> SEQUENCE: 212

```

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Val
          20           25           30
Gly Leu Asn Gly Gln Asp Gly Gly Val Gly Arg Lys Gly Met Thr Gly
          35           40           45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
          50           55

```

<210> SEQ ID NO 213

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Aquila chrysaetos canadensis*

<400> SEQUENCE: 213

```

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Val
          20           25           30
Gly Leu Asp Gly Gln Asp Gly Gly Val Gly Arg Lys Gly Met Thr Gly
          35           40           45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
          50           55

```

<210> SEQ ID NO 214

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Gallus gallus*

<400> SEQUENCE: 214

```

Arg Gly Asp Pro Gly Glu Ser Gly Ala Asp Asn Thr Gln Arg Gly Thr
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Gln Met Gly Glu Pro Gly Pro Ala
          20           25           30
Gly Gln Arg Gly Gln Asp Gly Gly Val Gly Arg Lys Gly Met Ala Gly
          35           40           45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
          50           55

```

<210> SEQ ID NO 215

<211> LENGTH: 59

<212> TYPE: PRT

-continued

<213> ORGANISM: *Picoides pubescens*

<400> SEQUENCE: 215

Arg	Gly	Asp	Pro	Gly	Asp	Ser	Gly	Val	Asp	Asn	Thr	Gln	Arg	Gly	Ser
1				5					10					15	

Arg	Gly	Gln	Lys	Gly	Glu	Ile	Gly	Pro	Met	Gly	Glu	Thr	Gly	Pro	Val
		20						25					30		

Gly	Leu	Glu	Gly	Gln	Asp	Gly	Gly	Val	Gly	Arg	Lys	Gly	Met	Thr	Gly
	35					40						45			

Arg	Arg	Gly	Pro	Ile	Gly	Val	Lys	Gly	Thr	Lys
	50				55					

<210> SEQ ID NO 216

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Meleagris gallopavo*

<400> SEQUENCE: 216

Arg	Gly	Asp	Pro	Gly	Asp	Ser	Gly	Ala	Asp	Asn	Thr	Gln	Arg	Gly	Thr
1				5					10					15	

Arg	Gly	Gln	Lys	Gly	Glu	Ile	Gly	Gln	Met	Gly	Glu	Pro	Gly	Pro	Ala
		20						25					30		

Gly	Gln	Arg	Gly	Pro	Asp	Gly	Gly	Val	Gly	Arg	Gln	Gly	Met	Ala	Gly
	35					40						45			

Arg	Arg	Gly	Pro	Ile	Gly	Val	Lys	Gly	Thr	Lys
	50				55					

<210> SEQ ID NO 217

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Merops nubicus*

<400> SEQUENCE: 217

Arg	Gly	Asp	Pro	Gly	Asp	Ser	Gly	Ala	Asp	Asn	Thr	Gln	Arg	Gly	Ser
1				5					10					15	

Lys	Gly	Gln	Lys	Gly	Glu	Ile	Gly	Pro	Met	Gly	Glu	Pro	Gly	Pro	Val
		20						25					30		

Gly	Leu	Asn	Gly	Gln	Asp	Gly	Gly	Val	Gly	Arg	Lys	Gly	Met	Thr	Gly
	35					40						45			

Arg	Arg	Gly	Pro	Ile	Gly	Val	Lys	Gly	Thr	Lys
	50				55					

<210> SEQ ID NO 218

<211> LENGTH: 57

<212> TYPE: PRT

<213> ORGANISM: *Colius striatus*

<400> SEQUENCE: 218

Arg	Gly	Asp	Pro	Gly	Asp	Ser	Gly	Ala	Asp	Asn	Thr	Gln	Arg	Gly	Ser
1				5					10					15	

Arg	Gly	Gln	Lys	Gly	Glu	Ile	Gly	Pro	Met	Gly	Glu	Pro	Gly	Pro	Ala
		20						25					30		

Gly	Leu	Asp	Gly	Gln	Asp	Gly	Gly	Val	Gly	Arg	Lys	Gly	Met	Thr	Gly
	35					40						45			

Arg	Arg	Gly	Pro	Ile	Gly	Val	Lys	Gly
	50				55			

<210> SEQ ID NO 219

<211> LENGTH: 59

-continued

<212> TYPE: PRT

<213> ORGANISM: *Apteryx australis mantelli*

<400> SEQUENCE: 219

```

Arg Gly Asp Pro Gly Glu Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Gly Ala
           20           25           30
Gly Ala Asn Gly Gln Asn Gly Gly Ile Gly Arg Lys Gly Met Ala Gly
           35           40           45
Arg Arg Gly Pro Ile Gly Ile Lys Gly Thr Lys
           50           55

```

<210> SEQ ID NO 220

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Aptenodytes forsteri*

<400> SEQUENCE: 220

```

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Ala
           20           25           30
Gly Leu Asp Gly Gln Asp Gly Gly Val Gly Arg Lys Gly Met Thr Gly
           35           40           45
Arg Arg Gly Leu Ile Gly Val Lys Gly Thr Lys
           50           55

```

<210> SEQ ID NO 221

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Columba livia*

<400> SEQUENCE: 221

```

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Ala
           20           25           30
Gly Leu Glu Gly Gln Glu Gly Gly Val Gly Arg Arg Gly Met Ala Gly
           35           40           45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
           50           55

```

<210> SEQ ID NO 222

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: *Patagioenas fasciata monilis*

<400> SEQUENCE: 222

```

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Ala
           20           25           30
Gly Leu Glu Gly Gln Glu Gly Gly Val Gly Arg Arg Gly Met Ala Gly
           35           40           45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
           50           55

```

<210> SEQ ID NO 223

-continued

<211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Pygoscelis adeliae*

<400> SEQUENCE: 223

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
 1 5 10 15
 Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Val
 20 25 30
 Gly Leu Asp Gly Gln Gly Gly Gly Val Gly Arg Lys Gly Met Thr Gly
 35 40 45
 Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
 50 55

<210> SEQ ID NO 224
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Calidris pugnax*

<400> SEQUENCE: 224

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
 1 5 10 15
 Arg Gly Pro Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Ala
 20 25 30
 Gly Leu Asp Gly Gln Asp Gly Gly Val Gly Arg Lys Gly Met Pro Gly
 35 40 45
 Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
 50 55

<210> SEQ ID NO 225
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Coturnix japonica*

<400> SEQUENCE: 225

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Thr
 1 5 10 15
 Arg Gly Gln Lys Gly Glu Ile Gly Gln Met Gly Glu Pro Gly Pro Ala
 20 25 30
 Gly Gln Arg Gly Pro Asp Gly Gly Val Gly Arg Gln Gly Met Ala Gly
 35 40 45
 Arg Arg Gly Thr Ile Gly Ile Lys Gly Thr Lys
 50 55

<210> SEQ ID NO 226
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Nipponia nippon*

<400> SEQUENCE: 226

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
 1 5 10 15
 Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Ala Pro Gly Pro Val
 20 25 30
 Gly Leu Gly Gly Glu Glu Gly Gly Val Gly Arg Lys Gly Met Thr Gly
 35 40 45
 Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
 50 55

-continued

<210> SEQ ID NO 227

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: Calypte anna

<400> SEQUENCE: 227

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
 1 5 10 15

Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Val
 20 25 30

Gly Val Asp Gly Gln Asp Gly Gly Val Gly Arg Lys Gly Met Thr Gly
 35 40 45

Arg Arg Gly Pro Ile Gly Pro Lys Gly Thr Lys
 50 55

<210> SEQ ID NO 228

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: Egretta garzetta

<400> SEQUENCE: 228

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
 1 5 10 15

Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Met
 20 25 30

Gly Leu Val Gly Gln Asp Gly Gly Val Gly Arg Lys Gly Met Thr Gly
 35 40 45

Arg Arg Gly Gln Ile Gly Val Lys Gly Thr Lys
 50 55

<210> SEQ ID NO 229

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: Serinus canaria

<400> SEQUENCE: 229

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Pro
 1 5 10 15

Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Asp Pro Gly Pro Ala
 20 25 30

Gly Pro Glu Gly Gln Asp Gly Gly Val Gly Arg Arg Gly Met Ala Gly
 35 40 45

Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
 50 55

<210> SEQ ID NO 230

<211> LENGTH: 59

<212> TYPE: PRT

<213> ORGANISM: Sturnus vulgaris

<400> SEQUENCE: 230

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
 1 5 10 15

Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Asp Pro Gly Pro Ala
 20 25 30

Gly Leu Glu Gly Gln Asp Gly Gly Val Gly Arg Arg Gly Met Pro Gly
 35 40 45

Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
 50 55

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<210> SEQ ID NO 231
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Taeniopygia guttata*

<400> SEQUENCE: 231

```

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Pro
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Asp Pro Gly Pro Ala
           20           25           30
Gly Leu Glu Gly Pro Asp Gly Glu Val Gly Arg Arg Gly Met Ala Gly
           35           40           45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
           50           55

```

<210> SEQ ID NO 232
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Lonchura striata domestica*

<400> SEQUENCE: 232

```

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Pro
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Asp Pro Gly Pro Ala
           20           25           30
Gly Pro Glu Gly Pro Asp Gly Glu Val Gly Arg Arg Gly Met Ala Gly
           35           40           45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
           50           55

```

<210> SEQ ID NO 233
 <211> LENGTH: 58
 <212> TYPE: PRT
 <213> ORGANISM: *Chaetura pelagica*

<400> SEQUENCE: 233

```

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Pro
1           5           10           15
Lys Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Glu
           20           25           30
Gly Val Ala Gly Gln Ala Gly Gly Ala Gly Arg Lys Gly Met Ala Gly
           35           40           45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr
           50           55

```

<210> SEQ ID NO 234
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Callipepla squamata*

<400> SEQUENCE: 234

```

Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Thr
1           5           10           15
Arg Gly Gln Lys Gly Glu Ile Gly Gln Met Gly Glu Pro Gly Pro Val
           20           25           30
Gly Gln Arg Gly Pro Asp Gly Gly Val Gly Arg Met Gly Met Ala Gly
           35           40           45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
           50           55

```

-continued

<210> SEQ ID NO 235
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Cuculus canorus*

<400> SEQUENCE: 235

```
Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Thr
1          5          10          15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Gln Ala
          20          25          30
Gly Pro Glu Gly Gln Glu Gly Gly Val Gly Arg Lys Gly Met Thr Gly
          35          40          45
Arg Arg Gly Ala Ile Gly Val Lys Gly Thr Lys
50          55
```

<210> SEQ ID NO 236
 <211> LENGTH: 58
 <212> TYPE: PRT
 <213> ORGANISM: *Lepidothrix coronata*

<400> SEQUENCE: 236

```
Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
1          5          10          15
Arg Gly Gln Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Pro Pro
          20          25          30
Gly Leu Lys Gly Gln Asp Gly Gly Val Gly Arg Lys Gly Met Thr Gly
          35          40          45
Arg Arg Gly Gln Ile Gly Val Lys Gly Thr
50          55
```

<210> SEQ ID NO 237
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Falco peregrinus*

<400> SEQUENCE: 237

```
Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Ser
1          5          10          15
Arg Gly Pro Lys Gly Glu Ile Gly Pro Met Gly Glu Pro Gly Leu Val
          20          25          30
Gly Leu Asp Gly Gln Asp Gly Gly Val Gly Arg Lys Gly Met Thr Gly
          35          40          45
Arg Arg Gly Gln Ile Gly Ile Lys Gly Thr Lys
50          55
```

<210> SEQ ID NO 238
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: *Melopsittacus undulatus*

<400> SEQUENCE: 238

```
Arg Gly Asp Pro Gly Asp Ser Gly Ala Asp Asn Thr Gln Arg Gly Pro
1          5          10          15
Arg Gly Gln Lys Gly Glu Ile Gly Gln Met Gly Glu Pro Gly Pro Val
          20          25          30
Gly Pro Asp Gly Gln Asn Gly Glu Val Gly Arg Gln Gly Met Ala Gly
          35          40          45
Arg Arg Gly Pro Ile Gly Val Lys Gly Thr Lys
```

-continued

50 55

<210> SEQ ID NO 239
 <211> LENGTH: 52
 <212> TYPE: PRT
 <213> ORGANISM: Gekko japonicus

<400> SEQUENCE: 239

Glu Lys Gly Asn Lys Gly Tyr Thr Gly Pro Leu Gly Pro Lys Gly Gln
 1 5 10 15

Pro Gly Arg Pro Gly Leu Gln Gly Pro Pro Gly Ala Thr Gly Arg Pro
 20 25 30

Gly Val Ser Arg Pro Gly Gln Arg Gly Pro Gln Gly Thr Pro Gly Leu
 35 40 45

Lys Gly Asn Lys
 50

<210> SEQ ID NO 240
 <211> LENGTH: 58
 <212> TYPE: PRT
 <213> ORGANISM: Anolis carolinensis

<400> SEQUENCE: 240

Lys Gly Gln Pro Gly Arg Pro Gly Val Pro Gly Ile Pro Gly Pro Arg
 1 5 10 15

Gly Glu Met Gly Leu Met Gly Pro Val Gly Asn Pro Gly Leu Ser Gly
 20 25 30

Val Pro Gly Asn Pro Gly Ile Pro Gly Ile Arg Gly Ser Pro Gly Phe
 35 40 45

Ser Gly Leu Lys Gly Arg Lys Gly Ser Ser
 50 55

<210> SEQ ID NO 241
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Python bivittatus

<400> SEQUENCE: 241

Arg Gly Asp Pro Gly Thr Ala Gly Thr Asp Ser Thr Gln Lys Gly Pro
 1 5 10 15

Lys Gly Gln Lys Gly Glu Leu Gly Pro Ala Gly Glu Ser Gly Arg Asp
 20 25 30

Gly Pro Arg Gly Asp Ala Gly Gly Ile Gly Arg Asp Gly Gly Leu Gly
 35 40 45

Arg Arg Gly Gln Pro Gly Ile Lys Gly Asn Lys
 50 55

<210> SEQ ID NO 242
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Protobothrops mucrosquamatus

<400> SEQUENCE: 242

Arg Gly Asn Pro Gly Ser Phe Gly Thr Asp Ser Ser Asp Arg Gly Leu
 1 5 10 15

Lys Gly Glu Lys Gly Glu Leu Gly Ser Ala Gly Glu Leu Gly Arg Asn
 20 25 30

Gly Pro Arg Gly Asp Ala Gly Gly Ala Gly Arg Asp Gly Ala Leu Gly
 35 40 45

-continued

Arg Arg Gly Pro Pro Gly Ile Lys Gly Asn Lys
50 55

<210> SEQ ID NO 243
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Ophiophagus hannah

<400> SEQUENCE: 243

Gly Lys Pro Gly Thr Pro Gly Phe Arg Gly Glu Arg Gly Leu Met Gly
1 5 10 15
 His Gly Gly Ser Pro Gly Leu Pro Gly Val Ser Gly Asn Pro Gly Ile
20 25 30
 Pro Gly Gly Arg Gly Ser Pro Gly Phe Pro Gly Ser Lys Gly Arg Lys
35 40 45
 Gly Ser Phe Gly Glu Lys Gly Glu Pro Gly Asp
50 55

<210> SEQ ID NO 244
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Thamnophis sirtalis

<400> SEQUENCE: 244

Arg Gly Asp Pro Gly Thr Phe Gly Thr Asp Ser Thr Asp Arg Gly Pro
1 5 10 15
 Lys Gly Glu Lys Gly Glu Leu Gly Pro Ala Gly Glu Leu Gly Arg Asp
20 25 30
 Gly Pro Arg Gly Asp Ala Gly Glu Ala Gly Arg Asp Gly Ser Leu Gly
35 40 45
 Lys Arg Gly Pro Pro Gly Ile Lys Gly Asn Lys
50 55

<210> SEQ ID NO 245
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Latimeria chalumnae

<400> SEQUENCE: 245

Arg Gly Asp Pro Gly Thr Pro Gly Thr Asn Asn Val Arg Arg Gly Pro
1 5 10 15
 Lys Gly Gln Lys Gly Glu Ile Gly Pro Gln Gly Asp Pro Gly Asn Asp
20 25 30
 Gly Thr Ser Gly Thr Gly Gly Pro Arg Gly Lys Gly Gly Ser Ala Gly
35 40 45
 Arg Arg Gly Pro Pro Gly Thr Lys Gly Glu Gln
50 55

<210> SEQ ID NO 246
 <211> LENGTH: 59
 <212> TYPE: PRT
 <213> ORGANISM: Rhincodon typus

<400> SEQUENCE: 246

Arg Gly Asp Pro Gly Glu Pro Gly Leu Thr Asn Ile Gln Lys Gly Asp
1 5 10 15
 Lys Gly Gln Arg Gly Asp Met Gly Leu Val Gly Glu Thr Gly Ala Ser
20 25 30
 Gly Arg Pro Gly Ser Pro Gly Ala Ser Gly Arg Arg Gly Thr Gln Gly
35 40 45

-continued

Arg Arg Gly Pro Pro Gly Ala Arg Gly Ser Lys
50 55

<210> SEQ ID NO 247
<211> LENGTH: 59
<212> TYPE: PRT
<213> ORGANISM: Callorhinchus milii

<400> SEQUENCE: 247

Arg Gly Asp Pro Gly Thr Ser Gly Thr Asp Asn Thr Gln Arg Gly Ile
1 5 10 15

Lys Gly Gln Lys Gly Asp Ser Gly Ile Ala Gly Asp Pro Gly Thr Asp
20 25 30

Gly Thr Pro Gly Gln Arg Gly Pro Leu Gly Lys Asn Gly Ala Thr Gly
35 40 45

Arg Arg Gly Ser Ala Gly Pro Arg Gly Asp Lys
50 55

<210> SEQ ID NO 248
<211> LENGTH: 54
<212> TYPE: PRT
<213> ORGANISM: Salmo salar

<400> SEQUENCE: 248

Arg Gly Asp Pro Gly Thr Thr Gly Gln Asp Asn Asn Val Ala Gly Pro
1 5 10 15

Lys Gly Asp Pro Gly Asp Val Gly Pro Val Gly Glu Pro Gly Asp Asp
20 25 30

Gly Lys Met Gly Gly Pro Gly Glu Pro Gly Arg Thr Gly Ser Asp Gly
35 40 45

Arg Arg Gly Pro Pro Gly
50

<210> SEQ ID NO 249
<211> LENGTH: 57
<212> TYPE: PRT
<213> ORGANISM: Xenopus laevis

<400> SEQUENCE: 249

Arg Gly Asp Pro Gly Glu Pro Gly Thr Asn Ser Asn Asn Gln Gly Pro
1 5 10 15

Lys Gly Leu Lys Gly Asn Ser Gly Arg Gln Gly Glu Thr Gly Arg Asp
20 25 30

Gly Val Gln Gly Lys Arg Gly Asp Asn Gly Pro Asn Gly Ser Gln Gly
35 40 45

Arg Arg Gly Pro Pro Gly Leu Lys Gly
50 55

<210> SEQ ID NO 250
<211> LENGTH: 51
<212> TYPE: PRT
<213> ORGANISM: Cyprinus carpio

<400> SEQUENCE: 250

Arg Gly Asp Pro Gly Asn Pro Gly Ile Asp Asn Asn Ser Arg Gly Pro
1 5 10 15

Lys Gly Asp Ile Gly Ser Pro Gly Ile Gln Gly Glu Pro Gly Gln Asp
20 25 30

Gly Leu Ser Gly Gln Ile Gly Asp Glu Gly Arg Pro Gly Pro Asn Gly

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35	40	45	
Arg Arg Gly			
50			
<210> SEQ ID NO 251			
<211> LENGTH: 529			
<212> TYPE: DNA			
<213> ORGANISM: Equus caballus			
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tttccccctt ctcctgctca tgcccccttc tggttttcag atcaggggtc aggtgatcga			120
ggggcgccag ctgccagggg tgaacatcaa gcctgtggtc aaggtcaccc cggccaggca			180
gaccaagcgg actcggattc acaggggaaa cagcccgctc ttcaacgagg tgggagacat			240
ggcgtttttag ggctggtagc ttggtgggcc ttccagattg ggagcaccgc gcagatacct			300
ggcaattctt tcagtttttg ttcatggcgc taactttggt ttgagagggtg tgccagggtcc			360
tgagtacgtt atctgaggaa ctggagagag ggttctagtt cttattcctg tcccgggctc			420
ctggtgctcc acatccctgt ctctctgtgg ggccagccac ccatgctgtc ctggagatga			480
caacctctga gaggtcaggg gtggaacacc ccagaacttg tcgagtcct			529
<210> SEQ ID NO 252			
<211> LENGTH: 529			
<212> TYPE: DNA			
<213> ORGANISM: Artificial Sequence			
<220> FEATURE:			
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide			
<400> SEQUENCE: 252			
ggtccccgag atttctggct ttctttgtgg cgaggtcggg ggaggtgggt gcctgactgt			60
tttccccctt ctcctgctca tgcccccttc tggttttcag atcaggggtc aggtgatcga			120
ggggcgccag ctgccagggg tgaacatcaa gcctgtggtc aaggtcaccc cggccaggca			180
gaccaagtgg actcggattc acaggggaaa cagcccgctc ttcaacgagg tgggagacat			240
ggcgtttttag ggctggtagc ttggtgggcc ttccagattg ggagcaccgc gcagatacct			300
ggcaattctt tcagtttttg ttcatggcgc taactttggt ttgagagggtg tgccagggtcc			360
tgagtacgtt atctgaggaa ctggagagag ggttctagtt cttattcctg tcccgggctc			420
ctggtgctcc acatccctgt ctctctgtgg ggccagccac ccatgctgtc ctggagatga			480
caacctctga gaggtcaggg gtggaacacc ccagaacttg tcgagtcct			529
<210> SEQ ID NO 253			
<211> LENGTH: 450			
<212> TYPE: DNA			
<213> ORGANISM: Equus caballus			
<400> SEQUENCE: 253			
actggttgca aactccaac tgtgatcact gacatattcc atgagtgtca ttttgtggct			60
gaaggtgtgt ttgcctggcc cggacagaaa ggaggggtgat ggggccttgg gtgacaggca			120
ctgaccaaag ctcttttttc taccgccgag ccggccatct accacattcc tggttttgag			180
gtaagtcttg ctcttccctc ttcttcttca aactcatggc ccgcctctgt gtgtttgcag			240
ccccctatga gctaggaagg gcagtcaggt gtataccacc accggcccgcc cttcagggat			300
ggggtgggtg agggccgggg tccctggcca gctcaggcct ttccgcctct tcttccctga			360

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gactcagact gtagggacct catggtcatc cagcccacgg gctctcagct gggctgcatg 420
ttagaatccc ttcggcagct tgaaaaatca 450

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<210> SEQ ID NO 254
<211> LENGTH: 450
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

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<400> SEQUENCE: 254
actggttgca aactccaac tgtgatcact gacatattcc atgagtgtca tttgtggct 60
gaaggtgtgt ttgcctggcc cggacagaaa ggagggtgat ggggccttgg gtgacaggca 120
ctgaccaaag ctcttttttc taccgccgag cgggccatct accacattac tggttttgag 180
gtaagtcttg ctcttccctc ttcttcttca aactcatggc ccgcctctgt gtgtttgcag 240
ccccctatga gctaggaagg gcagtcaggt gtataccacc accggcccgcc cttcagggat 300
ggggtgggtg agggccgggg tccttgccca gtcaggcctt tcccgccctt tcttccctga 360
gactcagact gtagggacct catggtcatc cagcccacgg gctctcagct gggctgcatg 420
ttagaatccc ttcggcagct tgaaaaatca 450

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<210> SEQ ID NO 255
<211> LENGTH: 499
<212> TYPE: DNA
<213> ORGANISM: Equus caballus

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<400> SEQUENCE: 255
ttcttttgaa ctctctccag tgacactttt agctgtaaaa tttgtgaac agattttctg 60
ctggccattc taaataacac aaaataacta tatatttgtt ttctttttta tactcactag 120
aataacttct ttatcgccctt ttatatatag gttgtattgc tgggaaaata caatgcacag 180
ggcttaggtt cagatcatga attaatgctg agatgtacca aaggacaaga atacgtcaaa 240
gtcgtcatgc aaaatgggcg aatgatggga gctgtcttaa ttggtgaaac cgatttagaa 300
gaaacatttg aaaacttgat tttaaaccag atgaatcttt cagcatatgg agaagatctg 360
ctggatccag atattgacat agaagattat ttgactaaa aaggtcattc caagaaccac 420
ataaagtccc aaataagaca aaaaagtcac acatcaataa agtaaatgat tgcactgatt 480
taatgatgac cacattgaa 499

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<210> SEQ ID NO 256
<211> LENGTH: 499
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
        polynucleotide

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ctggccattc taaataacac aaaataacta tatatttgtt ttctttttta tactcactag 120
aataacttct ttatcgccctt ttatatatag gttgtattgc tgggaaaata caatgcacag 180
ggcttaggtt cagatcatga attaatgctg agatgtacca aaggacaaga atacgtcaaa 240
gtcgtcatgc aaaatgggcg aatgatggga gctgtcttaa ttggtgaaac cgatttagaa 300

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gaaacatttg aaaacttgat tttaaaccag atgaatcttt cagcatatgg agaagatctg	360
ctgcatccag atattgacat agaagattat ttgactaaa aaggtcattc caagaaccac	420
ataaagttcc aaataagaca aaaaagtcac acatcaataa agtaaatgat tgcactgatt	480
taatgatgac cacattgaa	499

<210> SEQ ID NO 257

<211> LENGTH: 704

<212> TYPE: DNA

<213> ORGANISM: Equus caballus

<400> SEQUENCE: 257

agacctcggg cccatggtac gatgctcctt gttcccagac tctcctgtgc tggaggccag	60
agaaacaaa ctcacactgg ctgccaggg gccgaaggac catggaagtg gtctttgagg	120
ctggggagca tatgcggtca ggaacagagg gcccctgtaa gcccacctg tgtgcccag	180
aaggacgttt ccattgtgga acagaagcaa gccctaccga gaggagcttc ttctgggcag	240
aagttgtgct ctttagatat ttacgaaagc gcttgtaag ccaaaggctg tgggggcggg	300
cacgtaccta agtctagcgc acctcacgac agagtaccga ccttcctctg cctatccatc	360
ccttttctg tagtgtctcg tgcaatattc cagagttcct gctcttcttt gcagatgaaa	420
atcttacacc aattttcttc cccacgccag ggtctgcctg ggagtgatgg tgtctcggga	480
agtctggag aaccaggagg ggacgtgagt gtctcttaca gctttggagg gaggtggcgg	540
ggcctctgtg ctggtattta gcagggtgcc tccagatctt ccagacactg cctggggcgt	600
tgttaaccag caactctttg tttaaagaat ttaaagagat ttctgaaacc tttgtcacc	660
aaatcttgat gcttatggaa aacattcgtc cttattgttt gatc	704

<210> SEQ ID NO 258

<211> LENGTH: 704

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 258

agacctcggg cccatggtac gatgctcctt gttcccagac tctcctgtgc tggaggccag	60
agaaacaaa ctcacactgg ctgccaggg gccgaaggac catggaagtg gtctttgagg	120
ctggggagca tatgcggtca ggaacagagg gcccctgtaa gcccacctg tgtgcccag	180
aaggacgttt ccattgtgga acagaagcaa gccctaccga gaggagcttc ttctgggcag	240
aagttgtgct ctttagatat ttacgaaagc gcttgtaag ccaaaggctg tgggggcggg	300
cacgtaccta agtctagcgc acctcacgac agagtaccga ccttcctctg cctatccatc	360
ccttttctg tagtgtctcg tgcaatattc cagagttcct gctcttcttt gcagatgaaa	420
atcttacacc aattttcttc cccacgccag ggtctgcctg cgagtgatgg tgtctcggga	480
agtctggag aaccaggagg ggacgtgagt gtctcttaca gctttggagg gaggtggcgg	540
ggcctctgtg ctggtattta gcagggtgcc tccagatctt ccagacactg cctggggcgt	600
tgttaaccag caactctttg tttaaagaat ttaaagagat ttctgaaacc tttgtcacc	660
aaatcttgat gcttatggaa aacattcgtc cttattgttt gatc	704

<210> SEQ ID NO 259

<211> LENGTH: 11

<212> TYPE: DNA

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<213> ORGANISM: Equus caballus

<400> SEQUENCE: 259

ccaagcggac t 11

<210> SEQ ID NO 260

<211> LENGTH: 11

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 260

ccaagtggac t 11

<210> SEQ ID NO 261

<211> LENGTH: 11

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 261

ccaagtggac t 11

<210> SEQ ID NO 262

<211> LENGTH: 11

<212> TYPE: DNA

<213> ORGANISM: Equus caballus

<400> SEQUENCE: 262

accaggaatg t 11

<210> SEQ ID NO 263

<211> LENGTH: 11

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 263

accaggaatg t 11

<210> SEQ ID NO 264

<211> LENGTH: 11

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 264

accagtaatg t 11

What is claimed is:

1. A method of mating a horse comprising:

(a) detecting in a biological sample obtained from said horse, said sample comprising nucleic acids that include the coding regions for dysferlin (DYSF), polynucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) and collagen type VI alpha 3 chain (COL6A3):

(1) a thymine (T) or a cytosine (C) at nucleotide 1001 of SEQ ID NO: 1, wherein detection of the thymine indicates the presence of variant P5;

(2) an adenine (A) or cytosine (C) at nucleotide 1001 of SEQ ID NO:56, wherein detection of the adenine indicates the presence of variant P6;

(3) a cytosine (C) or a guanine (G) at nucleotide 1001 of SEQ ID NO:103, wherein detection of the cytosine indicates the presence of variant P8; and

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(4) a cytosine (C) or a guanine (G) at nucleotide 1001 of SEQ ID NO: 163, wherein detection of the cytosine indicates the presence of variant K1;
 wherein in each case the presence of the nucleotide can be inferred from detecting the nucleotide present at the complement thereof;
 wherein the presence of any one or more of said variants P5, P6, P8 and K1 in said nucleic acid is indicative of inherited equine myopathy in said horse;
 wherein said horse is not homozygous for the presence of P5, P6, P8 or K1;
 wherein said horse is heterozygous for at least one of P5, P6, P8 or K1; and
 (b) mating said horse with a mare or sire.

2. The method according to claim 1 wherein said mare or sire is free of P5, P6, P8 and K1.

3. The method according to claim 1 wherein said mare or sire is heterozygous for variant P5, P6, P8 or K1.

4. The method according to claim 1, further comprising: contacting the nucleic acids of the sample with at least one oligonucleotide probe to form a hybridized nucleic acid; and amplifying the hybridized nucleic acid.

5. The method according to claim 4, wherein:

Exon B of the equine dysferlin coding region (DYSF), as defined in SEQ ID NO: 5, or a portion thereof is amplified;

Exon I of the equine dysferlin coding region (DYSF), as defined in SEQ ID NO: 59, or a portion thereof is amplified;

Exon 12 of the equine pyridine nucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1), as defined in SEQ ID NO: 106, or a portion thereof is amplified; or

Exon 26 of the equine collagen type VI alpha 3 chain coding region (COL6A3), as defined in SEQ ID NO: 166, or a portion thereof is amplified.

6. The method according to claim 4, wherein the hybridized nucleic acid is amplified using polymerase chain reaction, strand displacement amplification, ligase chain reaction, or nucleic acid sequence-based amplification.

7. The method according to claim 4, wherein at least one oligonucleotide probe is immobilized on a solid surface or a semisolid surface.

8. The method according to claim 1, wherein the nucleic acid is genomic DNA.

9. The method according to claim 4, wherein the nucleic acid is genomic DNA.

10. The method according to claim 5, wherein the nucleic acid is genomic DNA.

11. The method according to claim 6, wherein the nucleic acid is genomic DNA.

12. The method according to claim 7, wherein the nucleic acid is genomic DNA.

13. The method according to claim 1 wherein the nucleic acid is RNA.

14. The method according to claim 4 wherein the nucleic acid is RNA.

15. The method according to claim 5 wherein the nucleic acid is RNA.

16. The method according to claim 6 wherein the nucleic acid is RNA.

17. The method according to claim 7 wherein the nucleic acid is RNA.

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18. The method according to claim 13, wherein the method comprises converting the RNA into DNA by reverse transcriptase.

19. The method according to claim 14, wherein the method comprises converting the RNA into DNA by reverse transcriptase.

20. The method according to claim 15, wherein the method comprises converting the RNA into DNA by reverse transcriptase.

21. The method according to claim 16, wherein the method comprises converting the RNA into DNA by reverse transcriptase.

22. The method according to claim 17, wherein the method comprises converting the RNA into DNA by reverse transcriptase.

23. A method for detecting a biomarker in a horse, the method comprising:

(a) obtaining a biological sample from said horse, the biological sample comprising nucleic acid that includes the coding regions for dysferlin (DYSF) corresponding to SEQ ID NOs: 2 and 57, polynucleotide-disulfide oxidoreductase domain-containing protein 1 (PYROXD1) corresponding to SEQ ID NO: 104 and collagen type VI alpha 3 chain (COL6A3) corresponding to SEQ ID NO: 164; and detecting:

(1) a thymine (T) or a cytosine (C) at nucleotide 264 of SEQ ID NO:2, wherein detection of the thymine indicates the presence of variant P5;

(2) an adenine (A) or cytosine (C) at nucleotide 448 of SEQ ID NO:57, wherein detection of the adenine indicates the presence of variant P6;

(3) a cytosine (C) or a guanine (G) at nucleotide 1311 of SEQ ID NO:104, wherein detection of the cytosine indicates the presence of variant P8; and

(4) a cytosine (C) or a guanine (G) at nucleotide 437 of SEQ ID NO: 164, wherein detection of the cytosine indicates the presence of variant K1

wherein in each case the presence of the nucleotide can be inferred from detecting the nucleotide present at the complement thereof;

wherein the presence of any one or more of said variants P5, P6, P8 and K1 in said nucleic acid is indicative of inherited equine myopathy in said horse;

wherein said horse is not homozygous for the presence of P5, P6, P8 or K1;

wherein said horse is heterozygous for at least one of P5, P6, P8 or K1; and

(b) mating said horse with a mare or sire.

24. The method according to claim 23 wherein said mare or sire is free from variants P5, P6, P8 and K1.

25. The method according to claim 24 wherein said mare or sire is heterozygous for variant P5, P6, P8 or K1.

26. The method of claim 23, further comprising: contacting the nucleic acid with at least one oligonucleotide probe to form a hybridized nucleic acid; and amplifying the hybridized nucleic acid.

27. The method of claim 26, wherein the hybridized nucleic acid is amplified using polymerase chain reaction, strand displacement amplification, ligase chain reaction, or nucleic acid sequence-based amplification.

28. The method of claim 26, wherein at least one oligonucleotide probe is immobilized on a solid surface or semisolid surface.

* * * * *